

Doctoral Thesis

Signal Processing in Phase-Domain All-Digital Phase-Locked Loops

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Kurzfassung

Die Implementierung von drahtlosen Kommunikationssystemen auf einem einzigen Chip mit einer einheitlichen Technologie benötigt digitale Realisierungen von ursprünglich analogen Komponenten, wie beispielsweise die Phasenregelschleife (Phase-Locked Loop, PLL). Sogenannte ADPLLs (All-Digital PLLs) werten die Zeitpunkte der Nulldurchgänge von Signalen anstatt deren Amplitudenwerte aus. Dadurch kann der Frequenz-Synthesizer gänzlich mit digitalen CMOS Schaltkreisen realisiert werden. Diese Dissertation analysiert ADPLLs und hebt die Signalverarbeitungsaspekte auf der Systemebene hervor. Ein Modell im z -Bereich und ein Mixed-Signal Modell werden für die Entwicklung von Signalverarbeitungsalgorithmen, für grundlegende Simulationen und für die Verhaltensanalyse entwickelt. Der Einfluss von Störquellen auf das Spektrum des Phasenrauschens wird analysiert und mit den Ergebnissen ereignisgesteuerter Simulationen verglichen. Während Oszillatorrauschen, Referenzrauschen und Frequenzquantisierung einschließlich Fehlerdiffusion und Rauschformung den Ausgangsrauschpegel erhöhen, erzeugen Phasenquantisierung und unerwünschte Kopplungseffekte innerhalb des Chips parasitäre Spektrallinien im Ausgangsspektrum. Zusätzlich wird das Verhalten einer bekannten ADPLL Architektur, welche einen periodisch ungleichförmigen Systemtakt verwendet um den Oszillortakt mit dem Referenztakt zu synchronisieren, ausgiebig studiert und mathematisch beschrieben. Basierend auf der Analyse der Spektrallinien durch Kopplungseffekte und der Analyse des zeitvarianten Verhaltens der ADPLL Architektur wird eine alternative ADPLL Architektur, welche den gleichförmigen Referenztakt direkt als Systemtakt nützt, vorgeschlagen. Die Vorteile liegen in einer reduzierten Schaltungskomplexität und in einer starken Verminderung der parasitären Spektrallinien im Ausgangsspektrum. Zuletzt wird eine ADPLL Erweiterung vorgeschlagen, welche Frequenzsprünge innerhalb nur einer Referenzperiode mit einer Auflösung besser als 20 ppm der Trägerfrequenz ermöglicht. Die vorgeschlagene Methode eignet sich für MB-OFDM-UWB (Multiband Orthogonal-Frequency-Division-Multiplexing Ultra-Wideband) Kommunikationssysteme, in denen die Zeit zwischen der Wiederholung eines Frequenzbandes sehr kurz und die Anforderung an das Phasenrauschen gering ist.

Abstract

The implementation of wireless transceivers on a single chip in a single technology requires digital realizations of traditional analog building blocks such as phase-locked loops (PLLs). All-digital PLLs (ADPLLs) utilize the zero crossings of signals instead of their amplitudes to realize the frequency synthesizer entirely in digital CMOS technology. This thesis analyzes ADPLLs and highlights the system-level signal processing aspects. A z -domain model and a mixed-signal model are used to develop signal processing algorithms, to perform high-level simulations, and to evaluate the performance of ADPLLs. The impact of imperfections on the output phase noise spectrum are analytically described and compared to event-driven simulation outcomes. Oscillator noise, frequency quantization noise with $\Sigma\Delta$ noise shaping, and reference clock jitter raise the output phase noise level, whereas phase quantization and injection pulling manifest themselves as spurs in the output phase noise spectrum. Furthermore, the behavior of a well-known phase-domain ADPLL architecture that employs a periodically time-varying system clock to synchronize the oscillator output clock and the reference clock is extensively investigated and mathematically described. Based on the analysis of injection pulling spurs and the analysis of the time-varying ADPLL operation, an alternative ADPLL architecture, that directly utilizes the uniform reference clock as system clock, is proposed. The advantages are a reduced complexity of the circuits and an improved performance concerning spurs in the phase noise spectrum. Finally, a fast frequency-hopping ADPLL is proposed that enables frequency hops within less than one reference cycle with a frequency resolution of less than 20 ppm of the synthesized frequency. The proposed ADPLL is suitable for Multiband Orthogonal-Frequency-Division-Multiplexing Ultra-Wideband (MB-OFDM-UWB) communication systems, where the time between the recurrence of a frequency band is short and the requirements on the phase noise are not demanding.

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List of Abbreviations

A/D	Analog-to-Digital
ADPLL	All-Digital Phase-Locked Loop
BJT	Bipolar Junction Transistor
C/D	Continuous-to-Discrete-Time
CMOS	Complementary Metal-Oxide-Semiconductor
CTFT	Continuous-Time Fourier Transform
D/A	Digital-to-Analog
D/C	Discrete-to-Continuous-Time
DDS	Direct Digital Synthesis
DCO	Digitally Controlled Oscillator
DL	Delay Line
DLL	Delay Locked Loop
DSP	Digital Signal Processing
ESD	Energy Spectral Density
FCC	Federal Communications Commission
FIR	Finite Impulse Response
FPC	Frequency-to-Phase Converter
FCW	Frequency Command Word
GaAs	Gallium Arsenide
GFSK	Gaussian Frequency Shift Keying
GSM	Global System for Mobile Communications
HBT	Heterojunction Bipolar Transistor
HDL	Hardware Description Language
IIR	Infinite Impulse Response
IT/S	Impulse-Train-to-Sequence
LF	Loop Filter
LFSR	Linear Feedback Shift Register
LO	Local Oscillator
LSB	Least Significant Bit
MAC	Metastability Avoidance Circuit
MASH	Multi-Stage Noise Shaping (Structure)
MB-OFDM-UWB	Multiband Orthogonal-Frequency-Division-Multiplexing Ultra-Wideband
MESFET	Metal Semiconductor Field Effect Transistor
NCO	Numerically Controlled Oscillator
PD	Phase Detector

List of Abbreviations

PFD	Phase-Frequency Detector
PLL	Phase-Locked Loop
PSD	Power Spectral Density
PVT	Process, Voltage, and Temperature
RF	Radio-Frequency
SDR	Software-Defined Radio
SiGe	Silicon-Germanium
SOI	Silicon on Insulator
SSMF	Single-Stage Multiple Feedback ($\Sigma\Delta$ modulator)
SQNR	Signal-to-Quantization-Noise Ratio
S/IT	Sequence-to-Impulse-Train
TDC	Time-to-Digital Converter
UWB	Ultra Wideband
VCO	Voltage Controlled Oscillator
VDL	Venier Delay Line
VHDL	(Very-High-Speed Integrated Circuits) HDL
VHDL-AMS	VHDL with analog and mixed-signal extensions
WSS	Wide-Sense Stationary
ZOH	Zero-Order Hold

Signals and Variables

The list of signals and variables is not complete. Any additional signals and variables will be explained within the context and therefore are of minor importance.

Variable Frequency

f_v	Variable or DCO output frequency
f_0	DCO free running frequency
f_v^{ss}	Desired, steady-state DCO output frequency
T_v	Reference period $1/f_v$
CKV	Clock signal of the DCO with frequency f_v
i	Time index of the reference signal, i.e., iT_v

Reference Frequency

f_r	Reference frequency
T_r	Reference period $1/f_r$
REF	Clock signal of the reference with frequency f_r
n	Time index of the reference clock, i.e., $n T_r$

Retimed Reference Frequency

f_{CKR}	Retimed reference frequency
T_{CKR}	Retimed reference period
CKR	Clock signal of the retimed reference with (average) frequency f_r
m	Time index of the retimed reference clock, i.e., $m T_{CKR}[m]$

Uniform Phase Signals

$N_r[n]$	Frequency command word, ratio $f_v^{ss}[n]/f_r$ or desired number of CKV periods in one REF period
$N_v[n]$	Actual frequency ratio $f_v[n]/f_r$ or actual number of CKV periods in one REF period
$\varphi_v[n]$	Variable phase
$\varphi_r[n]$	Reference phase

Non-Uniform Phase Signals

$N_{CKR}[m]$	Desired ratio $f_v^{ss}[m]/f_{CKR}[m]$ or desired number of CKV periods in one CKR period
$N_{CKV}[m]$	Actual frequency ratio $f_v[m]/f_{CKR}[m]$ or actual number of CKV periods (T_v) in one REF period (T_r)
$\varphi_{CKV}[m]$	Retimed variable phase

Signals and Variables

$\varphi_{CKR}[m]$	Retimed reference phase
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Loop Signals

φ_e	Phase error
$\overline{N_v}$	Loop filter output
$\overline{f_v}$	Normalization output and normalized DCO input
d	Digital tuning word
α	Proportional factor of the loop filter
ρ	Integral term of the loop filter
K	DCO gain
$\hat{K}$	DCO gain estimate

z -Domain Noise Signals

η_{DCO}	DCO or oscillator noise
η_{TDC}	TDC or phase-detector noise
η_{REF}	Reference noise

TDC

ε	Retiming error
ξ	Alternative timing error, $1 - \varepsilon$
Δt_{res}	TDC time resolution

Symbols and Operators

General Notation

$\lfloor \cdot \rfloor$	Floor operation
$\lceil \cdot \rceil$	Ceiling operation
$\delta(\cdot)$	Dirac delta distribution
$\delta[\cdot]$	Kronecker delta
$\mathcal{E}\{\cdot\}$	Expectation operator

Autocorrelation and Power Spectral Density

$r_x(\tau), r_x[l]$	Autocorrelation function (continuous-time and discrete-time)
$S_x(\omega), S_x(e^{j\omega T})$	Double-sided PSD (continuous-time and discrete-time)
$L_x(\omega), L_x(e^{j\omega T})$	Single-sided PSD (continuous-time and discrete-time)
$\mathcal{L}_x\{\Delta\omega\}$	Normalized single-sided PSD

Transfer Functions

H_{cl}	Closed loop frequency transfer function of the ADPLL
H_{ol}	Open loop frequency transfer function of the ADPLL
Φ_{cl}	Closed loop phase transfer function of the ADPLL
Φ_{ol}	Open loop phase transfer function of the ADPLL
$H_{\eta_{DCO}}$	Frequency DCO noise transfer function
$H_{\eta_{TDC}}$	Frequency DCO noise transfer function
$\Phi_{\eta_{DCO}}$	Phase DCO noise transfer function
$\Phi_{\eta_{TDC}}$	Phase TDC noise transfer function

Introduction

Phase-locked loops (PLLs) are an essential part of wireless communication systems, microprocessors, and many other applications. Often the PLLs are described colorfully as the “*heart of the system*” as it produces the beat for the digital logic on the chip [1]. The usage of a PLL is manifold, e.g., to generate a stable frequency, to recover a signal from a noisy communication channel, or to distribute a clock on a microprocessor. In telecommunications, the PLL is used for modulation/demodulation, and synchronization at different levels, i.e., carrier synchronization (frequency synthesis), clock synchronization (clock recovery), and symbol synchronization (software PLLs, delay locked loops (DLLs)) [2].

A PLL is a control circuit with negative feedback loop that generates an output signal with a certain relation to a reference signal. It consists of three basic functional blocks, i.e., a phase detector (PD), a loop filter (LF), and an oscillator, which might be a voltage controlled oscillator (VCO) or a digitally controlled oscillator (DCO). Basically PLLs are classified into four types [3]: analog PLLs, digital PLLs (DPLLs), all-digital PLLs (ADPLLs), and software PLLs. Although PLLs were originally purely analog devices, functional parts migrated step-by-step into the digital domain. Digital PLLs are hybrid systems, where some parts such as the phase detector and/or loop filter are implemented with digital circuits and some parts such as the oscillator are analog circuits. ADPLLs are classified by the use of only digital circuits without analog tuning¹.

In this thesis we focus on the signal processing aspects in ADPLLs for frequency synthesis in wireless communication systems.

1.1 Technology, Market, and Design Trends

The recent and ongoing explosive growth of radio-frequency (RF) wireless communication systems such as Global System for Mobile Communications (GSM) and Bluetooth requires low-cost/low-area, low-voltage, and low-power transceivers that can be fabricated in high volumes and achieve a high yield [6]. Furthermore, the market demands for increasingly complex handheld devices that offer multiple wireless standards and a constantly increasing amount of software functionality. The flexibility of the radio and the possibility to implement the entire system, i.e., the RF front-end and the baseband processor on a single chip using one technology to further decrease costs, favors

¹The categorization of PLLs in general and ADPLLs in particular is often misleading. ADPLLs for wireless communication systems are mixed-signal systems that contain digital and analog signals, see Sec. 2.2, and should not be confused with entirely digital systems such as direct digital synthesis (DDS) and PLLs that employ numerically controlled oscillators (NCOs) such as [4, 5].

1.2 Analog Phase-Locked Loops versus All-Digital Phase-Locked Loops

a digital intensive approach. The digital baseband and application processor is integrated in the most recent CMOS technology, where analog design is becoming increasingly difficult as the voltage headroom is shrinking (the supply voltage is decreasing and the threshold voltage is increasing as CMOS scales down), digitally intensive solutions of traditionally analog components are needed [7–9]. Moreover, experiments with deep-submicron CMOS logic revealed a novel design paradigm: “*In a highly scaled CMOS technology, time-domain resolution of a digital signal edge transition is superior to voltage resolution of analog signals*” [10]. Consequently, these technology trends indicate that the processing of discrete-amplitude continuous-time signals will soon outperform the processing of analog signals in CMOS. Thus, the market requirements and the technology trend concur and suggest a digital CMOS approach for wireless transceiver design. Furthermore, a digital intensive approach provides several additional advantages: improved reusability and portability, better testability, lower parameter variability, optimized power and area dissipation, higher degree of integration, and the use of automated computer-aided design tools [6].

Shifting analog operations into the digital domain, however, is no straight-forward task. While the functionality of the system remains the same, the emerging problems, the required methodologies, the simulation environments, and the analysis tools are different. On the one hand, in contrast to analog systems the complexity of these digital intensive mixed-signal systems has increased and requires a holistic design approach [11] including digital predistortion or postcorrection of analog imperfections to meet the specifications [12]. On the other hand, the use of digital signal processing enables new possibilities, which are beyond the scope and boundaries of analog systems.

1.2 Analog Phase-Locked Loops versus All-Digital Phase-Locked Loops

The background discussed in Sec. 1.1 has triggered the development and research on ADPLLs and reflects the novel design paradigm of system-on-chip design for mixed-signal RF radios [11, 12]. Its digital nature allows for a great degree of flexibility and its digital circuits resolve time quantities of clock transitions instead of voltage amplitudes. Moreover, the local oscillator used in the receiver and transmitter likewise, is one of the most critical components in mobile RF systems. It has to fulfill, on the one hand, the stringent constraints on implementation issues (mainly low-cost and low-power) and, on the other hand, the strict specifications of the wireless standard (phase noise, spurious emission, and locking time) [13, 14].

Besides the differences in the design methodology and the resulting advantages and disadvantages, as discussed above, traditional PLLs and ADPLLs have some conceptual differences. While traditional PLLs are inherently nonlinear systems [15], phase-domain ADPLLs aim to implement the linearized frequency domain model of PLLs and are, therefore, ideally linear systems. As a result, conventional PLLs, such as a charge-pump-based PLL, do not truly operate in the phase domain [16]. The phase modeling is a small-signal approximation that holds only in the close vicinity of the operational point, i.e., phase lock. A direct result of this shortcoming is false frequency locking, which requires additional measures to be avoided [16]. By contrast, the linear operation of ADPLLs enables theoretically the usage of a wide-bandwidth loop-filter and, hence, fast frequency locking.

Another key difference is in the representation of the reference and feedback signals. Traditional PLLs are edge based and convert the relation between the clock edges into an analog phase error signal, i.e., by “*converting the timed logic levels into analog quantities*” [17]. Phase-domain ADPLLs,

however, use digital representations of the reference signal and feedback signal and compare these digital signals in order to obtain a corresponding digital phase error signal. As a consequence, phase-domain ADPLLs do not control the relation between the edges of the reference clock and the edges of the synthesized clock, which is elaborated mathematically in Sec. 3.2.

Digital solutions have advantages due to the usage of digital circuits, that include their intrinsic immunity to noise, their area consumption and power dissipation. As a drawback ADPLLs suffer additionally from quantization effects and the crosstalk in the mixed-signal system. In particular, coupling effects between the analog oscillator and the digital circuits cause severe spurs in the output phase noise spectrum [18] and coupling effects of multiple oscillators on the chip [19]. Thus, the integration of a high-performance RF transceiver together with a large amount of digital logic gates on a single chip is a key challenge [13, 20].

1.3 Overview of All-Digital Phase-Locked Loops

In this section we provide a brief overview over ADPLL implementations from the literature. Early ADPLLs have typically employed a ring oscillator as DCO and have been used for less demanding applications, e.g., in clock recovery systems [21, 22] and integer- N frequency synthesis [23, 24]. A key advantage of integer- N ADPLL solutions was the wide multiplication range [25, 26]. An attractive architecture for integer- N frequency synthesis are bang-bang PLLs, which use a binary phase error signal [27–34]. For all-digital fractional- N frequency synthesizers designers often employ a similar architecture as for traditional fractional- N frequency synthesizers, as illustrated in Fig. 1.1 [35, 36]. Such architectures can be found in several variations for different applications [37–42], which differ in performance and complexity. In general, however, the architecture of fractional- N PLLs is shifted into the digital domain, where a time-to-digital converter (TDC) replaces the phase-frequency detector (PFD), a digital loop filter replaces the analog loop filter, and a DCO replaces the VCO. From a circuit-level point of view ADPLLs are digital systems, because all components can be implemented with digital CMOS without analog tuning, see Sec. 3.1. Nevertheless, from a signal processing point of view, ADPLLs are mixed-signal systems, where the DCO and the TDC are the interfaces between the analog and digital domain. The key advantage is that the analog signals are clock signals (discrete-amplitude and continuous-time) with its information encoded in the transitions, which can be effectively resolved by CMOS logic. Other architectures aim to reduce the complexity by avoiding TDCs on account for increased noise level and spurs [43, 44].

In this thesis we consider phase-domain ADPLLs that aim to digitally implement the phase-domain characteristics of PLLs instead of replacing analog components by a digital equivalent. The phase-domain operation was first proposed in [45, 46], significantly refined and analyzed for an ADPLL in [10, 16, 18, 47–61], and summarized in [6]. The details and advantages of the architecture are extensively discussed in Chapter 2.

Independent of the architecture, ADPLLs offer several advantages, which are difficult or simply not feasible in analog solutions. The digital nature of the loop filter allows a variable bandwidth that can be iteratively changed [55, 57, 62], dynamically adapted [63], or allows for more complex control mechanisms [64]. The complexity of the loop filters may be reduced by exploiting multi-rate filterbank theory [65, 66]. The digital tuning word enables search algorithm to increase the initial frequency acquisition [67–69] and directly perform two-point modulation [51, 70].

1.4 Scope of the Thesis

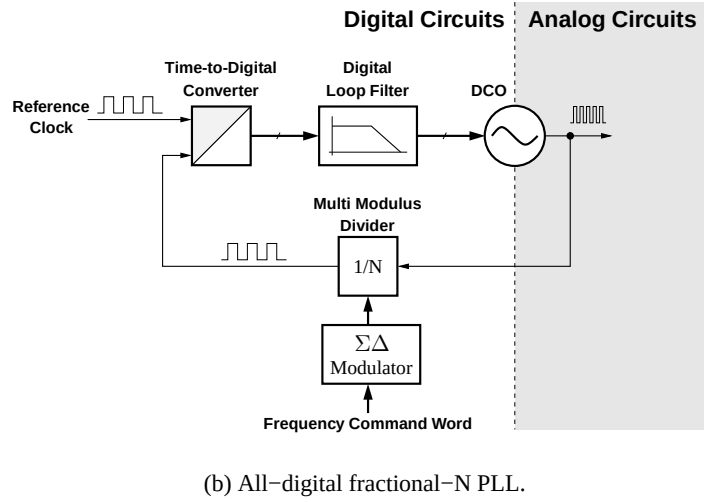
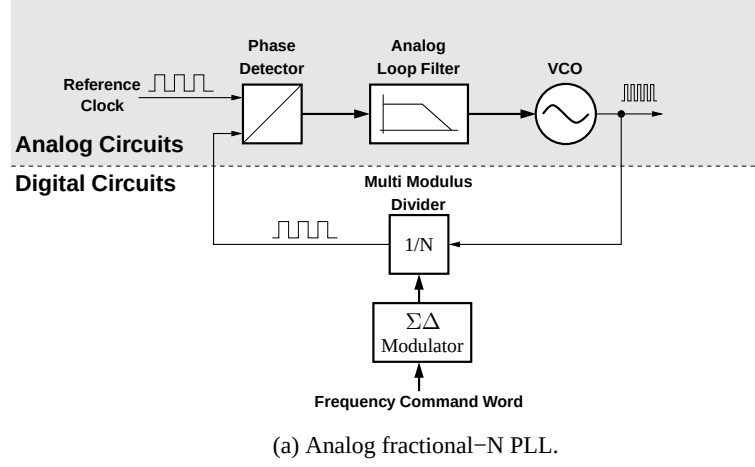


Figure 1.1 Comparison of an analog fractional- N frequency synthesizer architecture and all-digital fractional- N frequency synthesizer architecture [35, 36].

1.4 Scope of the Thesis

All-digital PLLs have attracted a lot of attention. However, designers have good experience with analog PLLs and their properties, but little knowledge of ADPLLs. The principle functionality of analog PLLs and ADPLLs are equivalent. However, their imperfections, problems, design procedure, and architecture are fundamentally different. The phase-domain ADPLL structure for instance often appears deterrent and too complex for analog designers, since it does not reflect the traditional PLL components. Therefore designers prefer architectures that are similar to their analog counterparts, cf. Fig. 1.1. Novel issues, such as quantization and synchronization of digital signals, have to be considered. On account of many practical advantages of ADPLLs, their complexity is in general higher compared to analog PLLs. That is, designers combine to have good analog circuit design skills with a deep system-level understanding.

While the number of implementations reported in the literature are rapidly increasing, the number of analytical papers with system-level descriptions or design procedures is scarce. However, to exploit the full possibilities of ADPLLs beyond the scope of traditional PLLs, e.g., by the aid of digital signal processing algorithms, an exact mathematical description is mandatory. Therefore one focus of this thesis is to provide a signal processing point of view on these mixed-signal system. On the one hand, we require a rather coarse abstraction of the system for algorithm design and, on the other hand, an exact model for performance evaluation. Thus, another focus of the thesis is modelling and evaluation of imperfections in ADPLLs in order to find structural improvements.

Moreover, the band switching requirements in frequency hopping wireless communication standards are becoming increasingly tight. In particular, in Multiband Orthogonal-Frequency-Division-Multiplexing Ultra-Wideband (MB-OFDM-UWB) the defined margin for the receiver and transmitter to switch between two bands is 9.5 ns, which is less than one reference cycle. Traditional solutions utilize more than one PLL to achieve this stringent constraint, which is not optimal in terms of area and power consumption. The digital nature of ADPLLs, however, allows to instantly change the digital tuning word. Thus, a fast hopping between two frequencies can be achieved by changing the digital tuning word such that the switching time is limited only by the DCO transient. The aim is to achieve a closed loop operation of a single ADPLL that achieves frequency hops within less than one reference cycle.

1.5 Outline of the Thesis and Main Contributions

In the following, we provide a chapter by chapter overview of the thesis and emphasize the main contributions of the author. The publications and patent applications from this work are listed.

Introduction

In Chapter 1 we give an introduction to PLLs and discuss technology and design trends in order to motivate the advantages of ADPLLs. The author has contributed to two papers that focus on the recent design paradigm in highly integrated systems

- F. Dielacher, C. Vogel, P. Singerl, S. Mendel, A. Wiesbauer, “A Holistic Design Approach for Systems on Chip,” in *Proceedings of the 22nd IEEE International SOC Conference (SOCC 2009)*, Belfast (Northern Ireland, UK), 9-11 September 2009, in press (invited paper) [11].
- C. Vogel, S. Mendel, P. Singerl, F. Dielacher, “Digital Signal Processing for Data Converters in Mixed-Signal Systems,” *e&i Elektrotechnik und Informationstechnik*, 2009, in press [12].

Model of a Phase-Domain ADPLL

In Chapter 2 we consider modeling and system-level properties of ADPLLs. We introduce a general uniform z -domain model that reflects the phase-domain operation of the system and the actual implementation, as the phase-domain ADPLLs implements the linearized phase-domain model of traditional PLLs. The digital signal representation using modulo arithmetics allows to overcome the unstable frequency-to-phase conversion without affecting the linear operation of the overall system. With the aid of the z -domain model important control system characteristics are derived. The existence of frequency signals, phase signals, and normalized frequency signals, which correspond to the slope of

1.5 Outline of the Thesis and Main Contributions

the phase signals, allows for evaluating the frequency and phase transfer functions. The analysis shows that phase-domain ADPLLs achieve frequency-lock and phase-lock simultaneously, which is a direct result from the phase-domain operation. Moreover, the noise transfer functions of the ADPLL, i.e., the TDC noise transfer function and the DCO noise transfer function are evaluated. The TDC noise transfer function is of low-pass nature, whereas the DCO noise transfer function appears as high-pass filter. The intrinsic trade-offs in PLLs among fast locking, that requires a wide loop bandwidth, good DCO noise suppression, that requires a wide loop bandwidth too, and TDC noise suppression, that requires a narrow loop bandwidth, are discussed. The second part of the chapter highlights the mixed-signal nature of the ADPLL by refining the z -domain model to a mixed-signal model. From a signal processing point of view, the DCO performs a digital-to-analog conversion and the TDC performs an analog-to-digital conversion.

- S. Mendel and C. Vogel, “A z -domain model and analysis of phase-domain all-digital phase-locked loops,” in *Proceedings of the IEEE Norchip Conference 2007*, Aalborg (Denmark), 19-20 November 2007 [71].

Architecture of a Phase-Domain ADPLL

In Chapter 3 we address ADPLL architectures and their analog building blocks. In the first part of this chapter we give an overview of DCO and TDC implementations from the literature and review the design challenges and resulting imperfections using a popular TDC architecture and a popular DCO architecture as examples. To overcome the coarse DCO frequency quantization, the digital tuning is dithered with a high-rate $\Sigma\Delta$ modulator output signal, which finally allows ADPLLs to compete with their analog counterparts. In contrast to 2nd order multi-stage noise shaping modulators, which produce significant spurs for static input sequences, we have proposed a single-stage multiple-feedback modulator, which produces no spurs on account of additional hardware complexity.

In the second part of this chapter we review the retimed reference architecture which uses a non-uniform reference clock to synchronize the slow reference clock and the fast DCO output clock. In contrast to the description given in the literature, our analysis focuses on the resulting periodically time-varying nature of the clock, phase, and normalized frequency signals in lock. The uniform z -domain model and the uniform mixed-signal models are refined to account for the non-uniform system clock for an exact mathematical description of the system and its time-varying operation. Moreover, the resulting practical problems, such as metastability and synchronization, and their solutions are also discussed.

In the third part, which is the main contribution of this chapter, we propose an alternative architecture that abandons the usage of the reference retiming mechanism and operates the entire ADPLL with the uniform reference clock. The synchronous reference architecture solves the synchronization between the reference clock and the DCO output clock with a dedicated metastability avoidance circuit. Although the uniform operation basically matches the z -domain model, a few minor differences in the operation are discussed.

- S. Mendel, C. Vogel, and N. Da Dalt, “Signal and timing analysis of a phase-domain all-digital phase-locked loop with reference retiming mechanism,” in *Proceedings of the 16th International Conference on Mixed Design of Integrated Circuits and Systems (MIXDES 2009)*, Lodz (Poland), 25-27 June 2009 (outstanding paper award) [72].

- S. Mendel, C. Vogel, and N. Da Dalt, “A phase-domain all-digital phase-locked loop architecture without reference clock retiming,” *IEEE Transaction on Circuits and Systems II - Express Briefs*, in press [73].
- S. Mendel and C. Vogel, “Frequency-to-phase converter with uniform sampling for all-digital phase-locked loops,” pending patent application, United States, 24 Sep. 2008.

Performance Analysis of a Phase-Domain ADPLL

In Chapter 4 we analyze the performance of phase-domain ADPLLs. We review the performance metrics from the literature and discuss the relation between phase and frequency spectra, single-sided and double-sided spectra, and baseband and passband spectra. We use this framework to compute the phase noise spectrum from discrete-time observations, i.e., rising and/or falling edges of clock signals. Furthermore, we can model noise on time-discrete observations and therefore perform simulations that compute only the time instances of the rising and/or falling edges of the desired clock signals. This concept is illustrated on the example of oscillator noise in combination with a framework to generate any desired colored noise. In the main contribution of this chapter, we derive the theoretical phase noise spectrum of all imperfections in closed form and compare it to the simulation results. In particular, oscillator noise, DCO quantization noise with $\Sigma\Delta$ dithering, reference clock jitter, and “linearized” TDC quantization noise is considered. Besides noise, the ADPLL suffers from quantization effects, which manifests as spurs in the output phase noise spectrum. Therefore, we have analyzed DCO quantization spurs and $\Sigma\Delta$ modulator spurs. Moreover, the periodic deterministic input signal to the TDC, derived in Chapter 3, invalidates a linearized noise model, which is used in the literature. Consequently the TDC quantization error manifests as spurs in the phase noise spectrum. These spurs are discussed in analogy to the spurs due to an ADPLL without TDC with the aid of a signal processing model. Besides quantization effects, analog coupling effects on the die are another cause of spurious tones. As a main contribution we have analyzed and categorized the spurs due to injection pulling. The spurs from simulation results match with the theory in their frequency locations and amplitudes. To this end, we have shown that an ADPLL with synchronous reference architecture significantly reduces the number of spurs in contrast to an ADPLL with retimed reference architecture from Chapter 3.

- S. Mendel, C. Vogel, and N. Da Dalt, “A phase-domain all-digital phase-locked loop architecture without reference clock retiming,” *IEEE Transaction on Circuits and Systems II - Express Briefs*, in press [73].

Enhanced Frequency Acquisition in a Phase-Domain ADPLL

In Chapter 5 we introduce a time-interleaved ADPLL operation for fast frequency-hopping applications. In particular, MB-OFDM-UWB requires a band switching within 9.5 ns, i.e., less than one reference cycle, and an accuracy of 20 ppm of the carrier frequency. The proposed solution exploits the digital nature of ADPLLs and the fact that the time between the recurrence of a frequency band is very short. Therefore, a single ADPLL simulates three frequency synthesizers with one physical system. By storing and restoring the state of the ADPLL for each frequency band we achieve frequency-lock and by adjusting the digital phase signals we achieve phase-lock. The performance of the proposed fast frequency-hopping ADPLL operation is evaluated and compared to the simulation

1.5 Outline of the Thesis and Main Contributions

results. The proposed frequency synthesizer achieves a very fast switching time and a spurious level, which is equivalent to a single ADPLL, on account of larger in-band phase noise.

- S. Mendel and C. Vogel, “Improved lock-time in all-digital phase-locked loops due to binary search acquisition,” in *Proceedings of the 14th IEEE International Conference on Electronics, Circuits and Systems (ICECS 2008)*, St. Julians (Malta), 1-3 September 2008, pp. 181-185 [74].
- S. Mendel, “Phase-lock in all-digital phase-locked loops,” pending patent application, United States, Oct. 2009.

Additional Scientific Work

The author has contributed to further publications which are outside the scope of this thesis, but within the area of mixed-signal systems. In particular, the author has worked on the digital enhancement of analog circuits with the aid of digital signal processing methods or more specific on the digital correction of channel mismatches in time-interleaved analog-to-digital converters.

- C. Vogel and S. Mendel, “A Flexible and Scalable Structure to Compensate Frequency Response Mismatches in Time-Interleaved ADCs,” in *IEEE Transactions on Circuits and Systems I: Regular Papers*, in press [75].
- C. Vogel, S. Saleem, and S. Mendel, “Adaptive Blind Compensation of Gain and Timing Mismatches in M-Channel Time-Interleaved ADCs,” in *Proceedings of the 14th IEEE International Conference on Electronics, Circuits and Systems (ICECS 2008)*, St. Julians (Malta), 1-3 September 2008, pp. 49-52 [76].
- S. Mendel and C. Vogel, “On the Compensation of Magnitude Response Mismatches in M-channel Time-interleaved ADCs,” in *Proceedings of the 2007 IEEE International Symposium on Circuits and Systems (ISCAS 2007)*, New Orleans (USA), 27-30 May 2007, pp. 3375-3378 [71].
- S. Mendel and C. Vogel, “A Compensation Method for Magnitude Response Mismatches in Two-Channel Time-Interleaved Analog-to-Digital Converters,” in *Proceedings of the 13th IEEE International Conference on Electronics, Circuits and Systems (ICECS 2006)*, Nice (France), 10-13 December 2006, pp. 712-715 [77].

2

Models of Phase-Domain All-Digital Phase-Locked Loops

This chapter introduces two system-level models, which illustrate the principal behavior, system-level properties, and design challenges of phase-domain ADPLLs. First, a comprehensive z -domain model is introduced [71]. The z -domain model for digital and all-digital PLLs or the s -domain model for analog PLLs is the classical analysis tool for linear or linearized PLLs. Although ADPLLs suffer from nonlinear quantization effects, the linear analysis gives relevant insight into its control system properties. In particular the steady state performance, i.e., noise suppression capability, and the transient behavior, i.e., overshoot and locking time, can be evaluated. Furthermore, the z -domain model relates the frequency, phase, and phase-slope signals with the basic phase-domain operation. Therefore the introduced model is sufficiently accurate for high-level simulations. Second, a mixed-signal model, that emphasizes the mixed-signal feedback path of the ADPLL, is introduced. Within the ADPLL signal processing chain a digital signal is converted into an oscillating signal with certain frequency and the phase from that oscillating signal is converted back into a digital signal. Accordingly we have a digital-to-frequency conversion, which can be seen as signal reconstruction, and an analog-to-digital-phase conversion, which can be seen as sampling of an analog signal. In contrast to the z -domain model, the mixed-signal model reveals the analog design challenges of phase-domain ADPLLs by highlighting the mixed-signal nature and its obstacles. Unlike other PLL models, which are a coarse abstraction and only valid under rigid constraints, both presented phase-domain ADPLL models are very accurate and reflect the behavior and the inherent imperfections of the system very well. The models are employed in Chapter 4 to evaluate the impact of all imperfections on the output phase noise spectrum.

The z -domain model and the related system-level analysis in Sec. 2.1 of this chapter are largely based on the paper [71], which has been refined and re-formatted for the purpose of this thesis.

2.1 z -Domain Model

For several decades experience concerning the design and analysis of analog PLLs has been built up, but until now only little research has been done on ADPLLs. The relevant literature is shortly summarized: An analysis of the frequency synthesizer introduced in [47] is presented in [16]. Although a z -domain model will offer the proper and more accurate description of a digital system, the analysis in [16] focuses on a linear s -domain approximation. Moreover, the entire design procedure of ADPLLs

2.1 *z*-Domain Model

in [78] is based on the analogy to the extensively analyzed analog charge-pump PLL, e.g., [17]. Design, concepts, and applications of ADPLLs are discussed in [79], and a design procedure of higher-order PLLs, analog and digital, is presented in [80]. The analysis of bang-bang PLLs is done with respect to the locking behavior in [81] and with respect to its nonlinear characteristics in [29].

Typically, designers use one model to analyze the performance of PLLs and a different model to simulate them. This shows that the principal models which are used for analysis are a strong abstraction of the actual implementations, which are used for simulation. Although in some aspects the presented *z*-domain model simplifies the architecture, i.e., quantization effects and the non-uniform reference clock of a popular implementation [48, 52, 54] are neglected¹, it matches the fundamental implementation very well, which makes it suitable for evaluating typical effects of ADPLLs, such as lock time, loop bandwidth, and noise behavior. While some imperfections, like quantization in Sec. 4.4 can easily be included into the *z*-domain model, other imperfections, like DCO dithering in Sec. 4.3 or injection pulling in Sec. 4.6 require a more sophisticated simulation environment, because the *z*-domain model operates on the slow reference frequency and higher frequency disturbances cannot be directly considered. Therefore ADPLLs are most often simulated on the rate of the DCO output clock in event-driven programs in the time-domain, e.g., [49, 60, 82–85]. Alternatively also high-level programming languages, such as C or Matlab can be used [86], which is especially convenient for algorithm development. A simulation focused on straightforward noise analysis is given in [87] and a Simulink/Matlab model of fractional-*N* frequency synthesizer is presented in [88].

In the following we will investigate the system-level properties of phase-domain ADPLLs. In contrast to the literature, which focuses mainly on the *s*-domain models or circuit level structures to describe PLLs or even ADPLLs, we introduce a *z*-domain model of phase-domain ADPLLs for two reasons. First, in contrast to other *z*-domain models such as in [16] the presented model reflects the actual operation and tasks of each building block, which makes it suitable for the design of signal processing algorithm or alternative analog components. Second, the transfer functions with respect to the output frequency and with respect to the output phase can be directly derived from the model, which makes it suitable for performance analysis. And third, the model is simple (as it operates on the rate of the reference clock) and accurate, which makes it suitable for high-level simulation.

2.1.1 Functionality of an All-Digital Phase-Locked Loop

Traditional PLLs contain three essential building blocks as shown in Fig. 1.1(a): a phase-detector (PD), a loop filter (LF), and a voltage controlled oscillator (VCO). The feedback mechanism ensures that the output phase of the VCO output signal is in phase-lock with a periodic reference signal, which is the input to the PLL. Thus, the phase detector compares the phases between reference and feedback signal. Phase detectors are typically nonlinear elements, such as mixers, XOR gates, or phase-frequency detectors (PFD). Consequently the phase error produces undesired high-frequency components, that require to be attenuated by the analog loop filter. The resulting output voltage controls the output frequency of the VCO. In steady state, i.e., when phase-lock is achieved, the frequency of the oscillator output signal matches exactly the frequency of the reference input signal. In case the synthesized frequency is required to be larger than the reference frequency, a frequency divider is included in the feedback path (integer-*N* PLL). That is, besides the reference clock (also reference frequency) the PLL has a second input signal, which is the frequency command word N_r .

¹Quantization effects are considered in Chapter 4 and the *z*-domain model is extended to a non-uniform *z*-domain model in Chapter 3.

In case the relation between synthesized frequency and reference frequency is a fractional number, i.e., N_r is a fractional number, the divider value N is dithered such that its average value matches N_r (fractional- N PLL).

ADPLLs aim to replace the analog components by building blocks, which can be effectively implemented in digital CMOS technology without analog tuning. A possible solution, that bases on the relation to traditional PLLs, is shown in Fig. 1.1(b). Its basic operation matches the analog fractional- N PLL in Fig. 1.1(a) and the description above. In this thesis, however, we focus on phase-domain ADPLLs [6, 10, 16, 18, 47–61]. The architecture, which is shown in Fig. 3.9, aims to implement the phase-domain description of traditional PLLs. The key principle is to convert the oscillator output signal, which may be of sinusoidal or rectangular waveform, into a digital phase signal. The phase detector compares this digital feedback phase to a digitally generated reference phase by means of a subtraction. A digital loop filter further attenuates noise and distortions and a DCO generates the desired output signal. As the architecture in Fig. 3.9 has to consider several practical issues, which are discussed extensively in Sec. 3.2, its structure is rather complex. Therefore we introduce in the following a sophisticated z -domain model to derive the characteristics of phase-domain ADPLLs. Similar to the operation of traditional PLLs, which are intrinsically nonlinear devices, we describe the phase-domain ADPLL in the frequency domain. Thus, the signals are not described in their time-domain waveforms, but with respect to their frequency and phase information. Since phase-domain ADPLLs aim to implement exactly the functionality of the phase-domain description of traditional PLLs, the model is accurate and can be used besides performance analysis for algorithm design and high-level simulations.

2.1.2 Phase-Domain Operation

Figure 2.1 shows the z -domain model of a phase-domain ADPLL. The model includes frequency and phase signals and, thus, reflects both frequency and phase behavior of the PLL. The sampling frequency of the discrete-time system is the slow reference frequency f_r . The time index n corresponds to the n th reference cycles, i.e., nT_r with $T_r = 1/f_r$ being a reference period. The feedback mechanism of the ADPLL controls the variable frequency $f_v[n]$ generated by the DCO so that in steady state it is in frequency and phase lock with the desired frequency $f_v^{ss}[n] = 1/T_v^{ss}[n]$. Note that the variable frequency $f_v[n]$ is constant over one reference period T_r and is updated with each reference cycle n . A frequency-to-phase converter (FPC) converts the variable frequency signal $f_v[n]$ into a digital phase signal $\varphi_v[n]$.

Contrary to traditional PLLs for fractional- N frequency synthesis, phase-domain ADPLLs do not use a division by $N_r[n]$ in the feedback path as shown in Fig. 1.1, but a multiplication with $N_r[n]$ in the reference path. Thus, the desired frequency $f_v^{ss}[n]$ is given as a multiple $N_r[n]$ of the reference frequency f_r produced by a crystal oscillator², i.e.,

$$f_v^{ss}[n] = N_r[n] f_r. \quad (2.1)$$

The frequency control word $N_r[n]$ is the ratio between the desired frequency $f_v^{ss}[n]$ and the reference frequency f_r or the number of *desired* DCO periods $T_v^{ss}[n]$ within one reference period T_r . It is in general a fractional number consisting of an integer part $N_r^{int}[n]$ and an fractional part $N_r^{frac}[n]$, i.e.,

$$N_r[n] = \frac{f_v^{ss}[n]}{f_r} = \frac{T_r}{T_v^{ss}[n]} = N_r^{int}[n] + N_r^{frac}[n]. \quad (2.2)$$

²The reference frequency is typically in the range between 10 and 40 MHz [6].

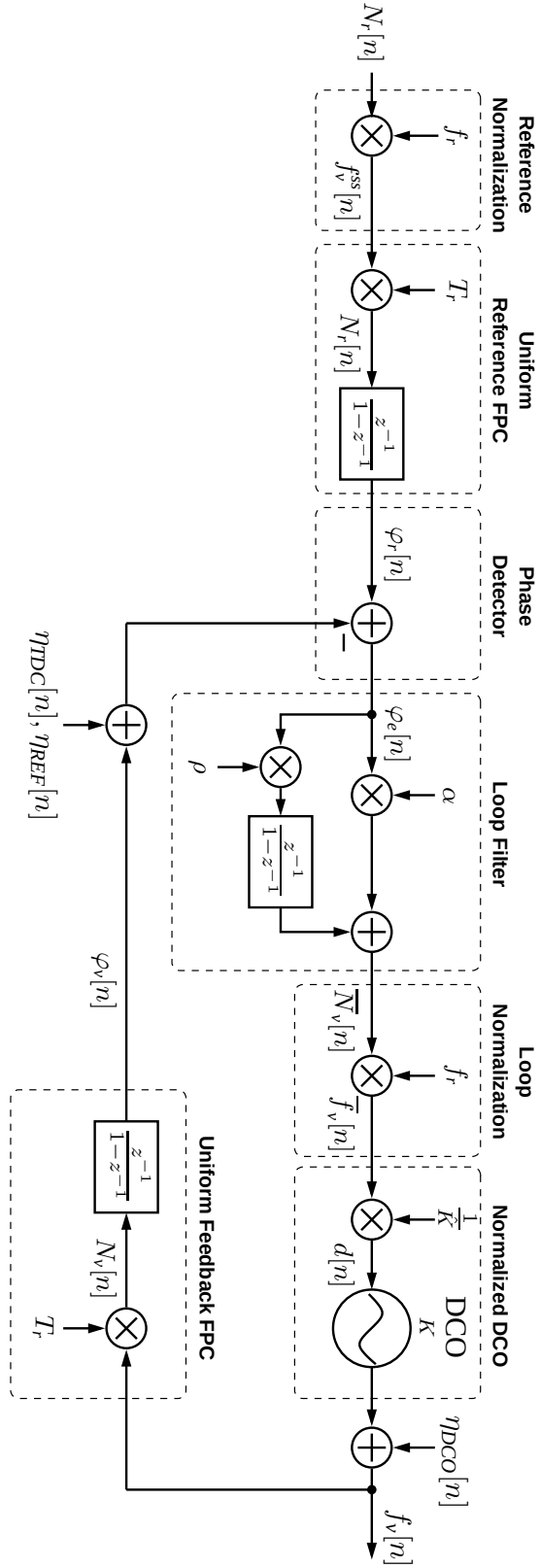


Figure 2.1 A z-domain model of a phase-domain all-digital phase-locked loop.

A second FPC converts the desired frequency $f_v^{ss}[n]$ into a digital reference phase signal $\varphi_r[n]$. Note that in practice the FPC in the feedback path and the FPC in the reference path are fundamentally different. While the reference phase signal $\varphi_r[n]$ can be directly generated by digital logic, the feedback FPC is a complex and demanding mixed-signal component, as elaborated in Chapter 3. The phase detector computes the difference between reference phase and variable phase through a digital subtraction. The resulting phase error $\varphi_e[n]$ is filtered by the digital loop filter and normalized by the reference frequency f_r and an estimate of the DCO gain $\hat{K}$. The resulting oscillator tuning word $d[n]$ is the input to the DCO. The feedback mechanism of the phase-locked loop controls the oscillator tuning word $d[n]$, so that the output of the DCO $f_v[n]$ becomes $f_v^{ss}[n]$. Hence in steady state, i.e., when lock is achieved, the tuning word $d[n]$ equals $d^{ss}[n]$ and the DCO output frequency $f_v[n]$ equals $f_v^{ss}[n]$.

In the following we investigate the key components of the phase-domain ADPLL in order to obtain their input-output relationship in form of transfer functions.

Digitally Controlled Oscillator Model

A digitally controlled oscillator, i.e., a digital-to-frequency converter (DFC), converts a digital input word $d[n]$ into an oscillating signal (typically a sine or a rectangular wave) with frequency $f_v[n]$. The frequency change corresponding to the change of the least significant bit (LSB) of $d[n]$ is the DCO gain K with the unit Hz/LSB. Thus, the frequency granularity of the DCO is limited to K Hz/LSB. This quantization is not included in this linear analysis but considered in Sec. 2.2 and analyzed in Sec. 4.3. The idealized digital-to-frequency conversion is modeled as a real valued gain factor K , i.e.,

$$\frac{f_v(z)}{d(z)} = K. \quad (2.3)$$

Within the thesis $x(z)$ denotes the z -transform of the discrete-time signal $x[n]$, e.g., $f_v(z)$ is the z -transform of $f_v[n]$. The DCO is an analog component and suffers, in practice, from process, supply voltage, and temperature (PVT) effects. Thus, the DCO is a nonlinear, time-varying system which needs to be controlled. Note that the DCO output frequency for $d[n] = 0$ is not zero, but

$$f_v(d[n])|_{d[n]=0} = f_0 \quad (2.4)$$

where f_0 is called the free running frequency. However, although $f_v[n] = Kd[n] + f_0$ is a more accurate description of the DCO, f_0 is neglected in this model, because it does not significantly affect the transfer function of the system. To be precise, in the model the free running frequency f_0 simply reduces the initial frequency offset to $f_v^{ss}[n] - f_0$ instead of $f_v^{ss}[n]$. Furthermore, in the model the DCO changes its frequency immediately in response to a change of the digital tuning word without any transient behavior. This model is only valid if the DCO settling time constant is much smaller than the reference clock period, which is typically fulfilled for most DCOs [89].

Frequency-to-Phase Converter Model

A frequency-to-phase converter (FPC) converts a frequency signal into a phase signal evaluated at discrete time instants nT_r . The phase signal of a sinusoidal signal with frequency $f(t)$ and phase $2\pi\varphi_0$, i.e., $\sin(2\pi\varphi(t))$, normalized by 2π is $\varphi(t) = \int_0^t f(\tau)d\tau + \varphi_0$. The FPC takes a sample after each

2.1 z-Domain Model

reference cycle n , corresponding to the time $t = nT_r$. Under the assumption that the variable frequency $f_v(t)$ is constant over the $(n-1)$ th reference period and $\varphi[n] = 0$ for $n < 0$ we get

$$\begin{aligned}\varphi[n] &= \varphi[n-1] + f[n-1]T_r \\ &= \varphi[n-1] + N[n-1]\end{aligned}\tag{2.5}$$

where

$$N[n] = \frac{f[n]}{f_r} = \frac{T_r}{T[n]}\tag{2.6}$$

is the slope of the phase signal $\varphi[n]$. The transfer function of the FPC in eqn. (2.5) in the z -domain is

$$\frac{\varphi(z)}{f(z)} = T_r \frac{z^{-1}}{1 - z^{-1}}.\tag{2.7}$$

The frequency-to-phase conversion requires one reference cycle, so that $\varphi[n]$ accumulates $N[n-1]$. While a stable system requires at least a delay of one sample z^{-1} in the feedback path, the delay in the reference path is added for consistency in Fig. 2.1. However, eqn. (2.5) shows that the FPC can be implemented as an accumulator with $\varphi[0] = \varphi_0$. Under the assumption that $f[n]$ and consequently $N[n]$ are constant over n , we can rewrite the phase signal eqn. (2.5)³ as

$$\varphi[n] = \varphi[n-1] + N = nN + \varphi_0\tag{2.8}$$

which can be assured in steady state. The frequency-to-phase conversion in the reference path can be simplified by substituting $f_v^{ss}[n]$ from eqn. (2.1) for $f[n]$ into eqn. (2.5). Hence, we can rewrite eqn. (2.5) for the reference FPC as⁴

$$\varphi_r[n] = \varphi_r[n-1] + N_r[n-1].\tag{2.9}$$

Thus, by accumulating $N_r[n]$ at the rate of the reference frequency we directly produce a reference phase signal by means of digital arithmetics. The feedback frequency-to-phase conversion is more challenging in practice, because the variable phase has to be extracted from the physical oscillation of the DCO output clock. One way to implement the feedback FPC is with the combination of an accumulator and a TDC, whose input signals are typically clock signals with rectangular waveforms, as discussed in Sec. 3.2. From a system-level perspective, however, we can express the variable phase signal $\varphi_v[n]$ in analogy to eqn. (2.9) as

$$\varphi_v[n] = \varphi_v[n-1] + N_v[n-1].\tag{2.10}$$

The signal $N_v[n]$ is the ratio between the actual DCO output frequency $f_v[n]$ and the reference frequency f_r or the number of *actual* DCO periods $T_v[n]$ within one reference period T_r . It is in general a fractional number consisting of an integer part $N_v^{int}[n]$ and a fractional part $N_v^{frac}[n]$, i.e.,

$$N_v[n] = \frac{f_v[n]}{f_r} = \frac{T_r}{T_v[n]} = N_v^{int}[n] + N_v^{frac}[n].\tag{2.11}$$

Note the analogy to the frequency command word $N_r[n]$ in eqn. (2.2).

³Note that in Sec. 3.2 we will discuss an ADPLL architecture that uses a non-uniform reference clock and we will refine the z -domain model accordingly in Fig. 3.11.

⁴The multiplication with f_r to generate the steady state frequency $f_v^{ss}[n]$ cancels the multiplication with T_r of the FPC in Fig. 2.1.

Digital Loop Arithmetic

The simplified mixed-signal paths, including the digital-to-frequency conversion within the DCO and the frequency-to-digital-phase conversion in the feedback path, of the model are analyzed in Sec. 2.2. By contrast, the arithmetic within the loop is purely digital. The phase error is given by the difference between the phase signals, i.e.,

$$\varphi_e[n] = \varphi_r[n] - \varphi_v[n]. \quad (2.12)$$

Substituting eqn. (2.9) and eqn. (2.10) into eqn. (2.12) results in

$$\varphi_e[n] = \varphi_e[n-1] + N_e[n-1] \quad (2.13)$$

with

$$N_e[n] = N_r[n] - N_v[n]. \quad (2.14)$$

The signal $N_e[n]$ is the slope of the phase error $\varphi_e[n]$. The operation in the phase-domain allows for a simple phase detector, which computes the phase error as a simple subtraction. The linear phase detector is a major advantage of phase-domain ADPLLs, because it does not produce harmonics [46], so that a proportional factor α can be used as a loop filter, i.e., a type-I PLL, which is

$$\frac{\overline{N}_v(z)}{\varphi_e(z)} = \alpha. \quad (2.15)$$

While a type-I PLL is superior concerning lock time it has inferior noise attenuation compared to higher order PLLs. In a type-II PLL an integral term ρ is added, i.e.,

$$\frac{\overline{N}_v(z)}{\varphi_e(z)} = \alpha + \rho \frac{z^{-1}}{1 - z^{-1}}. \quad (2.16)$$

Higher-order PLLs use additional integrators for improved noise suppression with the drawback of arising stability issues. The digital loop filter in ADPLLs enables the possibility to add additional digital finite impulse response (FIR) or infinite impulse response (IIR) filters [6, 53]. The output of the loop filter $\bar{f}_v[n]$ is multiplied with f_r , since an LSB integer change of $N_r[n]$ corresponds to a frequency change of f_r . In a further normalization step, $\bar{f}_v$ is normalized by $\hat{K}$ to make the loop ideally independent from PVT effects. The factor $\hat{K}$ is an estimate of the actual DCO gain K . If the estimate $\hat{K}$ equals the actual gain K the normalized DCO with input $\bar{f}_v$ and output f_v has a transfer function of unity.

2.1.3 Modulo Arithmetics

In principle, due to the pole at the unit circle $z = 1$ the transfer function of the FPC in eqn. (2.7) is unstable⁵. An unstable subsystem, however, would lead to unbounded output signals and therefore to unexpected behavior or even system failures. To avoid stability problems, the digital representation of the phase signals uses modulo arithmetics to restrict the phase signals to a certain amplitude range without affecting the overall system behavior [46, 52].

In the final implementation the phase signals $\varphi_r[n]$ and $\varphi_v[n]$ are digital quantities and their range of values is limited by their wordlength, i.e., the number of bits B used to represent the signal values. Typically the representation of the phase signals is split into an integer and a fractional part. For

⁵Stability requires all poles to be within the unit circle $|z| < 1$.

2.1 z-Domain Model

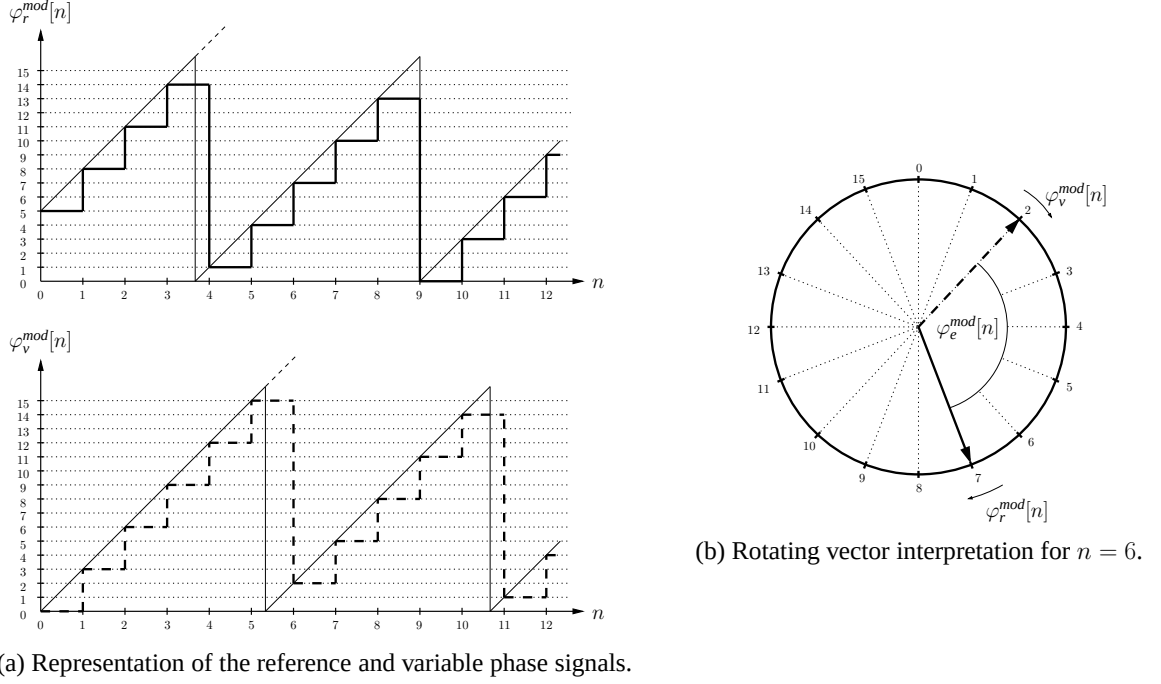


Figure 2.2 Illustration of the modulo arithmetics used to represent the phase signals to avoid unstable behavior for $N_v[n] = N_r[n] = 3$.

the sake of notational simplicity we do not distinguish between integer and fractional values here. The modulo arithmetics, however, apply for integer and fractional representations likewise. That is, while the signals $\varphi_r[n]$ and $\varphi_v[n]$ are unbounded and may increase linearly with time, their digital representations, denoted as $\varphi_r^{mod}[n]$ and $\varphi_v^{mod}[n]$, are restricted to the range between 0 and $2^B - 1$, i.e., using unsigned number representation, whereas the phase error representation $\varphi_e^{mod}[n]$ may be negative and is restricted from -2^{B-1} to $2^{B-1} - 1$, i.e., using 2's complement number representation. That is, the amplitude ranges of the phase signals are

$$\varphi_r^{mod}[n], \varphi_v^{mod}[n] \in \{0, 1, \dots, 2^B - 1\} \quad (2.17)$$

$$\varphi_e^{mod}[n] \in \{-2^{B-1}, \dots, -1, 0, 1, \dots, 2^{B-1} - 1\}. \quad (2.18)$$

As soon as the phase signals $\varphi_r^{mod}[n]$ and $\varphi_v^{mod}[n]$ exceed their limit of $2^B - 1$, they will experience an overflow effect. An accumulator with modulo- 2^B arithmetics, which discards its carry out bit when an overflow occurs, effectively limits the range. The digital representation of the phase signals with implicit modulo arithmetics can be expressed as

$$\varphi_r^{mod}[n] = \varphi_r[n] \bmod 2^B \quad (2.19)$$

and

$$\varphi_v^{mod}[n] = \varphi_v[n] \bmod 2^B. \quad (2.20)$$

Figure 2.2 shows an example for the modulo arithmetics. The phase signals $\varphi_r[n]$ and $\varphi_v[n]$ are ramp signals in lock with equal slopes of $N_v[n] = N_r[n] = 3$. Their representations $\varphi_r^{mod}[n]$ and $\varphi_v^{mod}[n]$ are limited to $B = 4$ bits and consequently transform the ramp signals into sawtooth signals with similar slopes. In this example, $\varphi_r^{mod}[n]$ is $\{5, 8, 11, 14, 1, 4, 7, \dots\}$ and $\varphi_v^{mod}[n]$ is $\{0, 3, 6, 9, 12, 15, 2, \dots\}$

and the difference between both signals, i.e. $\varphi_r^{mod}[n] - \varphi_v^{mod}[n]$, would be $\{5, 5, 5, 5, -11, -11, 5, \dots\}$ but $\varphi_r[n] - \varphi_v[n]$ should result in a constant phase error of $\varphi_e[n] = 5$. To overcome this ambiguity, the phase error $\varphi_e[n]$ itself is represented using 2's complement notation [46]

$$\varphi_e^{mod}[n] = ((\varphi_e[n] + 2^{B-1}) \bmod 2^B) - 2^{B-1}. \quad (2.21)$$

The interpretation of the phase error as 2's complement number, implicitly implements eqn. 2.21 without using additional hardware. However, the phase error representation $\varphi_e^{mod}[n]$ is always 5 for the example in Fig. 2.2 and, thus, matches the phase error $\varphi_e[n]$. Figure 2.2(b) gives an alternative interpretation of the modulo arithmetics [52]: the phase signals are represented as rotating vectors, where the smaller angle between them represents the phase error $\varphi_e^{mod}[n]$.

For the presented example the ranges of the phase signals are given by $\varphi_r^{mod}[n], \varphi_v^{mod}[n] \in \{0, \dots, 15\}$ and $\varphi_e^{mod}[n] \in \{-8, \dots, 7\}$. Note that the range of the phase error reflects the lock range of the ADPLL. Thus, in our example the ADPLL cannot distinguish between a phase error $\varphi_e[n]$ of -9 and 7 or -8 and 8 . This ambiguity may cause false locking and can be prevented by increasing the number of bits for the digital representation. However, the strong nonlinear behavior due modulo arithmetics becomes only evident when the bounds of $\varphi_e[n]$ in eqn. (2.21) are violated. Throughout the remainder of the thesis we assume that the phase error is represented with a sufficient number of bits and, therefore, neglect the modulo operation in the models.

2.1.4 Steady State

It is crucial to understand the locking process of phase-domain ADPLLs. For the following analysis we assume a linear DCO as in eqn. (2.3) and $\hat{K} = K$. Consequently, the locking process can be described as

$$\bar{f}_v[n] = f_v[n] \rightarrow f_v^{ss}[n], \quad (2.22)$$

$$\bar{N}_v[n] = N_v[n] \rightarrow N_r[n] \quad (2.23)$$

where the symbol $\rightarrow$ means converging to. That is, as $N_v[n]$ converges to $N_r[n]$ the slope of the phase error approaches zero (cf. eqn. (2.14)) and the phase error settles to a constant value (cf. eqn. (2.13)) denoted as $\varphi_e^{ss}[n]$, i.e.,

$$N_e[n] \rightarrow 0, \quad (2.24)$$

$$\varphi_e[n] \rightarrow \varphi_e[n] = \varphi_e[n-1] = \varphi_e^{ss}[n]. \quad (2.25)$$

In a type-I PLL the phase error is $\varphi_e^{ss}[n] = N_r[n]/\alpha$, because eqn. (2.15) and eqn. (2.23) must be fulfilled. In a type-II PLL the steady state phase error $\varphi_e^{ss}[n]$ will converge to zero, because the memory in the integral path of the loop filter will converge to $N_r[n]$.

2.1.5 Transfer Functions

In this section we use the derived input-output relationship of all blocks to derive an overall input-output transfer function. Furthermore, the s -domain approximation is derived, because many designs are based on the long experience with analog PLLs and s -domain models, e.g. [90]. Additionally, the impact of noise sources on the performance of the system is analyzed with respect to their transfer functions. Basically, we distinguish between three kinds of noise sources: DCO noise η_{DCO} , reference

2.1 z-Domain Model

noise η_{REF} , and TDC noise η_{TDC} . The loop arithmetic, i.e., loop filter and normalization by f_r and $\hat{K}$, is immune to external noise sources, such as thermal noise, due to its digital nature. The intrinsic quantization errors of digital arithmetics are typically insignificant in contrast to TDC and DCO quantization or noise sources such as oscillator and reference noise. The z -domain model gives useful insights how these noise sources affect the output phase of the ADPLL. A more detailed description on the properties of each noise source and their impact on the phase-noise power spectrum is given in Chapter 4.

ADPLL z-Domain Transfer Function

The open loop transfer function, i.e., when we assume $\varphi_e(z) = \varphi_r(z)$, of the discrete-time z -domain model of the type-II ADPLL in Fig. 2.1 with respect to the phase-domain signals is

$$\Phi_{ol}(z) = \frac{\varphi_v(z)}{\varphi_e(z)} = \frac{\alpha(z-1) + \rho}{(z-1)^2} \quad (2.26)$$

where we assumed again $\hat{K} = K$. Otherwise the coefficients α and ρ would be scaled by $K/\hat{K}$. The closed loop transfer function is

$$\Phi_{cl}(z) = \frac{\varphi_v(z)}{\varphi_r(z)} = \frac{\alpha(z-1) + \rho}{(z-1)^2 + \alpha(z-1) + \rho}. \quad (2.27)$$

Substituting $\rho = 0$ into eqn. (2.26) and eqn. (2.27) leads to a type-I ADPLL. It is interesting to note, that the input/output transfer functions are scaled versions of the phase transfer functions $\Phi_{ol}(z)$ eqn. (2.26) and $\Phi_{cl}(z)$ eqn. (2.27),

$$H_{ol}(z) = \frac{f_v(z)}{\varphi_e(z)} = f_r \frac{\alpha(z-1) + \rho}{(z-1)}, \quad (2.28)$$

$$H_{cl}(z) = \frac{f_v(z)}{N_r(z)} = f_r \Phi_{cl}(z). \quad (2.29)$$

Considering the reference frequency f_r as input to the system results in similar transfer functions, where the scaling changes from f_r to N_r in eqn. (2.28) and eqn. (2.29). The relationship between the closed loop frequency and phase transfer function is not surprising, because frequency information is converted into phase information so that both frequency and phase lock is achieved generally simultaneously. This fact is a key feature of phase-domain ADPLLs in contrast traditional PLLs. In traditional PLLs the linear s -domain approximation is only valid in the vicinity of the operational point f_v^{ss} due to its nonlinear characteristics and the limited input range of its phase detectors. Therefore PLLs are often have a two step acquisition. In a first frequency acquisition step it is ensured that the requirements of the phase detector are met, such that in the second phase acquisition step phase-lock can be achieved.

ADPLL s-Domain Transfer Function

Although the z -transform is the natural description of a discrete-time system it is common practice in the literature to approximate ADPLLs with a linear continuous-time system in the s -domain. Figure 2.3 shows the s -domain approximation of the z -domain model in Fig. 2.1. The accuracy of the approximation depends on the PLL bandwidth. The sampling rate f_r must be at least 10 times larger than the PLL bandwidth, is a rule of thumb [17] which is used in practice. In this case a truncated Taylor series expansion results in

$$z = e^{sT_r} \approx 1 + sT_r. \quad (2.30)$$

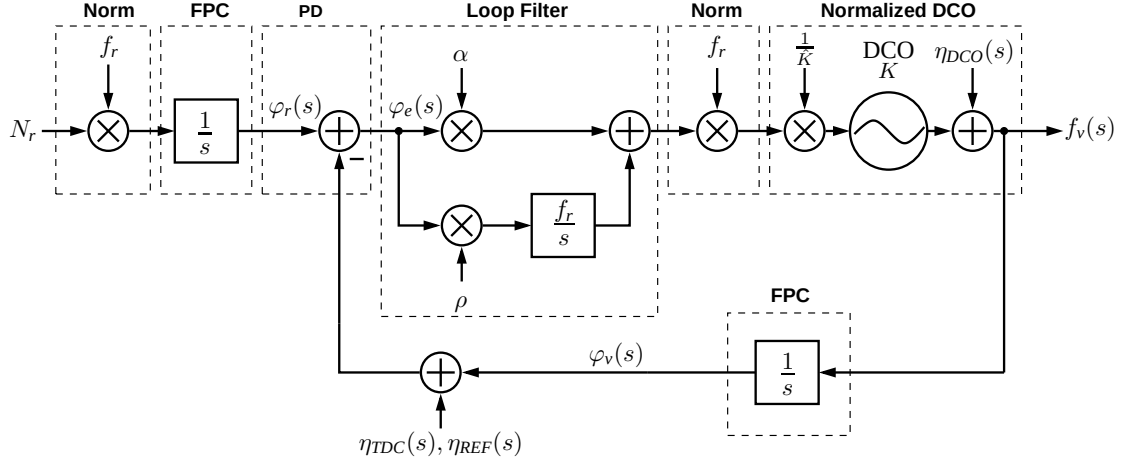


Figure 2.3 An s -domain model of a phase-domain all-digital phase-locked loop.

Substituting eqn. (2.30) into eqn. (2.26) gives the open loop s -domain transfer function, i.e.,

$$\Phi_{ol}(s) = \frac{\varphi_v(s)}{\varphi_e(s)} = \frac{\alpha f_r s + \rho f_r^2}{s^2} \quad (2.31)$$

and substituting eqn. (2.30) into eqn. (2.27) results in the closed loop s -domain transfer function, i.e.,

$$\Phi_{cl}(s) = \frac{\varphi_v(s)}{\varphi_r(s)} = \frac{\alpha f_r s + \rho f_r^2}{s^2 + \alpha f_r s + \rho f_r^2}. \quad (2.32)$$

Figure 2.4 shows the magnitude response of the transfer function $H_{cl}(z)$ and its analog approximation $H_{cl}(s)$ for three different loop filters. The reference frequency is chosen to be 13 MHz, corresponding to the gain of approximately 142 dB in Fig. 2.4, and $K = \hat{K}$. The s -domain transfer function is only accurate as long as the bandwidth is small compared to the sampling frequency f_r . Decreasing the loop filter parameter α and ρ increases the bandwidth of the type-II ADPLL. Figure 2.5 shows the corresponding step responses of the ADPLL transfer functions in the s and z -domains and simulation outcomes. A change of the frequency command word by one corresponds to a frequency offset of $f_r = 13$ MHz. The acquisition or lock time is the time until the ADPLL compensates the initial frequency offset with a certain steady state accuracy (e.g., 20 ppm). With increasing bandwidth the lock time is reduced but the overshoot is increased.

DCO Noise Transfer Function

Typically the oscillators are one of the major contributors to the output noise of a well-designed frequency synthesizer. A detailed description of oscillator noise can be found in Sec. 4.2. The presented model simplifies the continuous-time noise $\eta_{DCO}(t)$ to a discrete-time noise sampled at the rate of the reference frequency, i.e., $\eta_{DCO}[n]$. The oscillator frequency noise transfer function is

$$H_{\eta_{DCO}}(z) = \frac{f_v(z)}{\eta_{DCO}(z)} = \frac{(z-1)^2}{(z-1)^2 + \alpha(z-1) + \rho} \quad (2.33)$$

and the phase noise transfer function is

$$\Phi_{\eta_{DCO}}(z) = \frac{\varphi_v(z)}{\eta_{DCO}(z)} = T_r \frac{(z-1)}{(z-1)^2 + \alpha(z-1) + \rho}. \quad (2.34)$$

2.1 z-Domain Model

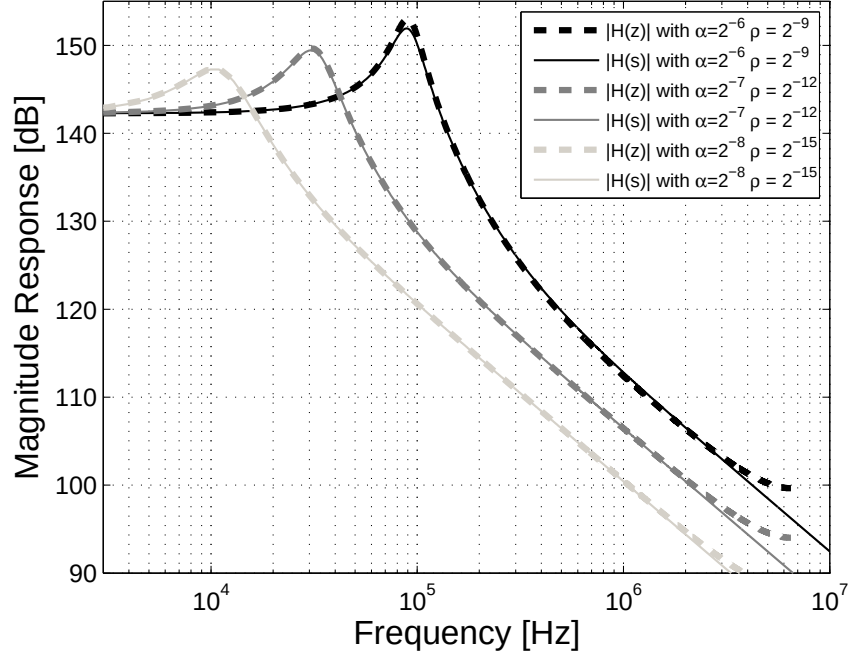


Figure 2.4 Magnitude response of a phase-domain ADPLL z -domain model $|H_{cl}(z)|$ and s -domain model $|H_{cl}(s)|$ for three different loop filters.

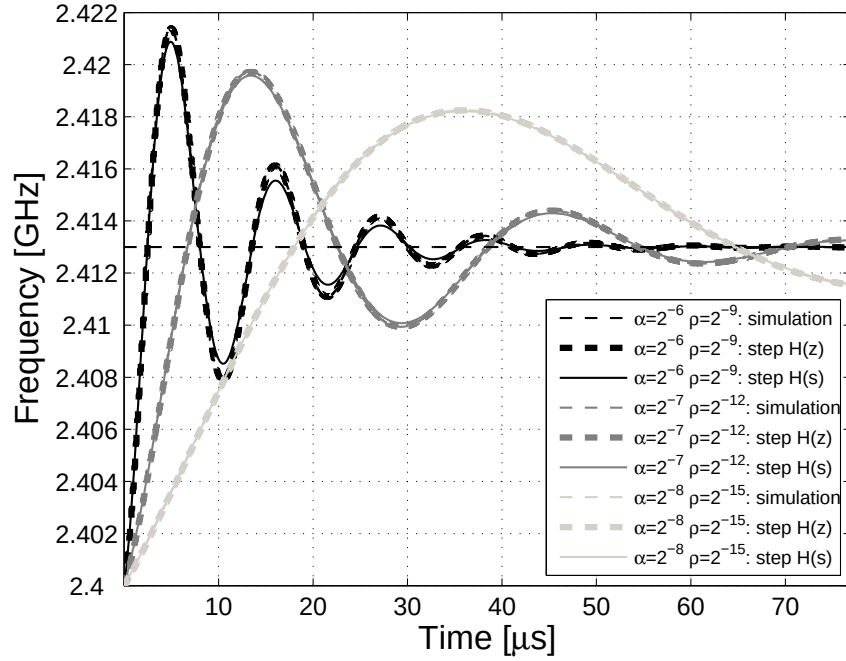


Figure 2.5 Step response of a phase-domain ADPLL with a numerical time-domain simulation, a z -domain model, and a s -domain model.

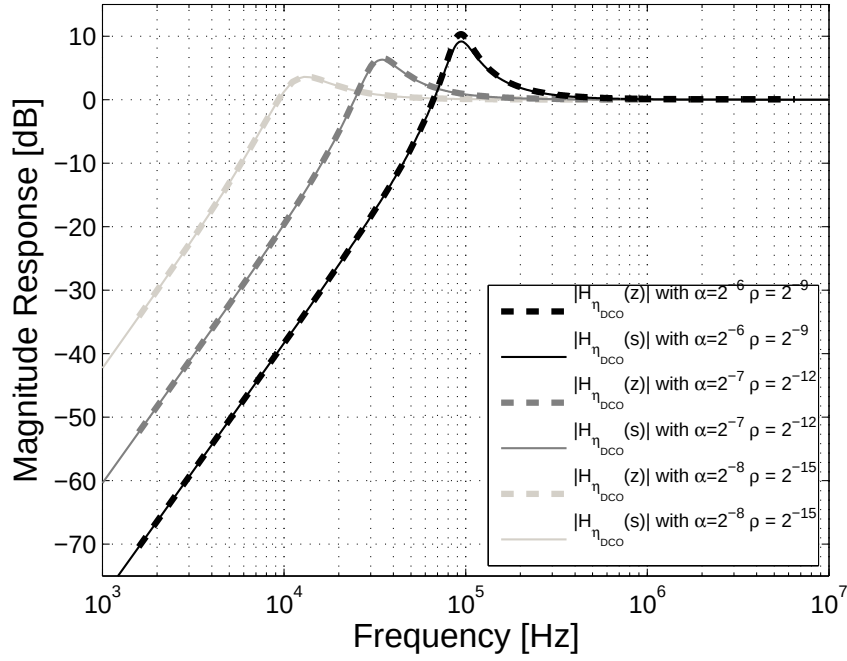


Figure 2.6 Magnitude response of the DCO noise frequency transfer function $H_{\eta_{DCO}}(z)$ and $H_{\eta_{DCO}}(s)$.

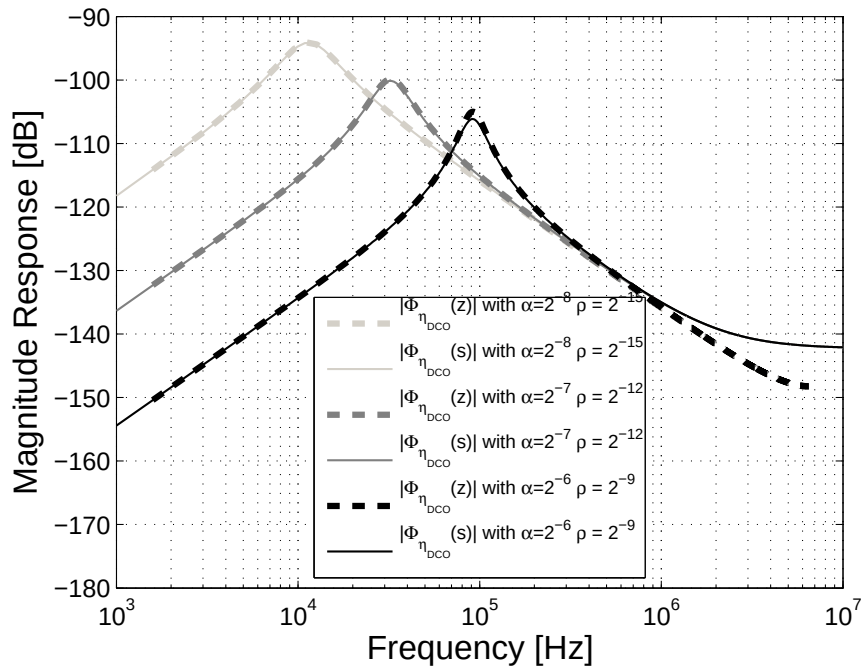


Figure 2.7 Magnitude response of the DCO noise phase transfer function $\Phi_{\eta_{DCO}}(z)$ (z -domain model) and $\Phi_{\eta_{DCO}}(s)$ (s -domain model).

2.1 z-Domain Model

Figure 2.6 and Fig. 2.7 show the magnitude response of DCO noise transfer function with respect to the output frequency $H_{\eta_{DCO}}(z)$ and with respect to the output phase $\Phi_{\eta_{DCO}}(z)$ for three different loop filters, respectively. The transfer function with respect to the output frequency $H_{\eta_{DCO}}(z)$ differs from the transfer function with respect to the output phase $\Phi_{\eta_{DCO}}$ by an additional 20 dB per decade increase due to the frequency-to-phase conversion, i.e., integration. The frequency transfer function $H_{\eta_{DCO}}(z)$ appears as high-pass (with +40 dB per decade increase for a type-II ADPLL) filter and the phase transfer function $H_{\eta_{DCO}}(z)$ as a band-pass filter. From a design perspective, it is interesting to note that with increasing bandwidth of the ADPLL the DCO noise is better suppressed.

Reference and TDC Noise Transfer Function

TDC noise $\eta_{TDC}[n]$ and reference noise $\eta_{REF}[n]$ affect the output frequency $f_v[n]$ via $H_{\eta_{TDC}}(z)$ and the output phase $\varphi_v[n]$ via $\Phi_{\eta_{TDC}}(z)$. On the one hand, with each reference cycle the feedback frequency-to-phase converter accumulates $N_v[n] = f_v[n]T_r$ eqn. (2.11). Any reference clock jitter on T_r will affect $N_v[n]$ and the measurement of $\varphi_v[n]$. With the aid of the mixed-signal analysis in Sec. 2.2 we provide a more sophisticated explanation on reference jitter and its effects on the system in Sec. 4.5. On the other hand, the fractional part of the variable phase $\varphi_v[n]$ is measured with a TDC. Unfortunately, the TDC suffers from quantization effects as discussed in Sec. 3.1.2. It is common practice to replace a quantizer by an additive, white, uniformly distributed noise source with a variance of $\Delta t_{res}^2/12$, where Δt_{res} is the resolution of the TDC [16]. In general, this linearization is not valid unless other noise sources break the deterministic pattern of the quantization process [91]. In a first approximation, however, we will consider the TDC as an additive noise source. A more sophisticated analysis of TDC quantization will be done in Sec. 4.4.

The reference and TDC transfer function with respect to variable frequency $f_v[n]$ is

$$H_{\eta_{TDC}}(z) = \frac{f_v(z)}{\eta_{TDC}(z)} = \frac{f_v(z)}{\eta_{REF}(z)} = -f_r \frac{\alpha(z-1)^2 + \rho(z-1)}{(z-1)^2 + \alpha(z-1) + \rho} \quad (2.35)$$

and with respect to variable phase $\varphi_v[n]$ is

$$\Phi_{\eta_{TDC}}(z) = \frac{\varphi_v(z)}{\eta_{TDC}(z)} = \frac{\varphi_v(z)}{\eta_{REF}(z)} = -\frac{\alpha(z-1) + \rho}{(z-1)^2 + \alpha(z-1) + \rho}. \quad (2.36)$$

Figure 2.8 and Fig. 2.9 show the magnitude responses of TDC noise transfer function with respect to the output frequency $H_{\eta_{TDC}}(z)$ and with respect to the output phase $\Phi_{\eta_{TDC}}(z)$ for three different loop filters. The ADPLL high-pass filters with +20 dB per decade the TDC noise $\eta_{TDC}[n]$ and reference noise $\eta_{REF}[n]$ with respect to the output frequency and low-pass filters with -20 dB per decade these noise contributions with respect to the output phase. The relation between $H_{\eta_{TDC}}(z)$ and $\Phi_{\eta_{TDC}}(z)$ is an additional attenuation of 20 dB per decade. In contrast to the DCO noise, the TDC noise is better attenuated with decreasing bandwidth of the ADPLL.

Trade-offs in ADPLLs

In fact, the loop filter with proportional factor α and integral factor ρ has to be designed to fulfill the following requirements: stability, transient behavior (lock time, overshoot), and steady state behavior (noise suppression). In PLLs, some of these requirements are contradicting. The intrinsic trade-off between lock time and noise suppression limits the performance in the one or the other direction. On the one hand a fast acquisition process (see Fig. 2.5 and Fig. 2.4) and a good DCO noise suppression

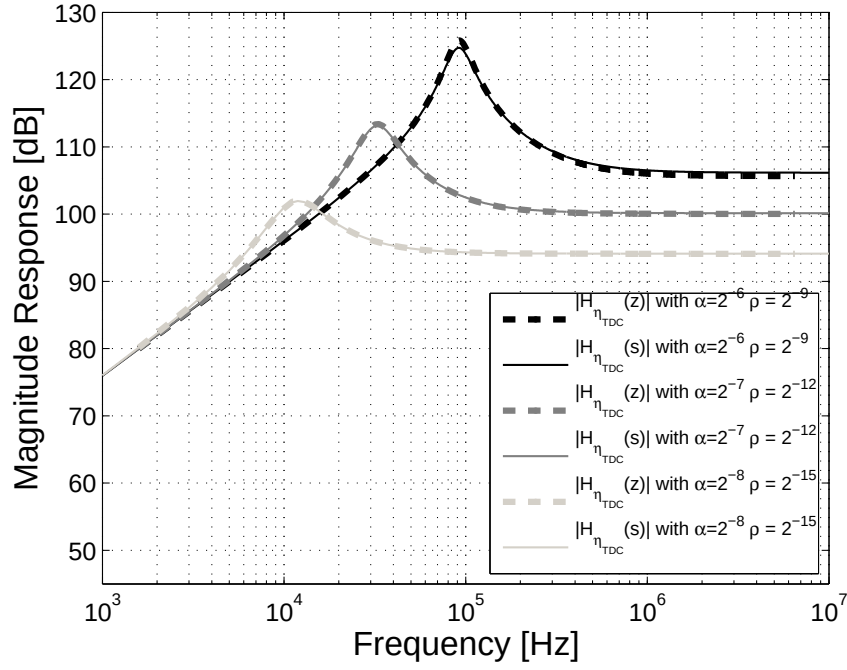


Figure 2.8 Magnitude response of the TDC noise frequency transfer function $H_{\eta_{TDC}}(z)$ (z-domain model) and $H_{\eta_{TDC}}(s)$ (s-domain model).

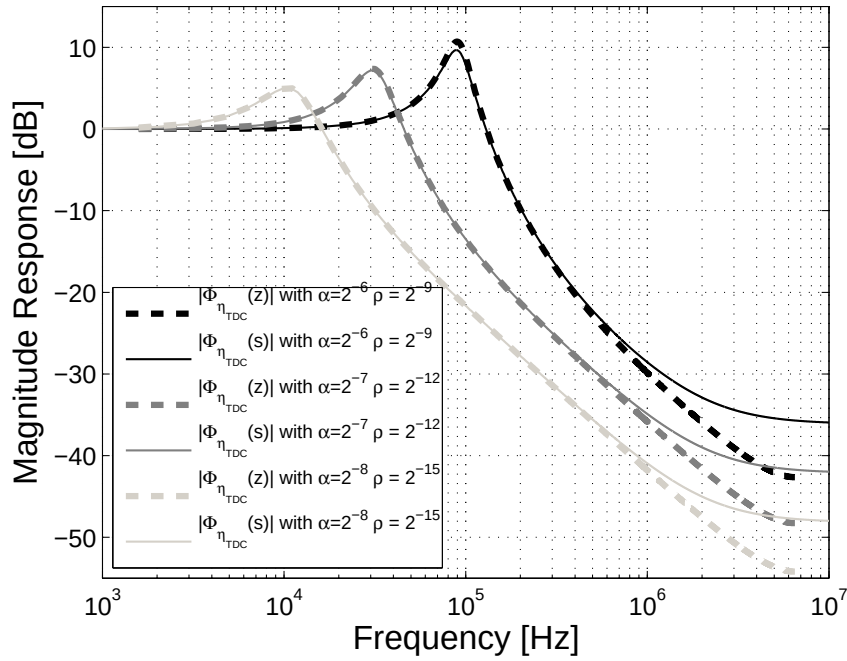


Figure 2.9 Magnitude response of the TDC noise phase transfer function $\Phi_{\eta_{TDC}}(z)$ (z-domain model) and $\Phi_{\eta_{TDC}}(s)$ (s-domain model).

2.2 Mixed-Signal Processing Model

require a large loop bandwidth (see Fig. 2.6 and Fig. 2.7), but on the other hand, good TDC and reference noise suppression require a narrow bandwidth (see Fig. 2.8 and Fig. 2.9). Typically the TDC quantization is the most critical imperfection in the system and defines the bandwidth in steady state. However, the noise suppression capability of the ADPLL can be enhanced by additional filtering. For instance, in wireless communication systems the stringent requirements cannot be fulfilled with a simple type-II ADPLL. Therefore additional infinite impulse response (IIR) filtering is used for further attenuation of noise [6, 53]. Additional IIR filtering provides better low-pass DCO noise suppression (e.g., -40dB/decade instead of -20 dB/decade in Fig. 2.7 [6]) and better high-pass TDC noise suppression (e.g., -40dB/decade instead of -20 dB/decade in Fig. 2.7 [6]). As a drawback the design becomes more complex, because the stability analysis is not straight forward anymore. A design example that is based on s -domain models is presented in [6].

Due to the digital nature of the loop arithmetics the performance of the system may be improved. For instance gear shifting exploits the mentioned trade-off by iteratively refining the bandwidth, as presented in [55, 57]. That is, during acquisition the bandwidth is large for fast locking and in steady state the bandwidth is narrow for good noise suppression.

2.2 Mixed-Signal Processing Model

Although their name may suggest otherwise, all-digital PLLs for wireless communication systems are mixed-signal systems. The name arises from the fact that all components can be integrated in CMOS technology without any analog tuning or passive components. Nano-scale CMOS technology is required to comply with the stringent requirements on high integration and costs. Furthermore, the digital intensive approach provides all additional advantages of digital circuits, see Sec. 1.1 for a detailed discussion, while analog design in CMOS is becoming increasingly difficult as the voltage headroom shrinks. Implementing typical analog building blocks as the PLL with digital logic allows for integrating the RF transmitter together with the digital baseband processor on a single chip, i.e., [13, 52, 54]. Actually, the stringent requirement on high integration has driven the development and research on ADPLLs.

In contrast to the z -domain abstraction, which models the mixed-signal feedback path entirely in the digital domain, the presented mixed-signal model emphasizes the existence of analog parts, digital parts, and the interfaces between them. Moreover the z -domain model operates entirely on the reference frequency, whereas the mixed-signal model converts the digital tuning word $d[n]$ into a continuous-time signal with frequency $f_v(t)$. From that oscillating signal its phase is extracted and sampled to produce a digital phase signal $\varphi_v[n]$. The variable phase $\varphi_v[n]$ can be compared to a digital reference phase in order to update the digital tuning word $d[n]$, as described in Sec. 2.1. In the following we will investigate the signal processing aspects of the mixed-signal chain and analyze the analogy of the digital-to-frequency conversion to a digital-to-analog (D/A) conversion and the analogy of frequency-to-digital-phase conversion to analog-to-digital (A/D) conversion.

2.2.1 Basic Principle

Figure 2.10 shows the mixed-signal model of a phase-domain ADPLL. While the digital part matches the z -domain model (Fig. 2.1 in Sec. 2.1), the mixed-signal path is more elaborated. Besides the

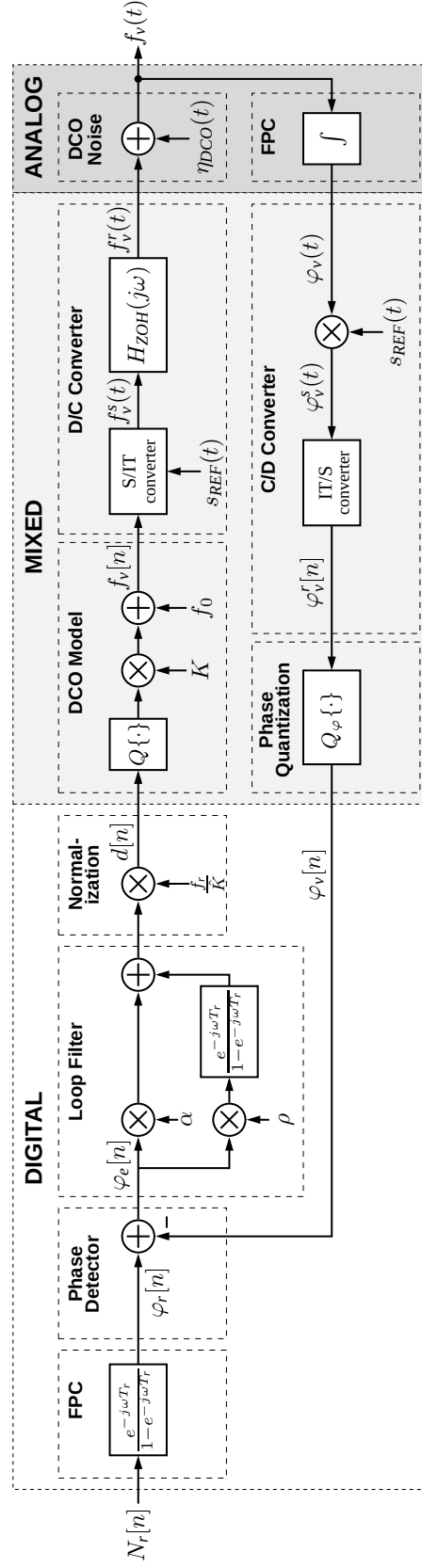


Figure 2.10 Mixed-signal model of phase-domain ADPLLs, where S/IT denotes sequence-to-impulse-train and IT/S denotes impulse-train-to-sequence.

2.2 Mixed-Signal Processing Model

oscillator noise⁶, the frequency quantization of the DCO and the phase quantization of the FPC are included to point out the relation to D/A and A/D conversion. An analysis of the quantization effects is provided in Sec. 4.3 for DCO quantization and Sec. 4.4 for TDC, or more general, frequency-to-phase converter quantization.

The feedback mechanism of the ADPLL controls the frequency $f_v[n]$ so that it settles in steady state on average at the desired frequency $f_v^{ss}[n]$, only disturbed by noise and distortions. While the frequency $f_v[n]$ is updated within each reference cycle n , the actual frequency signal is a continuous-time signal $f_v(t)$. In this sense, the DCO performs a signal reconstruction, i.e., a discrete-to-continuous-time (D/C) conversion [92]. Similar, in the feedback path, we want to extract the variable phase $\varphi_v[n]$ for each reference cycle n . Therefore the continuous-time phase $\varphi_v(t)$ is sampled to produce the discrete-time phase signal $\varphi_v^r[n]$, i.e., we perform a continuous-to-discrete-time (C/D) conversion [92]. For the C/D and D/C conversion we describe the reference clock as an impulse train, where each impulse corresponds to the time of a rising edge transition of the reference clock, i.e.,

$$s_{REF}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_r) \quad (2.37)$$

where $\delta(\cdot)$ is a Dirac Delta impulse. Reference clock jitter may be included in the impulse train, as shown in eqn. (4.65) in Sec. 4.5.

2.2.2 Digital-to-Frequency Conversion

The digital-to-frequency conversion performs a signal reconstruction, where the discrete-time signal $f_v[n]$ is converted to a continuous-time signal $f_v^r(t)$. To emphasize the analogy to a D/C converter we have included the frequency quantization to the DCO model, i.e.,

$$f_v[n] = Q\{d[m]\}K + f_0. \quad (2.38)$$

where $Q\{\cdot\}$ quantizes the tuning word $d[n]$ to an integer number. Note that in practice high-speed DCO dithering is used to increase the DCO resolution, see Sec. 3.1.1. The digital frequency signal $f_v[n]$ is the input to the D/C converter, where it is multiplied with the impulse train $s_{REF}(t)$ to yield

$$f_v^s(t) = \sum_{n=-\infty}^{\infty} f_v[n]\delta(t - nT_r). \quad (2.39)$$

The n th sample is now associated with the impulse at $t = nT_r$. Filtering the impulse train $f_v^s(t)$ with the reconstruction filter $h_{ZOH}(t)$ gives the reconstructed continuous-time signal $f_v^r(t)$, i.e.,

$$f_v^r(t) = f_v^s(t) * h_{ZOH}(t) = \sum_{n=-\infty}^{\infty} f_v[n]h_{ZOH}(t - nT_r). \quad (2.40)$$

In an ideal reconstruction system the weighted impulse train $f_v^s(t)$ is the input to an ideal low-pass filter with gain T_r and a cutoff frequency of π/T_r , which interpolates between the impulses of $f_v^s(t)$. By contrast, the DCO keeps the frequency $f_v[n]$ constant over the n th reference period and $h_{ZOH}(t)$ appears as a zero-order hold (ZOH) with a hold time of T_r and a gain of 1. Hence,

$$h_{ZOH}(t) = u(t) - u(t - T_r) = \begin{cases} 1 & 0 \leq t < T_r, \\ 0 & \text{otherwise.} \end{cases} \quad (2.41)$$

⁶Jitter of the reference clock can be included in $s_{REF}(t)$ in eqn. (2.37).

with $u(t)$ being the unit step function defined as

$$u(t) = \begin{cases} 1 & t \geq 0, \\ 0 & t < 0. \end{cases} \quad (2.42)$$

The effect of the ZOH becomes evident in the frequency domain. Applying the continuous-time Fourier transform (CTFT) to eqn. (2.40) results in

$$\begin{aligned} F_v^r(j\omega) &= \sum_{n=-\infty}^{\infty} f_v[n] H_{ZOH}(j\omega) e^{-j\omega T_r n} \\ &= H_{ZOH}(j\omega) \underbrace{\sum_{n=-\infty}^{\infty} f_v[n] e^{-j\omega T_r n}}_{F_v(e^{j\omega T_r})} \end{aligned} \quad (2.43)$$

where $F_v^r(j\omega)$ and $H_{ZOH}(j\omega)$ are the CTFTs of $f_v^r(t)$ and $h_{ZOH}(t)$, respectively. The reconstructed spectrum $F_v^r(j\omega)$ is a frequency scaled (i.e., ω is replaced by ωT_r) and scaled (by the frequency response of the filter $H_{ZOH}(j\omega)$) version of the digital spectrum $F_v(e^{j\omega T_r})$, which is the Fourier transform of $f_v[n]$. The Fourier transform of the ZOH in eqn. (2.41) is a sinc-shaped function, i.e.,

$$H_{ZOH}(j\omega) = \frac{\sin(\omega T_r/2)}{\omega/2} e^{j\omega T_r/2}. \quad (2.44)$$

Substituting eqn. (2.44) into eqn. (2.43) yields the reconstructed spectrum using a ZOH, i.e.,

$$F_v^r(j\omega) = F_v(e^{j\omega T_r}) \frac{\sin(\omega T_r/2)}{\omega/2} e^{j\omega T_r/2}. \quad (2.45)$$

In contrast to the ideal reconstruction with an ideal low-pass filter the frequency response of a ZOH is not bandlimited and will introduce spectral copies of the frequency spectrum $F_v(e^{j\omega T_r})$ at multiples of the reference frequency f_r . The mixed-signal model of the ADPLL illustrates the reconstruction of the analog signal $f_v(t)$ from its digital representation $f_v[n]$, which is used in Sec. 2.1.

2.2.3 Frequency-to-Digital-Phase Conversion

The counterpart to reconstruction is sampling of analog signals. In fact the mixed-signal feedback path extracts a digital phase signal $\varphi_v[n]$ from the analog frequency signal $f_v(t)$. In the model the frequency signal $f_v(t)$ is first converted into a phase signal $\varphi_v(t)$ through integration. The resulting signal is then sampled by a C/D converter to produce the digital phase signal $\varphi_v^r[n]$. Unfortunately the frequency-to-digital-phase conversion suffers from a coarse phase quantization, so that the actual output $\varphi_v[n]$ is a quantized version of $\varphi_v^r[n]$. In fact the quantization is not limited by the number of bits used to represent $\varphi_v[n]$, but by the resolution of the TDC⁷.

The continuous-time variable phase signal $\varphi_v(t)$ is given by integrating the continuous-time frequency signal $f_v(t)$, i.e.,

$$\varphi_v(t) = \int_{\tau=0}^t f_v(\tau) d\tau. \quad (2.46)$$

⁷Furthermore, in the implementation in Chapter 3 the frequency-to-phase conversion is basically split into an integer and a fractional part. Due to that partitioning we have to distinguish between phase quantization $Q_\varphi\{\cdot\}$ and TDC quantization $Q_{TDC}\{\cdot\}$. More details can be found in Chapter 4.

2.3 Summary and Discussion

In steady state we assume that the variable frequency is constant, i.e., $f_v(t) = f_v^{ss}$ such that eqn. (2.46) simplifies to

$$\varphi_v(t) = t f_v^{ss}. \quad (2.47)$$

The continuous time phase signal $\varphi_v(t)$ is multiplied by $s_{REF}(t)$ to produce a weighted impulse train signal $\varphi_v^s(t)$, i.e.,

$$\begin{aligned} \varphi_v^s(t) &= \varphi_v(t) s_{REF}(t) \\ &= \varphi_v(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_r) \\ &= \sum_{n=-\infty}^{\infty} \varphi_v(nT_r) \delta(t - nT_r) \end{aligned} \quad (2.48)$$

The impulse train $\varphi_v^s(t)$ is a continuous-time signal, that is zero except at integer multiples of T_r . The final digital sequence $\varphi_v^r[n]$, however, is a sequence of numbers, without information on its sampling period T_r [92]. The impulse train-to-sequence converter is mathematically expressed as

$$\varphi_v^r[n] = \varphi_v(nT_r). \quad (2.49)$$

By substituting eqn. (2.47) into eqn. (2.49) and using eqn. (2.2) we get the reconstructed digital variable phase signal in steady state, i.e.,

$$\varphi_v^r[n] = nT_r f_v^{ss} = nN_r. \quad (2.50)$$

Under the assumption that $f_v(t)$ converges to the steady state frequency f_v^{ss} , the digital variable phase signal $\varphi_v^r[n]$ is a signal with slope N_r . Thus, in steady state the variable phase matches the reference phase (by expressing the reference phase in eqn. (2.9) with the simplification given by eqn. (2.8) and setting the initial phase $\varphi_0 = 0$) resulting in a phase error $\varphi_e[n]$.

The mixed-signal model of the frequency-to-phase converter validates the z -domain simplifications in Sec. 2.1.2 and shows how the digital phase signal $\varphi_v^r[n]$ is extracted from the frequency signal $f_v(t)$. Additionally the phase quantization $Q_\varphi\{\cdot\}$ accounts for the limited resolution of the frequency-to-digital-phase converter.

2.3 Summary and Discussion

In this chapter, we have provided a system-level description of phase-domain ADPLLs. First, we have introduced a z -domain model and have evaluated the phase-domain operation, the locking behavior, and the noise transfer functions. Unlike traditional fractional- N PLLs, where the frequency domain model is a rather coarse abstraction, that is used to analyze the system-level characteristics, the presented z -domain model is very accurate, because the concept of phase-domain ADPLLs is to implement exactly this phase-domain operation. Besides the multiplication by N in the feedback path instead of the division by N in the reference path, a key difference between phase-domain ADPLLs and traditional fractional- N PLLs is the phase detector. In phase-domain ADPLLs the frequency signals are converted into digital phase signals by a frequency-to-phase converter and compared by a digital subtraction, whereas commonly the relation of rising clock edge events is measured by the phase detector. As a results, traditional PLLs do not operate “truly” in the phase domain, and effects like false locking are a direct result of this fact [50]. Due to this reason additional measurement

elements, such as a phase-frequency detector are required. The correlative, i.e., nonlinear nature of such phase detectors requires strong loop filtering to mitigate the resulting spurs. By contrast, the phase-domain ADPLL uses a linear phase detector, i.e., a subtraction that allows for a loop filter without integrator to speed-up the frequency acquisition process. The frequency-to-phase converter is in practice implemented as an accumulator, which is an unstable subsystem. We have shown, however, that this instability is circumvented by employing modulo arithmetics, which itself does not affect the linearity of the structure. Unlike the z -domain models from the literature [16] the presented model reflects the actual operation of each building block, such that the signals within the structure roughly match the signals in an actual ADPLL architecture, presented in Sec. 3.3. The simplicity of the model and its accuracy makes it suitable for high-level simulations and algorithm design.

In the second part of this chapter, we highlighted the signal processing aspect of ADPLLs, which are in fact mixed-signal systems, with analog components, digital components, and interfaces between them. The analysis of the chapter focuses on the digital-to-frequency conversion, which is in fact a D/C conversion, and the frequency-to-digital-phase conversion, which is in fact a C/D conversion. However, different to conventional sampling of analog signals, the signal information in ADPLLs is encoded in the zero-crossings and not the amplitude. In this sense, phase-domain ADPLLs are an excellent example of the novel paradigm in scaled CMOS, where time-resolution outperforms voltage resolution.

3

Architectures of Phase-Domain All-Digital Phase-Locked Loops

In this chapter the implementation challenges and two possible architectures of phase-domain ADPLLs are discussed. In the first part, popular DCO [50, 93] and TDC [94] implementations are reviewed. Both analog building blocks are crucial components, because their resolutions directly influence the performance of the entire system, and are implemented entirely in CMOS technology. The coarse resolution of the DCO can be improved by high rate dithering, which finally allows ADPLLs to compete or even outperform their analog counterparts. The second part of this chapter reviews and analyzes a popular phase-domain ADPLL architecture for wireless communication systems, which has been first introduced in [47], step by step studied and improved in [10, 16, 48, 51, 56, 57], and finally presented in [6]. The ADPLL in [6] allows for implementing the entire RF transmitter on a single chip fabricated in the most recent CMOS technology [13, 52, 54]. In contrast to the literature, e.g. [6, 52], our analysis will provide a different view on the system. In particular we explicitly elaborate the time-varying nature of the architecture. To synchronize the two clock domains, namely the fast DCO output clock domain and the slow reference clock domain, a third non-uniform clock, i.e., the retimed reference clock is introduced. In steady state, the retimed reference clock undergoes a periodic pattern, that depends on the relation between the DCO clock and the REF clock. The relation between the uniform z -domain model from Sec. 2.1 and the time-varying architecture is further analyzed by introducing a non-uniform z -domain model. In the third and final part of this chapter, we propose an alternative phase-domain ADPLL architecture, i.e., the synchronous reference architecture. In contrast to the retimed reference architecture, the entire system is directly clocked by the uniform reference clock. Hence, the resulting structure matches the uniform models from Chapter 2. The synchronization between the two clock domains (DCO clock and reference clock) is solved by a dedicated metastability avoidance circuit.

The analysis of the time-varying behavior of the retimed reference ADPLL architecture in Sec. 2.1 and the proposed reference synchronous ADPLL architecture were originally published in [72] and [73]. The work appears refined, partly extended, and re-formatted for in this chapter.

3.1 Design Challenges in an All-Digital Phase-Locked Loop

The key advantage of ADPLLs is that it can be entirely implemented in CMOS technology. Often the PLL is a part of the analog radio-frequency (RF) front-end, which is traditionally implemented

3.1 Design Challenges in an All-Digital Phase-Locked Loop

using Gallium Arsenide (GaAs) MESFET, Si-bipolar junction transistors (BJTs), III-V heterojunction bipolar junction transistors (HBTs), or Silicon-Germanium (SiGe) BJTs [8]. In general, due to a high yield, a high degree of integration, and low costs, a single chip CMOS solution, where the RF front-end and the baseband digital signal processing (DSP) are integrated on a single chip, is desired [8]. Implementing analog building blocks in CMOS, however, is a challenging task, because of the shrinking supply voltage, the shrinking voltage headroom, and the increased noise and interferer levels [13]. Summarizing, we can say that deep-submicron CMOS technology allows a high level of integration with the handicap of challenging analog design.

As described in Sec. 2.2, an ADPLL consists basically of two mixed-signal building blocks: the DCO and the TDC. The architecture of these components shall be digital intensive and avoid any analog tuning to exploit low-voltage deep-submicron CMOS technology. For both building blocks we find effective CMOS implementations in the literature. The most popular solutions will be discussed in the following.

3.1.1 Digitally Controlled Oscillator

The digitally controlled oscillator (DCO) of ADPLLs replaces the traditional voltage controlled oscillator (VCO) of common analog PLLs. While the VCOs output frequency is set by an analog voltage signal, the DCOs output frequency is adjusted by a digital tuning word $d[n]$. The main drawback of DCOs is a coarse frequency quantization given by the smallest possible frequency step within the DCO. Accordingly, the DCO gain K is defined as the frequency change in response to the change of the least significant bit of the tuning word, i.e., its unit is [Hz/LSB]. A DCO for wireless communications, or high-performance applications in general, is typically an LC oscillator with switchable capacitances, named varactors. The smallest possible frequency change of these varactors defines the frequency granularity of the DCO, i.e., the value of K . Table 3.1 summarizes the performance characteristics for different LC tank based DCOs. Especially a fine frequency resolution, which is 23 kHz for $f_{vss} = 2.4$ GHz in [93], corresponding to the smallest varactor capacitance, of 38 aF in [93], is desired. Recently an alternative varactor circuit that exploits opposite characteristics of PMOS and NMOS transistors and achieves in contrast to [93] a 10 times finer frequency granularity, has been proposed in [95]. As CMOS scales down the frequency resolution is expected to improve further with the drawback of increased varactor size variations. Thus, good device matching and a linear tuning curve will become increasingly difficult. Besides LC tank based DCOs, research is going on to develop alternative DCO architectures in order to further enhance the frequency granularity, e.g., [96]. Nevertheless, any state-of-the-art DCO embedded in an ADPLL has an insufficient frequency resolution for high-performance applications. To overcome this strong limitation, high-speed $\Sigma\Delta$ dithering is used to improve the frequency resolution and shift the quantization errors to higher frequencies, i.e., outside the band of interest [93].

Exemplary DCO Architecture

In the following we will review the challenges in DCO design based on the approach in [93], which was the first DCO implementation complying with the strict wireless communication standards of Bluetooth [10] and GSM [50]. The requirements on the DCO and VCO are comparable: a large tuning range, a low phase noise, a low spurious level, a good linearity of the tuning curve, a small silicon area, and little power consumption. The key challenge is to find an appropriate trade-off

Work	CMOS [μm]	Center [GHz]	Range [MHz]	Noise [dBc @ MHz]	Supply [V]	Current [mA]	Area [mm^2]	Resolution [kHz]	
[93]	0.13	2.4	600	-112@0.5	1.5	2.3	0.54	23	38
[97]	0.065	2.4	200	-125@1	1.2	4.8	0.1	20	n/a
[50]	0.065	3.6	900	-165@20	1.4	18	n/a	12	n/a
[98]	0.09	3.3	600	-118@1	1.2	2	n/a	5	10
[99]	0.13	3.95	1000	-118@1	2.5	2.3	0.19	200	n/a
[100]	0.065	10	1000	-102@1	1.1	3	0.025	1000	55
[95]	0.065	5	1000	-118@1	0.6	1.8	n/a	1.4	0.8

Table 3.1 DCO performance comparison.

between those contradictory requirements for a given specification. One trade-off is the frequency granularity K , which should be as small as possible to achieve a good phase noise performance, and the frequency tuning range, which should be as large as possible to cover the bandwidth of the standard. The bandwidth of communications standards is constantly increasing and the phase noise emission masks are becoming more stringent to achieve higher data rates. The tuning range of the oscillator is designed typically 20% larger than the band of interest to account for PVT changes [101]. On the one hand, implementing the DCO with binary weighted varactors, i.e., the smallest varactor has half the capacitance of the second smallest varactor and so on, faces serious matching problems [102] and results in a strongly nonlinear tuning curve causing large phase noise. Furthermore an LSB change of the tuning word may require a large number of bits to flip. Therefore many varactors need to be charged or discharged, which raises the phase noise. On the other hand, using unit weights for the varactors significantly improves the linearity with the drawback of unacceptable large die area.

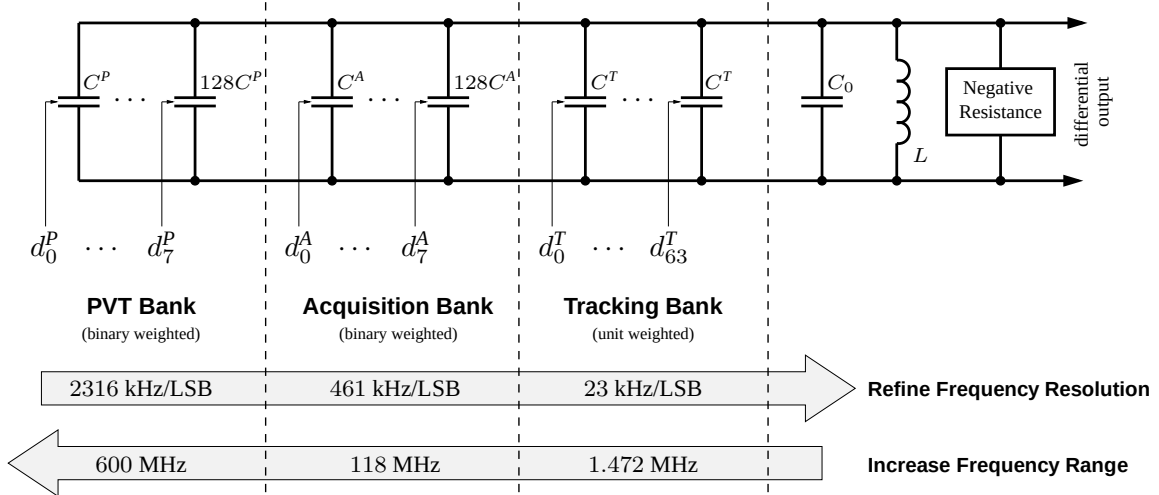


Figure 3.1 A DCO composed of an LC circuit with varactor banks [93].

To overcome this problem the narrowband nature of communication standards, such as Bluetooth and GSM, is exploited. From the Bluetooth band of 80 MHz only 320 kHz are required during the transmission for Gaussian frequency shift keying (GFSK). To be specific, in general a dynamic range

3.1 Design Challenges in an All-Digital Phase-Locked Loop

of 80 MHz should be supported, but in steady state only 320 kHz are needed for data modulation. The basic idea is to traverse several modes during the frequency acquisition process and iteratively refine the frequency resolution while reducing the frequency range. In each mode only varactors of a dedicated varactor bank are switched. Iterating to a finer mode by switching to the next bank freezes the tuning word of the current bank. The number of varactor banks is a designer decision. In [93], for instance, three modes have been used: PVT to adjust for changes in the environment, acquisition to select the frequency band, and tracking to perform the actual transmission, modulation, and tracking of changes in the environment. Figure 3.1 shows the DCO architecture with a separate varactor bank for each mode. That is, the PVT bank uses binary weight varactors ($C^P, 2C^P, \dots, 2^7 C^P$, i.e., 8 varactors representing 8 bits), has a coarse frequency resolution of 2316 kHz, and a wide dynamic range of approximately 600 MHz. The acquisition bank uses binary weight varactors ($C^A, 2C^A \dots \Delta 2^7 C^A$, i.e., 8 varactors representing 8 bits), has a frequency resolution of 461 kHz, and a dynamic range of 118 MHz. Finally, the tracking bank uses unit weighted varactors ($C^T, C^T, \dots, C^T$, i.e., 64 varactors representing 64 bits), has a fine frequency resolution of 23 kHz, and a narrow dynamic range of 1.472 MHz. Hence, this varactor bank implementation of the DCO circumvents the trade-off between frequency resolution and frequency range. In principle the bandwidth is incrementally refined during the locking process, which can be interpreted as a kind of gear shifting [55, 57].

Due to the two coarse varactor banks, i.e., PVT and acquisition, the dynamic range of the DCO is as large as required. Since during transmission only varactors in the tracking bank with good matching, i.e., good linearity, and fine frequency granularity are charged and discharged, the phase noise performance of the DCO is good. Additional care has to be taken, that the capacitive coupling effects due to varactor switching have as little impact as possible. A careful layout and a predefined switching order reduces this problem. Dynamic element matching may be used to further improve the linearity of the tuning curve [6, 93]. Note, however, that the final frequency granularity of 23 kHz in [93] is still too coarse for the stringent wireless communication standards. Finer frequency resolution is achieved by high-speed dithering of the DCO input signal, i.e., high-speed dithering some varactors of the tracking bank¹. Otherwise, the strong frequency quantization of the DCO will limit or even destroy the applicability of ADPLLs. The output frequency would slowly oscillate between two output values with a frequency difference of K with an average frequency corresponding to $f_v^{ss}[n]$, but significant spurs.

DCO Dithering

Figure 3.2 shows the structure for high-speed dithering of the DCO. Basically we trade a higher sampling rate for a finer frequency granularity. High-speed dithering of the oscillator tuning word $d[n]$ increases the DCO resolution, i.e., $f_v[n]$, during one reference cycle. In consequence the value of $N_v[n]$ and consequently the phase error $\varphi_e[n]$ are hardly affected by the DCO quantization. In this way the propagation of the quantization error through the system, which leads to large spurs, is prevented. On the one hand, the dither clock is in practice obtained from the DCO output clock by a clock divider with an integer division factor D . On the other hand, the dither frequency f_{dither} is by an arbitrary positive factor $L[n]$ larger than the reference frequency f_r , i.e.,

$$f_{dither}[n] = L[n] f_r = \frac{f_v^{ss}[n]}{D}. \quad (3.1)$$

¹To minimize parasitic coupling effects the high-speed dither sequence $d_{\Sigma\Delta}[l]$ operates a separated varactor bank in Fig. 3.2.

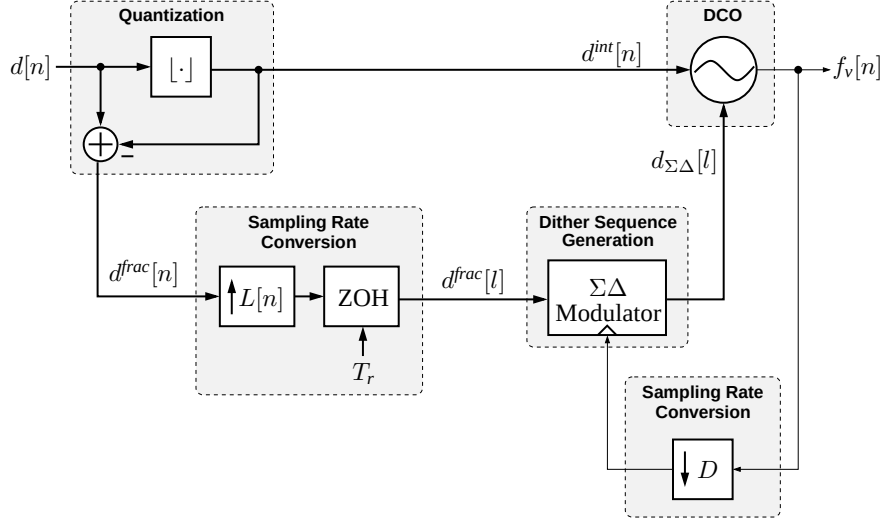


Figure 3.2 High-speed dithering of the finest DCO varactor bank.

The sampling rate conversion in Fig. 3.2 increases the sampling frequency by $L[n]$, which becomes with eqn. (3.1) and eqn. (2.1)

$$L[n] = \frac{f_v^{ss}[n]/D}{f_r} = \frac{1}{D} N_r[n]. \quad (3.2)$$

For notational simplicity we will assume that $L[n]$ is constant and an integer number so that we can write the n th reference cycle as $n = l/L$ with l being the time index for the dither sequence. During one reference cycle n , the signal value of $d^{frac}[l]$ is kept constant. Thus, the interpolation filter of the sampling rate conversion is a ZOH. In a hardware implementation, however, the sampling rate conversion is implicitly done by connecting $d^{frac}[n]$ directly to the modulator². To exemplify the relations we provide a numerical example for a Bluetooth ADPLL [93]. For a steady state frequency of $f_v^{ss}[n] = 2.4$ GHz, a reference frequency of $f_r = 13$ MHz, and a deviation factor of $D = 4$, we get $L \approx 46.15$ from eqn. (3.2) and a dither frequency $f_{dither} = 600$ MHz from eqn. (3.1).

The oscillator tuning word for the finest varactor bank is split into two parts, i.e., an integer part and a fractional part, which can be written as

$$d[n] = d^{int}[n] + d^{frac}[n]. \quad (3.3)$$

The integer part of the tuning word is given by

$$d^{int}[n] = Q\{d[n]\} = \lfloor d[n] \rfloor \quad (3.4)$$

where $\lfloor \cdot \rfloor$ denotes the floor operation. While the integer part $d^{int}[n]$ controls the varactors in the DCO directly, the fractional part $d^{frac}[n]$ is the input to a modulator. A modulator produces a high-speed integer sequence $d_{\Sigma\Delta}[l]$, whose average value is on average equal to $d^{frac}[n]$. Thus, we will describe the DCO output frequency in the following as

$$f_v[l] = (d^{int}[l/L] + d_{\Sigma\Delta}[l]) K. \quad (3.5)$$

²As the reference clock is asynchronous to the DCO clock and, therefore, also to the dither clock, synchronization issues between those domains have to be considered to avoid metastability.

3.1 Design Challenges in an All-Digital Phase-Locked Loop

1st Order $\Sigma\Delta$ Modulator

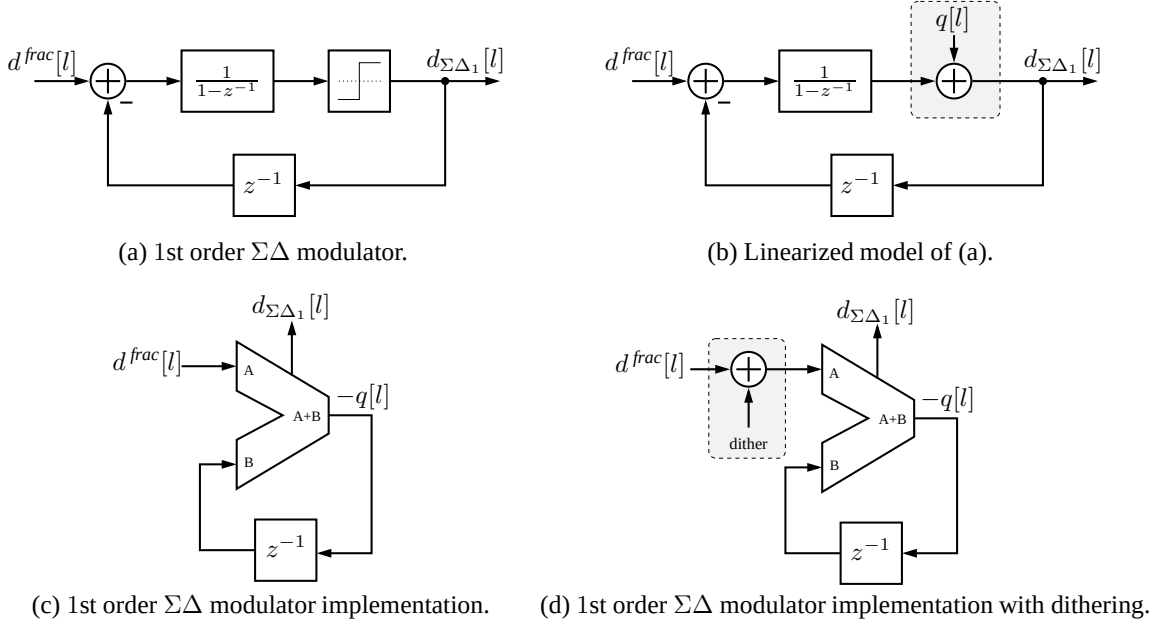


Figure 3.3 A 1st order $\Sigma\Delta$ modulator [103, 104].

The advantage of $\Sigma\Delta$ modulators [103, 104] is their noise shaping capability, i.e., the quantization error energy is shifted to higher frequencies. Figure 3.3 shows a 1st order $\Sigma\Delta$ modulator (a), its linearized z -domain model (b), a possible hardware implementation (c), and an enhanced 1st order $\Sigma\Delta$ modulator with additional randomization (d). The first-order $\Sigma\Delta$ modulator in Fig. 3.3(c) utilizes a simple adder that is used to accumulate $d^{frac}[l]$. The output signal $d_{\Sigma\Delta_1}[l]$ is one whenever an overflow occurs and zero otherwise. A $\Sigma\Delta$ modulator is a strongly nonlinear system, whose nonlinear analysis is difficult. Therefore, a linearized model, where the quantization error is modelled as an independent additive white Gaussian noise source $q[l]$, is used to describe the principle behavior. From the linearized model in Fig. 3.3(b) we get [103]

$$D_{\Sigma\Delta_1}(z) = D^{frac}(z) + \underbrace{(1 - z^{-1})}_{H_{\Sigma\Delta_1}(z)} Q(z) \quad (3.6)$$

where $D_{\Sigma\Delta_1}(z)$, $D^{frac}(z)$, and $Q(z)$ are the z -transforms of $d_{\Sigma\Delta_1}[l]$, $d^{frac}[l]$, and $q[l]$, respectively. Equation eqn. (3.6) reveals that the input $D^{frac}(z)$ passes unchanged through the system, while the quantization error signal $Q(z)$ gets high-pass filtered with $H_{\Sigma\Delta_1}(z) = (1 - z^{-1})$. In ADPLLs the in-band phase-noise and in-band spurs are the most critical figures of merit, while high-frequency noise is often less important. In other applications (such as ADCs) the high-pass noise shaped quantization error can be filtered by an additional low-pass filter to improve the signal-to-quantization-noise ratio (SQNR) [105].

Higher Order $\Sigma\Delta$ Modulator

The noise shaping capability of a $\Sigma\Delta$ modulator can be enhanced by cascading several 1st order $\Sigma\Delta$ modulators to a multi-stage noise shaping (MASH) structure, e.g., [6, 50]. Figure 3.4 depicts a 2nd

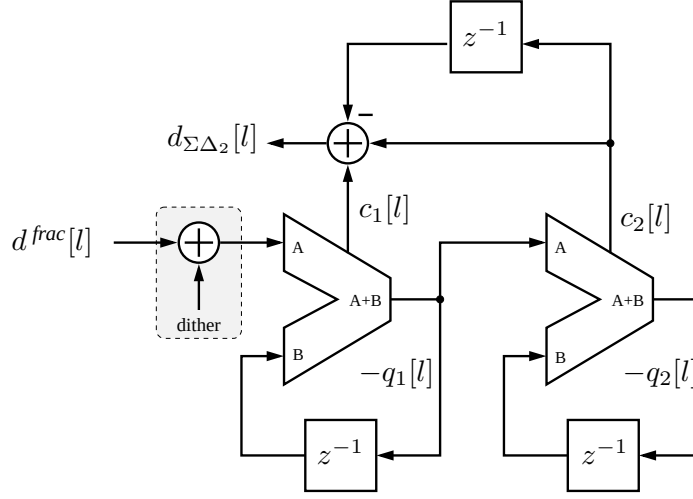


Figure 3.4 A 2nd order $\Sigma\Delta$ modulator with MASH 1-1 architecture [103] with optional input randomization.

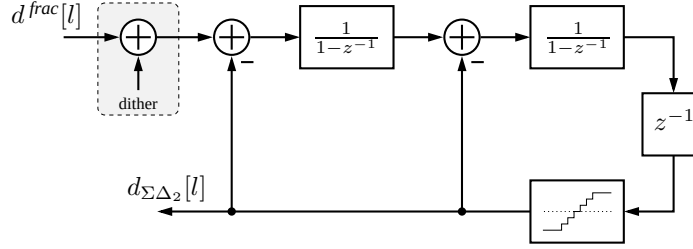


Figure 3.5 A 2nd order $\Sigma\Delta$ modulator with SSMF architecture [106] with optional input randomization.

order $\Sigma\Delta$ modulator with MASH 1-1 architecture. The output sequence $d_{\Sigma\Delta_k}(z)$ for a k th order $\Sigma\Delta$ modulator can be expressed as [103]

$$D_{\Sigma\Delta_k}(z) = D^{frac}(z) + \underbrace{(1 - z^{-1})^k}_{H_{\Sigma\Delta_k}(z)} Q(z). \quad (3.7)$$

Each stage adds a noise shaping factor of $(1 - z^{-1})$, i.e., additional 20 dB/dec. Note, however, that the output sequence of a $\Sigma\Delta$ modulator is not random but a deterministic sequence. For a frequency synthesizer the input to the modulator is a constant value, at most overlapped by some noise. Unless the noise destroys the deterministic pattern of the modulator output signal, large spurs appear in the output spectrum. Since the fast switching of the varactors consumes a lot of power, the number of varactors, i.e., the output levels of the modulator, shall be kept as small as possible. Basically there is a trade-off between complexity, number of output levels, noise shaping capability, and spurious behavior. A first-order MASH $\Sigma\Delta$ modulator for instance has low complexity, only two output levels, 20 dB/dec noise shaping, but many spurious tones in the output spectrum. Therefore, for DCO dithering typically 2nd or higher order modulators are used, e.g., [35].

In order to cope with spurs an enhanced $\Sigma\Delta$ MASH cell with additional randomization has been proposed in [6] and is shown in Fig. 3.3(d). The signal “dither” is a zero-mean (pseudo) random

3.1 Design Challenges in an All-Digital Phase-Locked Loop

sequence that aims to break the deterministic pattern of the $\Sigma\Delta$ modulator. In hardware a random binary sequence is generated with a linear feedback shift register (LFSR) and connected with an XOR gate to the LSB of the signal $d^{frac}[l]$. For higher-order MASH modulators it is sufficient to dither the first stage [6] to randomize any periodic output sequence with long periods, but this fails for short periods as shown in Sec. 4.3. Increasing the amplitude of the dither signal, i.e., dithering more bits than the LSB, increases the variability, but raises the noise level at the output of the modulator. Alternative $\Sigma\Delta$ modulator structures are not that prone to deterministic behavior [106]. Figure 3.5 shows a 2nd order single-stage multiple feedback (SSMF) $\Sigma\Delta$ modulator. In order to achieve stability the output levels have to be increased to a range of -2 to $+2$, i.e., having 5 output levels in contrast to 4 levels in the MASH 1-1 implementation [107]. Furthermore, the number of bits for the accumulator stages have to be increased to ensure stability. On account of this additional complexity we get an almost tone-free output signal, because the multiple feedback-paths and the large signal range effectively destroy the slow periodic behavior. In case some spurs are acceptable or the ADPLL uses 2-point modulation, i.e., the tuning word is not constant any more, alternative dithering schemes may be used. For instance a Ritchie structure similar to [108] may be used to reduce the number of output levels to 2, i.e., a single bit.

The effects of DCO quantization, $\Sigma\Delta$ dithering, and its noise shaping capability on the output phase noise are further discussed and evaluated in Sec. 4.3.

3.1.2 Time-to-Digital Converter

High-performance ADPLL implementations require a TDC to measure the phase error with high precision. However, the performance of a TDC suffers strongly from PVT and quantization effects. High-resolution TDCs, to comply with the strict communication standards for high-performance applications, and ADPLL structures that avoid the use of TDCs to reduce the power and area consumption, and design procedures for ADPLLs for less stringent specifications are open research issues. The requirements on the TDC, besides resolution, are specified in terms of power consumption, area consumption, and complexity. Similar to the DCO, the TDC should be implemented in CMOS technology without any analog tuning. Table 3.2 summarizes recent TDC implementations for different applications. A dedicated TDC solution for ADPLLs has been presented in [10] refined in [94, 109] and analyzed in [6, 16, 52, 110]. The TDC is basically a delay line (DL), where one delay, typically an inverter delay of approximately 20 ps, corresponds to the resolution of the TDC, denoted as Δt_{res} . Other TDC architectures use ring oscillators or delay locked loops (DLLs), where the time resolution corresponds to a buffer delay, e.g., [111–113]. To increase the TDC time resolution beyond one gate delay methods like pulse shrinking [114, 115] and Vernier delay lines (VDL) [116–119] utilize delay mismatches. While the delay mismatch may increase the resolution, the dynamic range of these TDC architectures is restricted. Some other TDC architectures overcome this limitation by combining a VDL with DLLs [120], combining a DL with a VDL [121, 122], combining a DL with pulse-shrinking [123, 124], using two-level [125], or multi-level DLs [126, 127]. Most of these approaches suffer from one or more of the following drawbacks: an analog intensive approach, a small dynamic range, a high power dissipation, a large silicon area, a poor variation robustness, and the need for calibration.

Recently the increased interest in ADPLLs has triggered several novel TDC concepts that demonstrate equal or even better performance in comparison to [94]. In [42, 128] a stochastic TDC is proposed that exploits, similar to VDLs, the (Gaussian) mismatch distribution of a set of latches.

Work	Technique	CMOS [μm]	Supply [V]	Power [mW]	Area [mm^2]	Res. [ps]	Range [ns]
[94, 109]	pseudo-diff DL	0.09	1.3	6.9/1.8	0.01	17 – 21	0.9
[114]	pulse shrinking	0.8	3 – 5.5	10	6	50	3200
[115]	pulse shrinking	0.8	5.0	20	32	6	368
[116]	VDL	0.7	5.0	n/a	10	30	0.25
[117]	VDL	0.13	n/a	n/a	0.03	2	0.5
[118, 119]	VDL	0.18	n/a	n/a	4	5	0.15
[113]	DLL	0.18	n/a	n/a	4.9	10 – 20	0.2 – 0.3
[112]	DLL	0.35	3.3	40	7.5	8.1	204000
[120]	DLL & VDL	0.18	1.8	46	0.6	28 – 38	8.45
[125]	2-level DL	0.35	3.0	50	0.6	24 – 30	4.6
[111]	DLL & DL	0.7	5	110	10.7	25	3
[126, 127]	multi-level DL	0.6	3.3	50	< 13.6	30	20
[122]	DL & VDL	0.09	1	1.0	0.6	1.25	560
[121]	DL & VDL	0.18	n/a	n/a	0.52	10	1.2
[123]	DL & pulse shrinking	0.13	1.5	1.09	0.018	24.5	5
[124]	DL & pulse shrinking	0.13	1.5	1.0	0.012	6.5	4.5
[42, 128]	stochastic	0.13	1.2	2.7	0.11	~ 0.7	n/a
[129–131]	interpolation	0.09	1.2	n/a	0.02	4.7	0.18
[34, 132, 133]	gated ring oscillator	0.13	1.5	2.2 – 21	0.04	~ 1.2	$\gg$
[134]	bang-bang DLL	0.09	1.0	8.1 – 16.5	0.0322	16	0.25 – 0.32

Table 3.2 TDC performance comparison.

The solution achieves a resolution of approximately 0.7 ps at the expense of slightly increased area compared with [94]. The approach in [129–131] uses fourfold local passive interpolation to achieve a sub-gate delay resolution of 4.7 ps and a good variation robustness. Another interesting TDC architecture was presented in [132, 133] and implemented in an ADPLL [34]. A ring oscillator consisting of 47 delay stages with a single shot precision of 6 ps and seven counters achieve a first order noise shaping effect by preserving its previous state, i.e., gating the ring oscillator. The high-pass shaped quantization noise can be filtered to further enhance the resolution such that the TDC can resolve a sinusoidal time signal with 1.2 ps amplitude. In [134] a digital bang-bang DLL that guarantees a constant resolution of 16 ps over process and temperature variations is used.

TDC Operation

In the following we will describe the TDC architecture from [94] in detail, because it has been successfully embedded in a phase-domain ADPLL [6, 10, 16, 52, 110]. The TDC is part of the frequency-to-digital-phase converter, where basically a counter measures the integer part and the TDC measures the fractional part of the feedback phase signal. To be precise, the TDC measures the relation between two different clock signals, denoted as REF and CKV, where the CKV clock is significantly faster than the REF clock. In Chapter 3 we will investigate two phase-domain ADPLL architectures: the retimed reference architecture and the synchronous reference architecture. Both systems require a TDC and measure the relation, i.e., the time between the rising edge events of CKV and REF, but different

3.1 Design Challenges in an All-Digital Phase-Locked Loop

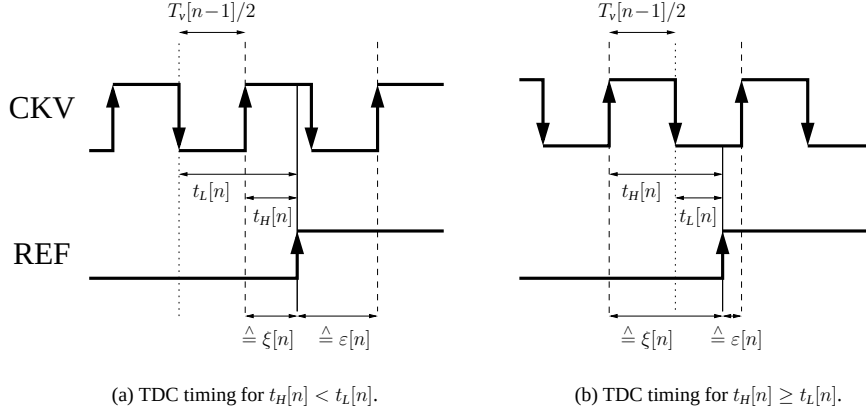


Figure 3.6 Basic TDC operation [94].

time quantities. In the retimed reference architecture the TDC measures $\varepsilon[n]^3$, which is the time between the rising edge of REF and the following rising edge of CKV normalized to one CKV period. In the synchronous reference architecture the TDC measures $\xi[n]$, which is the time t_H between the last rising edge of CKV before the rising edge of REF normalized to one CKV period, i.e.,

$$\xi[n] = \frac{t_H[n]}{T_v[n-1]}, \quad \xi[n] \in \{0, 1\}. \quad (3.8)$$

The normalization by $T_v[n-1]$ is due to the fact that the TDC measures a fractional part of the signal $N_v[n-1]$, which corresponds to the actual number of CKV cycles within one REF cycle, i.e., $T_r/T_v[n-1]$, cf. eqn. (2.11). For consistency we include the delay from eqn. (2.10) in eqn. (3.8) and the following derivations. The TDC basically has to perform two tasks: first to measure the value of $t_H[n]$ and, second, to normalize $t_H[n]$ to the CKV period $T_v[n-1]$. Since $t_H[n]$ lies between 0 and $T_v[n-1]$, the value of $\xi[n]$ is in the range between zero and one. Figure 3.6 illustrates the clock signals CKV and REF, the time signals $t_H[n]$ and $T_v[n-1]$, and the basic operation of the TDC. In Fig. 3.6 we distinguish between two cases: $t_H[n] < t_L[n]$ and $t_H[n] \geq t_L[n]$, where t_L is the time between the . While in both cases the calculation of $\xi[n]$ is similar the CKV period estimation varies, i.e.,

$$T_v[n-1]/2 = \begin{cases} t_H[n] - t_L[n] & \text{for } t_H[n] \geq t_L[n], \\ t_L[n] - t_H[n] & \text{for } t_H[n] < t_L[n]. \end{cases} \quad (3.9)$$

Under the assumption that $T_v[n-1]$ is constant over the $(n-1)$ th reference cycle, i.e., $T_v[n] = T_v[n-1]$, we can express $\varepsilon[n]$ by $\xi[n]$ and vice-versa as

$$\varepsilon[n] = \frac{T_v[n] - t_H[n]}{T_v[n-1]} = 1 - \xi[n]. \quad (3.10)$$

In the presented TDC structure the value of $\xi[n]$ is measured and $\varepsilon[n]$ is computed using eqn. (3.10), which makes the TDC suitable for both ADPLL architectures.

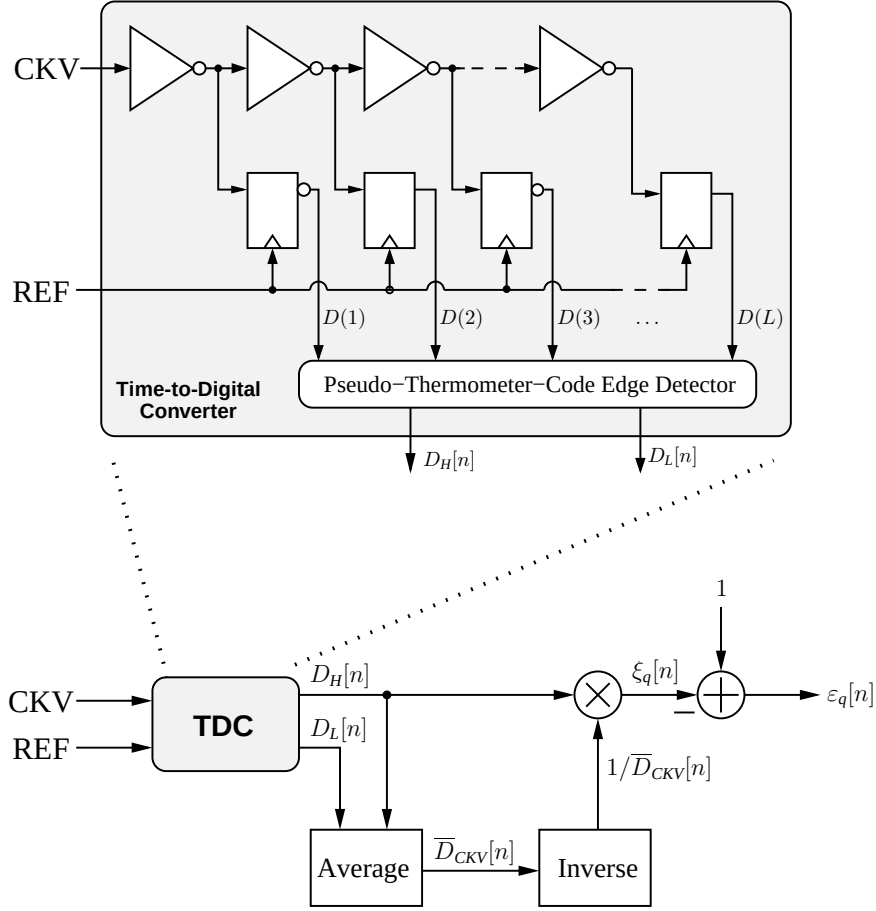


Figure 3.7 A time-to-digital converter architecture for ADPLLs from [94].

TDC Architecture

Figure 3.7 shows the investigated TDC architecture and Fig. 3.8 shows an exemplary timing diagram. The variable clock signal CKV is input to a delay line (typically implemented as a pseudo-differential inverter delay line) that produces delayed versions $D(l)$ for $l = 1, 2, \dots, L$ of the input signal. The delay of the inverters corresponds to the time resolution of the TDC Δt_{res} . The number of delay elements L must be chosen large enough, so that the delay line resolves at least one CKV cycle.

To measure the delay time the internal TDC signals $D(l)$ are sampled with the rising edge of REF. The index of the first 1 - 0 transition, denoted as digital signal $D_L[n]$, corresponds to the number of delay elements from the *falling* CKV edge to the rising REF edge. The index of the first 0 - 1 transition, denoted as digital signal $D_H[n]$, marks the number of delay elements from the *rising* CKV edge to the rising REF edge. The signals $D_H[n]$ and $D_L[n]$ are obtained from the L binary signals $D(l)$ by a pseudo-thermometer-code edge detector. The quantized values $D_H[n]$ and $D_L[n]$ represent

³The retimed reference architecture in Sec. 3.2 requires the value of $\varepsilon[n]$ on a different time bases, denoted as $\varepsilon[m]$. The principle operation, however, is not affected.

3.1 Design Challenges in an All-Digital Phase-Locked Loop

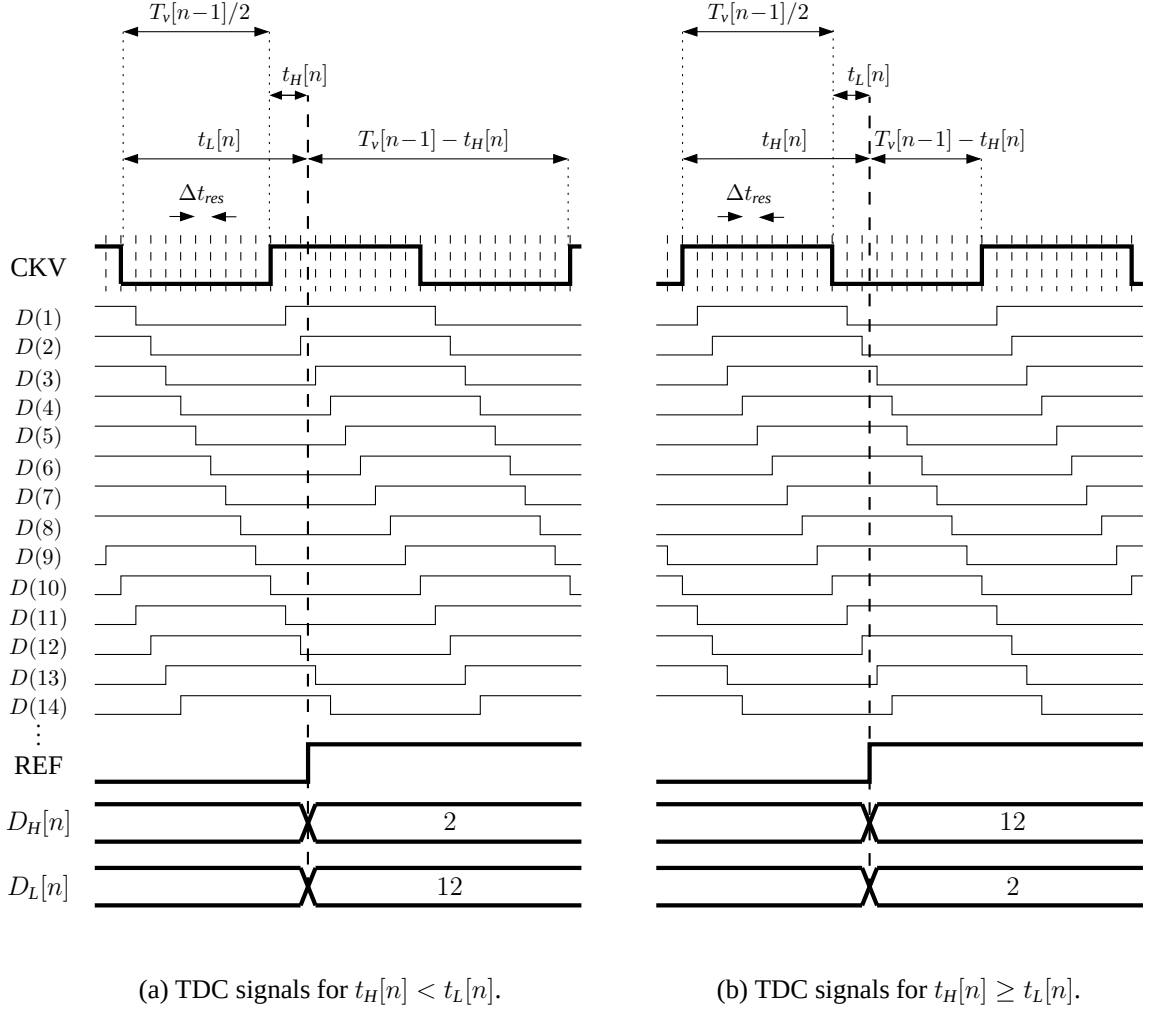


Figure 3.8 Timing diagram of the TDC architecture from [94].

the values of $t_H[n]$ and $t_L[n]$ with a resolution of Δt_{res} , i.e.,

$$\begin{aligned} D_H[n] &= \left\lfloor \frac{t_H[n]}{\Delta t_{res}} \right\rfloor, \\ D_L[n] &= \left\lfloor \frac{t_L[n]}{\Delta t_{res}} \right\rfloor \end{aligned} \quad (3.11)$$

Thus, the quantized observations of the TDC can be expressed as

$$\begin{aligned} Q_{TDC} \{t_H[n]\} &= D_H[n] \Delta t_{res}, \\ Q_{TDC} \{t_L[n]\} &= D_L[n] \Delta t_{res}. \end{aligned} \quad (3.12)$$

Replacing $t_H[n]$ and $t_L[n]$ by their digital representation $D_H[n]$ and $D_L[n]$ in eqn. (3.9), respectively, results in

$$\frac{D_{CKV}[n]}{2} = \begin{cases} D_H[n] - D_L[n] & \text{for } D_H[n] \geq D_L[n] \\ D_L[n] - D_H[n] & \text{for } D_H[n] < D_L[n]. \end{cases} \quad (3.13)$$

Thus, the half-period calculation is obtained through two quantized measurements and is therefore inaccurate. Since the value $T_v[n-1]$ occurs in the denominator in eqn. (3.8), a small deviation leads to a significant error. To increase the accuracy of eqn. (3.13) we average over several measurements⁴ of $\frac{D_{CKV}[n]}{2}$, i.e.,

$$\overline{D}_{CKV}[n] = \frac{2}{N_{avg}} \sum_{k=n-N_{avg}}^n \frac{D_{CKV}[k]}{2} \quad (3.14)$$

where N_{avg} is the number of REF cycles used for averaging. Thus, the CKV period measurement of the TDC can be expressed as

$$\overline{Q}_{CKV}\{T_v[n-1]\} = \overline{D}_{CKV}[n]\Delta t_{res}. \quad (3.15)$$

Finally the measured value $\xi_q[n]$ of $\xi[n]$ from eqn. (3.8) using eqn. (3.12) instead of $t_r[n]$ and eqn. (3.15) instead of $T_v[n]$ gives

$$\xi_q[n] = \frac{Q_{TDC}\{t_H[n]\}}{\overline{Q}_{CKV}\{T_v[n-1]\}} = \frac{D_H[n]\Delta t_{res}}{\overline{D}_{CKV}[n]\Delta t_{res}} = \frac{D_H[n]}{\overline{D}_{CKV}[n]}. \quad (3.16)$$

Note that the TDC is an analog component and that the delay line suffers from temperature and voltage drifts. Thus, the TDC resolution Δt_{res} is time-varying and would require some kind of calibration. Since in eqn. (3.16) both, $t_H[n]$ and $T_v[n-1]$ are expressed in multiples of the TDC resolution Δt_{res} any variations of the delay affect both values identically and therefore the terms Δt_{res} cancel each other in eqn. (3.16). Finally, the quantized value $\varepsilon_q[n]$ of $\varepsilon[n]$ is given by

$$\varepsilon_q[n] = 1 - \xi_q[n]. \quad (3.17)$$

An interesting approach to reduce the TDC quantization error has been proposed in [135]. The basic idea is the perform more than one TDC measurement within one reference cycle, by delaying one clock, such that the quantization error varies. In that sense the principle of oversampling is employed to improve the TDC performance.

3.2 Retimed Reference All-Digital Phase-Locked Loop Architecture

In this section we analyze a prominent phase-domain ADPLL architecture, which was introduced in [47], iteratively refined, e.g., [48, 52, 54], and finally presented in [6]. The key principle of the architecture is the phase-domain operation, which was first presented for a digital PLL in [46]. A cascade of a D/A converter and a VCO performs the digital-to-frequency conversion and an accumulator that counts the number of rising edges of the clock signal performs the frequency-to-digital-phase conversion [46]. In order to comply with the stringent requirements of wireless communication standards and to be able to implement the entire system using digital circuits, the structure has been essentially modified in two points. First, a DCO replaces the cascade of a D/A converter and a VCO with analog tuning and, second, the resolution of the frequency-to-digital-phase is enhanced beyond the resolution of the accumulator with the aid of a TDC. The price paid for the performance gain, see Sec. 4.4, is an increased complexity. Especially the synchronization between signals at the rate of the slow reference

⁴Note that averaging does not increase the resolution beyond the strong quantization of $D_H[n]$ and $D_L[n]$ unless the values of $t_H[n]$ and $t_L[n]$ vary sufficiently.

3.2 Retimed Reference All-Digital Phase-Locked Loop Architecture

frequency and signals at the rate of the fast ADPLL output frequency is demanding. In the retimed reference architecture this problem is solved by introducing a common non-uniform retimed reference clock. The resulting structure, its signals, and operation is discussed in the following. The relation to the uniform z -domain model from Chapter 2 is shown by introducing a non-uniform z -domain model. Finally, the mixed-signal model from Sec. 2.2 is adopted as well, to account for the non-uniform retimed reference clock.

3.2.1 Non-Uniform Structure

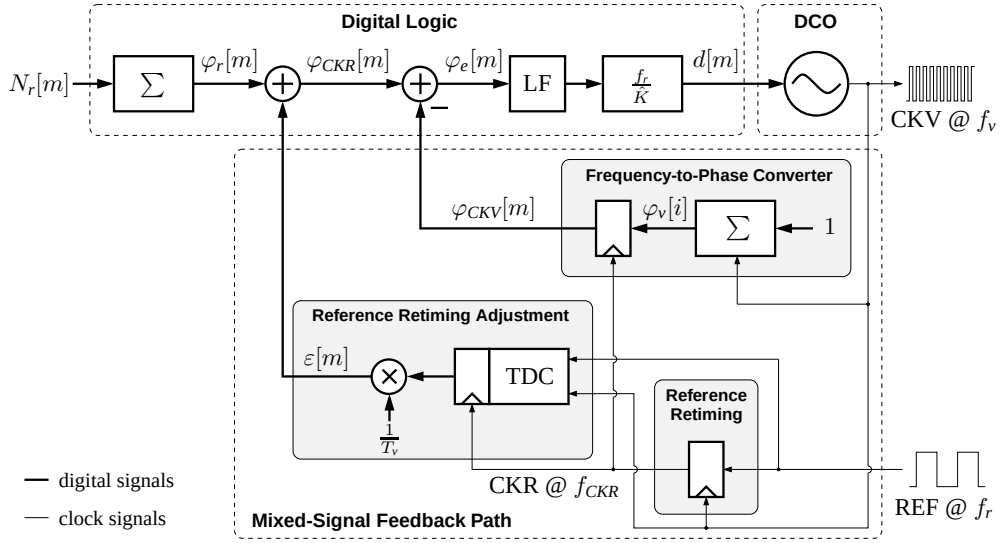


Figure 3.9 A phase-domain ADPLL architecture with retimed reference clock [48, 52, 54].

The phase-domain ADPLL architecture with reference retiming mechanism [48, 52, 54] is shown in Fig. 3.9. Apart from the sampling clock of the system, which is the retimed reference clock CKR instead of the reference clock REF, the architecture corresponds to the model in Fig. 2.1 from Sec. 2.1. In fact, the CKR clock is used to synchronize the fast CKV clock and the slow CKR clock by re-sampling REF with CKV as shown in the “reference retiming” block in Fig. 3.9 and more detailed in Fig. 3.12. The timing mechanism and the ADPLL operation are exemplarily demonstrated in Fig. 3.10 for $N_r[m] = 3\frac{3}{4}$. The retimed reference clock CKR with time index m is given by delaying the reference clock REF with frequency f_r and time index n to the next rising edge of the variable output clock CKV with frequency f_v and time index i . Table 3.3 summarizes the clock signals with corresponding frequencies and time indices. The non-uniform CKR clock is then used throughout the system and induces in consequence a time-varying reference phase $\varphi_{CKR}[m]$.

While the uniform reference phase $\varphi_r[m]$ can be computed as in eqn. (2.9) by using CKR instead of REF and time index m instead of n in eqn. (2.9), i.e.,

$$\varphi_r[m] = \varphi_r[m-1] + N_r[m-1] \quad (3.18)$$

the feedback path is more demanding in practice, because a digital phase signal needs to be extracted from the analog oscillations of the DCO output signal. One possibility is to accumulate 1 at the rate

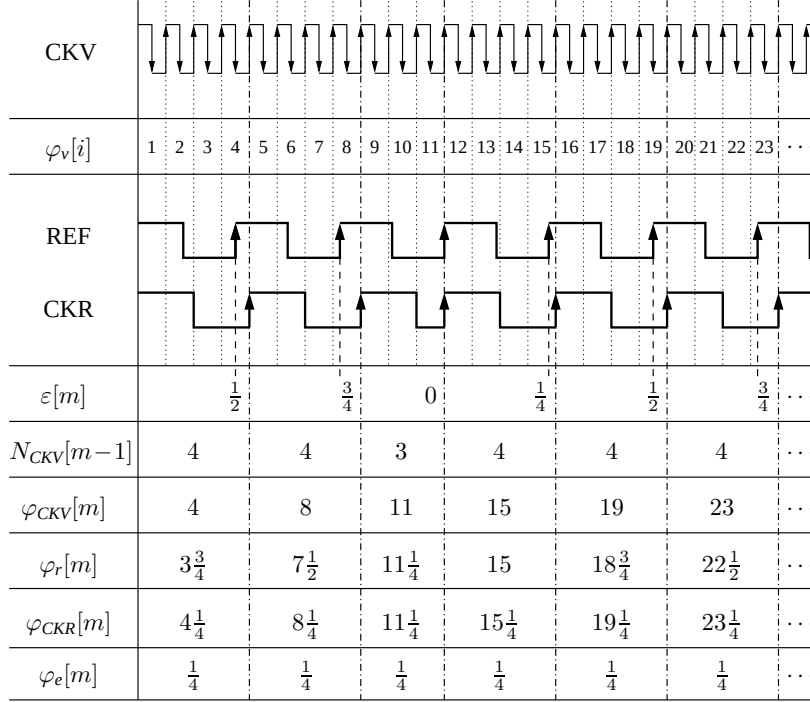


Figure 3.10 Example to illustrate the timing mechanism of the ADPLL in Fig. 3.9 for $N_r[m] = 3\frac{3}{4}$ in lock.

Signal name	Clock	Frequency	Time index
Reference frequency	REF	f_r	n
Variable frequency	CKV	f_v	i
Retimed reference frequency	CKR	f_{CKR}	m

Table 3.3 ADPLL clock signals.

of the DCO output clock CKV at frequency f_v [52], i.e.,

$$\varphi_v[i] = \varphi_v[i-1] + 1. \quad (3.19)$$

Thus, the feedback path accumulates 1 at the fast rate f_v , while the reference path accumulates $N_r[m-1]$ at the low rate f_r . Once both phase signals have equal slopes (corresponding to $N_r[m-1]$ and $N_v[m-1]$) lock is achieved. Unfortunately, the two phase-domain signals $\varphi_r[m]$ and $\varphi_v[i]$ cannot be directly compared without synchronization, because the signals have different clock domains reflected by two frequencies f_r and f_v . In practice, sampling the accumulator output signal $\varphi_v[i]$ with the reference clock REF would cause serious metastability problems [16]. Accessing the accumulator value $\varphi_v[i]$ at a time instant, where its internal state changes, results in a corrupted signal value and the ADPLL loses lock immediately. To overcome this problem, the non-uniform retimed reference clock CKR is introduced to synchronize the CKV and the REF clock domain. Hence, sampling the accumulator output $\varphi_v[i]$ with CKR yields a variable phase signal at the rate of CKR, i.e., $\varphi_{CKV}[m]$. Note, however, that the accumulator in eqn. (3.19) can only resolve an integer number of CKV cycles, so that the frequency-to-digital-phase conversion is limited in its resolution to $T_v^{ss}[n]$ in lock. This

3.2 Retimed Reference All-Digital Phase-Locked Loop Architecture

coarse quantization would lead to undesired periodic patterns in the time domain and spurs in the frequency domain. Therefore a TDC as described in Sec. 3.1.2 measures the unresolved quantity $\varepsilon[m]$ with higher resolution (see Tab. 3.2) and, in the ideal case, compensates this periodic disturbance. The retimed reference phase $\varphi_{CKR}[m]$ is given by

$$\varphi_{CKR}[m] = \varphi_r[m] + \varepsilon[m]. \quad (3.20)$$

The value $\varepsilon[m]$ is the time between the rising edge of the REF clock and following rising edge of the CKV normalized to one CKV period (a fractional part of a CKV cycle and $0 \leq \varepsilon[m] < 1$), cf. Fig. 3.12. The slope of the retimed reference phase signal $\varphi_{CKR}[m]$ from eqn. (3.20) is

$$\begin{aligned} N_{CKR}[m-1] &= \varphi_{CKR}[m] - \varphi_{CKR}[m-1] \\ &= N_r[m-1] + \varepsilon[m] - \varepsilon[m-1]. \end{aligned} \quad (3.21)$$

The phase error is the difference between both retimed phase signals, i.e.,

$$\varphi_e[m] = \varphi_{CKR}[m] - \varphi_{CKV}[m]. \quad (3.22)$$

The basic ADPLL operation and functionality can be extracted from the description above and the timing diagram in Fig. 3.10. However, for a better understanding and more insights into the structure, the retiming mechanism, and the underlying synchronization we will relate the time-varying implementation in Fig. 3.9 to the uniform z -domain model in Fig. 2.1 by introducing a non-uniform z -domain model.

3.2.2 Non-Uniform z -Domain Model

In the following we will describe the ADPLL implementation in Fig. 3.9 in analogy to the model in Fig. 2.1 and account for the non-uniform clock CKR. To obtain the non-uniform z -domain model depicted in Fig. 3.11 the uniform frequency-to-phase converter is replaced by a non-uniform frequency-to-phase converter with a non-uniform clock CKR reflected in the time-varying periods $T_{CKR}[m]$. Thus, we can rewrite the retimed reference phase $\varphi_{CKR}[m]$ as

$$\varphi_{CKR}[m] = \varphi_{CKR}[m-1] + N_{CKR}[m-1] \quad (3.23)$$

where the time-varying slope $N_{CKR}[m-1]$ corresponds to the *desired* number of CKV cycles in one CKR cycle⁵, i.e.,

$$N_{CKR}[m-1] = \frac{f_v^{ss}[m-1]}{f_{CKR}[m-1]} = \frac{T_{CKR}[m-1]}{T_v^{ss}[m-1]}. \quad (3.24)$$

Accordingly, the variable phase $\varphi_{CKV}[m]$ is given by

$$\varphi_{CKV}[m] = \varphi_{CKV}[m-1] + N_{CKV}[m-1] \quad (3.25)$$

where the time-varying value $N_{CKV}[m-1]$ corresponds to the *actual* number of CKV cycles in one CKR cycle⁶, i.e.,

$$N_{CKV}[m-1] = \frac{f_v[m-1]}{f_{CKR}[m-1]} = \frac{T_{CKR}[m-1]}{T_v[m-1]}. \quad (3.26)$$

⁵The constant slope $N_r[m-1]$ with time basis T_r in eqn. (2.2) is replaced by the time-varying slope $N_{CKR}[m]$ with bases $T_{CKR}[m]$.

⁶The (in lock) constant slope $N_v[n]$ with time-varying basis T_r in eqn. (2.11) is replaced by the (even in lock) time-varying slope $N_{CKV}[m]$ with time basis $T_{CKR}[m]$.

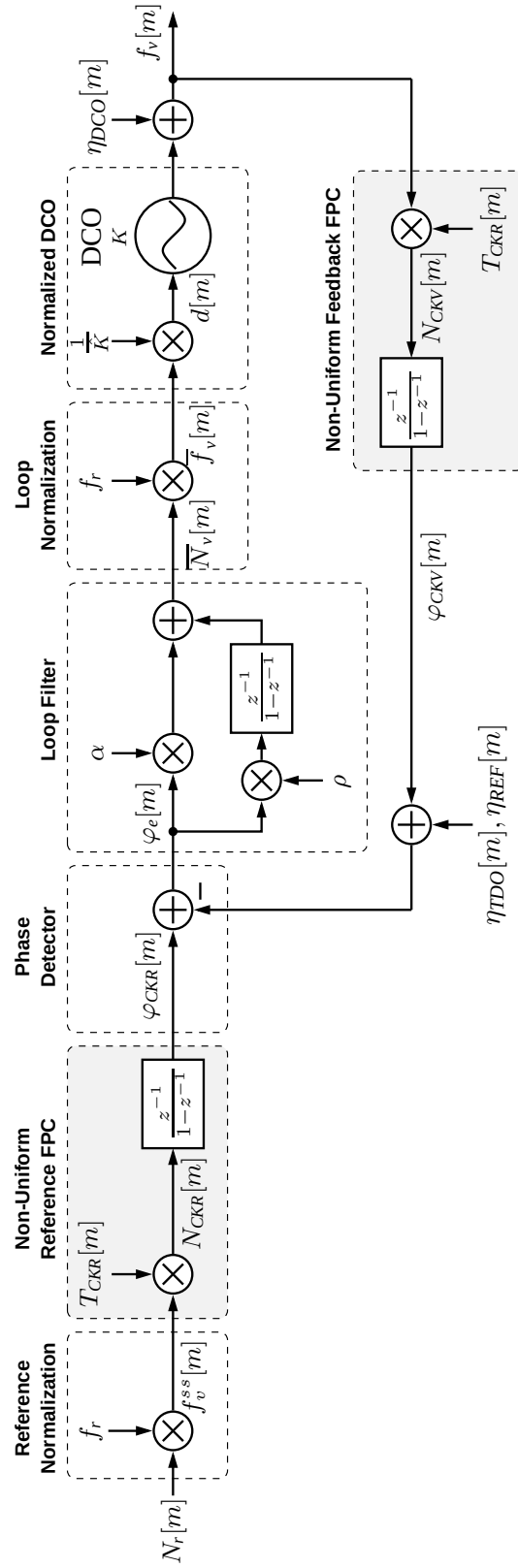


Figure 3.11 A non-uniform z-domain model of a phase-domain ADPLL.

3.2 Retimed Reference All-Digital Phase-Locked Loop Architecture

Thus, the novel frequency is based on the measured values of the elapsed retimed reference cycle. However, corresponding to eqn. (2.13) the phase error in eqn. (3.22) can be rewritten with eqn. (3.24) and eqn. (3.26) as

$$\varphi_e[m] = \varphi_e[m-1] + N_e[m-1]. \quad (3.27)$$

with

$$N_e[m-1] = N_{CKR}[m-1] - N_{CKV}[m-1]. \quad (3.28)$$

The feedback loop of the uniform model in Fig. 2.1 controls the digital tuning word $d[n]$, so that $N_v[n]$ converges to $N_r[n]$, e.g. [71] or Sec. 2.1.4. Similarly in the non-uniform model in Fig. 3.11 the value of $N_{CKV}[m]$ converges to a periodically time-varying reference $N_{CKR}[m]$ so that $N_e[m]$ in eqn. (3.27) is zero and the phase error $\varphi_e[m]$ is constant. In the following we will verify the non-uniform z -domain model by mapping the actual implementation in Fig. 3.9 into the framework of eqn. (3.23)-eqn. (3.27). The analysis shows essential details of the timing mechanism and the time-variant structure.

Acquisition

The variable phase signal $\varphi_{CKV}[m]$, its slopes $N_{CKV}[m-1]$, and the retiming error signal $\varepsilon[m]$ are basically extracted from the information encoded in the rising edges of the clock signals. In order to understand the synchronization and verify the definitions in the previous section we describe the clock signals as a sequence of time stamps corresponding to their rising edge events. For the following discussion we assume that all signals start synchronously at time 0. The time stamps of the reference clock REF are

$$t_r[m] = \sum_{l=1}^m T_r + t_0 = mT_r + t_0 \quad (3.29)$$

where t_0 is an arbitrary time offset to account for an initial phase between REF and CKV and $m \geq 0$. The time stamps of the variable clock CKV are

$$t_v[i] = \sum_{l=1}^i T_v[l-1] \quad (3.30)$$

with $i \geq 0$. Finally, the time stamps of the CKR clock are

$$t_{CKR}[m] = t_r[m] + \varepsilon[m]T_v[m-1] \quad (3.31)$$

where we have assumed, in consistence with the non-uniform z -domain model in Fig. 3.11, that $T_v[m]$ is constant over the m th CKR period. The period of CKR is simply the difference between two consecutive time stamps, i.e.,

$$T_{CKR}[m-1] = t_{CKR}[m] - t_{CKR}[m-1]. \quad (3.32)$$

Due to the retiming mechanism, which is illustrated in Fig. 3.12, the clocks CKR and CKV are synchronized, i.e., the m th rising edge time-stamp of CKR coincides with the $i[m]$ th CKV time-stamp, i.e.,

$$t_{CKR}[m] = t_v[i[m]]. \quad (3.33)$$

Sampling the high-rate accumulator value $\varphi_v[i]$ with the rising edge of CKR results in $\varphi_{CKV}[m]$, i.e.,

$$\varphi_{CKV}[m] = \varphi_v[i[m]] \Big|_{t_{CKR}[m]=t_v[i[m]]} = i[m] \quad (3.34)$$

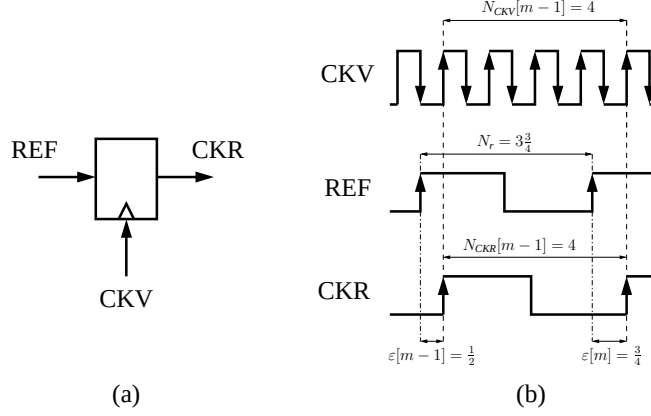


Figure 3.12 (a) Reference retiming mechanism and (b) the corresponding clock signals REF, CKV, and CKR in steady state for $N_r[m] = 3\frac{3}{4}$.

where we used the fact that $\varphi_v[i] = i$. In order to evaluate the value of $N_{CKV}[m]$ we express the retimed reference period in terms of CKV periods (by substituting eqn. (3.30) into eqn. (3.33) and the result into eqn. (3.32)), i.e.,

$$T_{CKR}[m-1] = \sum_{l=i[m-1]}^{i[m]} T_v[l-1] = (i[m] - i[m-1])T_v[m-1] \quad (3.35)$$

where we again assumed $T_v[l]$ to be constant over one CKR period. Substituting the variable period eqn. (3.35) into eqn. (3.26) and using eqn. (3.34) results in

$$\begin{aligned} N_{CKV}[m-1] &= \frac{T_{CKR}[m-1]}{T_v[m-1]} \\ &= i[m] - i[m-1] \\ &= \varphi_{CKV}[m] - \varphi_{CKV}[m-1]. \end{aligned} \quad (3.36)$$

in accordance with eqn. (3.25). Note that $\varphi_{CKV}[m]$ in eqn. (3.34) and consequently also $N_{CKV}[m-1]$ in eqn. (3.36) are integer numbers, which emphasizes the coarse granularity of the accumulator. By contrast, substituting eqn. (3.35) into eqn. (3.24) and using eqn. (3.36) yields

$$N_{CKR}[m-1] = N_{CKV}[m-1] \frac{T_v[m-1]}{T_v^{ss}[m-1]}. \quad (3.37)$$

The difference between the two slopes in eqn. (3.28) results with eqn. (3.37) in three cases

$$N_e[m-1] = N_{CKR}[m-1] \left(1 - \frac{f_v[m-1]}{f_v^{ss}[m-1]} \right) = \begin{cases} < 0 & \text{for } f_v[m-1] > f_v^{ss}[m-1], \\ = 0 & \text{for } f_v[m-1] = f_v^{ss}[m-1], \\ > 0 & \text{for } f_v[m-1] < f_v^{ss}[m-1]. \end{cases} \quad (3.38)$$

If the frequency $f_v[m-1]$ is too large, the value $N_e[m]$ will be negative, the phase error in eqn. (3.27) becomes smaller, and consequently the DCO output frequency is decreased, or vice versa. Consequently the DCO frequency is iteratively refined until the phase error is constant. The transient behavior, i.e., lock time and overshoot, depends on the loop filter characteristics.

3.2 Retimed Reference All-Digital Phase-Locked Loop Architecture

Steady State

In steady state the rising edge time stamps of the CKV clock from eqn. (3.30) simplify to

$$t_v[i] = iT_v^{ss}[i] \quad (3.39)$$

and the CKR clock from eqn. (3.31) to

$$t_{CKR}[m] = mT_r + t_0 + \varepsilon[m]T_v^{ss}[m]. \quad (3.40)$$

The retimed reference slope is obtained by substituting eqn. (3.40) into eqn. (3.35), the result into eqn. (3.24), and exploiting eqn. (2.2), i.e.,

$$N_{CKR}[m-1] = N_r[m-1] + \varepsilon[m] - \varepsilon[m-1] \quad (3.41)$$

in accordance with eqn. (3.21). The reference phase $\varphi_{CKR}[m]$ is given by accumulating $N_{CKR}[m-1]$ and solving the general recursive equation in eqn. (2.8) with $\varphi = \varphi_{CKR}$ under the assumption that $\varphi_r[0] = 0$, i.e.,

$$\begin{aligned} \varphi_{CKR}[m] &= \varphi_{CKR}[m-1] + N_r[m-1] + \varepsilon[m] - \varepsilon[m-1] \\ &= \underbrace{mN_r[m-1]}_{\varphi_r[m]} + \varepsilon[m] - \varepsilon[0]. \end{aligned} \quad (3.42)$$

in accordance with eqn. (3.23).

To evaluate the variable phase $\varphi_{CKV}[m]$ we express it as the synchronization index $i[m]$ eqn. (3.34), explicitly by substituting eqn. (3.40) and eqn. (3.39) into eqn. (3.33) and solving for $i[m]$, i.e.,

$$i[m] = \varphi_{CKV}[m] = \underbrace{mN_r[m-1]}_{\varphi_r[m]} + \varepsilon[m] + \underbrace{\frac{t_0}{T_v^{ss}[m]}}_{\varepsilon[0]} \quad (3.43)$$

where we have used eqn. (2.2). The quantity $\varepsilon[0]$ is the initial phase between CKR and CKV when lock is achieved. Combining eqn. (3.43) and eqn. (3.36) gives the variable phase in lock

$$N_{CKV}[m] = N_r[m-1] + \varepsilon[m] - \varepsilon[m-1] \quad (3.44)$$

which equals the retimed reference phase $N_{CKR}[m]$. Thus, the slope of the phase error $N_e[m]$ in eqn. (3.28) becomes zero and the phase error $\varphi_e[m]$ is constant. Figure 3.12 shows the generation of the retimed reference clock in lock and illustrates the relation given in eqn. (3.44). Note that the difference between $\varphi_{CKV}[m]$ from eqn. (3.43) and $\varphi_{CKR}[m]$ eqn. (3.23) is $\varepsilon[0]$. Thus, the initial phase difference between REF and CKV introduces an additional error in $N_e[1]$, because the value of $\varepsilon[0]$ is intrinsically captured in the variable phase $\varphi_{CKV}[1]$, but not considered, i.e., measured, in the retimed reference phase $\varphi_{CKR}[1]$. This error is insignificant, because it affects the first measurement only. Nevertheless, the initial phase difference $\varepsilon[0]$ shows that the slope of the phase signal is controlled and not the absolute phase between CKV and CKR. This highlights a key property of phase-domain ADPLLs. They control the relation between two phases, which are basically continuous time signals evaluated at discrete time stamps, while common PLLs control the relation between rising clock edges. The presented non-uniform z -domain model reflects the actual time-varying structure of the ADPLL and gives useful insights into its functionality and challenges, such as synchronization issues and the

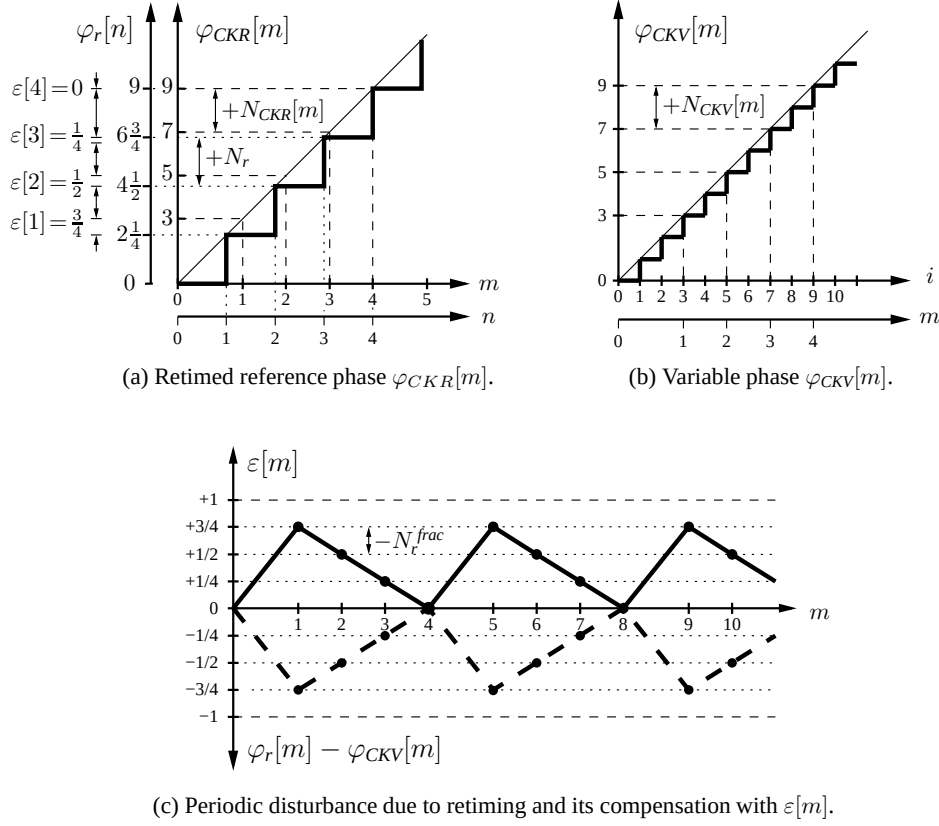


Figure 3.13 Illustration of the frequency-to-digital-phase conversion with reference retiming and $N_r[m] = 2\frac{1}{4}$ and $\varepsilon[0] = 0$.

feedback frequency-to-phase conversion by utilizing the edge events of clock signals. Furthermore, the model can be easily extended to include quantization effects and noise sources as described and analyzed in Chapter 4. However, all system-level properties, such as lock time and noise suppression capability, are best obtained from the uniform z -domain model in Fig. 2.1 in Sec. 2.1.2 due to the time-invariant transfer functions.

3.2.3 Signal and Timing Analysis

This section analyzes the periodic time-varying signal $N_{CKR}[m]$, the instantaneous period deviations of the CKR clock, and the retiming error signal $\varepsilon[m]$ depending on the frequency command word $N_r[m]$. For our further analysis we will assume ideal lock, i.e., the periodic pattern is not broken by any imperfections.

Retimed Reference Clock

In lock both slopes $N_{CKV}[m]$ and $N_{CKR}[m]$ are equal and will follow a certain pattern defined by the frequency command word $N_r[m]$. If the frequency command word is a rational number $N_r[m] = \frac{L}{M}$, where L and M are relatively prime integers, the pattern will be periodic with period M , i.e.,

$$N_{CKR}[m - M] = N_{CKR}[m] \text{ for } \forall m. \quad (3.45)$$

3.2 Retimed Reference All-Digital Phase-Locked Loop Architecture

$N_r^{frac}[m]$	L/M	Period M	Pattern of $N_{CKR}[m]$
0.1	1/10	10	AAAAAAAAAAB
0.3	3/10	10	AABAABAAAB
0.4	2/5	5	ABAAB
0.5	1/2	2	AB
0.6	3/5	5	ABABB
0.7	7/10	10	ABBABBABBB
0.8	4/5	5	ABBBB
0.9	9/10	10	ABBBBBBBBBB
0.25	1/4	4	AAAB
0.75	3/4	4	ABBB
0.333333	1/3	3	AAB
0.666666	2/3	3	ABB
0.125	1/8	8	AAAAAAAAAB
0.875	7/8	8	ABBBBBBBB

Table 3.4 Pattern for the retimed reference clock for different fractions $N_r^{frac}[m]$, where $A = N_r^{int}[m]$, $B = N_r^{int}[m] + 1$, and $N_r^{int}[m]$ is an integer.

For example, if $N_r[m]$ is $100\frac{1}{5}$ then $N_{CKR}[m]$ will be 100 for $m = 0 \dots 3$ and $N_{CKR}[m]$ will be 101 for $m = 4$ repeated periodically with $M = 5$. Averaging $N_{CKR}[m]$ over the period M results in $N_r[m]$. Table 3.4 shows the periodic pattern of $N_{CKR}[m]$ for various fractional frequency command words $N_r^{frac}[m]$. In lock the period $T_{CKR}[m]$ of the retimed reference clock CKR is directly connected with $N_{CKR}[m]$, i.e.,

$$T_{CKR}[m-1] = N_{CKR}[m-1]T_v^{ss}[m-1] = T_r + (\varepsilon[m] - \varepsilon[m-1])T_v^{ss}[m-1] \quad (3.46)$$

where we have used eqn. (3.41) and eqn. (2.2). Because $N_{CKR}[m]$ matches on average $N_r[m]$, the clock CKR periodically deviates by $(\varepsilon[m] - \varepsilon[m-1])T_v^{ss}[m-1]$ from the average period of T_r .

Retiming Error Signal

Figure 3.13 illustrates the generation of the reference phase signals $\varphi_r[n]$ and $\varphi_{CKR}[m]$ in (a), the variable phase signal $\varphi_{CKV}[m]$ in (b), their relation to $N_r[m]$, $N_{CKR}[m]$, and $N_{CKV}[m]$, and the resulting periodic disturbance and its compensation with $\varepsilon[m]$ in (c). The retiming mechanism extends each retimed reference period $T_{CKR}[m]$ to an integer multiple of one CKV period $T_v^{ss}[m]$, cf. Fig. 3.12, which can be seen as a ceiling operation to the variable phase signal

$$\varphi_{CKV}[m] = \lceil \varphi_v[m] \rceil \quad (3.47)$$

where $\lceil \cdot \rceil$ denotes the ceiling operator. In lock the variable phase $\varphi_{CKV}[m]$ converges to the retimed reference $\varphi_{CKR}[m]$ and $\varphi_v[m]$ converges towards $\varphi_r[m]$. Thus we can rewrite eqn. (3.43) in analogy to eqn. (3.20) as

$$\varphi_v[m] = \varphi_{CKV}[m] - \varepsilon[m] = mN_r[m] + \varepsilon[0] \quad (3.48)$$

end express the retiming error $\varepsilon[m]$ using eqn. (3.48) and eqn. (3.47) as

$$\begin{aligned}\varepsilon[m] &= \lceil \varphi_v[m] \rceil - \varphi_v[m] \\ &= \lceil mN_r + \varepsilon[0] \rceil - (mN_r + \varepsilon[0]) \\ &= 1 - (mN_r^{frac} + \varepsilon[0]) \bmod 1\end{aligned}\tag{3.49}$$

where mod denotes the modulo operator. As exemplified in Fig. 3.13(c), eqn. (3.49) reveals that $\varepsilon[m]$ follows a sawtooth oscillation with a maximum amplitude of 1 and a period of M samples. For $N_r^{frac}[m] < 0.5$ the sawtooth slope is negative $-N_r^{frac}[m]$ and for $N_r^{frac}[m] > 0.5$ the sawtooth slope is positive $N_r^{frac}[m]$. While the slope of the sawtooth signal can be predicted by

$$\varepsilon[m] = (\varepsilon[m-1] - N_r^{frac}[m]) \bmod 1\tag{3.50}$$

the actual values may differ due to different values of $\varepsilon[0]$. Note that $\varepsilon[0]$ corresponds to the initial relation between the reference clock REF and the DCO output clock CKV, when lock is achieved. The value of $\varepsilon[0]$ may be arbitrary, which emphasizes the fact that phase-domain ADPLLs control the digital phase signals and not the relation between two clock signals. Furthermore, different values for $\varepsilon[0]$ result in different quantization errors, which is especially critical for low fractional or integer channels as discussed in Sec. 4.4.

3.2.4 Non-Uniform Mixed-Signal Model

The ADPLL architecture with reference retiming mechanism uses the non-uniform reference clock CKR to sample (C/D conversion) the phase signal $\varphi_v(t)$ and to reconstruct (D/C conversion) the analog frequency signal $f_v(t)$. The general problem of non-uniformly sampled signals and reconstruction with a non-uniform clock are known from the theory of A/D [136] and D/A converters [76, 137, 138]. The periodically non-uniform clock produces undesired scaled and frequency shifted copies of the input signal. These spurious components impair the performance of the system. In the following we introduce a non-uniform mixed-signal model, depicted in Fig. 3.14, of the ADPLL with reference retiming to evaluate the impact of the non-uniform clock. We replace the uniform impulse train $s_{REF}(t)$ corresponding to the REF clock by a non-uniform impulse train $s_{CKR}(t)$ corresponding to the CKR clock in the mixed-signal model from Sec. 2.2. First, we describe the non-uniform impulse train $s_{CKR}(t)$, which follows a periodic pattern in steady state, in the frequency and in the time domain. Then, the effects on the sampling process, the introduced error, and its compensation with the aid of the TDC are discussed. Finally, the effects of signal reconstruction with a ZOH with periodically time-varying hold times on the output spectrum are elaborated.

Non-Uniform Reference Clock

Similar to the uniform reference clock REF in eqn. (2.37) we will describe the non-uniform reference clock CKR as an impulse train, where each impulse corresponds to a rising edge event of the CKR clock. Thus, each impulse is delayed by a certain amount of time, i.e.,

$$s_{CKR}(t) = \sum_{m=-\infty}^{\infty} \delta(t - (m + r[m])T_r)\tag{3.51}$$

where $r[m]$ are the relative time-offsets. To be specific, $r[m]$ is the time each impulse is delayed normalized to the reference period T_r . Since the relative delay of each impulse is given by $\varepsilon[m]T_v[m]$

3.2 Retimed Reference All-Digital Phase-Locked Loop Architecture

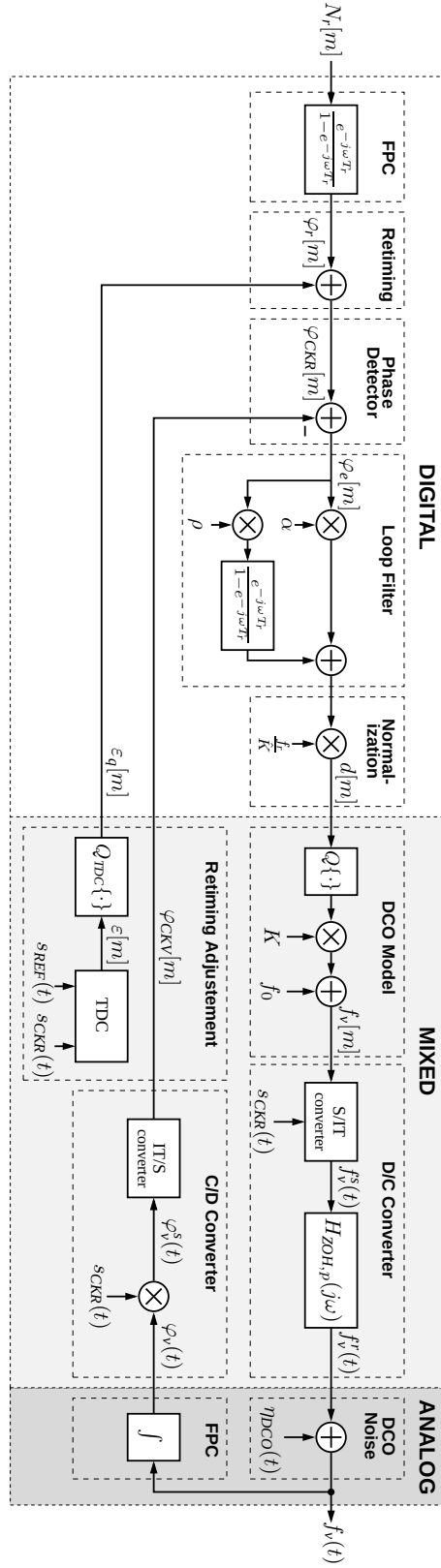


Figure 3.14 Non-uniform mixed-signal model of phase-domain ADPLLs with reference retiming.

we can express $r[m]$ as

$$r[m] = \frac{\varepsilon[m]T_v[m]}{T_r} = \frac{\varepsilon[m]}{N_v[m]} \quad (3.52)$$

where we used eqn. (2.11). In lock the retiming error $\varepsilon[m]$ follows a sawtooth signal, where the values are repeated periodically with M , see Sec. 3.3.3. Therefore we can rewrite the non-uniform impulse train eqn. (3.51) as a periodical non-uniform impulse train $s_{CKR}(t)$ with period M as

$$s_{CKR}(t) = \sum_{p=0}^{M-1} s_{CKR,p}(t). \quad (3.53)$$

with

$$s_{CKR,p}(t) = \sum_{k=-\infty}^{\infty} \delta(t - (kM + p + r[p])T_r) \quad (3.54)$$

and where $r[p]$ becomes

$$r[p] = \frac{\varepsilon[p]}{N_r[p]} \quad (3.55)$$

because $N_v[n]$ converges to $N_r[m]$. Note that the non-uniform impulse train $s_{CKR}(t)$ in eqn. (3.53) is decomposed into P uniform impulse trains $s_{CKR,p}(t)$ with relative time offsets of $(1+r[p]-r[p-1]) \times T_r$ with respect to each other. The Fourier transform of eqn. (3.53), which is derived in App. A.1, is

$$S_{CKR}(j\omega) = \frac{1}{MT_r} \sum_{p=0}^{M-1} \sum_{k=-\infty}^{\infty} 2\pi\delta\left(\omega - k\frac{2\pi}{MT_r}\right) e^{-jkp\frac{2\pi}{M}} e^{-jk\frac{2\pi}{M}r[p]}. \quad (3.56)$$

Non-Uniform Frequency-to-Digital-Phase Conversion

Since the periodically non-uniform clock CKR with its impulse train representation $s_{CKR}(t)$ is used, we can rewrite eqn. (2.48) using the general description of $s_{CKR}(t)$ from eqn. (3.51) as

$$\begin{aligned} \varphi_v^s(t) &= \varphi_v(t)s_{CKR}(t) \\ &= \varphi_v(t) \sum_{m=-\infty}^{\infty} \delta(t - (m + r[m])T_r) \\ &= \sum_{m=-\infty}^{\infty} \varphi_v((m + r[m])T_r) \delta(t - (m + r[m])T_r). \end{aligned} \quad (3.57)$$

After the impulse-train-to-sequence (IT/S) conversion we get

$$\varphi_v^r[m] = \varphi_v((m + r[m])T_r) \quad (3.58)$$

With the simplification of eqn. (2.47) and eqn. (2.8) for the steady state we can rewrite eqn. (3.58) as

$$\begin{aligned} \varphi_v^r[m] &= (m + r[m])T_r f_v^{ss} \\ &= mN_r + r[m]N_r \\ &= mN_r + \varepsilon[m] \\ &= \varphi_r[m] + \varepsilon[m] \end{aligned} \quad (3.59)$$

where we used eqn. (2.2) in the second line, eqn. (3.52) in the third line, and expressed $\varphi_r[m]$ according to eqn. (2.8) with $\varphi_0 = 0$ in the fourth line. The mixed-signal analysis is consistent with the derivation

3.2 Retimed Reference All-Digital Phase-Locked Loop Architecture

in Sec. 3.3.3. The non-uniform sampling of the feedback phase $\varphi_v(t)$ introduces a retiming error $\varepsilon[m]$. The retiming adjustment mechanism measures the value of $\varepsilon[m]$ and adds it to the reference phase $\varphi_v[m]$ as discussed in Sec. 3.2. From a signal processing point of view the ideal TDC operation can be expressed as integrating from the rising REF edge to the subsequent rising CKV edge, or equivalently rising CKR edge, and normalizing the outcome to $1/T_v[m]$. That is,

$$\frac{1}{T_v[m]} \int_{t=mT_r}^{mT_r+r[m]T_r} 1dt = \frac{1}{T_v[m]} r[m]T_r = \varepsilon[m] \quad (3.60)$$

where we used eqn. (3.52). Thus, the effects due to the non-uniform clock CKR are ideally compensated by the TDC measurement and the resulting phase error is ideally constant. In practice, however, quantization errors and noise, which is further attenuated by the loop filter, will be superimposed on that constant value. Therefore, we will consider the reconstruction of non-uniformly sampled signals with a coherently non-uniform clock. That is, the reconstruction process uses the same non-uniform clock as the sampling process.

Non-Uniform Digital-to-Frequency Conversion

In order to evaluate the coherently non-uniform reconstruction of non-uniformly sampled signals we consider the entire sampling and reconstruction system as a continuous-time system for the sake of notational simplicity. The closed loop operation, i.e., the fact that the input signal $f_v(t)$ is affected by the delayed version of the output signal $f_v^r(t)$ is neglected. The error due to the non-uniform sampling is ideally compensated by the TDC in the feedback path anyway. A time-domain description of the problem is

$$\begin{aligned} f_{v,p}^s(t) &= f_v(t) s_{CKR,p}(t) \\ &= f_v(t) \sum_{k=-\infty}^{\infty} \delta(t - (kM + p + r[p])T_r) \end{aligned} \quad (3.61)$$

where $f_v(t)$ is the underlying continuous-time signal, which is sampled to produce the intermediate digital signal $f_v[m]$ and reconstructed by M uniform impulse trains $s_{CKR,p}(t)$. The impulse train $f_v^s(t)$ contains the samples at non-uniform time stamps which are reconstructed at the same non-uniform time stamps by a periodically time-varying reconstruction filter $h_{ZOH,p}(t)$, i.e.,

$$f_v^r(t) = \sum_{p=0}^{M-1} f_v^{s,p}(t) * h_{ZOH,p}(t) \xrightarrow{\mathcal{F}} F_v^r(j\omega) = \sum_{p=0}^{M-1} F_{v,p}^s(j\omega) H_{ZOH,p}(j\omega) \quad (3.62)$$

where $F_v^r(j\omega)$, $F_{v,p}^s(j\omega)$, and $H_{ZOH,p}(j\omega)$ are the CTFTs of $f_v^r(t)$, $f_{v,p}^s(t)$, and $h_{ZOH,p}(t)$, respectively. The CTFT of eqn. (3.61) is given by

$$\begin{aligned} F_{v,p}^s(j\omega) &= \frac{1}{2\pi} (F_v(j\omega) * S_{CKR}(j\Omega)) \\ &= \frac{1}{2\pi} F_v(j\omega) * \frac{1}{MT_r} \sum_{k=-\infty}^{\infty} 2\pi \delta\left(\omega - k\frac{2\pi}{MT_r}\right) e^{-jk\frac{2\pi}{M}p} e^{-jk\frac{2\pi}{M}r[p]} \\ &= \frac{1}{MT_r} \sum_{k=-\infty}^{\infty} F_v\left(j\left(\omega - k\frac{2\pi}{MT_r}\right)\right) e^{-jk\frac{2\pi}{M}p} e^{-jk\frac{2\pi}{M}r[p]} \end{aligned} \quad (3.63)$$

where we used the CTFTs of $s_{CKR,p}(t)$ from eqn. (A.8) derived in App. A.1. The DCO, i.e., the reconstruction filter $h_{ZOH,p}(t)$, of the retimed reference architecture appears as a ZOH with variable

3.2.5 Implementation Issues in the Mixed-Signal Feedback Path

hold times, which repeat periodically in lock with period M . The hold time of the p th reference cycle is the period of the CKR clock $T_{CKR}[p]$, i.e.,

$$h_{ZOH,p}(t) = u(t) - u(t - T_{CKR}[p]). \quad (3.64)$$

The CTFT of eqn. (3.64) can be obtained by substituting eqn. (3.46) instead of T_r into eqn. (2.44), which results in

$$\begin{aligned} H_{ZOH,p}(j\omega) &= \frac{\sin\left(\frac{\omega}{2}T_{CKR}[p]\right)}{\frac{\omega}{2}} e^{-j\frac{\omega}{2}T_{CKR}[p]} \\ &= \frac{2 \sin\left(\omega(1+r[p]-r[p-1])\frac{T_r}{2}\right)}{\omega} e^{-j\omega(1+r[p]-r[p-1])\frac{T_r}{2}}. \end{aligned} \quad (3.65)$$

Finally, substituting eqn. (3.63) into eqn. (3.62) yields

$$F_v^r(j\omega) = \frac{1}{T_r} \sum_{k=-\infty}^{\infty} A_k(j\omega) F_v \left(j \left(\omega - k \frac{2\pi}{MT_r} \right) \right) \quad (3.66)$$

with

$$A_k(j\omega) = \frac{1}{M} \sum_{p=0}^{M-1} H_{ZOH,p}(j\omega) e^{-jk\frac{2\pi}{M}r[p]} e^{-jkp\frac{2\pi}{M}} \quad (3.67)$$

and where $F_v(j\omega)$ is the CTFT of $f_v(t)$ ⁷. Thus, the effects due to the coherently non-uniform reconstruction are $M-1$ spectral copies of the input spectrum $F_v(j\omega)$, weighted by $A_k(j\omega)$ and frequency-shifted by $k\frac{2\pi}{MT_r}$, for $k = 1, \dots, M-1$. In case the ADPLL is used for frequency synthesis the input signal $f_v(t)$ will be ideally a constant f_v^{ss} in steady state. For a constant input signal the reconstruction at non-uniform time stamps with non-uniform hold times does not impair the output signal. However, any noise and distortions that overlays f_v^{ss} will be replicated by the periodically non-uniform operation. Since $A_k(j\omega) \ll A_0(j\omega)$ for $k = 1, \dots, M-1$ generally holds, the effects due to non-uniform reconstruction can be neglected for frequency synthesis. In case the ADPLL is used for 2-point modulation [51], however, additional errors degrade the output signal quality.

3.2.5 Implementation Issues in the Mixed-Signal Feedback Path

The mixed-signal feedback path of the retimed reference ADPLL is in its practical implementation more complex than illustrated in Fig. 3.11 and Fig. 3.12. Two specific problems have to be considered: metastability and synchronization between the TDC and the accumulator. In digital circuits metastable states are signal values between the two well defined states of zero and one. Sampling a digital gate which is in a metastable state, may lead to an unusually long response time resulting in malfunctions or even system failures. In synchronous logic, metastability may be effectively avoided by maintaining the setup and hold time requirements of the flip-flops. In asynchronous systems, however, metastability is an omnipresent phenomenon. Synchronization appears at three points of the retimed reference architecture: in the TDC (as the inverter array with delays Δt_{res} is sampled by the asynchronous REF clock), in the feedback accumulator (as the accumulator value at rate of CKV is sampled by CKR), and in the retiming mechanism (as the CKV clock is sampled by the asynchronous REF clock).

⁷Substituting the periodically time-varying ZOH from eqn. (3.65) into eqn. (3.66) yields the same result as the IN-CON (input non-uniformly sampled and output coherently non-uniformly reconstructed) case in [138].

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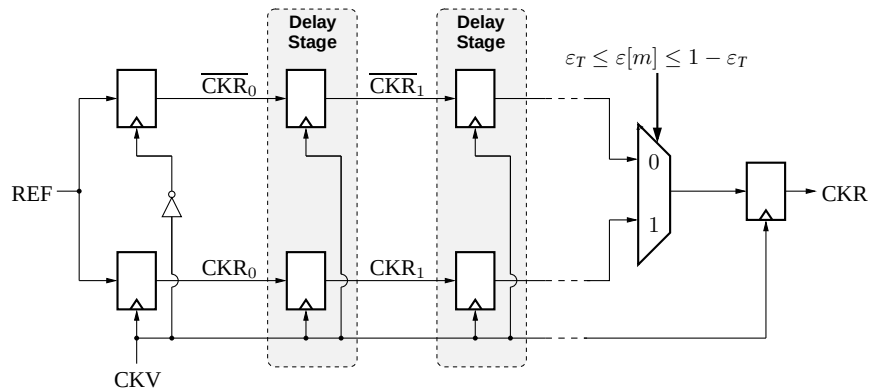


Figure 3.15 Reference retiming mechanism with metastability avoidance circuit.

Time-to-Digital Converter

Within the TDC the slow reference clock samples the intermediate signals $D(1), \dots, D(L)$, which are L delayed versions of the CKV clock. Due to the fast change of the intermediate signals and the asynchronous REF clock, metastability cannot be avoided. In case of metastability, the signal value D_H , cf. Fig. 3.8, is affected by an error of one due to the temperature code. Consequently the TDC time resolution lies in the interval between 0 and $2 \times \Delta t_{res}$, whereas the period estimation resolution lies in the interval between 0 and $3 \times \Delta t_{res}$, because D_L is affected by ± 1 as well. To overcome this problem, the probability of metastability can be reduced by minimizing the setup and hold times of the flip-flops. Shorter setup and hold time windows can be traded for increased propagation delays in CMOS flip-flops [53]. To capture the retiming error $\varepsilon[m]$ properly the metastability window can be reduced to 1 fs [139]. As a result the TDC flip-flops will have considerably longer propagation delays in comparison to the propagation delay of the accumulator, which makes the synchronization between the accumulator $\varphi_{CKV}[m]$ and the retiming error $\varepsilon[m]$ difficult and requires careful design.

Feedback Accumulator

In principle, the retiming mechanism is introduced to avoid metastability problems within the state of the feedback accumulator. The value of the signal $\varphi_v[i]$ at the rate of CKV is sampled by the CKR clock to produce $\varphi_{CKV}[m]$. Since the rising edges of CKV and CKR coincide, metastable states are prevented. The problem, however, is basically shifted to the retiming mechanism.

Retiming Mechanism

Sampling the reference clock REF by the asynchronous clock CKV may lead to metastable outputs. For clock signals metastability is even more critical, because the switching time of the output values of all logic gates driven by this clock becomes unpredictable. In such a worst case scenario the entire digital logic has to be reset. Retiming is a general solution to synchronize two clock signals with different frequencies [6]. Commonly several register stages are connected in serial, where each stage reduces the chance of metastability in an exponential fashion. Thus, a metastable state may be resolved in the current or the following CKV cycle. Although the states of the output clock CKR are well defined and metastability is avoided, a timing error of one CKV cycle, which is unacceptable for

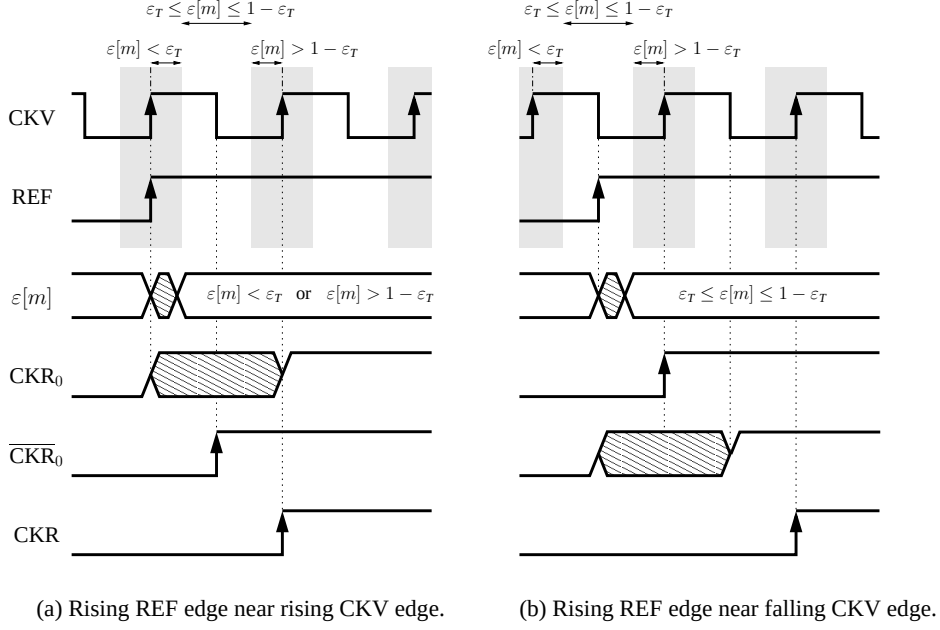


Figure 3.16 Exemplary timing diagram of the reference retiming circuit of Fig. 3.15.

ADPLLs, is potentially introduced. Therefore an enhanced solution is proposed in [6]. In essence two retimed clock signals CKR are produced, i.e., CKR_0 by over-sampling REF with CKV and $\overline{\text{CKR}}_0$ by over-sampling REF with an inverted version of CKV. In case the rising clock edges of REF and CKV happen to be close to each other, the relation between rising REF and falling CKV clock is approximately a half CKV cycle. Thus, either CKR_0 or $\overline{\text{CKR}}_0$ is in a well defined state. The final clock for the entire digital logic is chosen upon the TDC output. Since the TDC measures the relation between REF and CKV with a resolution of an inverter delay, the “better” clock can be chosen. In case the retiming error is in the range between $\frac{1}{4} < \varepsilon[m] < \frac{3}{4}$ the rising edge of REF is closer to the falling edge of CKV and otherwise closer to the rising edge of CKV. Metastability is circumvented for rising and falling REF edges, whereas $\varepsilon[m]$ measures the relation for the rising REF edge only. Therefore a midedge selector is used in practice, instead of $\varepsilon[m]$. Additional flip-flops are required to delay both clock signals until the TDC output is available, because the propagation delay of the TDC flip-flops is particularly long. Figure 3.15 shows the reference retiming mechanism with metastability avoidance enhancement and Fig. 3.16 illustrates the corresponding clock signals for both cases.

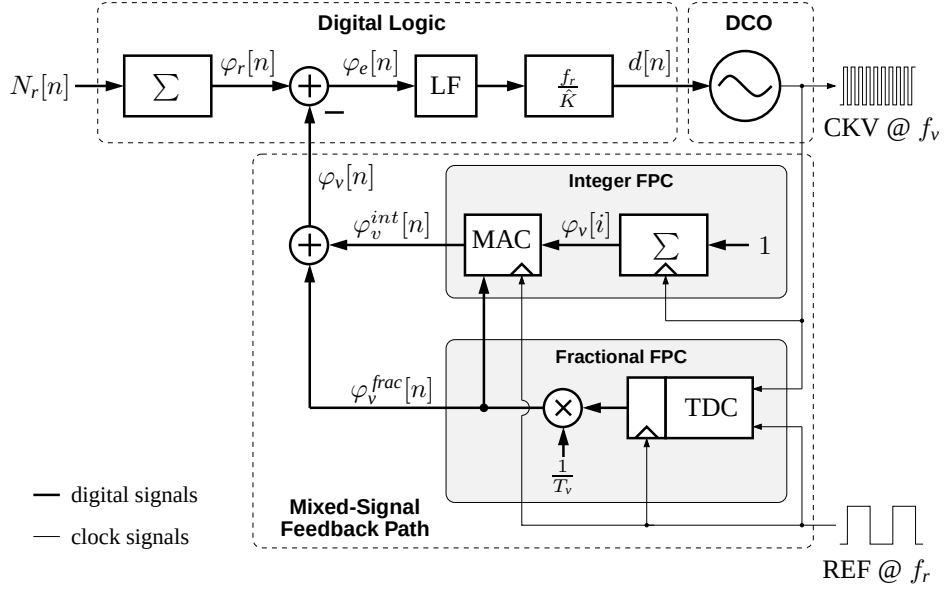


Figure 3.17 An ADPLL architecture with synchronous reference.

3.3 Synchronous Reference All-Digital Phase-Locked Loop Architecture

In this section we present an alternative phase-domain ADPLL architecture that avoids the reference retiming mechanism and operates the entire system on the uniform reference clock, i.e., synchronous with the reference. Besides the lower complexity of two as compared to three clock domains, a key feature of the proposed architecture is the improved performance concerning spurs. Injection pulling caused by substrate coupling effects introduces spurs at multiples of the reference clock in the output spectrum, see Sec. 4.6. Unfortunately, the periodically non-uniform retimed reference clock propagates the injected energy to additional frequencies and so called side-spurs appear in the output spectrum. Although careful analog design and a good isolation of the analog components from the digital circuitry may solve the problem of side-spurs, the presented solution avoids the generation of side-spurs in the first place. The impact of injection pulling on the output phase noise spectrum is further discussed in Sec. 4.6.

3.3.1 Uniform Structure

Figure 3.17 shows an alternative phase-domain ADPLL architecture that avoids retiming of the reference clock. In contrast to the retimed reference architecture in Sec. 3.2 the entire digital logic is clocked with the uniform reference clock REF⁸. While the principle structure of a phase-domain ADPLL remains, we present an alternative way to extract the phase information from the oscillating DCO output signal. Basically we implement the uniform model from Chapter 2, where we measure the values $\varphi_v[n]$ with slope $N_v[n]$ in the feedback path. The phase signals are identical to the uniform

⁸The synchronization between the CKV and the REF clock domains is done with a dedicated metastability avoidance circuit (MAC), refer to Sec. 3.3.5.

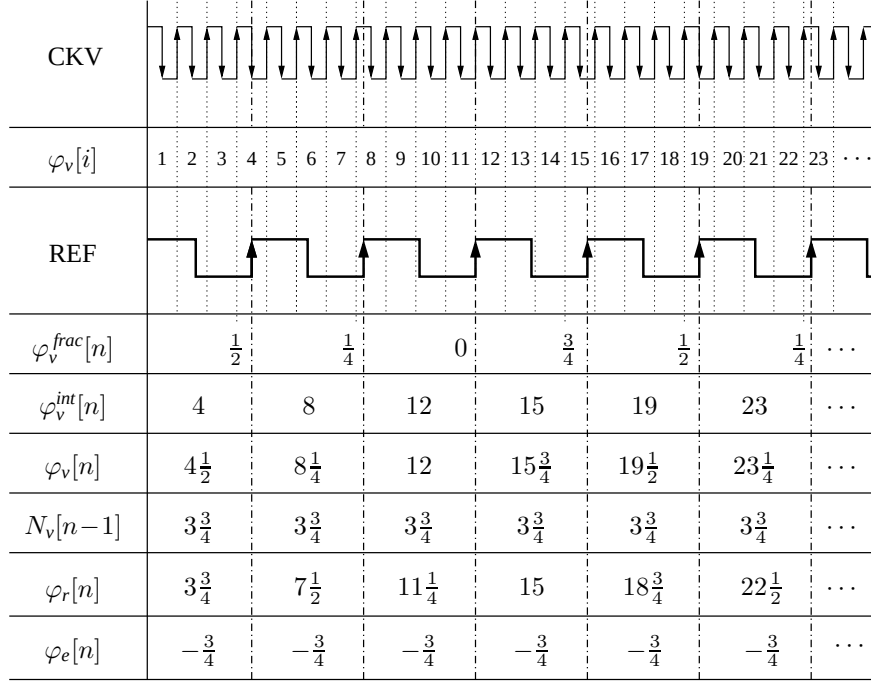


Figure 3.18 Synchronous reference timing mechanism in lock for $N_r[n] = 3\frac{3}{4}$.

z -domain model, but are repeated here for convenience. Accumulating $N_v[n]$ at the rate of the uniform reference frequency to the variable phase according to eqn. (2.10) leads to

$$\varphi_v[n] = \varphi_v[n-1] + N_v[n-1] \quad (3.68)$$

which can be compared without retiming to the uniform reference phase, which is given by

$$\varphi_r[n] = \varphi_r[n-1] + N_r[n-1]. \quad (3.69)$$

In steady state the value of $N_v[n]$ equals $N_r[n]$ and may be a fractional number consisting of an integer part $N_v^{int}[n]$ and a fractional part $N_v^{frac}[n]$. To measure the variable phase $\varphi_v[n]$ we split the computation into an integer part $\varphi_v^{int}[n]$ and a fractional part $\varphi_v^{frac}[n]$, i.e.,

$$\begin{aligned} \varphi_v[n] &= \varphi_v^{int}[n] + \varphi_v^{frac}[n] \\ &= \varphi_v^{int}[n-1] + N_v^{int}[n-1] + \varphi_v^{frac}[n-1] + N_v^{frac}[n-1]. \end{aligned} \quad (3.70)$$

While the principle operation matches eqn. (3.70), it is in practice more convenient to accumulate $N_v^{int}[n-1] + 1$ in the integer path and $N_v^{frac}[n-1] - 1$ in the fractional path. Consequently the integer part $\varphi_v^{int}[n]$ is determined by accumulating 1 at rate CKV and sampling its value with the reference clock REF (instead of CKR in the retimed reference architecture Fig. 3.9). As illustrated in Fig. 3.19, counting the number of rising edge events corresponds to the integer number of CKV periods $N_v^{int}[n-1]$ plus one, resulting in

$$\varphi_v^{int}[n] = \varphi_v^{int}[n-1] + (N_v^{int}[n-1] + 1). \quad (3.71)$$

3.3 Synchronous Reference All-Digital Phase-Locked Loop Architecture

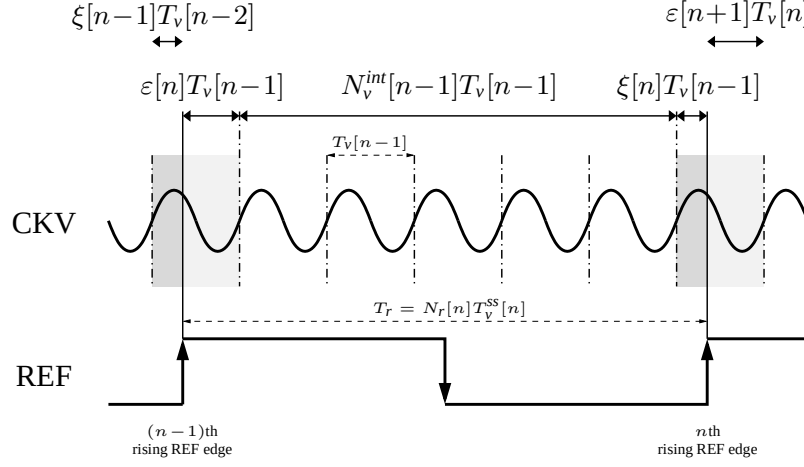


Figure 3.19 Relation between the reference clock REF and the variable clock CKV.

Determining the fractional part $\varphi_v^{frac}[n]$ is more difficult, because the clocks REF and CKV are not synchronized. Figure 3.19 illustrates the variable clock CKV for one reference cycle REF, where the fractional part $N_v^{frac}[n-1]$ consists of the fractional part $\varepsilon[n]$ at the beginning of one REF cycle and the fractional part $\xi[n]$ at the ending of one REF cycle, i.e.,

$$N_v^{frac}[n-1] = \varepsilon[n] + \xi[n] \quad (3.72)$$

where $\xi[n]$ is the time between the rising edge of CKV and the following rising edge of REF normalized by T_v , while $\varepsilon[n]$ is the time between the rising edge of REF and the following rising edge of the CKV clock normalized by T_v . One way to compute the fractional part $N_v^{frac}[n-1]$ is to measure $\varepsilon[n]$ and $\xi[n]$ with a time-to-digital converter (TDC). Alternatively we can exploit a different relation. Under the assumption $T_v[n] \approx T_v[n-1]$, we can relate $\varepsilon[n]$ and $\xi[n-1]$, cf. Fig. 3.19, i.e.,

$$\varepsilon[n] \approx 1 - \xi[n-1]. \quad (3.73)$$

The same relation as in eqn. (3.73) has also been used for the internal computation of $\varepsilon[n]$ within the TDC, i.e., see eqn. (3.10) in Sec. 3.1.2. However, in contrast to the retimed reference architecture the synchronous reference architecture updates the tuning word during one CKV cycle so that during the locking process the approximation in eqn. (3.73) introduces an insignificant error, which is getting smaller as $T_v[n]$ approaches $T_v^{ss}[n]$. Finally in steady state eqn. (3.73) will be exact. Accumulating $(N_v^{frac}[n] - 1)$ with eqn. (3.72) and eqn. (3.73) and solving the recursive relation yields the fractional part of the variable phase $\varphi_v^{frac}[n]$, i.e.,

$$\begin{aligned} \varphi_v^{frac}[n] &= \varphi_v^{frac}[n-1] + (N_v^{frac}[n-1] - 1) \\ &= \varphi_v^{frac}[n-1] + (\xi[n] - \xi[n-1]) \\ &= \varphi_v^{frac}[n-2] + (\xi[n] - \xi[n-1]) + (\xi[n-1] - \xi[n-2]) \\ &= \varphi_v^{frac}[n-2] + \xi[n] - \xi[n-2] \\ &\vdots \\ &= \varphi_v^{frac}[0] + \xi[n] - \xi[0] \end{aligned} \quad (3.74)$$

The initial value $\varphi_v^{frac}[0]$ is zero and the initial value of $\xi[0]$ is the relation between the clock signals CKV and REF, when lock is achieved. These constant values can be neglected, because a constant phase error will be compensated by the control loop. Hence we can simplify eqn. (3.74) to

$$\varphi_v^{frac}[n] = \xi[n]. \quad (3.75)$$

Figure 3.18 illustrates the synchronous reference timing mechanism in lock for $N_r[n] = 3\frac{3}{4}$. The differences to the retimed reference architectures is, besides the absence of the retimed reference clock, a simplified feedback path of the system, as discussed in the next section. However, the key advantage of the synchronous reference architecture in comparison to the retimed reference architecture is the use of a uniform reference clock for the entire system. In Chapter 4 we will show that the non-uniform reference clock CKR introduces spurs, denoted as side spurs, which are not evident when the entire ADPLL is clocked with the uniform reference clock REF.

For simulations a high-level time-domain simulation environment according to [86] has been used. Since the proposed synchronous reference architecture is slightly different to the retimed reference architecture, the model has been modified. The change of the tuning word $d[n]$ may occur at any time during a variable period $T_v[i]$ (shown in grey in Fig. 3.19). The period $T_v[i]$ is affected by two frequencies $f_v[n-1]$ corresponding to $d[n-1]$ and $f_v[n]$ corresponding to $d[n]$. The n -th CKV cycle is advanced to $\xi[n-1]T_v[n-1]$, when the positive edge of the REF clock triggers a change of the DCO frequency. In consequence, the oscillator completes its remaining cycle, i.e., $(1 - \xi[n-1])T_v[n]$, with the updated output frequency $f_v[n]$. Therefore, the DCO period $T_v[i]$ with an intermediate frequency change is modeled as

$$T_v[i] = \xi[n-1]T_v[n-1] + (1 - \xi[n])T_v[n]. \quad (3.76)$$

Transient effects like AM-to-PM noise conversion are not considered.

3.3.2 Relation to the Uniform z-Domain Model

The idea behind the synchronous reference architecture is to get close to the uniform z -domain model from Sec. 2.1. Thus the differences between the model and the actual implementation are minor ones. In contrast to the z -domain model, the implementation separates the feedback phase into an integer part $\varphi_v^{int}[n]$ and a fractional part $\varphi_v^{frac}[n]$. For a better understanding of the timing and the synchronization between the clock domains CKV and REF, we express the system with respect of its clock signals REF using $t_r[n]$ from eqn. (3.29) and CKV using $t_v[i]$ from eqn. (3.30). Although the following derivations provide a slightly different point of view, the essential results agree with Sec. 3.3.1.

The sampling process of the feedback accumulator by the reference clock REF can be expressed as

$$\varphi_v^{int}[n] = \varphi_v[i[n]] \Big|_{t_v[i[n]] \leq t_r[n]}.$$

A rising edge of REF may appear at any time during one CKV cycle. Hence, the last updated value of the accumulator value, denoted as $\varphi_v[i[n]]$, is sampled. The actual relation between CKV and REF can be described as

$$t_r[n] = t_v[i[n]] + \xi[n]T_v[i[n-1]] \quad (3.77)$$

where $i[n]$ corresponds the last rising edge of CKV before the n th rising edge of REF occurs. The sampled feedback accumulator value is an integer number and corresponds to the index $i[n]$, i.e.,

$$\varphi_v^{int}[n] = \varphi_v[i[n]] = i[n]. \quad (3.78)$$

3.3 Synchronous Reference All-Digital Phase-Locked Loop Architecture

Substituting eqn. (3.78) into eqn. (3.71) and solving for the slope $(N_v[n-1] + 1)$ gives

$$N_v^{int}[n-1] + 1 = \varphi_v^{int}[n] - \varphi_v^{int}[n-1] = i[n] - i[n-1]. \quad (3.79)$$

The result is not surprising and matches basically the analysis eqn. (3.36) with exception of the time index n instead of m . Expressing one reference period T_r in form of variable periods T_v gives

$$\begin{aligned} T_r &= t_r[n] - t_r[n-1] \\ &= t_v[i[n]] + \xi[n]T_v[i[n-1]] - t_v[i[n-1]] - \xi[n]T_v[i[n-2]] \\ &= \sum_{l=i[n-1]}^{i[n]} T_v[l-1] + \xi[n]T_v[i[n-1]] - \xi[n-1]T_v[i[n-2]] \\ &= (i[n] - i[n-1])T_v[n-1] + \xi[n]T_v[n-1] - \xi[n-1]T_v[n-2]. \end{aligned} \quad (3.80)$$

where we have used eqn. (3.77) in the second line, eqn. (3.30) in the third line, and assumed $T_v[l]$ to be constant over one REF cycle. Plugging eqn. (3.80) into the definition of $N_v[n-1]$ from eqn. (2.11) yields

$$N_v[n-1] = \frac{T_r}{T_v[n-1]} = \underbrace{i[n] - i[n-1]}_{N_v^{int}[n]+1} + \underbrace{\xi[n] - \xi[n-1]}_{\approx (N_v^{frac}[n-1]-1)} \frac{T_v[n-2]}{T_v[n-1]}. \quad (3.81)$$

Thus, the actually measured value slightly differs from the value $N_v[n]$ used in the uniform z -domain model. The measurement error of $\xi[n-1] \left(1 - \frac{T_v[n-2]}{T_v[n-1]}\right)$ arises from the approximation in eqn. (3.73), because the DCO frequency is changed during a variable period. Since typically $N_v \gg 1$ holds and the measurement of $\xi[n]$ is affected by quantization errors anyway, the systematic error during locking can be neglected. As $f_v[n]$ approaches $f_v^{ss}[n]$ the measurement of $N_v[n-1]$ becomes more and more accurate. Finally in lock the measurement of $N_v[n-1]$ is exact. That is, substituting $T_v^{ss}[n-1]$ instead of $T_v[n-1]$ into eqn. (3.81) gives the slope of the variable phase in lock, i.e.,

$$N_v[n-1] = \underbrace{i[n] - i[n-1]}_{N_v^{int}+1} + \underbrace{\xi[n] - \xi[n-1]}_{N_v^{frac}-1}. \quad (3.82)$$

In lock we can express the synchronization index $i[n]$ and consequently $\varphi_v[n]$ explicitly by evaluating eqn. (3.77). That is, plugging the steady state CKV time stamps from eqn. (3.39) and the REF time stamps from eqn. (3.29), T_v^{ss} instead of $T_v[n]$, into eqn. (3.77), and solving for $i[n]$ gives

$$i[n] = \varphi_v[n] = nN_r[n] + \underbrace{\frac{t_0}{T_v^{ss}}}_{\xi[0]} - \underbrace{\xi[n]}_{\varphi_v^{frac}[n]} \quad (3.83)$$

where we also used the definition of $N_r[n]$ from eqn. (2.2). The signal $\xi[0]$ corresponds to the initial relation between the reference clock REF and the variable clock CKV, when lock is achieved. Rewriting $\varphi_v[n]$ in lock with eqn. (3.70) and eqn. (3.83) gives

$$\varphi_v[n] = \varphi_v^{int}[n] + \varphi_v^{frac}[n] = nN_r[n] + \xi[0]. \quad (3.84)$$

Finally, substituting eqn. (3.83) into eqn. (3.82) proves that $N_v[n]$ matches $N_r[n]$ in steady state, i.e.,

$$N_v[n-1] = (N_r[n-1] - \xi[n] + \xi[n-1]) + (\xi[n] - \xi[n-1]) = N_r[n]. \quad (3.85)$$

3.3.3 Signal and Timing Analysis

Except for the TDC output signals $\varphi_r^{frac}[n]$ and $\xi[n]$, all signals are described in Sec. 2.1. Since the TDC has limited resolution, $\xi[n]$ will suffer from quantization effects. Consequently the signal characteristics of $\xi[n]$ will be important for the performance analysis in Chapter 4. In order to characterize the signal $\xi[n]$ in lock, we reinvestigate the integer phase in eqn. (3.77). The integer part of the variable phase $\varphi_v^{int}[n]$ corresponds to a previously updated value of $\varphi_v[i]$, which has integer accuracy. Thus, the integer part is obtained by rounding $\varphi_v[n]$ off, i.e.,

$$\varphi_v^{int}[n] = \lfloor \varphi_v[n] \rfloor \quad (3.86)$$

where $\lfloor \cdot \rfloor$ denotes the floor operation. Expressing the fractional part of the variable phase $\varphi_v^{frac}[n]$ in lock gives,

$$\begin{aligned} \varphi_v^{frac}[n] = \xi[n] &= \varphi_v[n] - \varphi_v^{int}[n] \\ &= \varphi_v[n] - \lfloor \varphi_v[n] \rfloor \\ &= (nN_r + \xi[0]) - \lfloor nN_r + \xi[0] \rfloor \\ &= (nN_r^{frac} + \xi[0]) \bmod 1 \end{aligned} \quad (3.87)$$

where we used eqn. (3.75) and eqn. (3.70) in the first line, eqn. (3.86) in the second line, and eqn. (3.84) in the third line. The description in eqn. (3.87) shows that $\xi[n]$ has similar signal characteristics as $\varepsilon[m]$ in Sec. 3.2.3 apart from an inverted slope, i.e., N_v^{frac} in eqn. (3.87) instead of $1 - N_v^{frac}$ in eqn. (3.49). The analysis shows that the absolute phase of the DCO output signal, i.e., the relation between REF and CKV, can be arbitrary due to arbitrary values of $\xi[0]$. The relative phase, i.e., the slope of the variable phase, will be controlled. Thus, the exact values of $\xi[n]$ cannot be predicted; the relative values, however, are ideally given by

$$\xi[n] = (\xi[n-1] + N_r^{frac}[n]) \bmod 1. \quad (3.88)$$

3.3.4 Relation to the Uniform Mixed-Signal Model

The synchronous reference architecture in Fig. 3.17 matches the uniform z -domain model in Fig. 2.1 and consequently its signal processing chain matches the uniform mixed-signal model in Fig. 2.10. The only difference is that the frequency-to-phase conversion is split into an integer part and a fractional part, where only the fractional part is quantized. As a result we have to distinguish between phase quantization $Q_\varphi\{\cdot\}$ that quantizes the entire phase signal and TDC quantization $Q_{TDC}\{\cdot\}$ that quantizes only the fractional part of the variable phase signal. While the quantization energy is equivalent, the resulting periodic pattern varies and produces spurs at different frequencies. The spurs due to TDC quantization are discussed in Sec. 4.4.

3.3.5 Implementation Issues in the Mixed-Signal Feedback Path

As discussed in Sec. 3.2.5, the reference retiming mechanism of the retimed reference ADPLL avoids metastability issues within the sampling process of the feedback accumulator signal $\varphi_v[i]$ by introducing a common retimed reference clock. By contrast, sampling the accumulator value $\varphi_v[i]$ with the asynchronous REF clock causes metastability problems when the rising edges of CKV and REF coincide. The ADPLL loses immediately lock in case of a corrupted accumulator value. To overcome

3.3 Synchronous Reference All-Digital Phase-Locked Loop Architecture

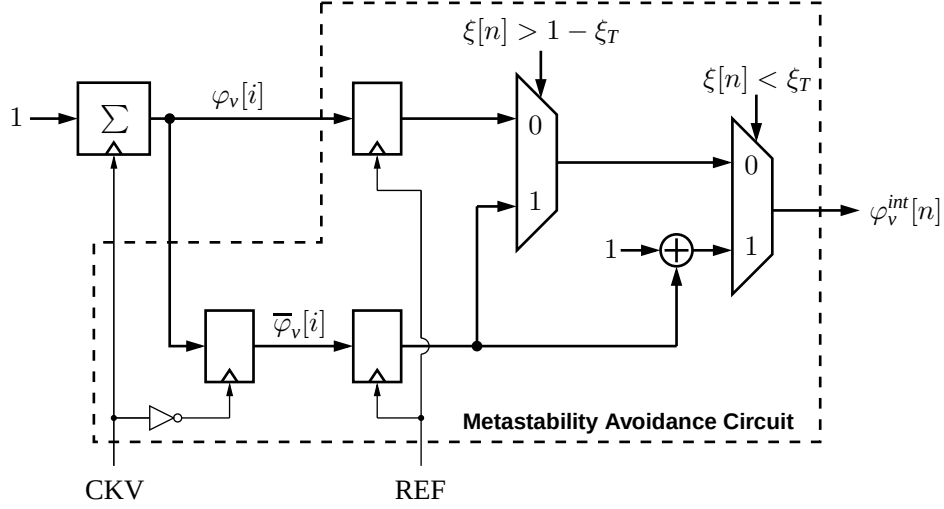


Figure 3.20 Metastability avoidance circuit (MAC).

this problem and guarantee undisturbed operation, we propose a sophisticated metastability avoidance circuit as shown as MAC block in Fig. 3.17. The solution basically uses the information of the TDC, which measures the relation between REF and CKV with high resolution, to choose between two signals: the accumulator output $\varphi_v[i]$, whose value is updated with CKV and an intermediate signal $\bar{\varphi}_v[i]$, whose value is updated with the falling edge of CKV. Hence the value $\xi[n]$ is used in a similar fashion as $\varepsilon[n]$ is used to avoid metastability within the retiming mechanism in Sec. 3.2.5. Basically we distinguish between three case, i.e.,

$$\varphi_v[m] = \begin{cases} \bar{\varphi}_v[i] + 1 & \text{for } \xi[n] < \xi_T \\ \varphi_v[i] & \text{for } \xi_T \leq \xi[n] \leq 1 - \xi_T \\ \bar{\varphi}_v[i] & \text{for } \xi[n] > 1 - \xi_T \end{cases} \quad (3.89)$$

where ξ_T is a certain threshold (e.g., $\xi_T = \frac{1}{4}$). Figure 3.20 shows the metastability avoidance circuit and Fig. 3.21 illustrated its basic principle for $\xi_T = \frac{1}{4}$. On the one hand, if the rising edge of REF appears close to the *falling* edge of CKV ($\xi_T \leq \xi[n] \leq 1 - \xi_T$ shown as white area in Fig. 3.21), the value of $\varphi_v[i]$ is well defined. On the other hand, if the rising edge of REF and *rising* edge of CKV happen to be close to each other (shown as grey area in Fig. 3.21), the well defined value of $\bar{\varphi}_v[i]$ can be used for further processing instead of the possibly corrupted value of $\varphi_v[i]$. As the value of $\bar{\varphi}_v[i]$ is updated at the falling CKV edge, i.e., when $\xi[n] = 0.5$ we have to distinguish between two cases: if $\xi[n]$ is close to 1 ($\xi[n] > 1 - \xi_T$) we will use $\bar{\varphi}_v[i]$, and if $\xi[n]$ is close to 0 ($\xi[n] < \xi_T$) we will use $\bar{\varphi}_v[i] + 1$, because the value of $\bar{\varphi}_v[i]$ has not been updated (cf. Fig. 3.21).

Besides the absence of the retimed reference clock, the proposed metastability avoidance circuit is another key advantage of the proposed synchronous reference architecture. Since the values of $\varphi_v[i]$ and $\bar{\varphi}_v[i]$ are sampled with the slow reference clock REF, their values are available during the entire reference cycle. Consequently the synchronization between the accumulator output and the TDC output is less complex in contrast to the retimed reference architecture. The drawback of the proposed solution may be additional power consumption due to re-sampling of the accumulator value with the falling edge of CKV. However, similar to the TDC, the metastability avoidance circuit is

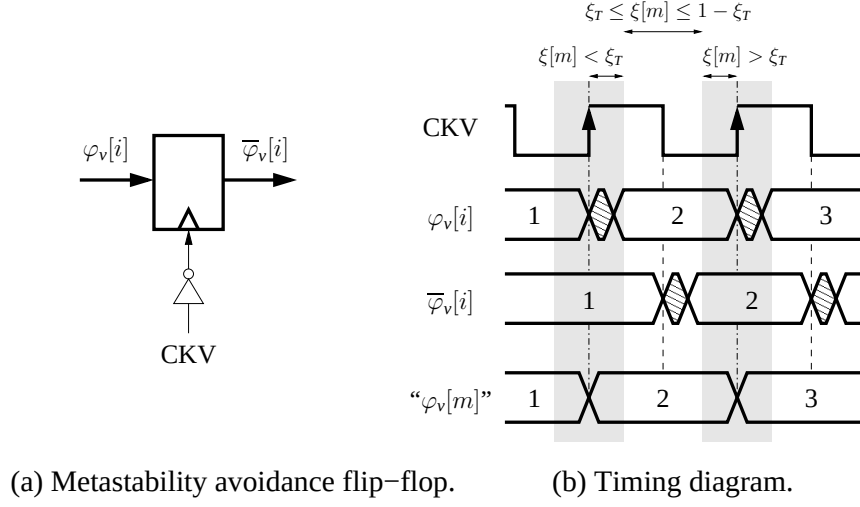


Figure 3.21 Basic metastability avoidance principle for $\xi_T = \frac{1}{4}$. The signal “ $\varphi_v[m]$ ” corresponds to the output of the metastability circuit, for the hypothetical case when the rate of REF is equal to the rate of CKV.

only needed for several CKV cycles during one REF cycle. Hence, the metastability avoidance circuit may be gated to reduce the power consumption. Furthermore, metastable states, i.e., $\xi[n]$, may be predicted from their previous values eqn. (3.88), so that the metastability circuit is not even required in each reference cycle.

3.4 Summary and Discussion

In this chapter, we discussed ADPLL architectures, their analog components, implementation issues, and design considerations. In the first part, we reviewed DCO and TDC architectures from the literature. The key feature of those analog components is their CMOS realization without any analog tuning. As analog CMOS design is challenging, the digital intensive approach seems to be the best choice in terms of complexity, design effort, area, and power consumption. The drawback of the utilization of digital logic are coarse quantization effects. In particular, the DCO suffers from a coarse frequency granularity, which is unacceptable for wireless communication standards. However, the DCO resolution can be enhanced by high rate dithering, were in essence a higher sampling rate is traded for a finer resolution. DCO dithering is a key concept in ADPLLs that allows them to compete with their analog counterparts. We have summarized DCO and TDC implementations and discussed important design considerations and trade-offs on the basis of two exemplary solutions.

In the second part of this chapter, we discussed a phase-domain ADPLL architecture. In contrast to the simplified models in Chapter 2, we have focused on the demanding mixed-signal feedback path, which converts the analog oscillation of the DCO output signal into a digital phase signal. The implementation utilizes a non-uniform retimed reference clock to synchronize the DCO output clock domain and the reference clock domain. The non-uniform operation has been extensively analyzed by re-adjusting the models from Chapter 2 to account for the non-uniform retimed reference clock. As a result, we have analytically described the periodic pattern of the reference clock in lock, its period, and the resulting phase signal characteristics as a parameter of the frequency command word.

3.4 Summary and Discussion

Furthermore, we have discussed the problem of metastability and its avoidance in the retimed reference ADPLL.

In the third part of this chapter, we proposed and analyzed an alternative ADPLL, i.e., the synchronous reference architecture, that operates the entire system with the uniform reference clock without any retiming. We have shown that the proposed operation matches the uniform models from Chapter 2 and the performance in terms of transfer functions is equivalent to the retimed reference architecture. Furthermore, as the periodically non-uniform retimed reference clock introduces a significant amount of spurs due to coupling effects on the chip, the proposed synchronous reference architecture reduces the spurious emissions, see Chapter 4. The synchronization between the two clock domains is solved by a sophisticated metastability avoidance circuit.

4

Performance Analysis of Phase-Domain All-Digital Phase-Locked Loops

A crucial performance characteristic of PLLs is their output phase noise, which is typically evaluated in the frequency domain. Within the loop many imperfect blocks cause noise or distortions, that contribute to the total output phase noise. A key advantage of ADPLLs is the immunity of digital circuits against noise. Consequently, noise sources are kept to a minimum. As a drawback of ADPLLs, besides these additive (linear) noise sources, quantization (nonlinear) effects are a major concern. Nonlinear effects typically introduce a periodic disturbance to the oscillator, which appears as spurs in the phase noise spectrum. Linear and nonlinear imperfections differ in their origin and their impact on the system. Therefore different analysis tools are required. In contrast to the analysis and characterization of linear influences, the analysis of nonlinear dynamics in feedback systems is very challenging. For this reason nonlinearities are often linearized and treated as an additive noise source with a certain noise distribution. For instance, quantization errors are modeled as white additive noise source with uniform distribution. This approach is only valid if the number of quantization levels is very large and the input signal space is large enough. In the literature often a hybrid evaluation consisting of a simplified theoretical analysis in combination with exhaustive simulations is used to capture nonlinear effects [89].

In this chapter we investigate the performance of ADPLLs under the influence of imperfections and its ability to mitigate them. For each noise source we derive the theoretical impact on the phase noise spectrum and compare it to simulation results. For the simulations we have used a high-level environment similar to [86]. Alternatively, event-driven VHDL simulators such as [49, 82, 83, 85] may be used.

Section 4.1 considers performance metrics from the literature, which are used to characterize phase noise. Within this analysis we show, on the one hand, how to model phase noise in discrete-time for simulation and modelling purposes, and, on the other hand, how to compute the phase noise spectrum from the clock edge events. Furthermore, we discuss the relation between frequency and phase noise spectra in the baseband and the passband. These relations are further discussed for oscillator noise in Sec. 4.2, which consists of phase jitter, phase wander, and flicker noise. The output spectrum due to DCO quantization with linearization exploiting $\Sigma\Delta$ noise shaping is extensively treated in Sec. 4.3. Section 4.4 considers the quantization effects of the frequency-to-phase converter, in general, and the TDCs, in particular. The analysis, which is based on the models of the previous chapters, show that the TDC cannot be considered as additive white noise source, because of its coarse quantization

4.1 Performance Metrics

and its repeating input sequence. With the aid of an open loop signal processing model, we give an overview of the frequency locations and amplitudes of TDC spurs. Section 4.5 covers jitter of the reference clock and its effect on the output spectrum. Finally, in Sec. 4.6 we analyze the mixed-signal nature of ADPLLs. The system suffers from undesired coupling effects between the digital part and the analog part. Especially the switching of the digital logic “pulls” the DCO output frequency. This periodic disturbance leads to undesired spurs in the phase noise spectrum. We derive a theoretical signal processing model to describe the spurs of the retimed reference architecture with a periodically non-uniform reference clock and the spurs of the synchronous reference architecture with a uniform reference clock. In accordance with the theory, simulation results emphasize the improved performance of the synchronous reference architecture as the spurious tones are significantly reduced [73].

The analysis of spurs due to injection pulling in Sec. 4.6 has been originally published in [73] and is extended and re-formatted for the purpose of this thesis.

4.1 Performance Metrics

In this section we introduce the phase noise power spectral density, which is typically used to characterize noise in PLLs [15, 49].

4.1.1 Power Spectral Density

Phase noise is a *random* process characterized through its statistical properties. Since a random signal $x(t)$ (similar to periodic signals) has *infinite* energy

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \infty \quad (4.1)$$

but *finite* average power

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt < \infty \quad (4.2)$$

the power spectral density (PSD) is a proper tool to describe its frequency distribution. The Wiener-Kinchin theorem states that the PSD is the Fourier transform of the autocorrelation function $r_x(\tau)$. Let $x(t)$ be a random process with *zero mean*, i.e.,

$$\mathcal{E}\{x(t)\} = 0 \quad \text{for all } t \quad (4.3)$$

where $\mathcal{E}\{\cdot\}$ denotes the expectation operator. The autocorrelation function $r_x(\tau)$ of $x(t)$ is given by [140]

$$r_x(\tau) = \mathcal{E}\{x(t)x(t+\tau)\}. \quad (4.4)$$

The assumption that the autocorrelation function depends only on the delay τ and the zero mean assumption implies that $x(t)$ is a *wide-sense stationary* (WSS) signal. Throughout the thesis we assume any random process to be wide-sense stationary. Finally, the two-sided PSD is defined as the Fourier transform of the autocorrelation function [140], i.e.,

$$S_x(\omega) = \int_{-\infty}^{\infty} r_x(\tau) e^{-j\omega\tau} d\tau. \quad (4.5)$$

The properties of the PSD are $S_x(\omega) \geq 0$ for all ω and for real-valued signals $x(t)$ the PSD is even symmetric, i.e., $S_x(\omega) = S_x(-\omega)$. Therefore, in electronic engineering in general and in the PLL

community in particular, only the positive frequencies are considered. The so called single-sideband PSD $L_x(\omega)$ is given by [15]

$$L_x(\omega) = 2S_x(\omega) \text{ for } \omega \geq 0. \quad (4.6)$$

The factor of two in eqn. (4.6) maintains the overall power of the signal $x(t)$, so that both descriptions double-sided PSD $S_x(\omega)$ and single-sided PSD $L_x(\omega)$ have equal power.

4.1.2 Phase Noise Spectrum

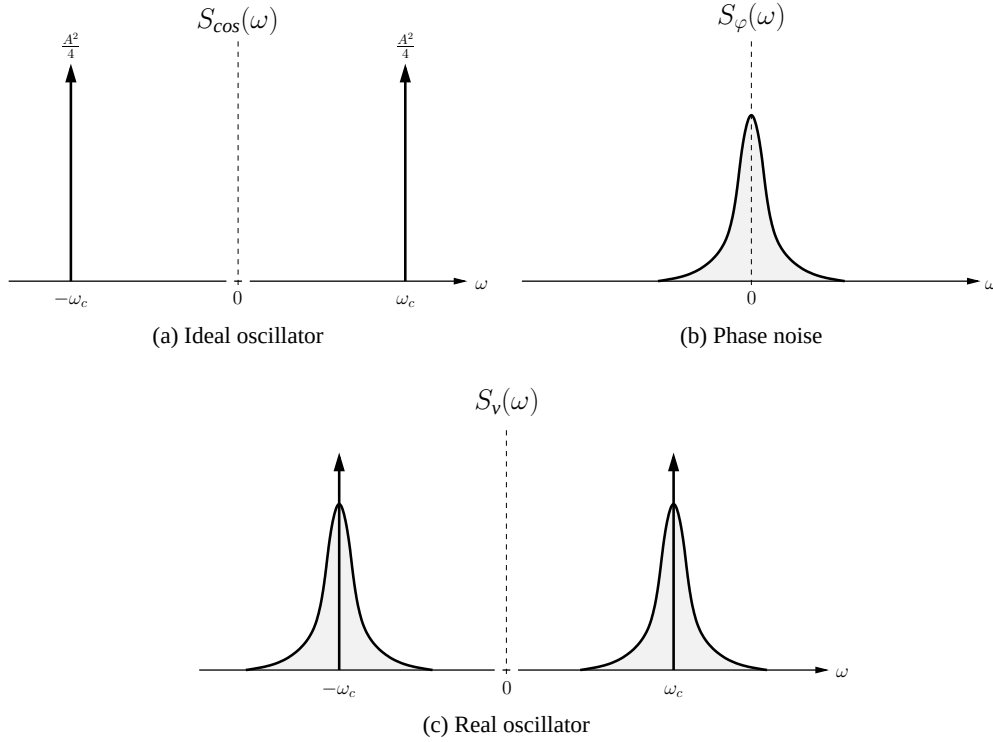


Figure 4.1 Double-sided PSD for (a) an ideal oscillator $S_{cos}(\omega)$, (b) phase noise $S_\varphi(\omega)$, and (c) an oscillator with phase noise $S_v(\omega)$.

The existence of single-sided, double-sided, passband, and baseband spectra often leads to confusion [15]. In this section we try to give a short overview of these spectra and their relation to each other. Therefore, we first investigate how the oscillator spectrum is generated. For the case of phase noise, we define the output signal of a frequency synthesizer as

$$v(t) = A \cos(\omega_c t + \varphi(t) + \varphi_0) \quad (4.7)$$

where A is the amplitude, $\omega_c = 2\pi f_v^{ss}$ is the angular frequency in steady state, $\varphi(t)$ is the phase of the signal, and φ_0 is a random but unknown initial value. In case of an ideal oscillator the phase $\varphi(t) = 0$ and the PSD of $v(t)$ is given by [49]

$$S_{cos}(\omega) = \frac{A^2}{4} [2\pi\delta(\omega + \omega_c) + 2\pi\delta(\omega - \omega_c)]. \quad (4.8)$$

4.1 Performance Metrics

The power of the carrier using Parseval's relation is

$$\int_{-\infty}^{\infty} |\cos(\omega_c t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{\cos}(\omega) d\omega = \frac{A^2}{2}. \quad (4.9)$$

In practice, however, the oscillator suffers from several imperfections and $\varphi(t)$ becomes a zero-mean random process, commonly called “phase noise”. Since the amplitude A is typically controlled by mechanisms that largely suppress fluctuations (limiter circuits), amplitude noise is significantly smaller than phase noise and is therefore generally neglected [49]. Due to the random variations the power of the oscillator spreads into nearby frequencies, such that the phase noise appears as a “skirt” around the oscillator frequency. For small fluctuations, i.e., $|\varphi(t)| \ll 1$, the output signal $v(t)$ in eqn. (4.7) can be approximated as

$$v(t) \approx A \cos(\omega_c t + \varphi_0) - A \sin(\omega_c t + \varphi_0) \varphi(t) \quad (4.10)$$

and the PSD of the right part in eqn. (4.10) becomes (see App. A.2)

$$S_v(\omega) = S_{\cos}(\omega) + \frac{1}{2\pi} (S_{\cos}(\omega) * S_{\varphi}(\omega)). \quad (4.11)$$

Figure 4.1 shows the relation of the double-sided PSD of an ideal oscillator in (a), the double-sided PSD of the phase noise in (b), and the resulting double-sided PSD of an real oscillator in (c). While the double-sided spectra are mathematically easier to deal with, in practice single-sided spectra are used to characterize phase noise. In the PLL community typically a normalized version of the single-sided passband PSD $L_v(\omega)$ plotted in a log-log scale is used. To be specific, the single-sided PSD $\mathcal{L}_{\varphi}\{\Delta\omega\}$ is the noise power in a 1 Hz bandwidth in a single sideband at frequency offset $\Delta\omega$ to the carrier frequency ω_c normalized to the carrier power in [dBc/Hz] [49], i.e.,

$$\mathcal{L}_{\varphi}\{\Delta\omega\} = 10 \log_{10} \left(\frac{\text{noise power in 1-Hz BW at } \omega_c + \Delta\omega}{\text{carrier power}} \right) \text{ in [dBc/Hz]}. \quad (4.12)$$

Practically $\mathcal{L}_{\varphi}\{\Delta\omega\}$ can be computed from the single-sided or double-sided passband spectra $L_v(\omega)$ and $S_v(\omega)$ (respectively), i.e., [15]

$$\begin{aligned} \mathcal{L}_{\varphi}\{\Delta\omega\} &= \frac{L_v(\Delta\omega + \omega_c)}{A^2/2} \\ &= \frac{S_v(\Delta\omega + \omega_c)}{A^2/4} \text{ for } \Delta\omega > 0. \end{aligned} \quad (4.13)$$

Note that the normalized single-sided passband spectrum $\mathcal{L}_{\varphi}\{\Delta\omega\}$ is related to the single-sided and double-sided phase noise baseband spectra $L_{\varphi}(\omega)$ and $S_{\varphi}(\omega)$, respectively, as illustrated in Fig. 4.2. In order to further analyze eqn. (4.13) we first derive the spectrum $L_v(\omega + \omega_c)$ by shifting the single-sided passband spectrum $L_v(\omega)$ to the baseband

$$L_v(\omega + \omega_c) = \frac{1}{2\pi} (L_v(\omega) * 2\pi\delta(\omega + \omega_c)). \quad (4.14)$$

Equivalently we can obtain $L_v(\omega + \omega_c)$ from the passband spectrum $S_v(\omega)$ by shifting the negative-frequency part of $S_v(\omega)$ to the right by ω_c , shifting the positive-frequency part to the left by ω_c , and adding both parts up [15]. That is,

$$\begin{aligned} L_v(\omega + \omega_c) &= \frac{1}{2\pi} ([u(\omega)S_v(\omega)] * 2\pi\delta(\omega + \omega_c)) + \frac{1}{2\pi} ([u(-\omega)S_v(\omega)] * 2\pi\delta(\omega - \omega_c)) \\ &= u(\omega + \omega_c)S_v(\omega + \omega_c) + u(-\omega + \omega_c)S_v(\omega - \omega_c) \end{aligned} \quad (4.15)$$

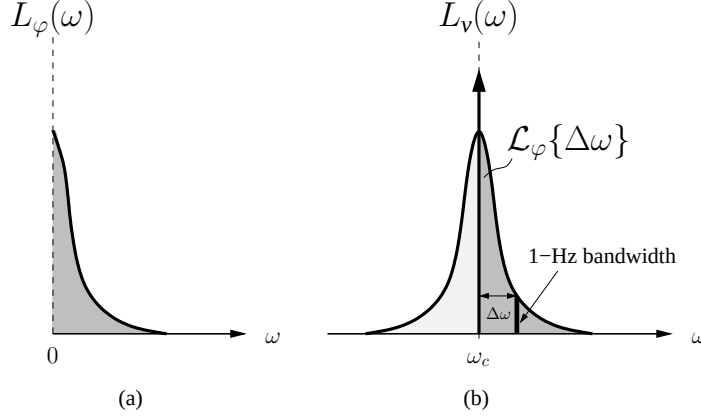


Figure 4.2 (a) Single-sided baseband phase noise PSD $L_\varphi(\omega)$ and (b) normalized passband PSD $\mathcal{L}_\varphi\{\Delta\omega\}$.

where $u(\omega)$ is the step function, i.e.,

$$u(\omega) = \begin{cases} 1 & \omega \geq 0, \\ 0 & \omega < 0. \end{cases} \quad (4.16)$$

With eqn. (4.11) and eqn. (4.8) the spectrum $L_v(\omega + \omega_c)$ from either eqn. (4.14) or eqn. (4.15) results in

$$L_v(\omega + \omega_c) = \frac{A^2}{2} 2\pi\delta(\omega) + \frac{A^2}{2} S_\varphi(\omega). \quad (4.17)$$

Substituting eqn. (4.17) into eqn. (4.13) shows the relation between the normalized single-sided passband spectrum $\mathcal{L}_\varphi\{\Delta\omega\}$ to the single-sided and double-sided baseband phase noise spectra $L_\varphi(\omega)$ and $S_\varphi(\omega)$ (respectively), i.e.,

$$\mathcal{L}_\varphi\{\Delta\omega = \omega\} \approx \frac{L_\varphi(\omega)}{2} = S_\varphi(\omega) \text{ for } \omega > 0. \quad (4.18)$$

The involved spectra are also depicted in Fig. 4.2. In eqn. (4.18) we want to emphasize the fact that these derivations are based on the approximation in eqn. (4.10). Often the single-sided baseband spectrum $\frac{L_\varphi(\omega)}{2}$ is mixed up with the single-sided normalized passband spectrum $\mathcal{L}_\varphi\{\Delta\omega\}$, which is problematic in case eqn. (4.10) is violated [15, Sec. 7.2.4].

4.1.3 Frequency Noise Spectrum

Although the phase noise spectrum is commonly used in the context of PLLs, the frequency noise spectrum provides an equivalent description of the noise process, which is sometimes more convenient and intuitive. The phase is related by integration to the frequency. Consequently the phase noise PSD can be converted into frequency noise PSD by differentiation (+20 dB/dec), or vice versa, by integration (−20 dB/dec), i.e.,

$$S_\varphi(\omega) = \frac{S_\omega(\omega)}{\omega^2}. \quad (4.19)$$

In analogy to $\mathcal{L}_\varphi\{\Delta\omega\}$ we define an equivalent normalized single-sided frequency spectrum $\mathcal{L}_\omega\{\Delta\omega\}$, which can be approximated by

$$\mathcal{L}_\omega\{\Delta\omega = \omega\} \approx \frac{L_\omega(\omega)}{2} = S_\omega(\omega) \text{ for } \omega > 0 \quad (4.20)$$

4.1 Performance Metrics

where $L_\omega(\omega)$ is the single-sided frequency noise PSD according to eqn. (4.6).

4.1.4 Computing the Phase/Frequency Noise Spectrum

In the investigated ADPLL architecture the phase information is basically extracted from the zero-crossing of the clock signals. In order to evaluate the PSD of the phase noise we consider correspondingly only the zero crossings of the DCO output signal. For this reason the actual DCO output waveform, i.e., sinusoidal or rectangular, is irrelevant. For modelling purposes it is therefore sufficient to compute the zero-crossing, e.g., [49, 86], which reduces the computational burden significantly. The simulation environment captures the time instances of the rising and falling edge events of CKV. We use the information of the CKV output clock time instances to estimate the baseband frequency noise spectrum $S_\omega(\omega)$ and transform it into the baseband phase noise spectrum $S_\varphi(\omega)$ using eqn. (4.19).

Including a noise signal $\Delta t_v[i]$ to the description of CKV in steady state eqn. (3.39) gives

$$t_v[i] = iT_v^{ss} + \Delta t_v[i]. \quad (4.21)$$

Accordingly, the noisy CKV periods are

$$T_v[i] = t_v[i+1] - t_v[i] = T_v^{ss} + \Delta T_v[i] \quad (4.22)$$

with

$$\Delta T_v[i] = \Delta t_v[i+1] - \Delta t_v[i] \quad (4.23)$$

being the period deviation. Extending the CKV period $T_v^{ss}[i]$ by $\Delta T_v[i]$ leads to an reduction of the desired CKV frequency f_v^{ss} by the frequency deviation Δf_v

$$f_v[i] = f_v^{ss}[i] - \Delta f_v[i]. \quad (4.24)$$

The frequency deviation can be expressed by the period deviation ΔT_v as [49, 60]

$$\Delta f_v[i] = f_v^{ss} - f_v[i] = \frac{1}{T_v^{ss}} - \frac{1}{T_v^{ss} + \Delta T_v[i]} = \frac{\Delta T_v[i]}{T_v^{ss}(T_v^{ss} + \Delta T_v[i])}. \quad (4.25)$$

Note that time-domain modelling of ADPLLs in [49, 60] uses a linearization of eqn. (4.25), i.e.,

$$\Delta f_v[i] \approx \frac{\Delta T_v[i]}{(T_v^{ss})^2}. \quad (4.26)$$

For the case of GSM, the linear approximation in eqn. (4.26) is accurate (approximation error is less than 26 ppm) as long as the maximal frequency deviation $\Delta f_v[i]$ is less than 10 MHz [60]. The angular frequency deviation $\Delta\omega[i]$ is

$$\Delta\omega[i] = 2\pi\Delta f_v[i]. \quad (4.27)$$

With eqn. (4.27) the discrete-time baseband double-sided PSD $S_\omega(e^{j\omega T_v^{ss}})$ and the discrete-time baseband single-sided PSD $L_\omega(e^{j\omega T_v^{ss}})$ can be estimated using discrete-time equivalents of eqn. (4.4), eqn. (4.5) and eqn. (4.6). There are several optimized algorithms in the literature to estimate the PSD from an observation vector. Within this thesis we use the Welch method [140, Sec. 2.7.2] which is implemented in the MATLAB function `pwelch`. Note that the discrete-time PSD and the continuous-time PSD are related by

$$S_\omega(e^{j\omega T_v^{ss}}) = \frac{S_\omega(\omega)}{T_v^{ss}}. \quad (4.28)$$

Therefore we have to normalize the discrete-time spectrum by the sampling frequency, which is in this case equal to the steady state frequency f_v^{ss} . The normalized single-sided frequency noise PSD $\mathcal{L}_\omega\{\Delta\omega = \omega\}$ is given by eqn. (4.20) and the normalized single-sided phase noise PSD $\mathcal{L}_\varphi\{\Delta\omega = \omega\}$ can be extracted from the frequency noise PSD using eqn. (4.19) and eqn. (4.18).

In App. A.3, an alternative method to compute directly the phase noise spectrum $\mathcal{L}_\varphi\{\Delta\omega = \omega\}$ from the CKV time instances $t_v[i]$ in eqn. (4.21) is discussed.

4.1.5 Phase/Frequency Noise Transfer Functions

In general the relation between the spectral density of the input $S_x(\omega)$ and the output $S_y(\omega)$ is given by

$$S_y(\omega) = S_x(\omega) |H(\omega)|^2 \quad (4.29)$$

where $H(\omega)$ is the transfer function of a linear time-invariant system from x to y . In case $H(\omega)$ is real valued, the relation in eqn. (4.29) holds for single-sided PSDs as well. The additive noise sources of DCO noise η_{DCO} , TDC noise η_{TDC} , and reference noise η_{TDC} are shown in Fig. 2.1. The impact on the output phase noise PSD $\mathcal{L}_\varphi\{\omega\}$ can be computed using the transfer functions $\Phi_{\eta_{DCO}}(z) = \frac{\varphi_v(z)}{\eta_{DCO}(z)}$ from eqn. (2.34) and $\Phi_{\eta_{TDC}}(z) = \frac{\varphi_v(z)}{\eta_{TDC}(z)}$ from eqn. (2.36) in combination with eqn. (4.29). The z -transform is related to the frequency response by

$$H(\omega) = H(z)|_{z=e^{j\omega T_r}}. \quad (4.30)$$

The analog spectrum $H(\omega)$ is periodic with the sampling frequency f_r of the discrete-time system $H(e^{j\omega T_r})$.

4.2 Oscillator Noise

In this section we consider the characteristics of the noise contributed by the oscillator. In ADPLLs the DCO with an digital tuning curve replaces the traditional VCO with an analog tuning curve. Therefore we have to distinguish between two imperfections: oscillator noise and quantization errors. Oscillator noise has special properties, which have been extensively studied in the literature, e.g., [49, 60, 141–157]. A general analysis of phase noise is presented in [141]. The special properties of oscillator noise [142, 143], the phase noise properties of CMOS oscillators [144, 145], AM-to-PM noise conversion in DCOs [146], the exact phase noise spectrum of DCOs with dithering [147], flicker noise [148], nonlinear phase noise [149], predicting phase noise in distributed oscillators [150, 151], noise properties in PLLs [152], and the locking conditions of analog PLLs [153] have been considered. Besides the analysis of phase noise, exact modelling of phase noise is crucial in the ADPLL/PLL design procedure. Thus, modelling of phase noise in fractional- N PLLs [154], modelling of colored noise [155], time dependent modelling of phase noise in VHDL-AMS [156], and event-driven phase noise modelling [49, 60] have been addressed. The quantization effects of the DCO are often neglected, because $\Sigma\Delta$ dithering moves the quantization errors to higher, i.e., less critical frequencies. Since the frequency-to-phase converter samples the phase signal in the investigated ADPLL structure, the frequency quantization noise is folded back to in-band phase noise [157].

4.2 Oscillator Noise

4.2.1 Theory

In the following we discuss the properties of oscillator noise, show how to model it in a discrete-time simulation environment, and derive the theoretical output phase and frequency noise spectra.

Frequency-Domain

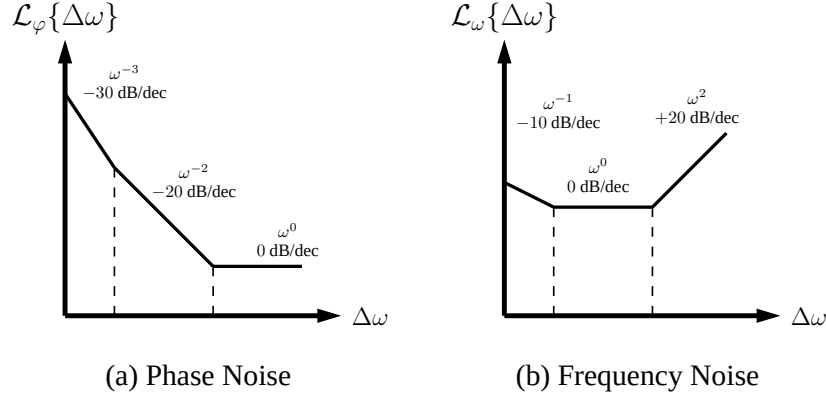


Figure 4.3 Single-sided PSD of (a) phase noise and (b) frequency noise in log-log scale.

Among the large amount of publications the most prominent reference on oscillator phase noise is probably Leeson's model [158]. The oscillator model consists of an amplifier and a resonator embedded in a positive feedback loop. The theoretical phase noise, produced by two independent sources, matches well with almost all physical oscillators [15]. An extended mathematical analysis and a discussion about “hidden parameters” in Leeson's model are presented in [159] and [160], respectively. Leeson's analysis has shown that the oscillator phase noise basically can be split in three dominant regions. Figure 4.3(a) shows the typical phase noise characteristics of the normalized single-sided phase noise spectrum $\mathcal{L}_\varphi\{\Delta\omega\}$, where the phase noise in dBc/Hz is plotted against the frequency deviation from the carrier $\Delta\omega$ in a logarithmic scale. The phase noise traverses three regions: ω^{-3} a flicker noise with a slope of -30 dB/dec (or -9 dB/octave), ω^{-2} a noise wander with a slope of -20 dB/dec (or -6 dB/octave), and ω^0 a noise jitter with 0 dB/dec. Alternatively the phase noise can be expressed as frequency noise, which is given by differentiating the phase noise spectrum, i.e., adding $+20$ dBc/dec. Accordingly, the three regions shown in Fig. 4.3(b) are: ω^{-1} a $1/f$ flicker noise with a slope of -10 dB/dec, ω^0 a frequency jitter with 0 dB/dec, and ω^2 with a slope of $+20$ dB/dec. The source of flicker noise in the ω^{-3} phase noise region is manifold, such as impurities in a conductive channel, and is an open research issue. The ω^{-2} phase noise region may be interpreted as frequency jitter, whereas ω^0 phase noise region may be interpreted as phase jitter.

Time-Domain

A time-domain description of oscillator noise is well suited for modelling and simulation of ADPLLs. Thus, the properties of jitter, wander, and flicker for a given specification have been extensively analyzed in [49, 60]. In particular, it was shown how to model jitter, wander, and flicker noise on the CKV time-stamps, i.e., on $\Delta t_v[i]$ in eqn. (4.21). That is, the disturbance $\Delta t_v[i]$ on the CKV

time-stamps consists of three contributors with different properties, i.e.,

$$\Delta t_v[i] = \Delta t_{jitter}[i] + \sum_{l=0}^i \Delta T_{wander}[l] + \sum_{l=0}^i h[l] * \Delta T_{flicker}[l] \quad \text{for } i > 0. \quad (4.31)$$

Equivalently the noise can be expressed as a period disturbance $\Delta T_v[i]$ as in eqn. (4.22), i.e., substituting eqn. (4.31) into eqn. (4.23) yields

$$\Delta T_v[i] = (\Delta t_{jitter}[i] - \Delta t_{jitter}[i-1]) + \Delta T_{wander}[i] + h[i] * \Delta T_{flicker}[i] \quad (4.32)$$

where the filter $h[i]$ is used to generate the desired properties of flicker noise. Each of the three noise types affects the CKV time instants and CKV periods in a different manner. Jitter is the instantaneous deviation of a CKV clock edge $\Delta t_{jitter}[i]$, whereas noise wander is an instantaneous CKV period deviation $\Delta T_{wander}[i]$. Flicker noise is a filtered period deviation $h[i] * \Delta T_{flicker}[i]$, where the filter $h[i]$ adds additional -10 dB/dec. The time-domain relations in eqn. (4.31) or eqn. (4.32) are used to model oscillator noise in event-driven [49] or high-level simulations [86]. Therefore, the random variables $\Delta t_{jitter}[i]$, $\Delta T_{wander}[i]$, and $\Delta T_{flicker}[i]$ are chosen from zero-mean normal distributions with variances $\sigma_{\Delta t}^2$, $\sigma_{\Delta T}^2$, and $\sigma_{\Delta 1/T}^2$, respectively. To achieve the desired slope in the phase noise spectrum, the white noise is filtered accordingly. In particular wander noise is accumulated to attenuate the noise by -20 dB/dec, flicker noise is accumulated and filtered with $h[i]$ to achieve -30 dB/dec, and phase jitter does not need additional shaping. A generic filter design procedure (based on [49]) to achieve -10 dB/dec or any other attenuation factor is presented in App. A.4.3. The normalized phase noise spectrum $\mathcal{L}_\varphi\{\Delta\omega\}$ can then be directly extracted from the noisy time signal $t_v[i]$ eqn. (4.21) as discussed in App. A.3 or via the frequency spectrum $\mathcal{L}_\omega\{\Delta\omega\}$ and the frequency signal $f_v[i]$ as shown in Sec. 4.1.4.

Alternatively to time deviation in eqn. (4.31) and period deviation in eqn. (4.32), we can model the noise process as frequency deviation signal $\Delta f_v[i]$ in eqn. (4.24), i.e.,

$$\Delta f_v[i] = \Delta f_{jitter}[i] + \Delta f_{wander}[i] + h[i] * \Delta f_{flicker}[i] \quad (4.33)$$

where the frequency noise concerning jitter $\Delta f_{jitter}[i]$, wander $\Delta f_{wander}[i]$, and flicker $\Delta f_{flicker}[i]$ are random signals with zero-mean normal distribution and variances $\sigma_{\delta f}^2$, $\sigma_{\Delta f}^2$, and $\sigma_{\Delta 1/f}^2$, respectively. Note that the nonlinear relation between frequency and period in eqn. (4.25) will affect the distribution of the random variable. To be specific, if the period deviation is a random variable with normal distribution, then the frequency deviation, will certainly *not* have a normal distribution. Since phase noise is defined in the frequency domain its time-domain distribution is not as important to us, we model the frequency deviations also as Gaussian noise. Since the phase noise is very small compared to one CKV period, the phase noise variance σ_t^2 can be approximated by the frequency noise variance σ_f^2 or vice versa, i.e.,

$$\sigma_t^2 \approx \frac{\sigma_f^2}{(f_v^{ss})^4} \quad (4.34)$$

as shown in App. A.5.

Again another possibility to model oscillator noise, is to exploit the uniform z -domain model in Fig. 2.1 or the non-uniform z -domain models in Fig. 3.11. In contrast to the phase noise model in eqn. (4.31) and the frequency noise model in eqn. (4.33), which are of the rate of the DCO output frequency f_v^{ss} , the z -domain models operate on the rate of the reference frequency f_r . The frequency noise of the oscillator is considered in $\eta_{DCO}[n]$ and can be modelled with the special characteristics

4.2 Oscillator Noise

of oscillator noise, i.e., jitter, wander, and flicker, in a similar fashion. According to the law of power addition of independent identically distributed random variables the frequency variance on rate f_r has to be scaled by [49, 71]

$$\sigma_{f_r}^2 = \frac{\sigma_f^2}{N_r} \quad (4.35)$$

where σ_f^2 corresponds to the frequency noise variance on rate f_v^{ss} and $\sigma_{f_r}^2$ is the frequency noise on rate f_r . As the oscillator noise is modelled in the digital domain with a sampling frequency of r , the phase noise spectrum can be obtained to $f_r/2$. The digital noise signal would be input to the zero-order hold of the signal reconstruction process and undergoes an additional sinc-shaped attenuation, which becomes evident at frequencies close to $f_r/2$. Oscillator phase noise is primarily critical in the in-band and the modelling error for high frequencies may be neglected.

Jitter, Wander, and Flicker

In the following we provide an overview of oscillator noise modelling, its relation in the frequency and time domain, and its relation between phase and frequency noise. For each contributor, i.e., jitter, wander, and flicker, the exact spectra $\mathcal{L}_\omega\{\Delta\omega\}$ and $\mathcal{L}_\varphi\{\Delta\omega\}$ are elaborated in the Appendix. Jitter is considered in App. A.4.2, wander is treated in App. A.4.1, and flicker is discussed in App. A.4.3. Table 4.1 summarizes the results and Fig. 4.4 illustrates the relation between the individual modelling steps. The noise generation of each contributor follows a similar pattern. First we produce a random white noise sequence from a zero-mean normal distribution with certain variance, which is σ_t^2 for the phase noise model and σ_f^2 for the frequency noise model. The generalized phase noise variance σ_t^2 becomes $\sigma_{\Delta t}^2$ for jitter, $\sigma_{\Delta T}^2$ for wander, and $\sigma_{\Delta 1/T}^2$ for flicker. Correspondingly, the generalized frequency noise variance σ_f^2 becomes $\sigma_{\delta f}^2$ for jitter, $\sigma_{\Delta f}^2$ for wander, and $\sigma_{\Delta 1/f}^2$ for flicker. In case the CKV period is significantly larger than the noise, we can use the approximation in eqn. (4.34) to convert a phase noise variance into a frequency noise variance, and vice versa. The initial power spectral densities for phase noise $\mathcal{L}_t\{\Delta\omega\}$ (which becomes $\mathcal{L}_{\Delta t}\{\Delta\omega\}$ for jitter, $\mathcal{L}_{\Delta T}\{\Delta\omega\}$ for wander, and $\mathcal{L}_{\Delta 1/T}\{\Delta\omega\}$ for flicker) and the frequency noise $\mathcal{L}_f\{\Delta\omega\}$ (which becomes $\mathcal{L}_{\delta f}\{\Delta\omega\}$ for jitter, $\mathcal{L}_{\Delta f}\{\Delta\omega\}$ for wander, and $\mathcal{L}_{\Delta 1/f}\{\Delta\omega\}$ for flicker) are white, i.e., similar to the jitter example in App. A.4.2.

The generated phase noise signal $\Delta t_v[i]$ (which becomes $\Delta t_{jitter}[i]$ for jitter, $\Delta t_{wander}[i]$ for wander, and $\Delta t_{flicker}[i]$ for flicker) or frequency noise signal $\Delta f_v[i]$ (which becomes $\Delta f_{jitter}[i]$ for jitter, $\Delta f_{wander}[i]$ for wander, and $\Delta f_{flicker}[i]$ for flicker) is shaped by a filter. The filter depends on the model (phase or frequency noise) and the noise type (jitter, wander, or flicker). In any case, the colored noise signal is added to the ideal signal iT_v^{ss} (phase noise) or f_v^{ss} (frequency noise). In case of phase noise we can directly compute the phase noise spectrum $\mathcal{L}_\varphi\{\Delta\omega\}$ as described in App. A.3 and in case of frequency noise we can directly compute $\mathcal{L}_\omega\{\Delta\omega\}$ as described in Sec. 4.1.4. The resulting spectra are related by $(f_v^{ss})^4/\Delta\omega^2$, where the term $(f_v^{ss})^4$ is due to the different variances and the term $1/\Delta\omega^2$ is due to the integration. Note that in the phase domain model we have to account for the normalization by the CKV period T_v^{ss} in eqn. (A.12) by multiplying the power spectral density with $(1/T_v^{ss})^2$.

The presented framework gives a good overview of oscillator phase/frequency noise and can be easily extended to model any colored noise.

Noise Type				
		Jitter	Wander	Flicker
Phase Noise	Model	$\Delta t_{jitter}[i] \sim \mathcal{N}(0, \sigma_{\Delta t}^2)$	$\Delta t_{wander}[i] \sim \mathcal{N}(0, \sigma_{\Delta T}^2)$	$\Delta t_{flicker}[i] \sim \mathcal{N}(0, \sigma_{\Delta 1/T}^2)$
	Variance	$\sigma_{\Delta t}^2 = \frac{\mathcal{L}_\varphi\{\Delta\omega\}}{(2\pi)^2 f_v^{ss}}$	$\sigma_{\Delta T}^2 = \frac{\mathcal{L}_\varphi\{\Delta\omega\}}{(2\pi)^2 (f_v^{ss})^3} (\Delta\omega)^2$	$\sigma_{\Delta 1/T}^2 = \frac{\mathcal{L}_\varphi\{\Delta\omega\}}{(2\pi)^2 (f_v^{ss})^3} (\Delta\omega)^3$
	Regions	ω^0 with 0 dB/dec	ω^{-2} with -20 dB/dec	ω^{-3} with -30 dB/dec
	PSD	$\mathcal{L}_\varphi\{\Delta\omega\} = (2\pi\sigma_{\Delta t})^2 f_v^{ss}$	$\mathcal{L}_\varphi\{\Delta\omega\} = (2\pi\sigma_{\Delta T})^2 (f_v^{ss})^3 \left(\frac{1}{\Delta\omega}\right)^2$	$\mathcal{L}_\varphi\{\Delta\omega\} = (2\pi\sigma_{\Delta 1/T})^2 (f_v^{ss})^3 \left(\frac{1}{\Delta\omega}\right)^3$
Frequency Noise	Model	$\Delta f_{jitter}[i] \sim \mathcal{N}(0, \sigma_{\delta f}^2)$	$\Delta f_{wander}[i] \sim \mathcal{N}(0, \sigma_{\Delta f}^2)$	$\Delta f_{flicker}[i] \sim \mathcal{N}(0, \sigma_{\Delta 1/f}^2)$
	Variance	$\sigma_{\delta f}^2 \approx \sigma_{\Delta t}^2 (f_v^{ss})^4$	$\sigma_{\Delta f}^2 \approx \sigma_{\Delta T}^2 (f_v^{ss})^4$	$\sigma_{\Delta 1/f}^2 \approx \sigma_{\Delta 1/T}^2 (f_v^{ss})^4$
	Regions	ω^2 with +20 dB/dec	ω^0 with 0 dB/dec	ω^{-1} with -10 dB/dec
	PSD	$\mathcal{L}_\omega\{\Delta\omega\} = \frac{(2\pi\sigma_{\delta t})^2}{(f_v^{ss})^3} (\Delta\omega)^2$	$\mathcal{L}_\omega\{\Delta\omega\} = \frac{(2\pi\sigma_{\Delta f})^2}{f_v^{ss}}$	$\mathcal{L}_\omega\{\Delta\omega\} = \frac{(2\pi\sigma_{\Delta 1/f})^2}{f_v^{ss}} \left(\frac{1}{\Delta\omega}\right)$

Table 4.1 Oscillator noise overview.

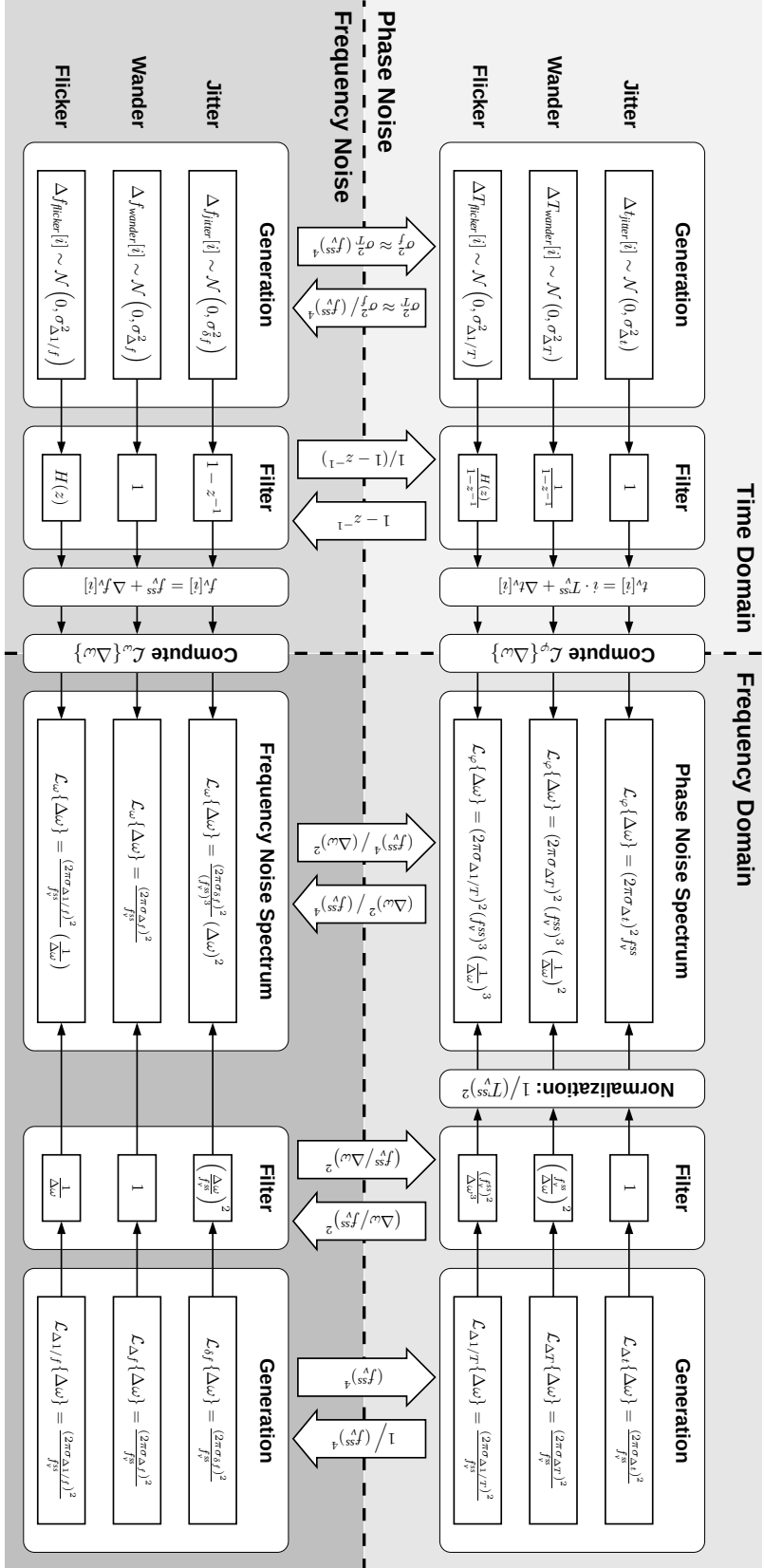


Figure 4.4 Relation between jitter, wander, and flicker in the time and the frequency domains interpreted as phase and frequency noise.

Theoretical Spectra

Combining the individual contributors for frequency or phase spectra, i.e., jitter, wander, and flicker noise, yields the oscillator frequency noise spectrum $\mathcal{L}_\omega^{ol}\{\Delta\omega\}$ or oscillator phase noise spectrum $\mathcal{L}_\varphi^{ol}\{\Delta\omega\}$, where the superscript ol denotes open loop. The theoretical output spectra are given by

$$\mathcal{L}_\omega^{cl}\{\Delta\omega\} = \mathcal{L}_\omega^{ol}\{\Delta\omega\} |H_{\eta_{DCO}}(\Delta\omega)|^2 \quad (4.36)$$

and

$$\mathcal{L}_\varphi^{cl}\{\Delta\omega\} = \mathcal{L}_\varphi^{ol}\{\Delta\omega\} |\Phi_{\eta_{DCO}}(\Delta\omega)|^2 = \mathcal{L}_\varphi^{ol}\{\Delta\omega\} |H_{\eta_{DCO}}(\Delta\omega)|^2 \quad (4.37)$$

where we have used the property from eqn. (4.29) and where cl denotes closed loop. The frequency responses $H_{\eta_{DCO}}(\Delta\omega)$ and $\Phi_{\eta_{DCO}}(\Delta\omega)$ are related to the z -domain transfer functions $H_{\eta_{DCO}}(z)$ in eqn. (2.33) and $\Phi_{\eta_{DCO}}(z)$ in eqn. (2.34) by eqn. (4.30).

4.2 Oscillator Noise

Frequencies	
Reference frequency f_r	13 MHz
Steady state frequency f_v^{ss}	2.413 MHz
Oscillator Phase Noise $\mathcal{L}_\varphi\{\Delta\omega\}$	
Jitter	-150 dBc/Hz
Wander	-112 dBc/Hz @ 500 Hz
Flicker	-125 dBc/Hz @ 1 MHz

Table 4.2 Oscillator noise - setup.

Phase Noise Standard Deviation	
Jitter $\sigma_{\Delta t}$	~ 103 fs
Wander $\sigma_{\Delta T}$	~ 11 fs
Flicker $\sigma_{\Delta 1/T}$	~ 11892 fs
Phase Noise PSD	
Jitter $\mathcal{L}_{\Delta t}\{\Delta\omega\}$	-337.61 dBc/Hz
Jitter $\mathcal{L}_{\Delta T}\{\Delta\omega\}$	-375.03 dBc/Hz
Jitter $\mathcal{L}_{\Delta 1/T}\{\Delta\omega\}$	-324.35 dBc/Hz
Frequency Noise Standard Deviation	
Jitter $\sigma_{\delta f}$	~ 600 kHz
Wander $\sigma_{\Delta f}$	~ 64 kHz
Flicker $\sigma_{\Delta 1/f}$	~ 69 MHz
Frequency Noise PSD	
Jitter $\mathcal{L}_{\delta f}\{\Delta\omega\}$	-37.70 dB
Jitter $\mathcal{L}_{\Delta f}\{\Delta\omega\}$	-18.27 dB
Jitter $\mathcal{L}_{\Delta 1/f}\{\Delta\omega\}$	-50.96 dB
Oscillator Frequency Noise $\mathcal{L}_\omega\{\Delta\omega\}$	
Jitter	26 dB @ 100 MHz
Wander	18.26 dB
Flicker	10.96 dB @ 1 MHz

Table 4.3 Oscillator noise - results using Tab. 4.2.

4.2.2 Simulation Results

The simulation setup and resulting internal parameters and properties are summarized in Tab. 4.2 and Tab. 4.3, where we used the relation from Tab. 4.1. Figure 4.5 and Fig. 4.6 show the generated open loop frequency noise spectrum $\mathcal{L}_\omega^{ol}\{\Delta\omega\}$ and open loop phase noise spectrum $\mathcal{L}_\varphi^{ol}\{\Delta\omega\}$, respectively. In both plots we have included the spectra of the individual noise contributors, i.e., jitter, wander, and flicker noise. The generation of flicker noise is demonstrated step-by-step for the same specification in App. A.4.3. The deviation between the expected and simulated outcomes in Fig. 4.5 and Fig. 4.6 concerning flicker (from 1 MHz upwards) and jitter (near 1 GHz) are discussed in App. A.4.3 and App. A.4.2, respectively. Nevertheless, the overall oscillator noise matches well with the phase noise modelling specifications in Tab. 4.2 and the resulting frequency noise properties in Tab. 4.3. Note that both oscillator noise modelling techniques, i.e., as phase noise using eqn. (4.32) or as frequency noise using eqn. (4.33), yield equivalent simulation outcomes.

The corresponding closed loop frequency noise spectrum $\mathcal{L}_\omega^{cl}\{\Delta\omega\}$ and phase noise spectrum $\mathcal{L}_\varphi^{cl}\{\Delta\omega\}$ of the ADPLL are shown in Fig. 4.7 and Fig. 4.8 for three different loop filters, respectively. The theoretical performance agrees with the simulation outcomes.

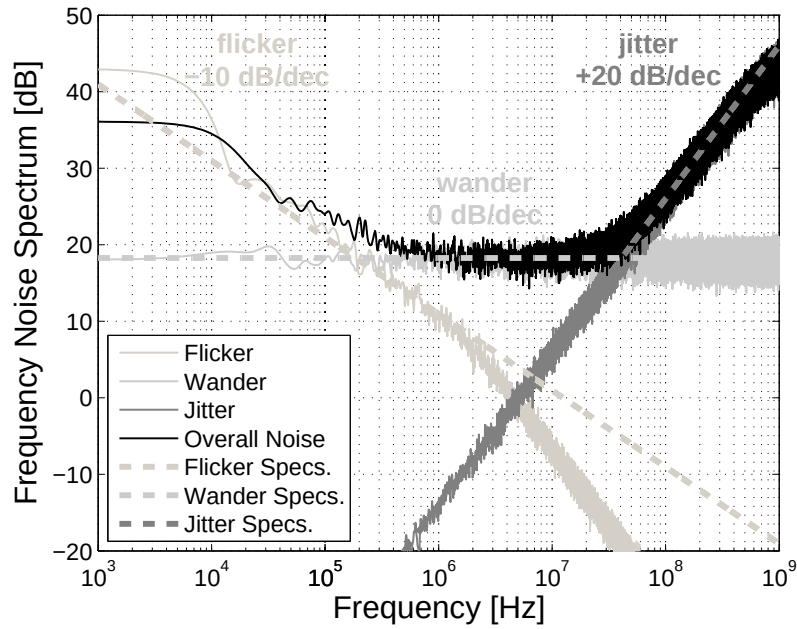


Figure 4.5 Generated oscillator frequency noise consisting of flicker with -10 dB/dec, wander with 0 dB/dec, and jitter with +20 dB/dec.

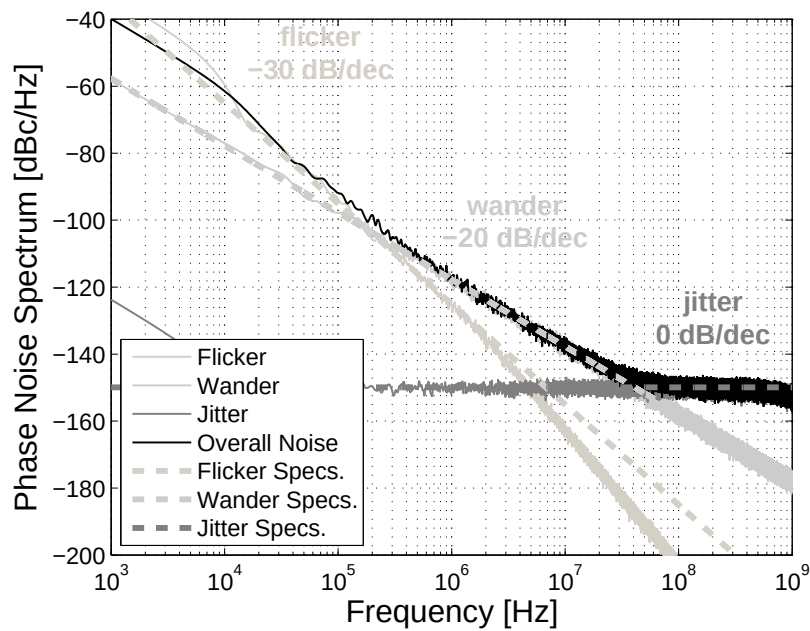


Figure 4.6 Generated oscillator phase noise consisting of flicker with -30 dB/dec, wander with -20 dB/dec, and jitter with 0 dB/dec.

4.2 Oscillator Noise

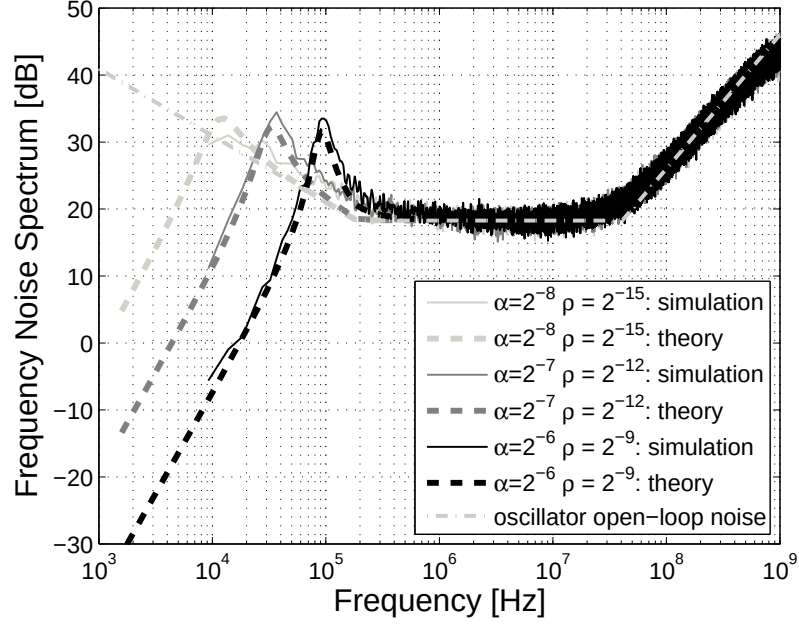


Figure 4.7 ADPLL output frequency noise spectrum of oscillator noise consisting of flicker, wander, and jitter for three different loop filters.

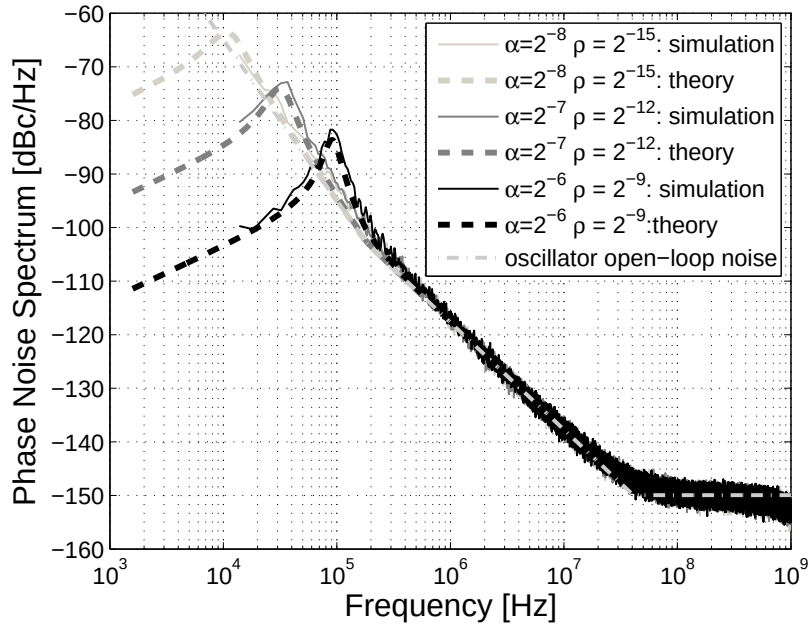


Figure 4.8 ADPLL output phase noise spectrum of oscillator noise consisting of flicker, wander, and jitter for three different loop filters.

4.3 Digitally Controlled Oscillator Quantization Noise with $\Sigma\Delta$ Noise-Shaping

In ADPLLs the DCO is a necessary key component. Unfortunately, its discrete tuning curve with rather coarse quantization, see Tab. 3.1, is one of the most severe imperfections in ADPLLs. Fortunately, dithering of the oscillator tuning word is an effective solution to overcome this problem. As discussed in Sec. 3.1.1 $\Sigma\Delta$ dithering serves two purposes at once. First, the frequency resolution during one reference cycle (denoted as $f_v[n]$) is increased. This prevents the instantaneous quantization error to propagate into the next reference cycle, which would lead to significant spurs. And second, $\Sigma\Delta$ dithering achieves noise shaping so that the quantization error energy is shifted to higher frequencies, i.e., out of the band of interest. Nevertheless, DCO dithering is a challenging task, because the $\Sigma\Delta$ modulator is a nonlinear circuit and its output signal is not random at all. Especially for static input signals, the $\Sigma\Delta$ modulator produces low-periodic sequences that introduce significant spurs and impair the performance. For DCO dithering, the modulator input $d^{frac}[l]$ is constant at least for L samples eqn. (3.2) and varies insignificantly over the reference periods as the loop filter is narrow in steady state. An additional randomization effect is achieved by dithering, i.e., randomizing the input of the modulator by adding a zero-mean random signal, as shown in Fig. 3.3(d).

4.3.1 Theory

In the following we will derive the output phase noise spectrum for a “linearized” DCO, a DCO with white dithering, a DCO with $\Sigma\Delta$ noise shaping, and a DCO with $\Sigma\Delta$ noise shaping and additional randomization.

“Linearized” DCO Spectrum

In order to evaluate the performance of the ADPLL, the DCO quantization is often linearized [6, 50, 147, 157]. Techniques like high-speed dithering can break the deterministic pattern due to the nonlinearity and validate the linearity assumption. Without additional circuitry the DCO itself will *not* produce a noise-like output spectrum. The ADPLL appears like a first-order $\Sigma\Delta$ modulator with the loop-filter after the integrator. Due to the narrow-bandwidth of the loop filter the tuning word and correspondingly the output frequency will exhibit a very slow oscillation around the desired value. The output phase noise spectrum exhibits severe spurs at low frequencies, which makes an ADPLL with coarse quantization impractical for wireless applications.

For the “linearized” analysis we treat the DCO as an additive white noise source with zero mean and variance

$$\sigma_q^2 = \frac{K^2}{12} \quad (4.38)$$

where K is the DCO gain, cf. Sec. 2.1.2 or Sec. 3.1.1. Following the steps in App. A.4.2, we can compute the phase noise spectrum $\mathcal{L}_q\{\Delta\omega\}$ as [50]

$$\mathcal{L}_q\{\Delta\omega\} = \frac{(2\pi\sigma_q)^2}{f_r}. \quad (4.39)$$

The linearized DCO quantization noise is held constant over one REF cycle, which appears as a zero-order hold, and is attenuated by the DCO noise transfer function. Consequently the ADPLL

4.3 Digitally Controlled Oscillator Quantization Noise with Noise-Shaping

frequency noise spectrum is

$$\mathcal{L}_\omega\{\Delta\omega\} = |H_{\eta_{DCO}}(\Delta\omega)|^2 |H_{ZOH}(\Delta\omega)|^2 \mathcal{L}_q\{\Delta\omega\} \quad (4.40)$$

and the ADPLL phase noise spectrum is

$$\mathcal{L}_\varphi\{\Delta\omega\} = \mathcal{L}_\omega\{\Delta\omega\} \left| \frac{1}{\Delta\omega} \right|^2. \quad (4.41)$$

DCO Spectrum with White Dithering

In order to achieve a “linearized” DCO spectrum, we can dither the digital tuning word with white noise. Except that the quantization error is spread over the frequency range from 0 to f_{dither} instead of 0 to f_s , the case of white DCO dithering agrees with the “linearized” DCO analysis. Consequently, the noise spectrum

$$\mathcal{L}_{q,\Sigma\Delta}\{\Delta\omega\} = \frac{(2\pi\sigma_q)^2}{f_{dither}} \quad (4.42)$$

is reduced by the factor $f_r/f_{dither} = 1/L$ in comparison to eqn. (4.39). Since white dithering is performed at the rate of f_{dither} , the ZOH becomes

$$H_{ZOH,\Sigma\Delta}(\omega) = \frac{\sin(\omega T_{dither}/2)}{\omega/2} e^{j\omega T_{dither}/2}. \quad (4.43)$$

The ADPLL phase noise output spectrum with white DCO dithering is

$$\mathcal{L}_\varphi\{\Delta\omega\} = |H_{\eta_{DCO}}(\Delta\omega)|^2 |H_{ZOH,\Sigma\Delta}(\Delta\omega)|^2 \mathcal{L}_{q,\Sigma\Delta}\{\Delta\omega\} \left| \frac{1}{\Delta\omega} \right|^2. \quad (4.44)$$

DCO Spectrum with $\Sigma\Delta$ Noise Shaping

The output spectrum of an ADPLL with DCO quantization noise shaping consists of two noise contributions [157]

$$\mathcal{L}_\varphi\{\Delta\omega\} = \mathcal{L}_\varphi^A\{\Delta\omega\} + \mathcal{L}_\varphi^B\{\Delta\omega\}. \quad (4.45)$$

The first contribution $\mathcal{L}_\varphi^A\{\Delta\omega\}$ is dominant in the out-of band region and arises from the noise shaping of the $\Sigma\Delta$ modulator. Hence, the phase noise spectrum contribution of the $\Sigma\Delta$ modulator is the quantization noise given by eqn. (4.42), shaped with the frequency response $H_{\Sigma\Delta}(\Delta\omega)$ according to eqn. (3.7), filtered with the ZOH from eqn. (4.43), and converted in the phase domain, i.e.,

$$\mathcal{L}_\varphi^A\{\Delta\omega\} = |H_{\Sigma\Delta}(\Delta\omega)|^2 |H_{ZOH,\Sigma\Delta}(\Delta\omega)|^2 \mathcal{L}_{q,\Sigma\Delta}\{\Delta\omega\} \left| \frac{1}{\Delta\omega} \right|^2. \quad (4.46)$$

The second contribution $\mathcal{L}_\varphi^B\{\Delta\omega\}$ is the *residual* quantization error of the $\Sigma\Delta$ modulator, which is dominant in the in-band region. The $\Sigma\Delta$ modulator produces a high-rate integer sequence, whose average value matches asymptotically the fractional input signal. As a result of the sampling process in the feedback path, the frequency is evaluated over each reference cycle, see Sec. 2.2 or Sec. 2.1. Rewriting the variable output frequency with dithering eqn. (3.5) at the rate of the reference frequency and assuming for notational simplicity that L is an integer number, we get

$$f_v[n] = \left(d_{int}[n] + \frac{1}{L} \sum_{l=0}^{L-1} d_{\Sigma\Delta}[nL + l] \right) K. \quad (4.47)$$

$\Sigma\Delta$ Order	Variance $\sigma_{q,res}^2$	Scope of applicability
1	1/6	integer L
1	$\sim 1/6$	fractional $L \gg 1$
2	1/3	$L > 2$
3	1	$L > 3$

Table 4.4 Variance of the accumulate-and-dump from [157].

The residual frequency quantization error is given by

$$\Delta f_v[n] = f_v[n] - f_v^{ss}[n] = \delta d[n] K \quad (4.48)$$

with

$$\delta d[n] = \frac{1}{L} \sum_{l=0}^{L-1} d_{\Sigma\Delta}[nL + l] - d^{frac}[n]. \quad (4.49)$$

It follows that the residual DCO quantization error depends on the decimated moving average of the $\Sigma\Delta$ modulator output in eqn. (4.49). This operation is known as accumulate-and-dump [157]. The sub-sampling process of the frequency-to-phase conversion from f_{dither} to f_r folds the residual DCO quantization spectrum. Therefore, the residual DCO quantization error appears as error at the phase detector, which becomes evident by substituting eqn. (4.47) into eqn. (2.11), i.e.,

$$N_v[n] = \left(d_{int}[n] + \frac{1}{L} \sum_{l=0}^{L-1} d_{\Sigma\Delta}[nL + l] \right) \frac{K}{f_r}. \quad (4.50)$$

As the feedback phase accumulates $N_v[n]$ at the rate of the reference frequency, the DCO quantization error directly impairs the phase error by bypassing the integration. Substituting eqn. (4.50) into eqn. (2.14) gives the instantaneous error at the phase detector, i.e.,

$$N_e[n] = N_v[n] - N_r[n] = \left(\frac{1}{L} \sum_{l=0}^{L-1} d_{\Sigma\Delta}[nL + l] - d^{frac}[n] \right) \frac{K}{f_r}. \quad (4.51)$$

With $\sigma_{q,res}^2$ being the variance of the sum $\sum_{l=0}^{L-1} d_{\Sigma\Delta}[nL + l]$ in eqn. (4.51) the noise spectrum at the phase detector due to DCO quantization with dithering becomes with eqn. (3.1)

$$\mathcal{L}_{q,res}\{\Delta\omega\} = \frac{\left(2\pi\sigma_{q,res}\frac{K}{f_{dither}} \right)^2}{f_r}. \quad (4.52)$$

In [157] the value of $\sigma_{q,res}^2$ for 1st, 2nd, and 3rd order $\Sigma\Delta$ dithering are computed analytically and are summarized in Tab. 4.4. The impact of the folded DCO quantization noise is given by

$$\mathcal{L}_{\varphi}^B\{\Delta\omega\} = |H_{\eta_{TDC}}(\Delta\omega)|^2 |H_{ZOH}(\Delta\omega)|^2 \mathcal{L}_{q,res}\{\Delta\omega\} \left| \frac{1}{\Delta\omega} \right|^2. \quad (4.53)$$

The overall phase noise due to DCO quantization with $\Sigma\Delta$ noise shaping is given by plugging eqn. (4.46) and eqn. (4.53) into eqn. (4.45), i.e.,

$$\begin{aligned} \mathcal{L}_{\varphi}\{\Delta\omega\} &= |H_{\Sigma\Delta}(\Delta\omega)|^2 |H_{ZOH,\Sigma\Delta}(\Delta\omega)|^2 \mathcal{L}_{q,\Sigma\Delta}\{\Delta\omega\} \left| \frac{1}{\Delta\omega} \right|^2 \\ &\quad + |H_{\eta_{TDC}}(\Delta\omega)|^2 |H_{ZOH}(\Delta\omega)|^2 \mathcal{L}_{q,res}\{\Delta\omega\} \left| \frac{1}{\Delta\omega} \right|^2. \end{aligned} \quad (4.54)$$

DCO Spectrum with $\Sigma\Delta$ Noise Shaping - Spurs

A $\Sigma\Delta$ modulator is a deterministic circuit, whose output sequence is perfectly predictable. In particular static or periodic input signals are problematic, because they introduce periodic output sequences. Consider for instance the input $d^{frac}[l] = 0.5$. The first-order $\Sigma\Delta$ modulator output in Fig. 3.3(c) produces a rectangular sequence with constant duty cycle of 0.5, which has harmonics at odd multiples of $f_{dither} \times 0.5$. In general, the periodic pattern of the $\Sigma\Delta$ output signal $d_{\Sigma\Delta_1}$ equals the denominator B of the fractional part of the tuning word, i.e., $d^{frac}[n] = A/B$, where A and B are relatively prime integer numbers. The spurs due to $\Sigma\Delta$ noise shaping manifest themselves at multiples of $f_{dither} \times 1/B$. Thus, 1st order high-speed $\Sigma\Delta$ dithering shifts the quantization noise to higher frequencies, but fails to randomize its output stream. Higher-order modulators have better noise shaping characteristics but still suffer from spurs for low periodic sequences, e.g., fractional numbers with small numerator such as $1/2$ or $2/5$.

DCO Spectrum with $\Sigma\Delta$ Noise Shaping and Randomization

In order to eliminate any periodic pattern of the $\Sigma\Delta$ modulator the input sequence $d^{frac}[l]$ is modified by a zero-mean random signal. The additionally introduced noise affects the tuning word with an amplitude of $2^{\#bits} \times LSB$. The noise spectrum becomes

$$\mathcal{L}_{dither}\{\Delta\omega\} = \frac{(2\pi 2^{\#bits} LSB K)^2}{12} \frac{1}{f_{dither}} \quad (4.55)$$

and contributes to the output phase noise by passing through the DCO noise transfer function, the zero-order hold at the rate of the dither sequence, and the frequency-to-phase conversion, i.e.,

$$\mathcal{L}_{\varphi}^C\{\Delta\omega\} = |H_{\eta_{DCO}}(\Delta\omega)|^2 |H_{ZOH,\Sigma\Delta}(\Delta\omega)|^2 \mathcal{L}_{dither}\{\Delta\omega\} \left| \frac{1}{\Delta\omega} \right|^2. \quad (4.56)$$

The output of the DCO quantization noise with $\Sigma\Delta$ noise shaping and additional randomization is given by adding $\mathcal{L}_{\varphi}^C\{\Delta\omega\}$ from eqn. (4.56) to eqn. (4.54), i.e.,

$$\begin{aligned} \mathcal{L}_{\varphi}\{\Delta\omega\} &= |H_{\Sigma\Delta}(\Delta\omega)|^2 |H_{ZOH,\Sigma\Delta}(\Delta\omega)|^2 \mathcal{L}_{q,\Sigma\Delta}\{\Delta\omega\} \left| \frac{1}{\Delta\omega} \right|^2 \\ &\quad + |H_{\eta_{TDC}}(\Delta\omega)|^2 |H_{ZOH}(\Delta\omega)|^2 \mathcal{L}_{q,res}\{\Delta\omega\} \left| \frac{1}{\Delta\omega} \right|^2 \\ &\quad + |H_{\eta_{DCO}}(\Delta\omega)|^2 |H_{ZOH,\Sigma\Delta}(\Delta\omega)|^2 \mathcal{L}_{dither}\{\Delta\omega\} \left| \frac{1}{\Delta\omega} \right|^2. \end{aligned} \quad (4.57)$$

4.3.2 Simulation Results

Table 4.5 shows the simulation setup and Tab. 4.6 provides the resulting numerical values for the individual noise terms and different experiments. Figure 4.9 and Fig. 4.10 show the ADPLL output spectrum with DCO $\Sigma\Delta$ noise shaping using a MASH 1-1 topology and a CKV clock division ratio of $D = 4$ and $D = 8$, respectively. Increasing the dithering frequency by a factor of two improves the noise gain $\mathcal{L}_{q,\Sigma\Delta}\{\Delta\omega\}$ by 3 dB, whereas $\mathcal{L}_{q,res}\{\Delta\omega\}$ is improved by 6 dB, because f_{dither} appears quadratically in eqn. (4.52). Note that the residual quantization noise appears at the phase detector and consequently is shaped by the TDC noise transfer function, which has a larger gain factor, so that the in-band phase noise is dominated by the residual quantization noise. A larger dithering frequency,

Frequencies	
Reference frequency f_r	13 MHz
Steady state frequency f_v^{ss}	2.413 MHz
DCO	
Free running frequency f_0	2.4 GHz
DCO resolution K	~ 23.07 kHz
Steady state tuning word d^{ss}	563.5
Loop Filter	
Proportional factor α	2^{-6}
Integration factor ρ	2^{-9}
CKV Clock Divider	
D (Fig. 4.9)	4
D (Fig. 4.10, 4.11, 4.12, 4.13)	8
$\Sigma\Delta$ Modulator	
MASH 1-1 (Fig. 4.9, 4.10, 4.11, 4.12)	
2nd order SSMF (Fig. 4.13)	
Randomization	
none (Fig. 4.9, 4.10, 4.13)	
1 bit with LSB= 2^{-8} (Fig. 4.11)	
4 bits with LSB= 2^{-8} (Fig. 4.12)	

Table 4.5 DCO quantization - setup.

Theoretical Spectrum	
“Linearized” DCO $\mathcal{L}_q\{\Delta\omega\}$ eqn. (4.39)	21.3 dB
2nd Order $\Sigma\Delta$ Noise Shaping for D = 4	
Oversampling factor L	~ 46.4
Dither frequency f_{dither}	603.25 MHz
White dithering $\mathcal{L}_{q,\Sigma\Delta}\{\Delta\omega\}$ eqn. (4.42)	4.6 dB
Residual quantization $\mathcal{L}_{q,res}\{\Delta\omega\}$ eqn. (4.52)	-151.3 dB
1 bit randomization $\mathcal{L}_{dither}\{\Delta\omega\}$ eqn. (4.55)	-37.5 dB
4 bit randomization $\mathcal{L}_{dither}\{\Delta\omega\}$ eqn. (4.55)	-19.5 dB
2nd Order $\Sigma\Delta$ Noise Shaping for D = 8	
Oversampling factor L	~ 23.2
Dither frequency f_{dither}	301.625 MHz
White dithering $\mathcal{L}_{q,\Sigma\Delta}\{\Delta\omega\}$ eqn. (4.42)	7.6 dB
Residual quantization $\mathcal{L}_{q,res}\{\Delta\omega\}$ eqn. (4.52)	-145.3 dB
1 bit randomization $\mathcal{L}_{dither}\{\Delta\omega\}$ eqn. (4.55)	-34.5 dB
4 bit randomization $\mathcal{L}_{dither}\{\Delta\omega\}$ eqn. (4.55)	-16.4 dB

Table 4.6 DCO quantization - results.

however, raises the power consumption considerably and should be avoided. The simulation setup uses a worst case scenario concerning spurious tones, as the fractional tuning word is chosen to be $1/2$. The output signal of the MASH 1-1 modulator is very regular and results in significant spurs at odd harmonics of half the dithering frequency, i.e., $d^{frac} \times f_{dither}$, cf. Fig. 4.9 and Fig. 4.10. In [6] a random sequence is added to the input of the MASH 1-1 modulator to reduce the amplitude of the spurs. Fig. 4.11 and Fig. 4.12 show the output phase noise spectrum for $D = 8$ with 1 bit and 4 bits randomization signal. Using a single bit does not increase the phase noise considerably, but also fails to reduce the spurs. Using 4 bits, however, reduces the spur amplitudes, but raises the noise level significantly. As discussed in Sec. 3.1.1 the cascaded MASH 1-1 architecture behaves worse than alternative $\Sigma\Delta$ architectures. Figure 4.13 shows the simulation outcome of a second order single-stage multiple feedback architecture from Fig. 3.5. Although the simulation setup is equivalent to Fig. 4.10, no spurs are evident in the spectrum. The small amount of pseudo noise at the input of the $\Sigma\Delta$ modulator was sufficient to effectively randomize the deterministic behavior. In case the loop filter bandwidth is chosen narrower, a single bit pseudo-random signal is sufficient to avoid any spurs in the output spectrum.

4.3 Digitally Controlled Oscillator Quantization Noise with Noise-Shaping

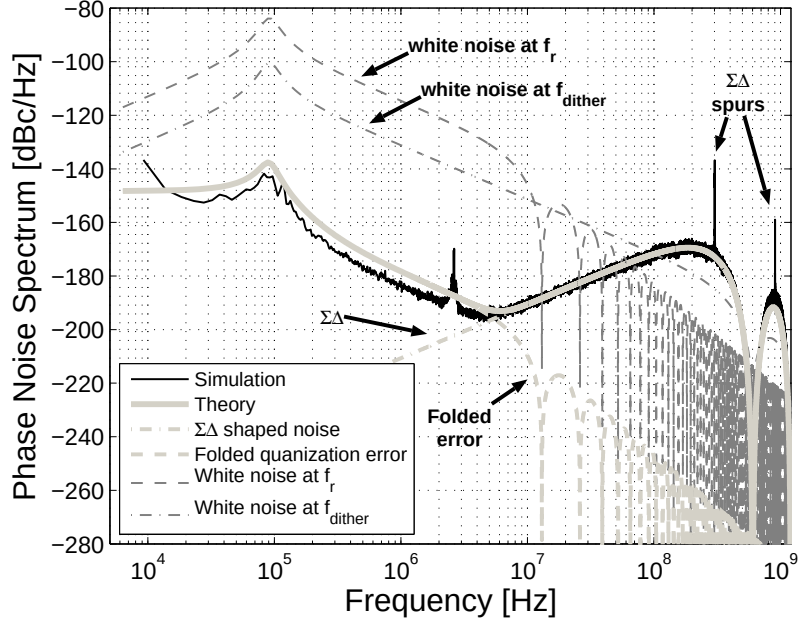


Figure 4.9 ADPLL output phase noise spectrum of the DCO quantization error with MASH 1-1 noise shaping and $D = 4$.

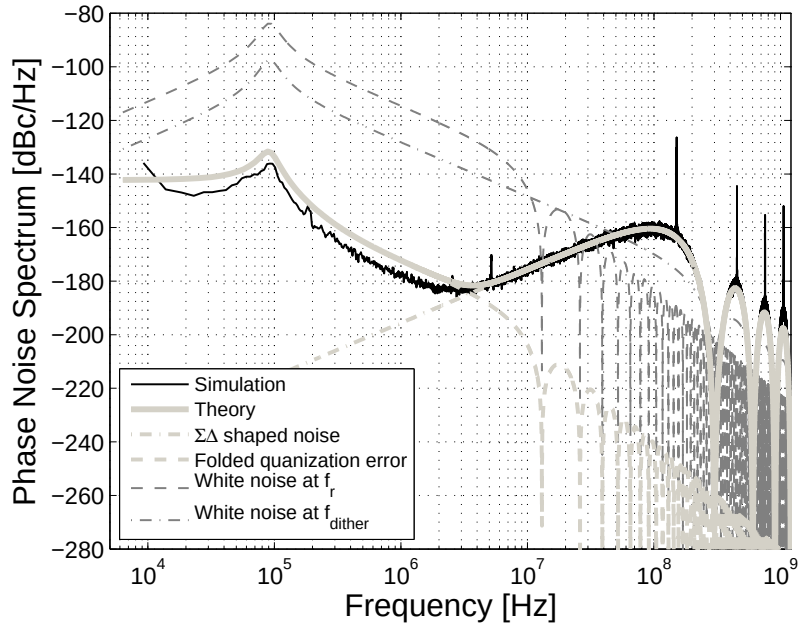


Figure 4.10 ADPLL output phase noise spectrum of the DCO quantization error with MASH 1-1 noise shaping and $D = 8$.

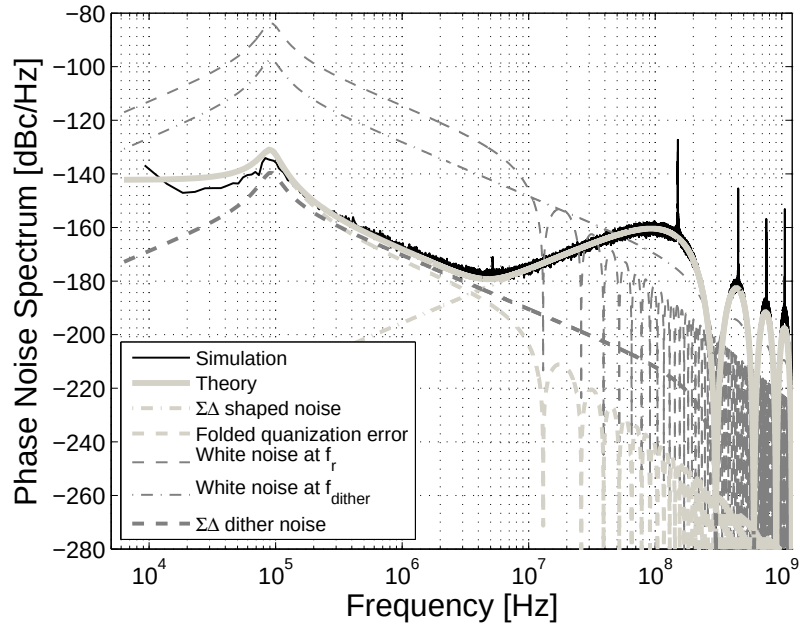


Figure 4.11 ADPLL output phase noise spectrum of the DCO quantization error with MASH 1-1 noise shaping and $D = 8$ and 1 bit randomization.

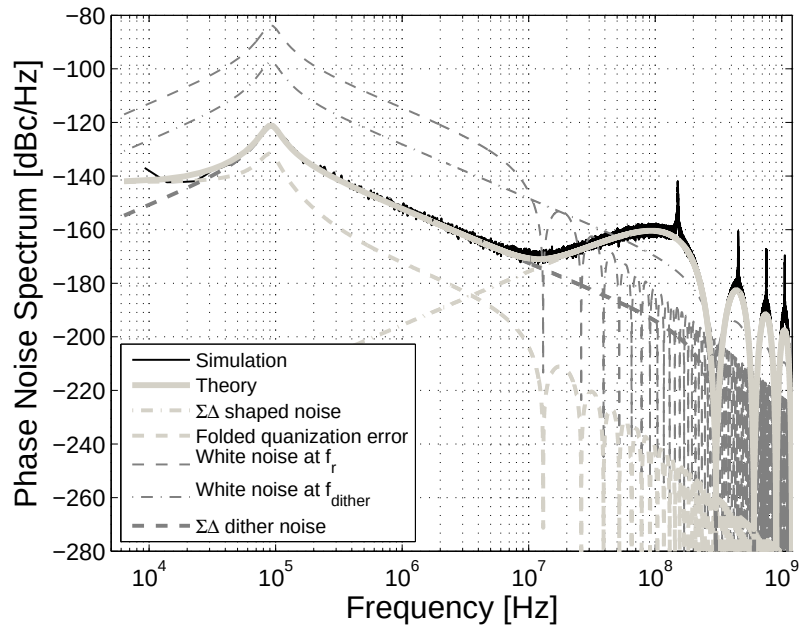


Figure 4.12 ADPLL output phase noise spectrum of the DCO quantization error with MASH 1-1 noise shaping and $D = 8$ and 4 bits randomization.

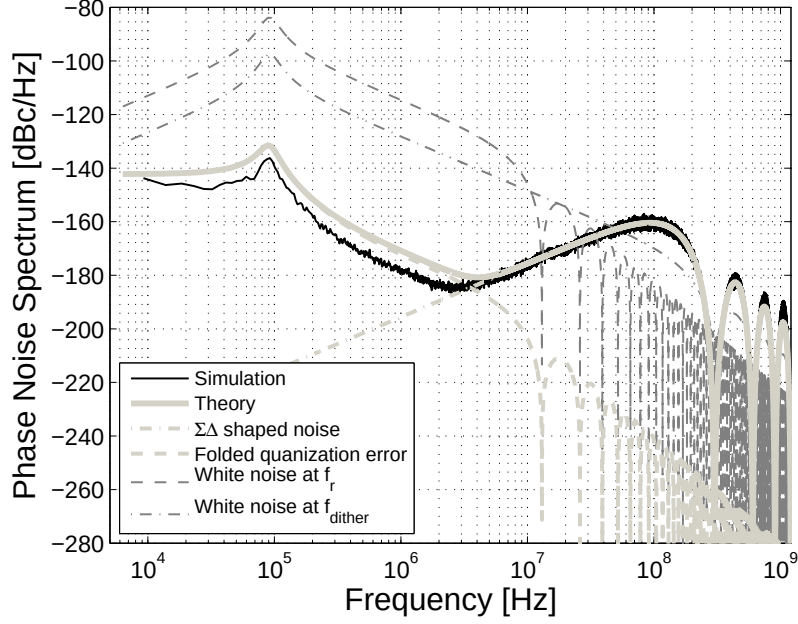


Figure 4.13 ADPLL output phase noise spectrum of the DCO quantization error with SSMF noise shaping and $D = 8$.

4.4 Frequency-to-Phase Converter Quantization

The quantization of the digital phase is a major imperfection of ADPLLs. The output phase noise of the ADPLL is dominated by DCO noise and TDC noise. As the TDC noise transfer function is conflicting with the DCO transfer function, i.e., wide bandwidth improves the DCO noise suppression but increases the TDC noise, and vice versa, the TDC resolution limits the performance of ADPLLs. Furthermore, a linearization of the TDC quantization is not valid, because of the coarse quantization levels of $\Delta t_{res}/T_v^{ss}$ and the low diversity of the input to the TDC, i.e., $\varepsilon[m]$ from eqn. (3.49) for the retimed reference architecture¹. As a result, the phase error will follow a periodic oscillation, which produces large in-band spurs. Typically an additional IIR filter with a narrow bandwidth, with the drawback of worse DCO noise suppression, is used to mitigate TDC spurs.

To evaluate the TDC performance and the resulting periodic patterns, the nonlinear dynamics of the ADPLL has to be investigated [161–165]. Several conditions complicate the nonlinear analysis: First, the signal $\varepsilon[m]$ depends on the frequency command word N_r , its initial value $\varepsilon[0]$, and a nonlinear modulo operation, cf. eqn. (3.49). Second, the TDC performs two different quantizations, i.e., the instantaneous time quantization $Q_{TDC}\{\cdot\}$ in the numerator of eqn. (3.16) and the DCO period quantization $\bar{Q}_{TDC}\{\cdot\}$, in the denominator of eqn. (3.16), which averages over N_{avg} quantized values of a CKV period. On the one hand, several parameters, such as N_r , $\varepsilon[0]$, N_{avg} , the TDC resolution, which depends on the time resolution Δt_{res} and the steady state frequency f_v^{ss} , and the loop filter

¹The retiming error $\varepsilon[m]$ from eqn. (3.49) for the reference retimed architecture and the fractional error $\xi[n]$ from eqn. (3.87) for the synchronous reference architecture have similar signal characteristics in steady state. The only difference is the sign of the sawtooth slope, which is $-N_r^{frac}$ for $\varepsilon[m]$ and N_r^{frac} for $\xi[n]$. In the following we will use the terminology for the retimed reference architecture. The analysis, however, is valid for the synchronous reference architecture as well.

influence the impact of TDC quantization on the output phase noise. On the other hand, several nonlinearities, i.e., $Q_{TDC}\{\cdot\}$, $\overline{Q}_{TDC}\{\cdot\}$, and the modulo operation, complicate the analysis.

In the following, however, we will use the analysis of the retiming error signal to explain the most severe spurs in the output phase noise spectrum.

4.4.1 Theory

We introduce a simple open loop signal model and compare the output of the ADPLL simulation to the output of the simplified model. First, we analyze an ADPLL without TDC, where the time quantization level of the frequency-to-phase converter corresponds to one CKV period T_v^{ss} or the integer resolution of the accumulator with respect to $N_v[n]$. Second, we refine the previously introduced model to analyze TDC quantization spurs.

Accumulator Quantization Spurs

In many applications the performance requirements are not as stringent as in wireless communication systems. Therefore, to save area and power consumption, designers aim to implement ADPLLs without TDCs, i.e., [44]. As discussed in Chapter 3, the coarse quantization of the accumulator, which corresponds to an integer accuracy of the frequency-to-phase conversion, introduces a periodic error. Under ideal conditions this error would correspond to the retiming signal $\varepsilon[m]$ given by eqn. (3.49). The dynamics of the loop is given by the loop filter parameters. In case of a narrow bandwidth the propagation of the error through the system can be neglected. Figure 4.14 shows the open loop model used to describe the output spectrum of the ADPLL with accumulator spurs, where we have used the uniform reference clock REF. The sawtooth signal of the retiming error can be rewritten as a Fourier series, i.e.,

$$\varepsilon[m] = \frac{1}{2} - \frac{1}{\pi} \sum_{k=1}^{\infty} \frac{1}{k} \sin\left(\frac{2\pi k}{M} m\right), \quad (4.58)$$

where $N_r^{frac} = L/M$, where L and M are prime integer numbers. As the signal is sampled with f_r , we observe harmonics at

$$f_{spur}(k) = \frac{2\pi f_r}{M} k \quad \text{for } k = 1, \dots, M-1. \quad (4.59)$$

Note that higher order harmonics are folded back to harmonic frequencies in the fundamental interval between 0 and f_r . The fundamental spur is located at $f_{spur}(k=L)$, i.e., $2\pi f_r \frac{L}{P} = 2\pi f_r N_r^{frac}$, but the retiming error signal produces M spurs, which are equally spaced between 0 and f_r . Thus, small values of M are critical, because the quantization error concentrates on a few spurs with large amplitude. Furthermore low fractional numbers close to 0 and large fractional values close to 1 are critical, because of large in-band spurs.

TDC Quantization Spurs

To investigate the impact of TDC quantization we extend the open-loop model in Fig. 4.14 by a quantizer as shown in Fig. 4.15, where the ADPLL output phase spectrum is impaired by the TDC quantization error $q_\varepsilon[m]$. Thus, we write the quantized TDC output signal $\varepsilon_q[m]$ as a sum of the ideal retiming error signal $\varepsilon[m]$ and a quantization error $q_\varepsilon[m]$, i.e.,

$$\varepsilon_q[m] = \varepsilon[m] + q_\varepsilon[m]. \quad (4.60)$$

4.4 Frequency-to-Phase Converter Quantization

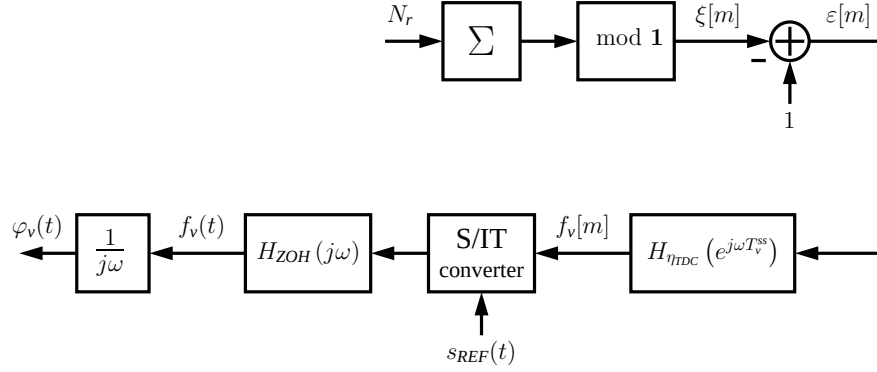


Figure 4.14 Open-loop accumulator quantization model.

The spectrum of the TDC quantization error $q_\varepsilon[m]$ has spurs on the same frequency bins as the sawtooth shaped input signal $\varepsilon[m]$. The signal $\varepsilon[m]$ distributes its signal energy among M equally spaced frequencies between 0 and f_r , i.e., on harmonics of f_r/M . Nonlinearities propagate signal energy to harmonics of the input signal. For a sawtooth shaped signal the quantization energy and the resulting spurs will appear at locations given by eqn. (4.59) but with significantly lower amplitude, i.e., $\Delta t_{res}/T_v^{ss}$ instead of 1.

Note, that in the open loop model we consider the instantaneous time-quantization $Q_{TDC}\{\cdot\}$ only, whereas the period quantization $\overline{Q}_{CKV}\{\cdot\}$ is neglected. For low values of M or integer channels the period quantization, however, will become the dominant quantization effect².

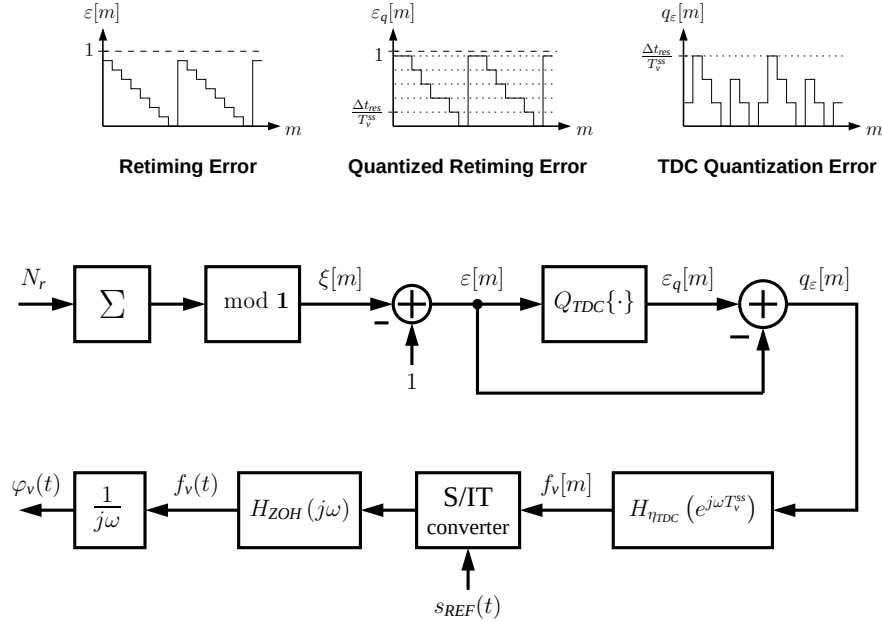


Figure 4.15 Open-loop TDC quantization model.

²After finalizing this part of the thesis the author became aware of an independent publication [166], where equivalent results have been derived.

“Linearized” TDC Quantization Noise

In the literature, e.g., [6], the TDC quantization effect is “linearized” in order to evaluate its effect on the output phase noise spectrum. Thus, TDC quantization is treated as a additive white gaussian noise with variance

$$\sigma_\varepsilon^2 = \frac{\Delta t_{res}^2}{12(T_{ss}^v)^2}. \quad (4.61)$$

The fact that the TDC performs a ceiling operation eqn. (3.11), i.e., the mean value is not zero, can be neglected as the closed loop will compensate that constant mean value. The “linearized” TDC quantization noise is white between 0 and f_r , i.e.,

$$\mathcal{L}_\varepsilon\{\Delta\omega\} = \frac{(2\pi\sigma_\varepsilon)^2}{f_r}. \quad (4.62)$$

“Linearized” TDC quantization affects the output frequency noise spectrum by

$$\mathcal{L}_\omega\{\Delta\omega\} = |H_{\eta_{TDC}}(\Delta\omega)|^2 |H_{ZOH}(\Delta\omega)|^2 \mathcal{L}_\varepsilon\{\Delta\omega\} \quad (4.63)$$

and the output phase noise spectrum by

$$\mathcal{L}_\varphi\{\Delta\omega\} = |H_{\eta_{TDC}}(\Delta\omega)|^2 |H_{ZOH}(\Delta\omega)|^2 \mathcal{L}_\varepsilon\{\Delta\omega\} \left| \frac{1}{\Delta\omega} \right|^2. \quad (4.64)$$

It is necessary to use $H_{\eta_{DCO}}(z)$ instead of $\Phi_{\eta_{DCO}}(z)$, because $H_{ZOH}(\omega)$ affects the frequency signal $f_v(t)$.

4.4.2 Simulation Results

First we determine the phase noise spectrum caused by accumulator quantization, where we simulate an ADPLL without TDC and, second, we perform simulations with a TDC to additionally evaluate its quantization effects. In order to capture the entire spur energy we plot the phase noise in dBc instead of dBc/Hz.

Accumulator Spurs

Reference
Reference frequency f_r 13 MHz
Frequency Command Word
$N_r = 186\frac{1}{8}$ in Fig. 4.16
$N_r = 186\frac{1}{9}$ in Fig. 4.17
$N_r = 186\frac{1}{100}$ in Fig. 4.18
$N_r = 186\frac{1}{101}$ in Fig. 4.19

Table 4.7 Accumulator spurs - setup.

“Linearized” Accumulator Noise	
Accumulator resolution Δt_{ref}	T_v^{ss}
Standard deviation σ_ε using eqn. (4.61)	$\frac{1}{12}$
PSD $\mathcal{L}_\varepsilon\{\Delta\omega\}$ using eqn. (4.62)	-65.97 dB
First Accumulator Spur	
$f_{spur}(1)$ for $M = 8$ using eqn. (4.59)	1.625 MHz
$f_{spur}(1)$ for $M = 9$ using eqn. (4.59)	~ 1.4444 MHz
$f_{spur}(1)$ for $M = 100$ using eqn. (4.59)	0.13 MHz
$f_{spur}(1)$ for $M = 101$ using eqn. (4.59)	~ 0.1287 MHz

Table 4.8 Accumulator spurs - results.

4.4 Frequency-to-Phase Converter Quantization

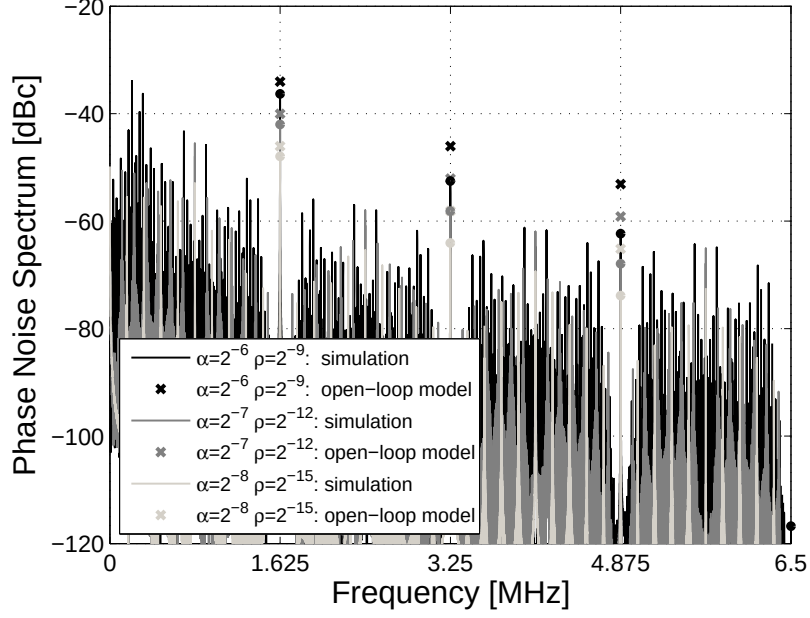


Figure 4.16 Output phase noise spectrum of an ADPLL with $N_r = 186\frac{1}{8}$ and three different loop filters without TDC.

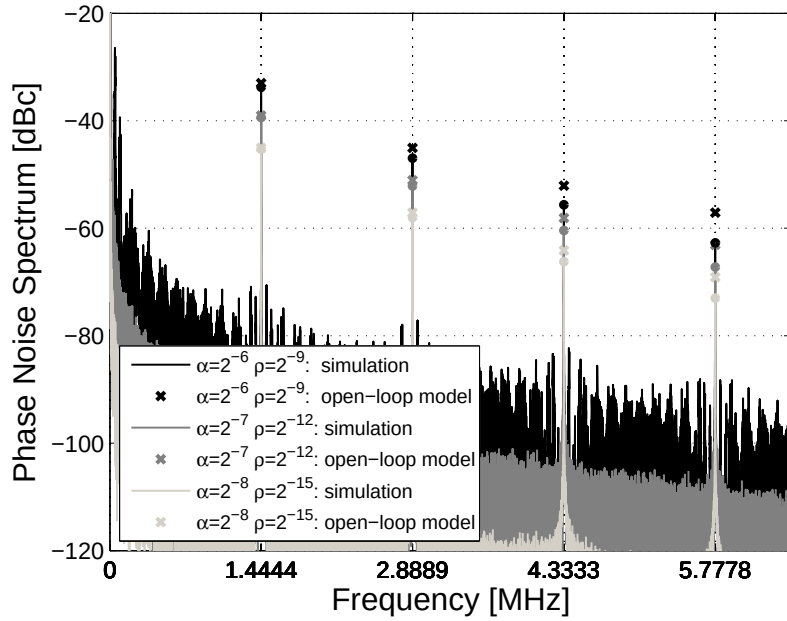


Figure 4.17 Output phase noise spectrum of an ADPLL with $N_r = 186\frac{1}{9}$ and three different loop filters without TDC.

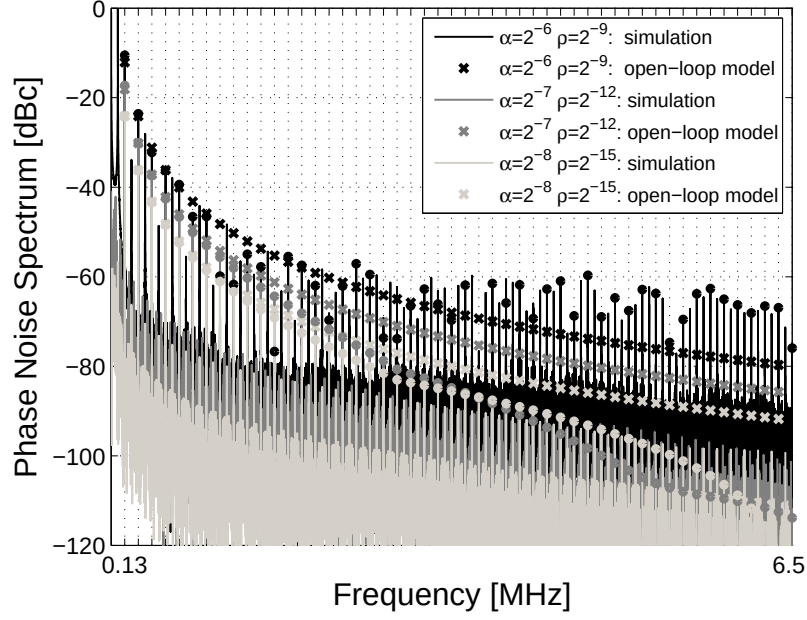


Figure 4.18 Output phase noise spectrum of an ADPLL with $N_r = 186 \frac{1}{100}$ and three different loop filters without TDC.

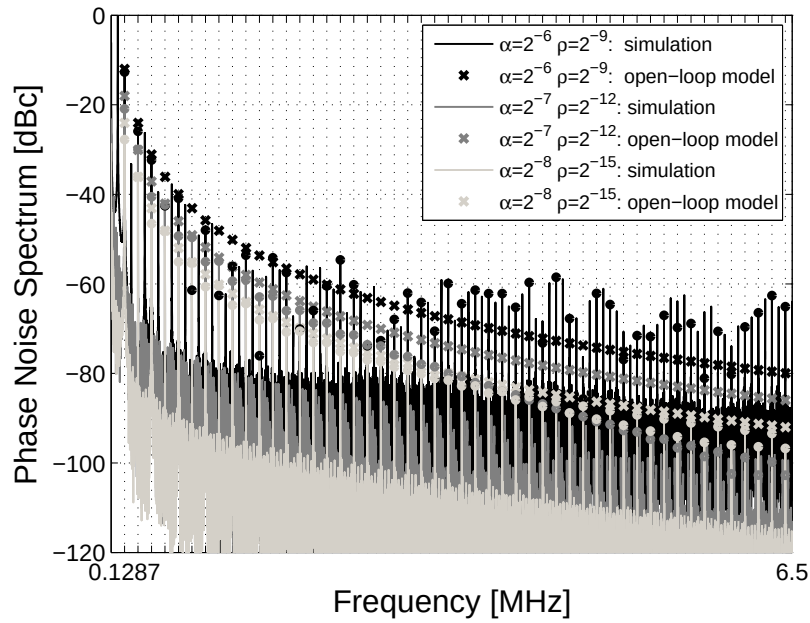


Figure 4.19 Output phase noise spectrum of an ADPLL with $N_r = 186 \frac{1}{101}$ and three different loop filters without TDC.

4.4 Frequency-to-Phase Converter Quantization

To evaluate the spurs of an ADPLL without TDC, we have performed a large number of simulations with different fractional command words. For a rather narrow loop filter the simulation outcome matches well with the open loop model in Fig. 4.14. However, to completely explain the dynamics of the system a theoretical approach is required.

The simulation setup is summarized in Tab. 4.7. The numerical results are summarized in Tab. 4.8 and depicted in Fig. 4.16 to Fig. 4.19 for M equal to 8, 9, 100, and 101. In all plots we have included the “linearized” spectrum (dark grey) and the expected spurs (light gray crosses) from the open loop model. For the chosen loop filter we have made the following observation, which are well represented by the four plots:

- For the coarse quantization and the deterministic signal characteristics any linearization, i.e., also for the in-band range, is invalid.
- For a wider loop filter the dynamics of the ADPLL become evident and additional spurs appear in the spectrum, which cannot be explained with the open loop model.
- For odd M the open loop model matches the simulated ADPLL quite well. The spurs caused by frequency-to-phase conversion are concentrated on the frequency locations, which are predicted by the open loop model, cf. Fig. 4.17 and Fig. 4.19.
- For even M and M larger than 10, we observe additional spurs which are mainly dominant at multiples of $f_{spur}(0)/2$, cf. Fig. 4.18.
- For even M and M smaller than 10, the nonlinear dynamics become evident at several spurs, cf. Fig. 4.16. In case of $M = 2$ or $N_r^{frac} = 0.5$ the loop dynamics are dominant and the open loop model is invalid.

We can conclude that an ADPLL without TDC introduces significant spurs in the phase noise spectrum and that further investigation of the non-linear loop dynamics is an important open research issue.

TDC Spurs

Reference			
Reference frequency f_r	13 MHz	“Linearized” TDC Noise	
Frequency Command Word N_r		Standard deviation σ_ε using eqn. (4.61)	~ 0.0139
$186\frac{1}{8}$ in Figs. 4.20 and 4.21		PSD $\mathcal{L}_\varepsilon\{\Delta\omega\}$ using eqn. (4.62)	-92.30 dBc/Hz
$186\frac{1}{9}$ in Fig. 4.22		First TDC Spur	
$186\frac{1}{100}$ in Fig. 4.23		$f_{spur}(1)$ for $M = 8$ using eqn. (4.59)	1.625 MHz
$186\frac{1}{101}$ in Fig. 4.24		$f_{spur}(1)$ for $M = 9$ using eqn. (4.59)	~ 1.4444 MHz
TDC Quantization		$f_{spur}(1)$ for $M = 100$ using eqn. (4.59)	0.13 MHz
Time quantization $Q_{TDC}\{\cdot\}$	$\Delta t_{res} = 20$ ps	$f_{spur}(1)$ for $M = 101$ using eqn. (4.59)	~ 0.1287 MHz
Period quantization $\overline{Q}_{TDC}\{\cdot\}$	$N_{avg} = 128$		

Table 4.10 TDC noise - results.

Table 4.9 TDC noise - setup.

Table 4.9 shows the TDC quantization simulation setup and Tab. 4.10 summarizes the numerical results. The spur locations predicted by the open loop model agree with the ADPLL without TDC,

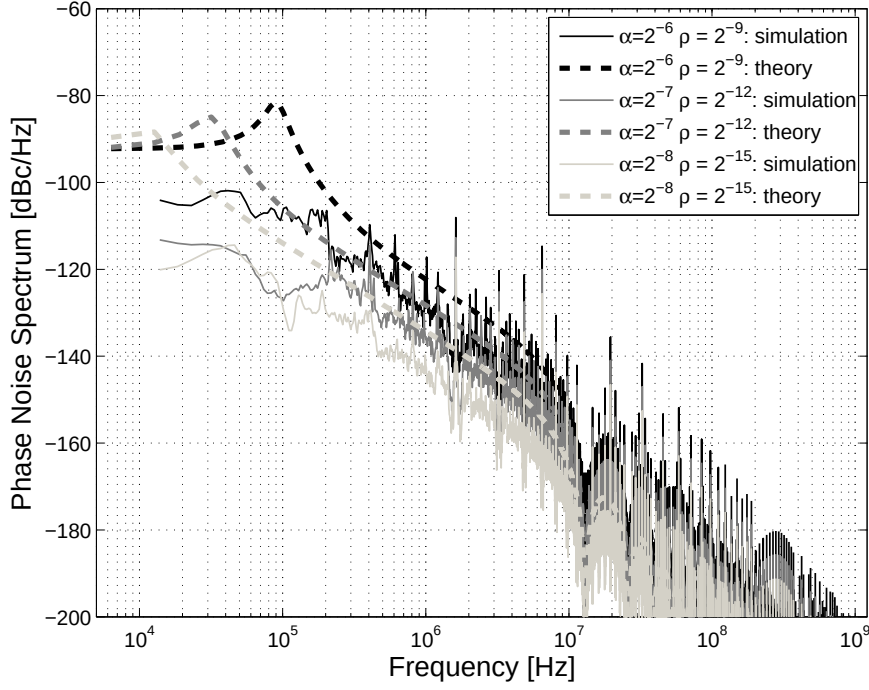


Figure 4.20 Output phase noise spectrum in dBc/Hz of an ADPLL with TDC and $N_r = 186\frac{1}{8}$ for three different loop filters.

but spur amplitudes are significantly reduced. Figure 4.20 depicts an exemplary output phase noise spectrum for $M = 8$, which emphasizes that the linear assumption is invalid. That is, the output spectra, shown for $M = 8, 9, 100$, and 101 in Fig. 4.21 to Fig. 4.24, respectively, contain severe spurs. In all plots we have compared the ADPLL simulation results with the results of the open loop model. While the spur locations agree with the open loop model, the spur amplitudes deviate slightly. However, this analysis may be sufficient for designers to estimate the impact of TDC quantization for a given specification. In practice the quantization energy distribution among the M spurs will depend on the nonlinearity in the tuning curve of the TDC, the value of the initial phase $\varepsilon[0]$, and the quantization of the period estimation $\bar{Q}_{TDC}\{\cdot\}$. Especially for small values of M the period estimation is rather coarse, because $\varepsilon[m]$ theoretically exhibits only M different values.

4.4 Frequency-to-Phase Converter Quantization

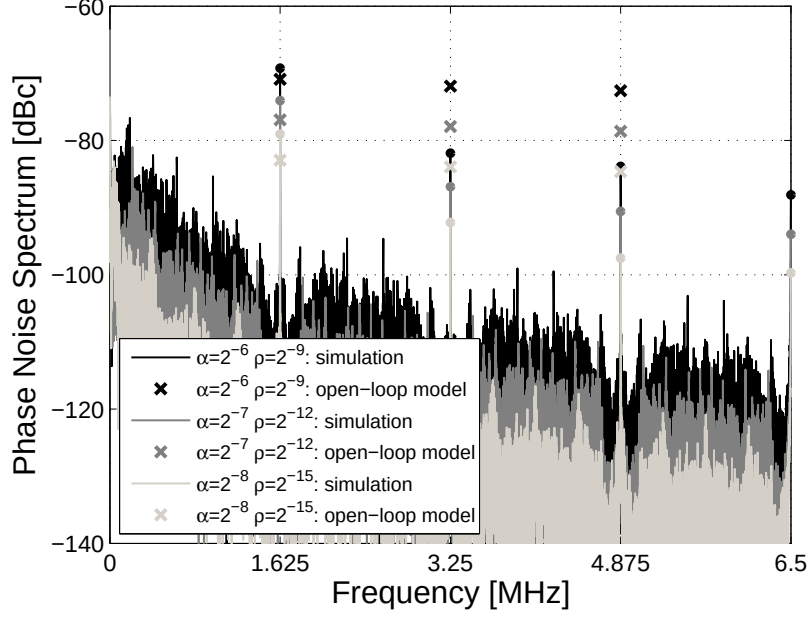


Figure 4.21 Output phase noise spectrum with TDC and $N_r = 186\frac{1}{8}$ for three different loop filters.

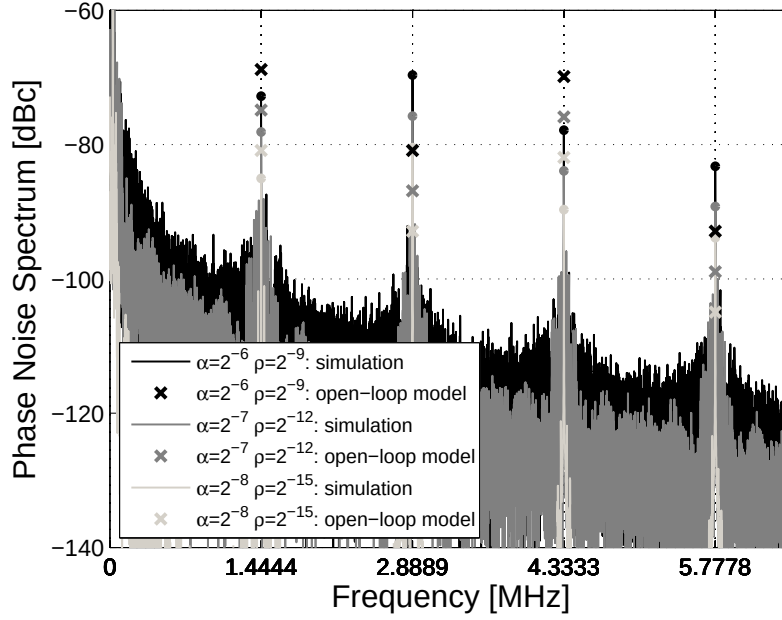


Figure 4.22 Output phase noise spectrum with TDC and $N_r = 186\frac{1}{9}$ for three different loop filters.

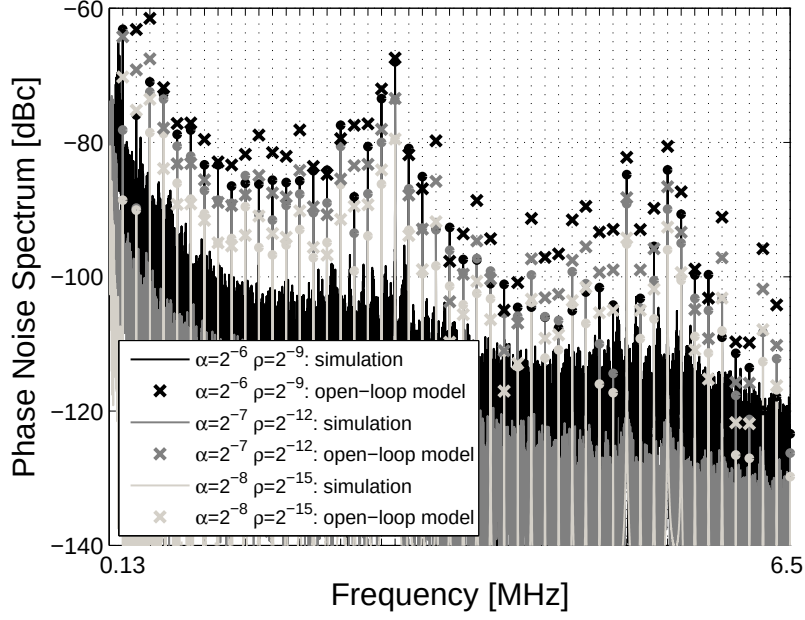


Figure 4.23 Output phase noise spectrum with TDC and $N_r = 186 \frac{1}{100}$ for three different loop filters.

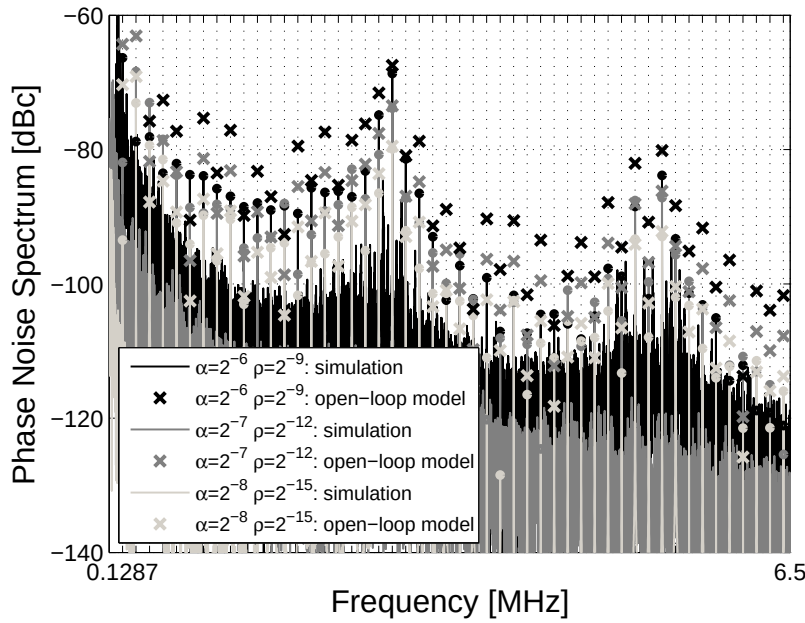


Figure 4.24 Output phase noise spectrum with TDC and $N_r = 186 \frac{1}{101}$ for three different loop filters.

4.5 Reference Noise

In the mixed-signal model in Fig. 2.10, the noise of the reference clock affects the frequency-to-digital phase conversion and the digital-to-frequency conversion. When the ADPLL is used as frequency synthesizer, the desired frequency f_v^{ss} is constant in steady state. Fortunately, any noise of the sampling clock has no impact on a constant output signal. Consequently we have to consider only the sampling process within the ADPLL by rewriting eqn. (2.48) including reference clock jitter Δt_{REF} as

$$\begin{aligned}
 \varphi_v^s(t) &= \varphi_v(t) s_{REF}(t) \\
 &= \varphi_v(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_r - \Delta t_{REF}) \\
 &= \sum_{n=-\infty}^{\infty} \varphi_v(nT_r + \Delta t_{REF}) \delta(t - nT_r - \Delta t_{REF}).
 \end{aligned} \tag{4.65}$$

The digital phase signal becomes

$$\begin{aligned}
 \varphi_v^r[n] &= \varphi_v(nT_r + \Delta t_{REF}) \\
 &= (nT_r + \Delta t_{REF}) f_v^{ss} \\
 &= nN_r + \Delta t_{REF} f_v^{ss}
 \end{aligned} \tag{4.66}$$

where we used eqn. (2.2) and eqn. (2.47). For the retimed reference architecture, we can employ the non-uniform mixed-signal model in Fig. 3.14. By rewriting the TDC operation eqn. (3.60), which is defined as the integral between the rising REF edge and the subsequent CKV edge and is not affected by noise of the reference clock, we obtain

$$\frac{1}{T_v^{ss}} \int_{t=mT_r+\Delta t_{REF}}^{mT_r+r[m]T_r} 1dt = \frac{1}{T_v^{ss}} (r[m]T_r - \Delta t_{REF}) = \varepsilon[m] - \Delta t_{REF} f_v^{ss}. \tag{4.67}$$

As expected, reference noise affects both architectures equally with $\Delta t_{REF} f_v^{ss}$.

4.5.1 Theory

Reference clock jitter is similar to the linearized TDC quantization scenario. Thus, assuming the reference noise signal Δt_{REF} is normally distributed with zero-mean and variance σ_{REF}^2 its spectrum is white (see App. A.4.2 for a similar example), i.e.,

$$\mathcal{L}_{REF}\{\Delta\omega\} = (2\pi\sigma_{REF})^2 f_r. \tag{4.68}$$

In analogy to eqn. (4.61) we define

$$\sigma_{\varepsilon, REF}^2 = \frac{\sigma_{REF}^2}{(T_v^{ss})^2} \tag{4.69}$$

so that the TDC error spectrum $\mathcal{L}_{\varepsilon}\{\Delta\omega\}$ is given by substituting $\sigma_{\varepsilon, REF}$ for σ_{ε} into eqn. (4.62). The ADPLL frequency and phase spectra are given by eqn. (4.63) and eqn. (4.64), respectively.

Frequencies	
Reference frequency f_r	13 MHz
Steady state frequency f_v^{ss}	2.413 MHz
Reference Clock Jitter	
REF spectrum $\mathcal{L}_{REF}\{\Delta\omega\}$	-140 dBc/Hz

Table 4.11 REF noise - setup.

Reference Clock Jitter	
Std. σ_{REF} using eqn. (4.68)	~ 4.4142 ps
Std. $\sigma_{\varepsilon, REF}$ using eqn. (4.69)	~ 0.0106
PSD $\mathcal{L}_{\varepsilon}\{\Delta\omega\}$ using eqn. (4.62)	-94.67 dBc/Hz

Table 4.12 REF noise - results.

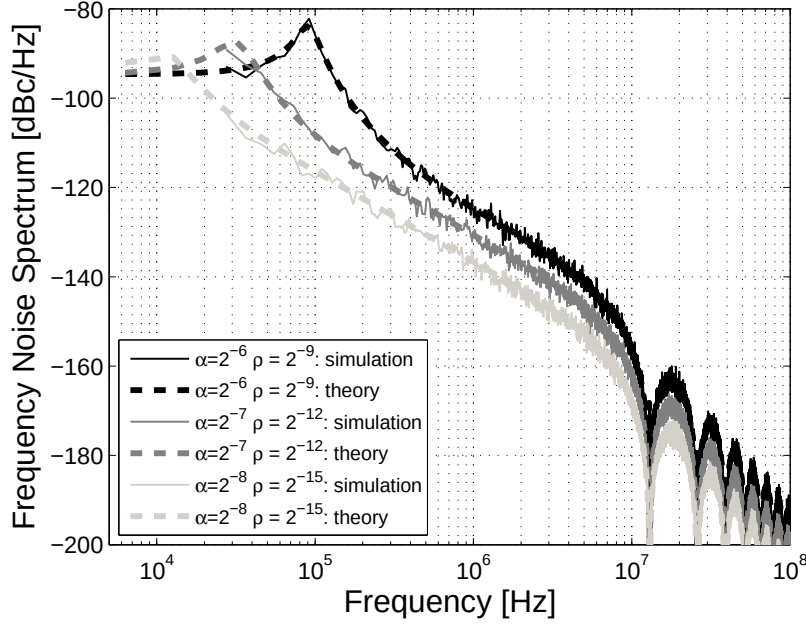


Figure 4.25 ADPLL output phase noise spectrum of reference clock jitter for three different loop filters.

4.5.2 Simulation Results

Figure 4.25 shows the phase noise spectrum of the simulation outcomes with the setup in Tab. 4.11 for three different loop filters. The jitter of the reference clock was set to -140 dBc/Hz, that corresponds to a standard deviation of $\sigma_{REF} = 4.41$ ps. At the phase detector reference jitter appears with a standard deviation of $\sigma_{\varepsilon, REF} \approx 0.0106$ and a noise level of -94.67 dB, see Tab. 4.12. By comparing $\mathcal{L}_{REF}\{\Delta\omega\}$ in eqn. (4.68) to $\mathcal{L}_{\varepsilon}\{\Delta\omega\}$ eqn. (4.62) we can state that the phase noise due to reference clock jitter is less than the “linearized” TDC quantization if

$$\mathcal{L}_{REF}\{\Delta\omega\} \leq \left(2\pi \frac{\Delta t_{res}}{\sqrt{12}}\right)^2 f_r \quad (4.70)$$

is fulfilled. For a TDC quantization of 20 ps the equivalent reference clock jitter is -136.88 dBc/Hz.

4.6 Injection Pulling

The mixed-signal nature of the ADPLL causes severe practical problems [18, 19]. Especially parasitic coupling between the digital circuitry and the LC oscillator, which share the same substrate, power, and ground planes [18], causes a periodic disturbance of the oscillator. The rising and falling edges of the non-uniform retimed reference clock trigger the massive digital logic, which “pulls” the digitally controlled oscillator (DCO) output frequency. This effect produces undesired spurs in the output phase noise spectrum. In [18] the spurs are separated into reference spurs and side spurs. Reference spurs are caused by injection pulling, whereas side spurs are caused by injection pulling and the periodically time-varying retimed reference clock. Furthermore, it was reported in [18] that some measured side spurs have been close to the margins of the GSM specification. Reducing the coupling on the die through careful circuit design mitigates these spurs; however, in Sec. 3.3 we have presented an alternative architecture that solves the problem of side spurs on system-level. In this section we analyze the generation of spurs due to injection pulling, classify them, and compare the spurious tones of the retimed reference architecture from Sec. 3.2 with the synchronous reference architecture from Sec. 3.3.

4.6.1 Theory

This section analyzes the generation of injection pulling spurs, their frequencies, amplitudes, and dependence on the frequency command word N_r . Basically, we can distinguish between three kinds of spurs: reference spurs, side spurs, and injection mismatch spurs. Reference spurs are caused by a periodic injection pulling signal which is due to parasitic coupling effects on the die. The periodic non-uniform retimed reference clock CKR generates side spurs by shifting energy of the reference spurs to additional frequencies. Different gains of the injected frequency among rising and falling CKR edges lead to injection mismatch spurs.

General Injection Pulling Model

Injection pulling changes the DCO output frequency $f_v(t)$ over a period shorter than a reference cycle. As the frequency-to-phase conversion in the feedback path of an ADPLL is basically an integration over one reference cycle, the error due to injection pulling appears as constant factor in the phase error. The loop compensates this factor and, consequently, the analysis of injection pulling can neglect the feedback mechanism, which leads to the open loop model shown in Fig. 4.26. The output signal $f_v(t)$ can be decomposed into an ideal signal $\tilde{f}_v(t)$, which is constant in lock, and an undesired injection pulling signal $f_{ip}(t)$. The non-uniform impulse train $s(t)$ represents the rising and falling edge events of the periodically non-uniform retimed reference clock and is decomposed into P sampling functions $s_p(t)$ at a P times lower rate. With each Dirac impulse of $s_p(t)$ the injection pulling generation filter produces an injection pulse. The 2-periodic impulse response $h_{\langle p, 2 \rangle}(t)$, where $\langle p, 2 \rangle$ stands for p modulo 2, accounts for different pulse shapes of the rising and falling CKR edges. To evaluate the effects of the injection pulling signal $f_{ip}(t)$ on the output frequency $f_v(t)$ we apply the injection pulling model to the retimed reference architecture. First we describe the P uniform impulse trains $s_p(t)$, whose sum represent the periodically non-uniform retimed reference clock, and then convolve $s_p(t)$ with the injection pulling generation impulse responses $h_{\langle p, 2 \rangle}(t)$.

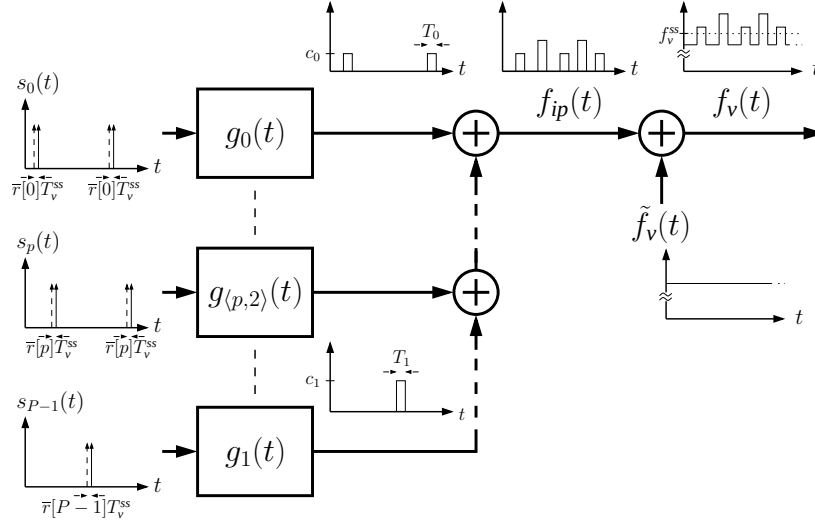


Figure 4.26 Injection pulling model: The non-uniform impulse train, corresponding to the rising and falling edges of the retimed reference clock $s(t)$, is decomposed into P sampling functions $s_p(t)$. The 2-periodic impulse response $h_{(p,2)}(t)$ models the shape of the frequency injection pulse for rising and falling edges, i.e., $h_{(p,2)} = h_0(t)$ for p is even and $h_{(p,2)} = h_1(t)$ for p is odd.

Retimed Reference Clock

The signal $s(t)$ in Fig. 4.27 describes the rising and falling edge events of the clock signal CKR (solid) as a train of Dirac impulses $\delta(t)$. As illustrated in Fig. 4.27, the time shifts between the non-uniform impulse train (solid) and the uniform impulse train with constant period $T_r/2$ (dashed) are given by $\varepsilon[m]T_v^{ss}$ for the rising CKR edges and $\bar{\varepsilon}[m]T_v^{ss}$ for the falling CKR edges. For the sake of brevity we introduce relative time offsets $\bar{r}[p]$, which are the deviations normalized to the uniform period $T_r/2$, i.e.,

$$\bar{r}[p] = \begin{cases} \frac{2}{N_r} \varepsilon[p/2] & \text{for } p \text{ is even,} \\ \frac{2}{N_r} \bar{\varepsilon}[(p-1)/2] & \text{for } p \text{ is odd} \end{cases} \quad (4.71)$$

where we used $T_r = T_v^{ss} N_r$. Since $\varepsilon[m]$ and $\bar{\varepsilon}[m]$ are periodic with M , the non-uniform sampling train $s(t)$ with the relative time offsets $r[p]$ is periodic with $P = 2M$, i.e.,

$$s(t) = \sum_{p=0}^{P-1} s_p(t) \quad (4.72)$$

with

$$s_p(t) = \sum_{l=-\infty}^{\infty} \delta\left(t - (lP + p + \bar{r}[p]) \frac{T_r}{2}\right). \quad (4.73)$$

Note that the M relative time offsets $\bar{r}[p]$ of the impulse train $s_{CKR}(t)$ in eqn. (4.71), which consider only the rising edge events, are related to the P time offsets $\bar{r}[p]$ by $r[p] = \frac{1}{2}\bar{r}[2p]$ for $p = 0, \dots, M-1$.

4.6 Injection Pulling

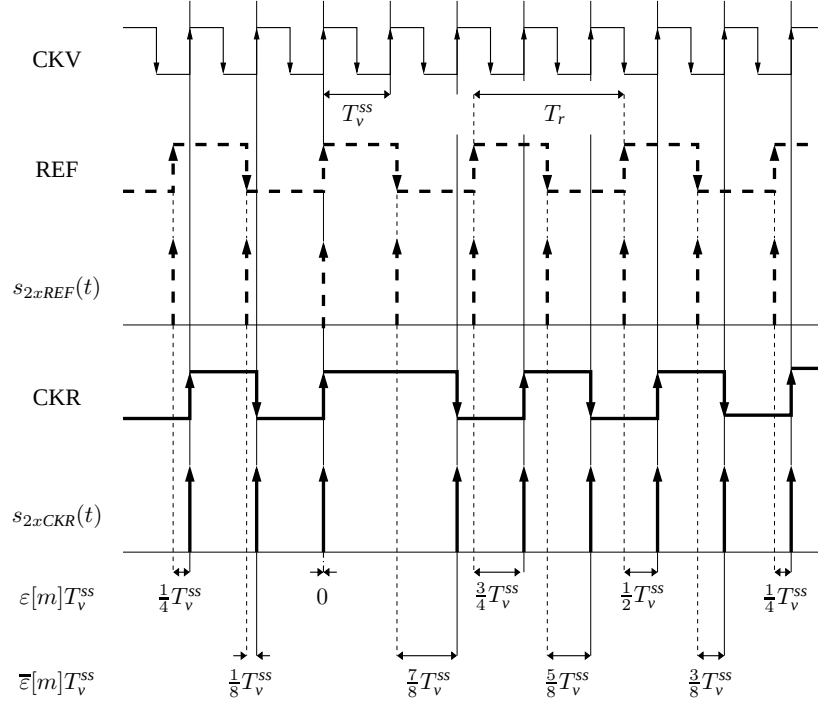


Figure 4.27 Reference clock REF, variable clock CKV, retimed reference clock CKR, and the impulse train $s(t)$ corresponding to the rising and falling edges of the uniform clock REF (dashed) and non-uniform clock CKR (solid) in steady state for $N_r = 2\frac{1}{4}$.

Injection Pulling Signal

The injection pulling signal $f_{ip}(t)$ is given by the sum of the convolutions of the impulse responses $g_{\langle p,2 \rangle}(t)$ with the non-uniform impulse trains $s_p(t)$ in eqn. (4.72), i.e.,

$$f_{ip}(t) = \sum_{p=0}^{P-1} \sum_{l=-\infty}^{\infty} g_{\langle p,2 \rangle} \left(t - (lP + p + \bar{r}[p]) \frac{T_r}{2} \right). \quad (4.74)$$

After applying the continuous-time Fourier transform (CTFT) to eqn. (4.74) and some simplifications, we obtain

$$F_{ip}(j\Omega) = \frac{2}{T_r} \sum_{k=-\infty}^{\infty} 2\pi\delta \left(\omega - k \frac{2 \times 2\pi f_r}{P} \right) A_k(j\omega) \quad (4.75)$$

with

$$A_k(j\omega) = \frac{1}{P} \sum_{p=0}^{P-1} G_{\langle p,2 \rangle}(j\omega) e^{-j\omega \bar{r}[p] \frac{T_r}{2}} e^{-jkp \frac{2\pi}{P}}. \quad (4.76)$$

and $G_{\langle p,2 \rangle}(j\omega)$ is the CTFT of $g_{\langle p,2 \rangle}(t)$. The spectrum $F_{ip}(j\omega)$ in eqn. (4.75) allows for describing the frequency locations and the amplitudes of the spurs caused by injection pulling. Based on this description, we can classify the spurs into reference spurs, side spurs, and injection mismatch spurs. An example of such an output spectrum (converted into the phase domain by integration, i.e., $F_{ip}(j\omega)/(j\omega)$) is shown in Fig. 4.29. Basically, the spectrum consists of $P = 2M$ spurs uniformly spaced between 0 and $2f_r$, which are periodically repeated with $2f_r$. The amplitudes of the spurs

are given by the frequency response $A_k(j\omega)$, which depends on the time offsets $\bar{r}[p]$ and the impulse responses $g_{\langle p,2 \rangle}(t)$. Reference spurs are introduced by injection pulling, are located at multiples of $(2f_r)$, and are weighted by $A_0(j\omega)$. Side spurs are caused by the time-shifts $\bar{r}[p]$ of the injection pulling signal due to the non-uniform clock CKR. They appear at $k\frac{2f_r}{P} + l(2f_r)$ weighted by $A_k(j\omega)$ with $k = 1 \dots P-1$ and $l = 1 \dots \infty$. Injection mismatch spurs are determined by the periodicity (with period two) of the impulse responses $g_{\langle p,2 \rangle}(t)$. They are located at $f_r + l(2f_r)$ weighted by $A_{P/2}(j\omega)$ with $l = 1 \dots \infty$. Note that injection mismatch spurs and side spurs contribute energy to $A_{P/2}(j\omega)$.

When we assume that injection pulling appears with a uniform clock signal, the time offsets $\bar{r}[p]$ are zero and we can simplify eqn. (4.75) to

$$F_{ip}(j\omega) = \frac{2}{T_r} \sum_{k=-\infty}^{\infty} \left(2\pi\delta(\omega - k\pi 4f_r) A_0(j\omega) + 2\pi\delta(\omega - 2\pi f_r - k4\pi f_r) A_{P/2}(j\omega) \right) \quad (4.77)$$

and eqn. (4.76) to

$$\begin{aligned} A_0(j\omega) &= \frac{1}{2} (G_0(j\omega) + G_1(j\omega)), \\ A_{P/2}(j\omega) &= \frac{1}{2} (G_0(j\omega) - G_1(j\omega)). \end{aligned} \quad (4.78)$$

Thus, all side spurs vanish. Only the reference spurs with the weight $A_0(j\omega)$ and the injection mismatch spurs with the weight $A_{P/2}(j\omega)$ remain in the spectrum. In case the injection pulling signal is identical for rising and falling edges, the 2-periodic impulse response $g_{\langle p,2 \rangle}(t)$ simplifies to a non-periodic impulse response $g(t)$ and the weight $A_{P/2}(j\omega)$ becomes zero. We can conclude that an ADPLL with uniform clock, i.e., the synchronous reference architecture in Sec. 3.3, does not generate side spurs.

4.6.2 Model

Note that four CKV cycles within one REF cycle (shown in dark and light grey in Fig. 4.28) are affected by the transition of the injection pulling frequency. Similar to a tuning word update, the injection pulling pulse may start and end at any time within one CKV cycle and, therefore, has to be modeled accordingly.

In accordance with the injection pulling model in [18] we have modeled $g_{\langle p,2 \rangle}(t)$ as an rectangular function

$$g_{\langle p,2 \rangle}(t) = c_{\langle p,2 \rangle} (u(t) - u(t - T_{\langle p,2 \rangle})) \quad (4.79)$$

where $u(t)$ is the continuous-time unit step function, $T_{\langle p,2 \rangle}$ is the injection pulling time, and $c_{\langle p,2 \rangle}$ is a gain factor.

4.6.3 Simulation Results

We have simulated an ADPLL with retimed reference architecture and an ADPLL with synchronous reference architecture with the configuration in Table 4.13. Note that the simulation setup with $N_r = 632/5$ agrees with the simulation in [18, Fig. 7], where the actual output frequency is the DCO output divided by two. Expressing the phase noise in dBc/Hz and using the same parameters for the computation of the phase noise yields the same result as in [18, Fig. 7]. The simulated results

4.6 Injection Pulling

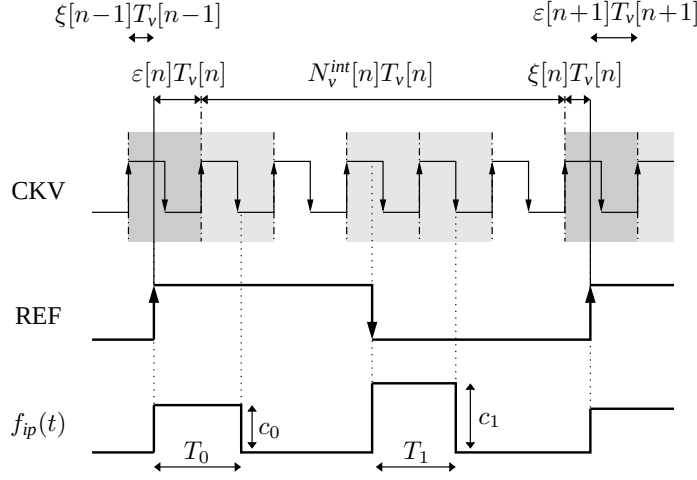


Figure 4.28 Relation between the reference clock REF, the variable clock CKV, and the injection pulling signal $f_{ip}(t)$.

Frequencies	
Reference frequency f_r	26 MHz
Frequency Command Word N_r	
$N_r = 63 \frac{2}{5}$ in Fig. 4.29 and Fig. 4.30	
$N_r = 63 \frac{3}{10}$ in Fig. 4.31 and Fig. 4.32	
Injection Pulling	
Impulse response $g_{\langle p,2 \rangle}(t)$	$c_{\langle p,2 \rangle} (u(t) - u(t - T_{\langle p,2 \rangle}))$
Injection pulling period $T_{\langle p,2 \rangle}$	$12 \times T_v^{ss}$
Injection gain $c_{\langle p,2 \rangle}$	$c_0 = 56 \text{ kHz}, c_1 = 64 \text{ kHz}$

Table 4.13 Injection pulling - setup.

Steady State Frequency
$f_v^{ss} = 1.6484 \text{ GHz}$ in Fig. 4.29 and Fig. 4.30
$f_v^{ss} = 1.6458 \text{ GHz}$ in Fig. 4.31 and Fig. 4.32

Table 4.14 Injection pulling - results.

of both architectures are compared with the values of $A_k(j\Omega)$ converted into the phase domain by integration, i.e., $A_k(j\Omega)/(j\Omega)$. The number of spurs, their frequency location, and their amplitudes match with the theory given by eqn. (4.75) for the retiming architecture and given by eqn. (4.77) for the synchronous reference architecture. In both architectures the reference spurs $A_0(j\Omega)/(j\Omega)$ and the injection mismatch spurs $A_{P/2}(j\Omega)/(j\Omega)$ have similar amplitudes. To mitigate the reference spurs the coupling effects on the die and simultaneously the injected frequency (the gain $g_{\langle p,2 \rangle}$ of the impulse response $g_{\langle p,2 \rangle}(t)$ in the model in Fig. 4.26) have to be reduced. Matching the load on both edges, i.e., eliminating the mismatches of the pulses, avoids the injection mismatch spurs in the synchronous reference architecture. In the retimed reference architecture the injection mismatch spurs will only reduce their amplitude to the level of the side spurs, because of the remaining deviation $r[P/2]$. Finally, the side spurs of the retimed reference architecture appear at up to -100 dBc in Fig. 4.29 and Fig. 4.31. By contrast, no side spurs are evident in the phase noise of the synchronous reference architecture in Fig. 4.30 and Fig. 4.32.

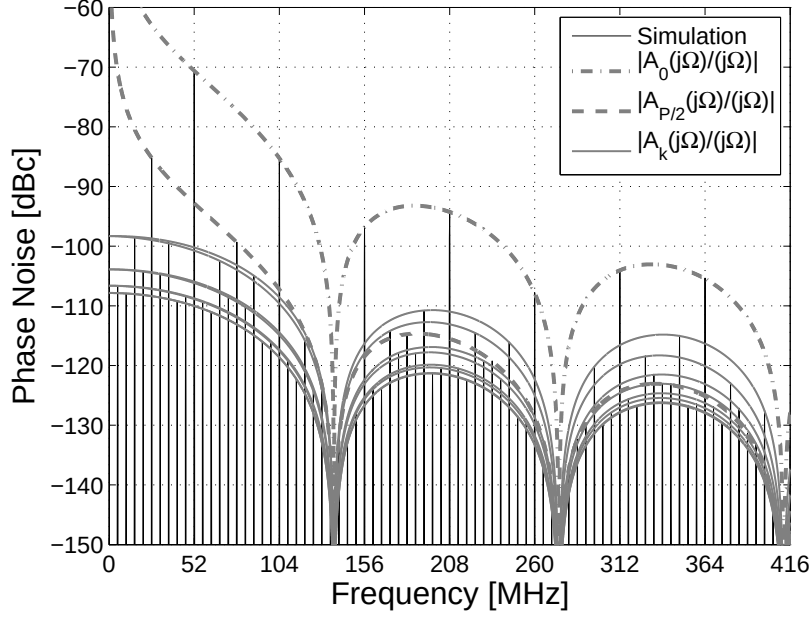


Figure 4.29 Phase noise of the reference retimed ADPLL with injection pulling and a frequency command word of $N_r = 63\frac{2}{5}$, i.e., a period of $M = 5$.

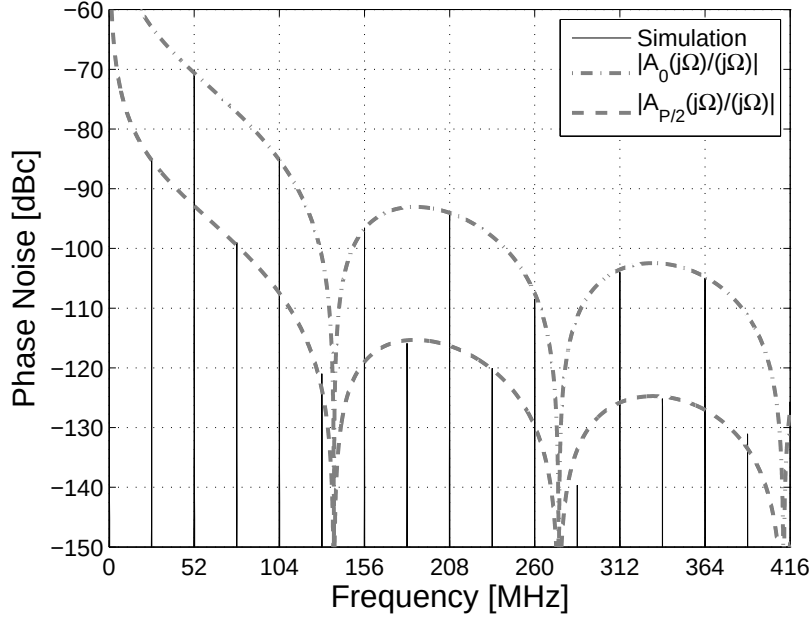


Figure 4.30 Phase noise of the synchronous reference ADPLL with injection pulling a frequency command word of $N_r = 63\frac{2}{5}$, i.e., a period of $M = 5$.

4.6 Injection Pulling

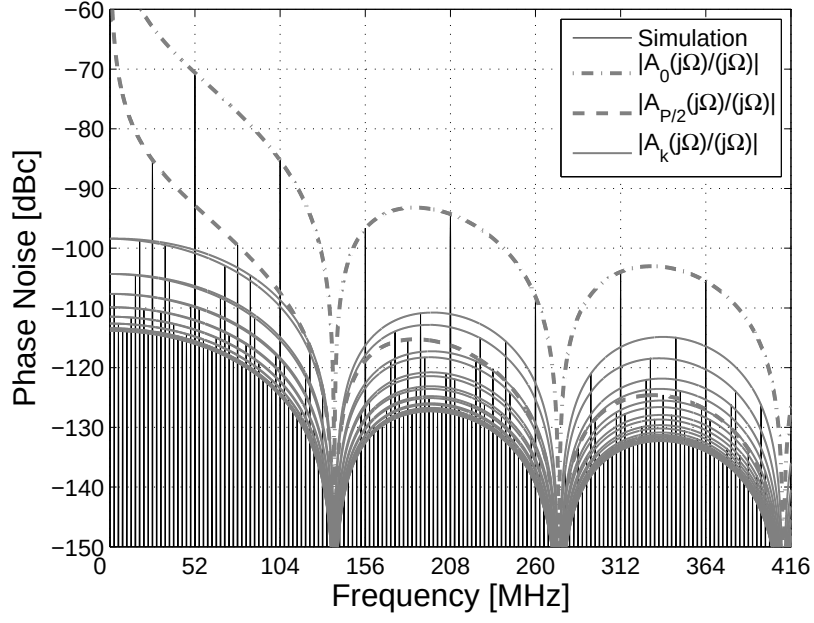


Figure 4.31 Phase noise of the reference retimed ADPLL with injection pulling and a frequency command word of $N_r = 63\frac{3}{10}$, i.e., a period of $M = 10$.

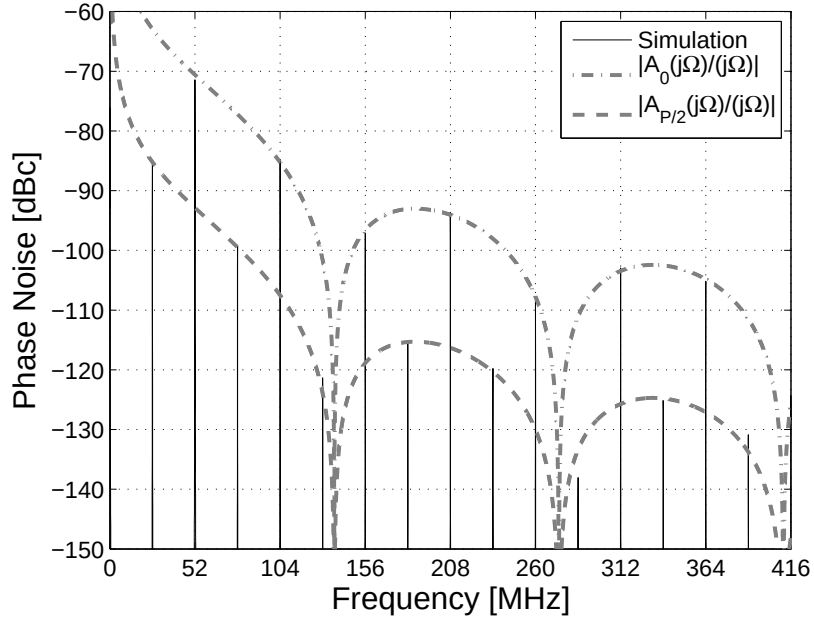


Figure 4.32 Phase noise of the synchronous reference ADPLL with injection pulling a frequency command word of $N_r = 63\frac{3}{10}$, i.e., a period of $M = 10$.

4.7 Summary and Discussion

In this chapter, we performed a thorough performance analysis of ADPLLs. For each imperfection we have investigated its effect on the ADPLL output phase and have compared the theory to the simulation results. In a first step, however, we have reviewed the phase noise power spectral density, which is used in the PLL community as performance metrics. As a result of the analysis, we have, on the one hand, shown how to compute the phase and frequency noise spectra from the zero-crossings of clock signals and, on the other hand, how to model phase noise on the zero-crossings of the clock signals in a high-level simulation environment. The approach is exemplified for oscillator phase noise, where we have presented a rigorous time-domain framework to model any colored noise according to certain specifications in the phase noise spectrum. Moreover, we have described the effects of reference clock jitter on the output phase noise of an ADPLL frequency synthesizer.

In ADPLLs the DCO suffers not only from oscillator noise, but also from its coarse frequency resolution of its discrete tuning curve. Therefore we have described the ADPLL output spectrum, considering the DCO frequency granularity, with a “linearized” DCO, a DCO with white dithering, and a DCO with $\Sigma\Delta$ noise shaping. The spurs due to the deterministic MASH $\Sigma\Delta$ modulator output signal were described and circumvented by a proposed single-stage multiple feedback $\Sigma\Delta$ modulator, that effectively destroy any deterministic pattern for an output phase noise spectrum without spurious tones.

Besides the DCO, the TDC is the second critical component within the ADPLL. In contrast to the literature, we have shown that the strong quantization of the TDC cannot be considered as linearized additive noise source. Thus, the ADPLL output spectrum due to TDC quantization is not white and contains strong spurs. Therefore, we have presented an open loop signal processing model that predicts the frequency location and amplitudes of these TDC spurs under certain conditions. In fact the TDC is the key bottleneck of ADPLLs, because its imperfection is mainly responsible for the in-band phase noise and its spurs require strong loop filtering. Improved TDC architectures with finer resolution and methods that achieve a white TDC noise spectrum are open research issues. Especially a TDC that achieves a noise shaping effect is desired, because the loop filter would attenuate the high-pass shaped quantization noise and increase the performance drastically.

The last part of this section considers coupling effects within the mixed-signal part of the ADPLL. The noise due to the digital circuits affects the frequency of the analog oscillator. With each rising and falling edge of the system clock, the DCO output frequency is “pulled”. This effect is known as injection pulling. In order to analyze the impact of injection pulling on the output phase, we have presented an open loop signal processing model. With the aid of this model, we have classified the spurs into three categories: reference spurs due to injection pulling, side spurs due to the non-uniform reference clock, and injection mismatch spurs due to different impacts among rising and falling edges. In accordance with the theory, simulation results have shown that the synchronous reference architecture reduces the number of spurious tones by eliminating the class of side spurs. However, coupling effects are another severe problem in ADPLLs and should be further investigated.

5

Fast Frequency-Hopping Phase-Domain All-Digital Phase-Locked Loops

In this chapter we will propose a method to perform a frequency hop within one reference cycle with an instantaneous frequency offset of less than 20 ppm of the synthesized frequency. Therefore we first exploit the digital nature of an ADPLL by storing and restoring its state for each frequency band, and, second, utilize its phase domain characteristics, that allows for changing the frequency and adjust the phase signal accordingly in a digital manner. The presented method fulfills the stringent switching requirements of 9.5 ns for Multiband Orthogonal-Frequency-Division-Multiplexing Ultra-Wideband (MB-OFDM-UWB) after an initial acquisition process. To sum up, we propose a look-up table (LUT) based method to preserve the state of the ADPLL over a short period where the ADPLL synthesizes a different output frequency. Storing and restoring the state of the ADPLL includes the phase error (or equivalently the tuning word) and any memory in the system (delay line within the TDC or loop filter memory for higher order ADPLLs). The initial frequency acquisition and the frequency tracking, however, is slowed down in contrast to a regular ADPLL operation with the same parameters.

In the following we will briefly review the MB-OFDM-UWB specification with a focus on frequency synthesis. The key requirements, challenges, and solutions from the literature are discussed in Sec. 5.1. In Sec. 5.2 we extend the ADPLL description to include frequency hopping. The proposed algorithm is split into two parts. The method to achieve frequency-lock is considered in Sec. 5.3 and the method to achieve phase-lock is described in Sec. 5.4. Section 5.5 analyzes the output-phase noise performance of the ADPLL, Sec. 5.6 presents the simulation results, and in Sec. 5.7 we conclude the chapter with a short summary and discussion.

5.1 Multiband Orthogonal-Frequency-Division-Multiplexing Ultra-Wideband

The Federal Communications Commission (FCC) allows a spectrum of 7500 MHz frequency range for unlicensed use of commercial ultra-wideband (UWB) communication systems as long as their power spectral densities fulfill certain criteria, e.g., at least below -41.25 dBm in a 1 MHz bandwidth. Among many different possible approaches, such as pulse-based radio, MB-OFDM-UWB has gained a lot of support by the industry. The OFDM based approach for UWB systems seems promising

5.1 Multiband Orthogonal-Frequency-Division-Multiplexing Ultra-Wideband

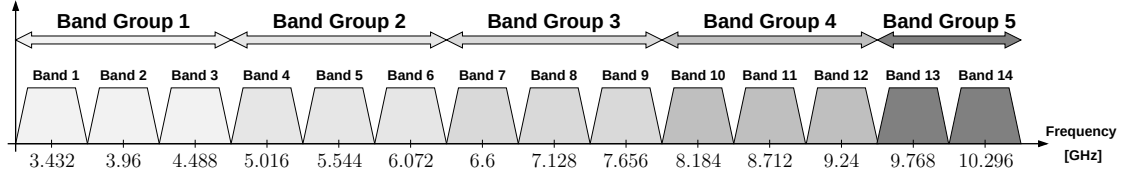


Figure 5.1 Frequency bands and frequency band groups for MB-OFDM-UWB.

to support high data rates over short distances. In MB-OFDM-UWB the entire frequency range of 7500 MHz is split into several band groups, which themselves are split into 3 or 2 frequency bands of 528 MHz as shown in Fig. 5.1. The carrier frequency of the k th frequency band is given by

$$f_c(k) = 2904 + 528 k \quad [\text{MHz}]. \quad (5.1)$$

In order to support multiple access, to avoid narrow-band interferences, and to increase the diversity of the channels, time-frequency codes are employed. Figure 5.2 illustrates the concept of time-frequency codes exemplified with frequency band group 1. Within the frequency band group the system hops among the frequency bands in a fixed periodic pattern. The carrier frequency is changed with a rate of 3.2 MHz which leads to a duration of one frequency band of 312.5 ns. Each band consists of a cyclic prefix of 60.6 ns to avoid inter-symbol-interference, an OFDM symbol of 242.4 ns, and a guard interval of 9.5 ns that enables the receiver and the transmitter to switch between the bands. Each frequency band has a bandwidth of 528 MHz and uses a 128 point IFFT for OFDM modulation. That is, the OFDM sub-carriers are spaced in a 4.125 MHz grid and are modulated with quadrature pulse shift keying (QPSK). For more details on MB-OFDM-UWB refer to [167, 168] and the references therein.

The most challenging requirement on the frequency synthesizer in MB-OFDM-UWB systems is the short switching time of less than 9.5 ns for frequency hops of 528 MHz and more. Due to the wide spacing of the subcarriers of 4.125 MHz and the usage of QPSK the in-band phase noise is not critical, cf. [169]. The use of only 2 bits for modulation reduces the requirements on the transmitter/receiver in general, and the synthesizer phase noise in particular. On the contrary, out-of-band spurs of the synthesizer may violate the strict FCC emission regulation mask and are therefore more critical.

The frequency planning of MB-OFDM-UWB allows for a straightforward frequency synthesizer, that generates the desired carrier frequencies with a single ADPLL and several frequency mixer stages [168]. This solution has advantages in complexity and enables a very fast switching of a few nanoseconds, as shown in the implementation in [170]. However, frequency mixers are analog building blocks and suffer from their nonlinear tuning curves that introduce significant spurs in the output phase noise spectrum. To reduce the number of spurs, the number of mixers is reduced by employing two PLLs, e.g., [171–173]. The transceiver in [174] uses three PLLs, which are operated simultaneously to produce a quasi spur-less spectrum on account of complexity, die size, and power consumption. In [175] two PLLs, which are operated in an interleaved mode, are used. While the first PLL generates the frequency for the current frequency band, the second PLL locks to the frequency of the successive frequency band within 312.5 ns. An interesting approach has been proposed in [176], where a single ADPLL without mixers is used to generate the center frequencies for MB-OFDM-UWB. In principle, the operation consists of two modes, i.e., an acquisition mode and transmission mode. During acquisition the ADPLL is operated in closed loop to determine and store the digital tuning word for each frequency band.

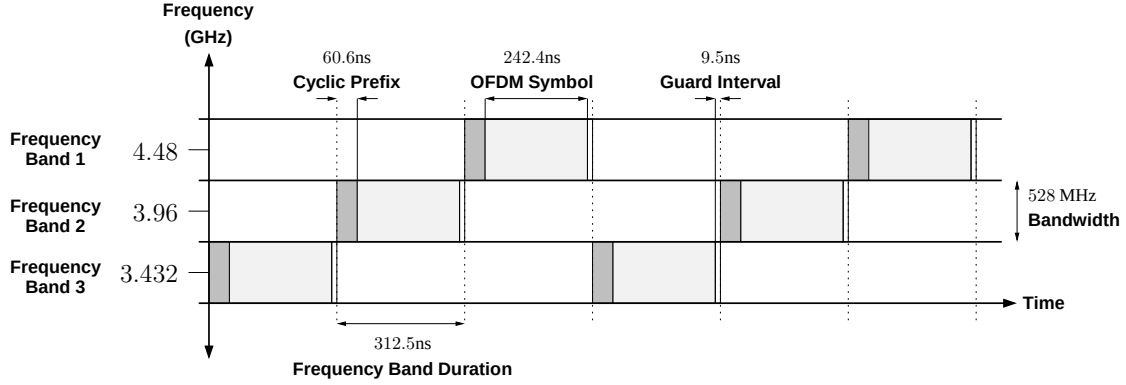


Figure 5.2 An example of time-frequency codes used for MB-OFDM-UWB systems [168].

During transmission the ADPLL is operated in open loop by reloading the stored tuning words. In that way the switching time of the ADPLL is only limited by the switching time of the DCO. However, the drawback of that method is the open loop mode during transmission. First, the oscillator's in-band phase noise is not suppressed and, second, changes in the environment cannot be tracked. While the noise may be acceptable, the inability of the ADPLL to track drifts causes problems, because the required frequency accuracy of 20 ppm cannot be guaranteed. In particular, on-chip self-heating will degrade the ADPLL performance and could lead to a violation of the specifications.

5.2 A Frequency Hopping All-Digital Phase-Locked Loop

In the following we will extend the ADPLL description to account for a frequency-hopping environment. In such a system the desired frequency and therefore also the frequency command word changes each K_r th retimed reference cycle, i.e.,

$$f_v^{ss}[k] = N_r[k] f_r \quad (5.2)$$

where k denotes the index of the actual frequency band, i.e.,

$$k = \left\lfloor \frac{m}{K_r} \right\rfloor. \quad (5.3)$$

The time-frequency codes of MB-OFDM-UWB repeat periodically with a certain period [167]. Without loss of generality and to simplify the notation, we will assume that the code period matches the number of frequency bands K_b with

$$k_b = (k \bmod K_b) + 1 \in [1, K_b]. \quad (5.4)$$

where k_b is the index of the current frequency band and is in the range from 1 to K_b . The framework, however, is suitable for any other time-frequency codes as well. Figure 5.3 shows the retimed reference architecture from Fig. 3.9, where we have included the frequency hopping aspect with the time index k . Furthermore, we have slightly redrawn the structure by first computing $N_e[m]$ given by eqn. (3.28) and then computing $\varphi_e[m]$ by accumulating $N_e[m]$ as in eqn. (3.27)¹. The characteristics of the

¹In contrast to the analysis in the previous sections we neglect the delay in $N_r[m]$, $N_{CKR}[m]$, $N_{CKV}[m]$, and $N_e[m]$ for notational simplicity.

5.3 Instantaneous Frequency-Lock

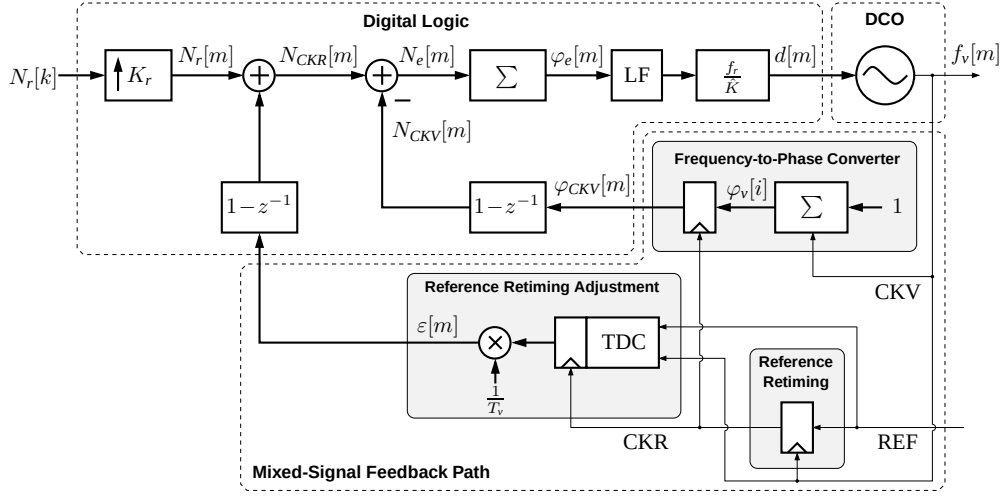


Figure 5.3 A retimed reference ADPLL for frequency-hopping applications.

modified structure is identical, but beneficial in explaining and evaluating the proposed fast-frequency hopping algorithm.

5.3 Instantaneous Frequency-Lock

The proposed fast frequency-hopping ADPLL method aims to simulate K_b PLLs with one physical system. We operate a single ADPLL in a time-interleaved operation, where we restore the state of the ADPLL to its previous condition of the same frequency band. Hence, we exploit the short time between the reoccurrence of one frequency band, which corresponds in the worst case to 4 frequency bands, i.e., $1.25 \mu\text{s}$. Drifts during that time are supposed to be rather small in contrast to the required accuracy. However, as we operate three ADPLLs in a time-interleaved mode, the bandwidth has to be chosen accordingly wider, see Sec. 5.3.2.

In the proposed implementation we store and restore the phase error $\varphi_e[m]$ of the ADPLL for each frequency band in a LUT as illustrated in Fig. 5.4. In the first cycle of a frequency hop, i.e., k_b changes, the phase error $\varphi_e^{\text{in}}[m]$, which corresponds to the old frequency band, is updated and stored in the LUT. Simultaneously the signal **restore** is set to one, such that the phase error $\varphi_e(k_b)$, which corresponds to the actual frequency band, is restored from the LUT. Figure 5.5 shows an example for the proposed method, with $K_b = 3$ and $K_r = 4$. Each fourth retimed reference cycle, i.e., when a frequency hop occurs, the phase error is updated according to eqn. (3.28) and eqn. (3.27) and stored in the LUT as indicated as grey sample in Fig. 5.5. At the same retimed reference cycle the stored phase error which corresponds to the novel frequency band is reloaded as indicated as bold sample in Fig. 5.5.

5.3.1 System Memory and TDC Modification

Any memory in the system, e.g., the loop filter for higher order ADPLLs or any additional IIR filters, has to be stored and restored as the phase error. In particular, as discussed in Sec. 3.1.2, the TDC uses a moving average filter with a memory of N_{avg} taps to improve the accuracy of the DCO

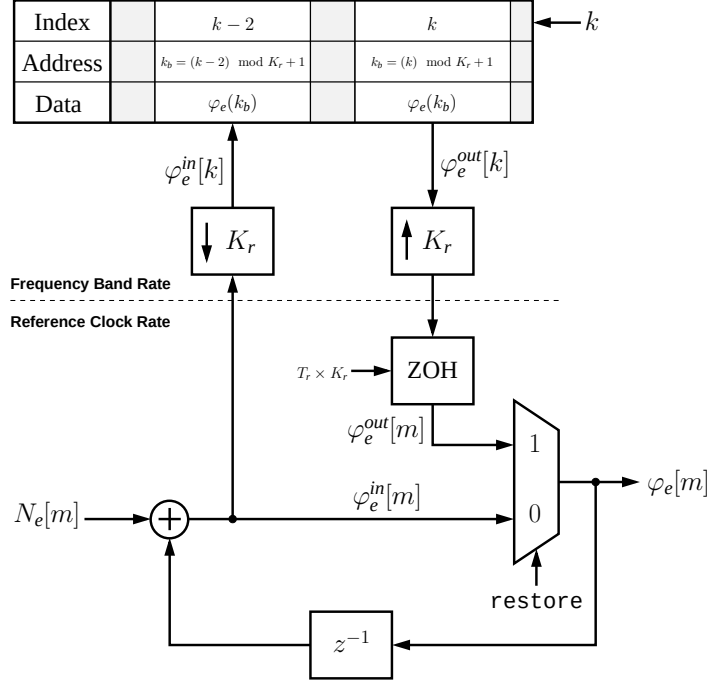


Figure 5.4 Look-up table based phase error preservation.

period measurement. To account for the fast-hopping environment, we use K_r delay lines instead of one, as shown in Fig. 5.6. Additionally, the measurements from other bands may be combined to further enhance the resolution of the DCO period estimation. For example with $N_r[m] = N_r(k_b)$ we can include the DCO period estimation of frequency band $k_b + 1$ in the computation of $\overline{D}_{CKV}[m]$ corresponding to band k_b by

$$\overline{D}_{CKV}[m] = \overline{D}_{CKV}(k_b) + \overline{D}_{CKV}(k_b + 1) \frac{N_r(k_b)}{N_r(k_b + 1)}. \quad (5.5)$$

5.3.2 Initial Frequency Acquisition

In the following we compare the initial frequency acquisition process of the proposed interleaving technique to the regular frequency acquisition of an ADPLL. Let $h_u[m]$ be the discrete-time step response derived from the ADPLL transfer function in eqn. (2.29), then the corresponding continuous-time step response of the DCO output frequency is given by

$$h_u(t) = \sum_{m=0}^{\infty} h_u[m] h_{ZOH}(t - mT_r) \quad (5.6)$$

with $h_{ZOH}(t)$ being a ZOH with a hold time of T_r^2 . The ADPLL output frequency signal depends on the frequency step, i.e., $f_v^{ss} - f_0$, and the step response $h_u[m]$,

$$f_v(t) = f_0 + (N_r - N_0)h_u(t) \quad (5.7)$$

²The retimed reference actually has a ZOH with periodically time-varying hold-times $T_{CKR}[m]$ as discussed in Sec. 3.2, which is neglected in eqn. (5.6) for simplicity.

5.3 Instantaneous Frequency-Lock

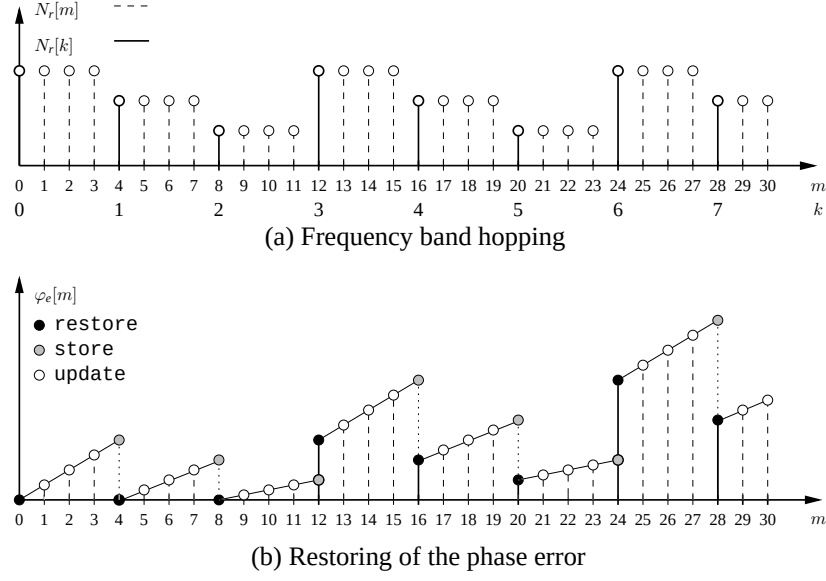


Figure 5.5 Look-up table based phase error preservation example.

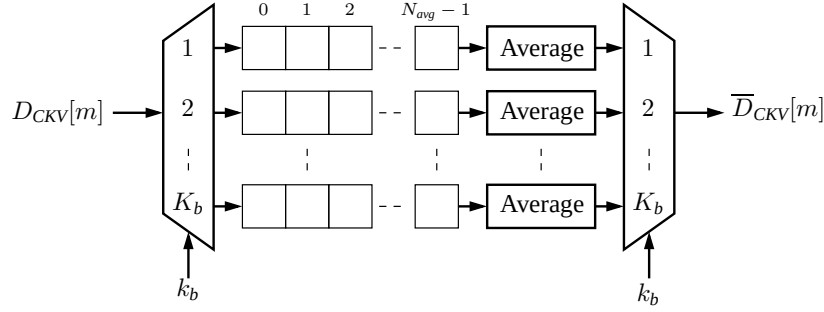


Figure 5.6 TDC extension to store and restore the CKV period estimation for each frequency band.

with

$$N_0 = \frac{f_0}{f_r}. \quad (5.8)$$

Since the ADPLL with instantaneous frequency-lock operates all K_b bands in an interleaved manner, where each band lasts for K_r retimed reference cycles, the frequency acquisition process for one band is slowed down in contrast to eqn. (5.6). In order to relate the acquisition process of the time-interleaved operation to the regular operation we define the step response $\bar{h}_u(t)$. Therefore, for each frequency band of the fast hopping ADPLL, the locking process over time can be approximated by

$$\bar{h}_u(t) = \sum_{m=0}^{\infty} h_u[m] h_{ZOH} \left(t - m \frac{T_r}{K_b} \right). \quad (5.9)$$

That is, the locking time is increased by a factor of K_b , because each frequency band is operated only $(1/K_b)$ th of the time.

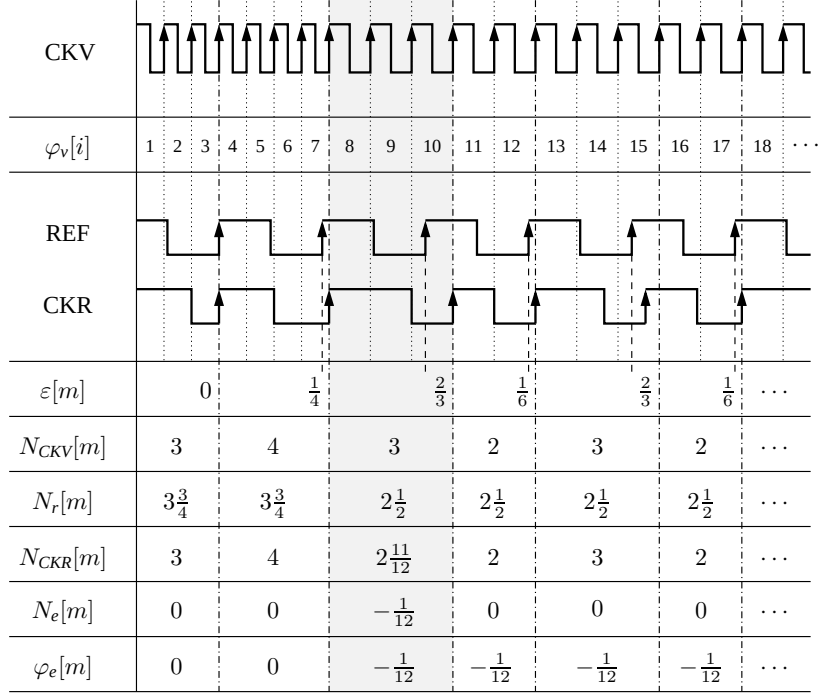


Figure 5.7 ADPLL signals with a frequency hop after the second retimed reference cycle and $\varepsilon[m-1] = 3/4$.

5.4 Instantaneous Phase-Lock

Even in a first-order ADPLL it is not straightforward to change the DCO output frequency $f_v[m]$, by restoring a certain phase error $\varphi_e[m]$ without disturbing the frequency-to-phase conversion in closed loop operation. In order to avoid disturbances, we have to modify the digital phase signals. We first describe the cause of the perturbation and present a straight forward open/closed loop method as a possible solution. Finally, we refine the method by utilizing additional measurements to operate the ADPLL continuously in closed loop.

5.4.1 Problem Statement

Figure 5.7 illustrates the arising problem. Assuming the ADPLL is in lock, the frequency command word is $N_r[m]$ and the feedback path produces $N_{CKV}[m]$. The retiming error $\varepsilon[m]$ is differentiated and added to the frequency command word to produce $N_{CKV}[m]$ according to eqn. (3.21) such that the phase error $\varphi_e[m]$ is constant in first-order ADPLLs or zero in higher order ADPLLs. After the second retimed reference cycle the ADPLL switches its desired frequency, i.e., the frequency command word changes from $N_r[m] = 3\frac{3}{4}$ to $N_r[m] = 2\frac{1}{2}$. The stored phase error is restored from the LUT and injected to the system. However, even under the assumption that reloaded phase error leads immediately to the ideal DCO output frequency, an instantaneous error affects $N_e[m]$. This error, denoted as $\Delta\varepsilon[m]$, remains in the system until the feedback mechanism compensates it, which can be a time consuming process over several reference cycles.

The source of the error $\Delta\varepsilon[m]$ is the memory in the differentiator and the fact that the retiming

5.4 Instantaneous Phase-Lock

error $\varepsilon[m]$ is normalized to the DCO period, cf. eqn. (3.10). With the frequency hop, the intrinsic normalization of the TDC operation corresponds to the previous instead of the current frequency band. Thus, the error $\Delta\varepsilon[m]$ is given by

$$\Delta\varepsilon[m] = \varepsilon[m-1] \frac{N_r[m]}{N_r[m-1]} \quad (5.10)$$

which is exactly $-\frac{1}{12}$ for the example in Fig. 5.7. Under the assumption that the DCO transient can be neglected, e.g., innarrow-band applications, we can digitally adjust the phase signals to achieve immediate lock by computing $N_{CKR}[m]$ as

$$N_{CKR}[m] = N_r[m] + \varepsilon[m] - \varepsilon[m-1] \frac{N_r[m]}{N_r[m-1]}. \quad (5.11)$$

5.4.2 Open/Close Loop Mechanism

In case of wide-band applications, such as MB-OFDM-UWB with a frequency hop of 528 MHz or more, the DCO transient cannot be neglected. That is, the assumption in eqn. (3.26) that $f_v[m]$ is constant over one retimed reference cycle is not valid anymore. As a result, the value of $N_{CKV}[m]$ is corrupted by an unknown error, which is introduced by the integration over the DCO transient. The desired number of CKV cycles $N_{CKR}[m]$ in eqn. (3.24) does not agree with the actual number of CKV cycles $N_{CKV}[m]$ in eqn. (3.26) during the CKR cycle after the hop, such that eqn. (3.21) is not a correct reference.

One solution to overcome this problem is to use a hybrid of open loop and closed loop operation. Hence, in the first retimed reference cycle after a frequency hop, depicted as grey area in Fig. 5.7, we open the loop, such that the corrupted value of $N_e[m]$ cannot affect the system. In the second retimed reference cycle after a frequency hop the loop is closed again. Under the assumption that the DCO transient is shorter than one retimed reference cycle, the value $\varepsilon[m]$ can be accurately measured by the TDC in the first retimed reference cycle and used in the second retimed reference cycle to compute $N_{CKR}[m]$ according to eqn. (3.21). Consequently we operate the system for two retimed reference cycles with the same phase error and tuning word. The locking time is therefore additionally degraded, i.e.,

$$\bar{h}_u(t) = \sum_{m=0}^{\infty} h_u[m] h_{ZOH} \left(t - nT_r \frac{K_r - 1}{K_r K_b} \right). \quad (5.12)$$

The factor $K_r - 1$ is the number of retimed reference cycles used in closed loop for one frequency band and the factor $K_r K_b$ is the total number of retimed reference cycles within one frequency pattern. Unfortunately, simulations have shown that operating the system twice with the same phase error introduces a kind of drifting effect, which degrades the phase noise performance. Therefore an advanced solution that achieves a continuous closed loop operation is desired.

5.4.3 Extended Closed-Loop Operation

To operate the ADPLL in closed loop over a frequency hop, a measurement from the first retimed reference cycle after the frequency hop is required. Unfortunately, the DCO transient, which should be shorter than 9.5 ns to comply with the MB-OFDM-UWB specification, will impair the measurement of $N_{CKV}[m]$. However, to overcome this problem we will consider only the second half of this retimed

reference cycle, where we assume that the DCO transient is shorter than $T_r/2$. Figure 5.8 illustrates the relation between $N_{CKV/2}[m]$ and $N_{CKR/2}[m]$, which are used instead of $N_{CKV}[m]$ and $N_{CKR}[m]$ in the first retimed reference cycle after a frequency hop. On the one hand, the number of *actual* CKV cycles during one half CKR period $N_{CKV/2}[m]$ can be measured by

$$N_{CKV/2}[m] = \varphi_{CKV}[m] - \bar{\varphi}_{CKV}[m] \quad (5.13)$$

where $\bar{\varphi}_{CKV}[m]$ is the feedback accumulator value sampled with the falling edge of CKR. On the other hand, the *desired* number of CKV cycles during one half CKR period $N_{CKR/2}[m]$ is given by

$$N_{CKR/2}[m] = \frac{N_r[m]}{2} + \varepsilon[m] - \bar{\varepsilon}[m] \quad (5.14)$$

where $\bar{\varepsilon}[m]$ is the time between the falling REF edge to the consecutive rising CKV edge normalized to one CKV period, as defined in Sec. 4.6. The quantity $\bar{\varepsilon}[m]$ has to be measured by the TDC. That implies, that in the first retimed reference cycle after a frequency hop the TDC has to be operated twice, i.e., for the falling and rising REF edge. Consequently the delay line in Fig. 3.8 has to be sampled also with the falling REF edge, which requires an additional array of flip-flops in Fig. 3.7.

Finally, the phase error in the first cycle after a frequency hop is given by

$$\varphi_e[m] = \varphi_e[m-1] + N_{e/2}[m] \quad (5.15)$$

with

$$N_{e/2}[m] = N_{CKR/2}[m] - N_{CKV/2}[m]. \quad (5.16)$$

By considering a half retimed reference cycle, the locking and tracking behavior of the ADPLL will be affected. In fact using $2N_{e/2}[m]$ in eqn. (5.15) will produce a step response as defined in eqn. (5.9), but, as we measure $\varepsilon[m]$ and $\bar{\varepsilon}[m]$, both with a time-resolution of Δt_{res} , the TDC quantization error on $N_{e/2}[m]$ is twice as large as on $N_e[m]$. To keep the TDC quantization noise as small as possible, we use eqn. (5.15) on account of a slower acquisition and tracking performance. Note that we can modify the PLL bandwidth to enhance the loop dynamics, while the TDC quantization noise impairs the in-band phase noise and cannot be filtered due to the low-pass nature of the TDC noise transfer function, see Sec. 2.1.5. In consequence, each fourth retimed reference cycle we have a differently scaled error, which affects the bandwidth of the ADPLL and results in a time-varying system. Nevertheless, the step response of a time-interleaved ADPLL with closed loop operation using eqn. (5.15) can be approximated by

$$\bar{h}_u(t) = \sum_{m=0}^{\infty} h_u[m] h_{ZOH} \left(t - nT_r \frac{K_r - \frac{1}{2}}{K_r K_b} \right) \quad (5.17)$$

5.5 Phase Noise Performance Analysis

Although storing and restoring of the ADPLL state introduces no additional error in a noiseless scenario, it propagates the noise on the phase error signal during the next occurrence of the frequency band in a real world scenario. In lock, the phase error signal $\varphi_e[m]$ will jitter around a mean value that corresponds to the desired value $\varphi_e^{ss}[m]$, which would be 0 in a type-II ADPLL, cf. Sec. 2.1.4. That is,

$$\varphi_e[m] = \varphi_e^{ss}[m] + \eta_{Hop}[m] \quad (5.18)$$

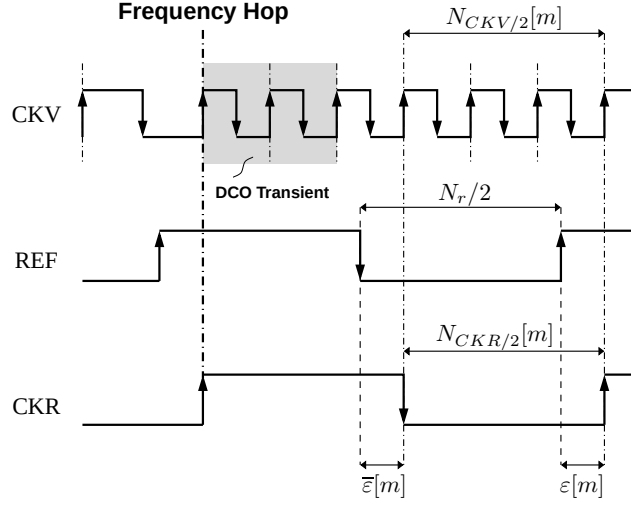


Figure 5.8 Relation between $N_r[m]/2$, $N_{CKV/2}[m]$, and $N_{CKR/2}[m]$.

where $\eta_{Hop}[m]$ is the noise on the phase error signal before a frequency hop occurs. Figure 5.9 illustrates the impact of phase error imperfection on the output phase noise in the time-interleaved operation. Each time a frequency hop occurs, $\varphi_e^{ss}[m]$ and the error on the phase detector $\eta_{Hop}[m]$ is stored. By reloading the state of the ADPLL for the k -th frequency band, the error signal $\eta_{Hop}[k]$ is reloaded likewise. Due to the fast hopping the error $\eta_{Hop}[k]$ will not be compensated by the feedback mechanism of the ADPLL and the error in the successive occurrence of the same band is therefore $\eta_{Hop}[k-1] + \eta_{Hop}[k]$. Thus, the hopping error accumulates at the rate of the frequency band, where $H_{Int}(\Delta\omega)$ is the frequency-response of the accumulator. The frequency rate conversion is done with a ZOH, i.e., $H_{ZOH,K_r}(\omega)$ with a hold time of f_r/K_r , as the error appears as constant over one frequency band. Due to the closed loop operation the hopping error is filtered by the loop with the TDC noise transfer function $H_{TDC}(\Delta\omega)$, filtered by the ZOH of the reconstruction $H_{ZOH}(\omega)$, and converted into the phase domain by $\frac{1}{\omega}$. That is, the error introduced by the frequency hopping is

$$\mathcal{L}_{Hop}\{\Delta\omega\} = \mathcal{L}_\eta\{\Delta\omega\} |H_{Int}(\Delta\omega)|^2 |H_{ZOH,K_r}(\Delta\omega)|^2 |H_{TDC}(\Delta\omega)|^2 |H_{ZOH}(\Delta\omega)|^2 \left| \frac{1}{\Delta\omega} \right|^2 \quad (5.19)$$

where $\mathcal{L}_\eta\{\Delta\omega\}$ is the noise spectrum of $\eta_{Hop}[k]$. Any imperfection in the system contributes to the noise spectrum $\mathcal{L}_\eta\{\Delta\omega\}$. In state-of-the art phase-domain ADPLLs, however, the TDC quantization is considered as the most severe imperfection at the phase detector. In the following we will therefore only consider TDC quantization. Thus, the spectrum $\mathcal{L}_\eta\{\Delta\omega\}$ becomes

$$\mathcal{L}_{\eta_{Hop}}\{\Delta\omega\} = \frac{(2\pi\sigma_{\eta_{Hop}})^2}{f_r/K_r}. \quad (5.20)$$

with

$$\sigma_{\eta_{Hop}}^2 = \frac{\Delta t_{res}^2}{3(T_v^{ss})^2}. \quad (5.21)$$

In contrast to eqn. (4.62), the spectrum eqn. (5.20) spreads its uniformly distributed energy over the frequency band rate f_r/K_r instead of f_r . Moreover, in contrast to eqn. (4.61), the variance $\sigma_{\eta_{Hop}}^2$ accounts for the ceiling operation. In closed loop operation the TDC will compensate for the mean

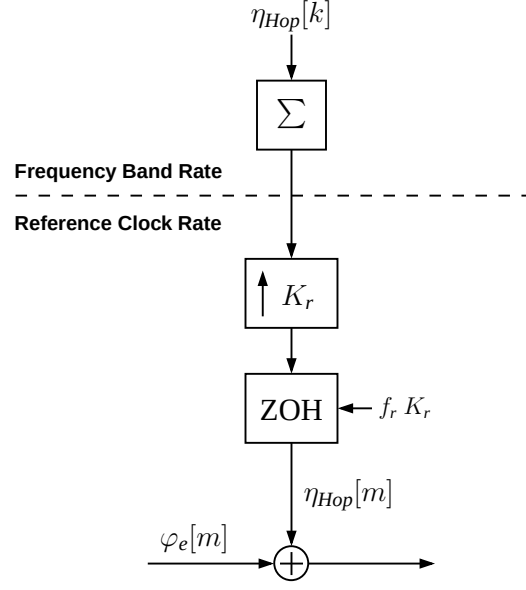


Figure 5.9 Model of the frequency-hopping error due to the time-interleaved operation.

value of the TDC quantization error, which is actually not zero-mean, i.e., in the interval $-\Delta t_{res}/T_v^{ss}$ to 0, see Sec. 4.4.1. Due to the time-interleaved operation the mean value is not compensated and the variance becomes eqn. (5.21). Moreover, eqn. (5.20) is independent of the reference frequency f_r , because the duration of one frequency band, i.e., $\frac{K_r}{f_r}$ is determined by the standard. Thus, increasing the reference frequency will not improve $\mathcal{L}_{\eta_{Hop}}\{\Delta\omega\}$.

The output phase noise spectrum of a fast frequency-hopping ADPLL with time-interleaved operation is the sum of all imperfections. In particular, the hopping error $\mathcal{L}_{Hop}\{\Delta\omega\}$ given by eqn. (5.20) in the in-band region and the oscillator noise, see Sec. 4.2, in the out-of-band region will be dominant.

5.6 Simulations

MB-OFDM-UWB		
Number of bands	$K_b = 3$	
Frequency band 1	$f_v^{ss}[1] = 3.432 \text{ GHz}$	
Frequency band 2	$f_v^{ss}[2] = 3.96 \text{ GHz}$	
Frequency band 3	$f_v^{ss}[3] = 4.488 \text{ GHz}$	
ADPLL		
Reference frequency	$f_r = 12.8 \text{ MHz}$	
REF cycles per band	$K_r = 4$	
Oscillator Noise		
Setup	A	B
Jitter $\sigma_{\delta f}$	2380 fs	238 fs
Wander $\sigma_{\Delta T}$	106 fs	10.6 fs
Flicker ³ $\sigma_{\Delta T, 1/T}$	1875 fs	187.5 fs
TDC Quantization		
Setup	A	B
Resolution Δt_{res}	20 ps	1 ps
Averaging N_{avg}	128	128

Table 5.1 ADPLL - setup.

Oscillator Noise		
	A	B
Jitter $\mathcal{L}_{\Delta t}\{\Delta\omega\}$	-141.15 dBc/Hz	-121.15 dBc/Hz
Wander $\mathcal{L}_{\Delta T}\{\Delta\omega\}$	-113.43 dBc/Hz	-93.43 dBc/Hz
Flicker $\mathcal{L}_{\Delta 1/T}\{\Delta\omega\}$	-128.47 dBc/Hz	-108.47 dBc/Hz
TDC Quantization Noise		
	A	B
$\mathcal{L}_\varepsilon\{\Delta\omega\}$ in eqn. (4.62)	-115.19 dBc/Hz	-89.17 dBc/Hz
Hopping Noise		
	A	B
$\mathcal{L}_{Hop}\{\Delta\omega\}$ in eqn. (5.20)	-103.15 dBc/Hz	-77.13 dBc/Hz

Table 5.2 ADPLL - results for 3.432 GHz.

5.6 Simulations

To evaluate the proposed time-interleaved ADPLL for fast frequency-hopping applications we use two different noise setups³ considering oscillator noise and TDC quantization. The simulation setup is summarized in Tab. 5.1 and the results are summarized for the first frequency, i.e., $f_v^{ss}[k] = 3.432$ GHz, in Tab. 5.2. For higher frequencies the phase noise is slightly worse.

Figure 5.10 depicts the locking process of a fast frequency-hopping ADPLL for MB-OFDM-UWB with time-interleaved operation for three different loop filters. Each band is operated for $K_r = 4$ reference cycles in an interleaved manner. Due to the time-interleaved operation the locking process for each band is slowed down by approximately the number of bands $K_b = 3$ in comparison to an ADPLL in regular operation. Figure 5.10 shows that the theoretical step responses (dashed) given by eqn. (5.17) agree with the simulation results (solid). The initial frequency acquisition may be accelerated by techniques from the literature, such as gear shifting [55, 57, 62], reference feedforward compensation [177], or binary search acquisition [68, 74].

Figure 5.11 with noise setup A and Fig. 5.12 with noise setup B show the output phase noise spectrum $\mathcal{L}_\varphi\{\Delta\omega\}$ of the first frequency band, i.e., $f_v^{ss}[1] = 3.432$ GHz. The phase noise has been computed by first combining the output frequency vectors $f_v[i]$, which correspond to the same frequency band, and then using the method described in Sec. 4.1.4. In both plots the simulation results (solid and bold) are compared to the theoretical output spectrum (dashed and bold) for three different loop filters ($\alpha = 2^{-6}$, $\rho = 2^{-9}$ in black, $\alpha = 2^{-7}$, $\rho = 2^{-12}$ in grey, and $\alpha = 2^{-8}$, $\rho = 2^{-15}$ in light grey). The theoretical output phase noise has two major components: the hopping noise spectrum

³The flicker noise filter matches the example in App. A.4.3.

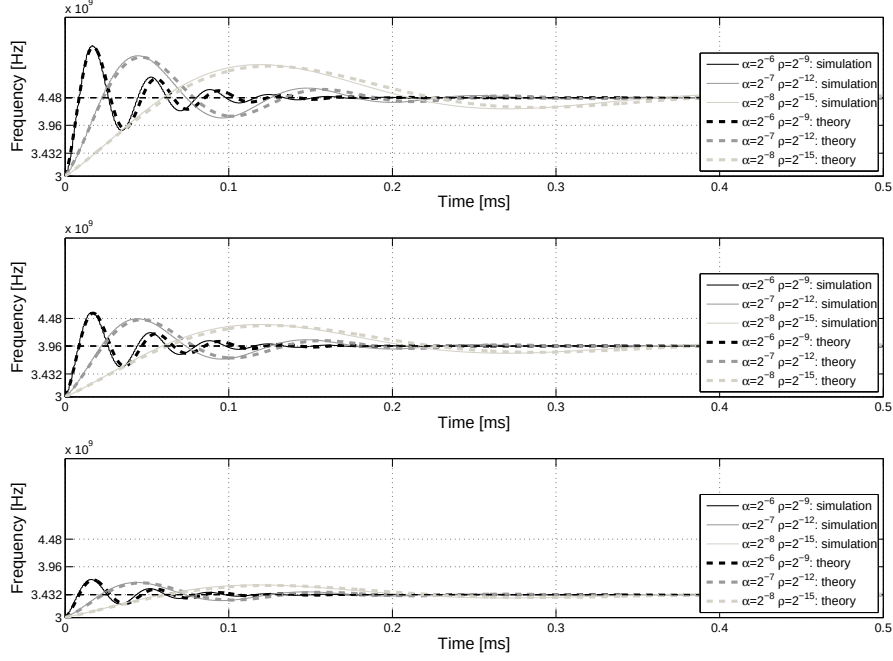


Figure 5.10 Locking process of a fast frequency-hopping ADPLL with time-interleaved operation.

$\mathcal{L}_{Hop}\{\Delta\omega\}$ (solid), which is dominant in the in-band region, and oscillator noise (dashed), which is dominant in the out-of-band region. While the out-of-band noise appears similar to regular ADPLL operation, the frequency hopping method introduces in-band phase noise that decays with 20 dB/dec and is identical for all loop filters. Reducing the oscillator noise by a factor of 10, i.e., by 20 dB from noise setup A to noise setup B, and increasing the TDC time resolution from 20 ps to 1 ps, reduces the hopping noise substantially, because the noise on the phase error signal $\varphi_e[m]$ is downscaled. Besides the dominant imperfection of TDC quantization, also DCO wander contributes to the ripple on the phase error signal. The time-interleaved ADPLL operation fails to filter oscillator wander, which appears to be the limiting factor for the proposed method. However, the phase noise is still better than the phase noise of an uncontrolled oscillator, i.e., an open loop operation as proposed in [176]. On the one hand, the closed loop operation controls the desired frequency and compensates voltage and temperature drifts and, on the other hand, in-band flicker noise is suppressed.

Figure 5.13 with noise setup A and Fig. 5.14 with noise setup B show the frequency accuracy of the fast frequency hopping ADPLL in steady state for all three frequency bands and three different loop filters. The signal $f_v[m]$ corresponds to the average DCO output frequency over one retimed reference cycle. Both plots show that the the output frequency jitters around its desired value for all frequency bands, such that lock is achieved. However, in Fig. 5.13 the noise is too large to fullfill the instant lock margin of ± 20 ppm for all loop-filter types, whereas the noise in Fig. 5.14 is not problematic.

5.6 Simulations

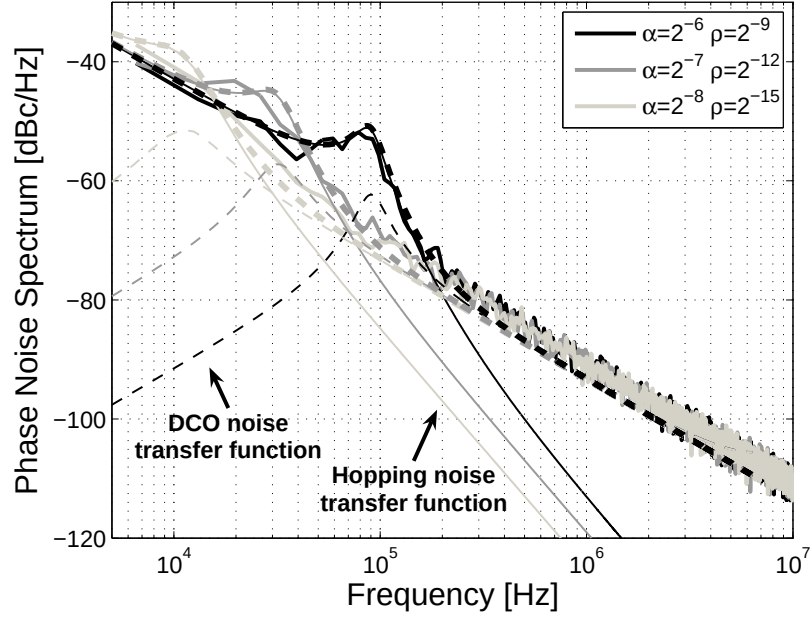


Figure 5.11 Phase noise of the fast frequency-hopping ADPLL and synthesized frequency 3.432 GHz for noise setup A.

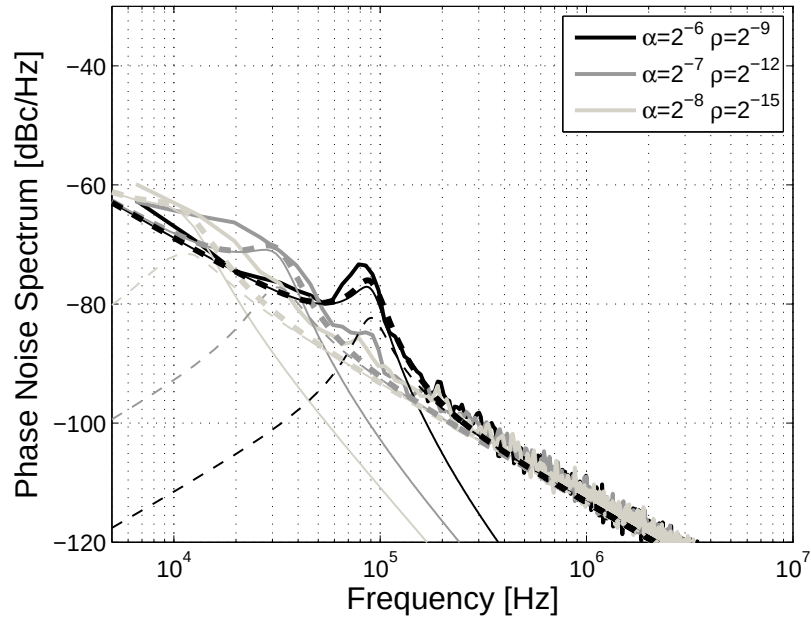


Figure 5.12 Phase noise of the fast frequency-hopping ADPLL and synthesized frequency 3.432 GHz for noise setup B.

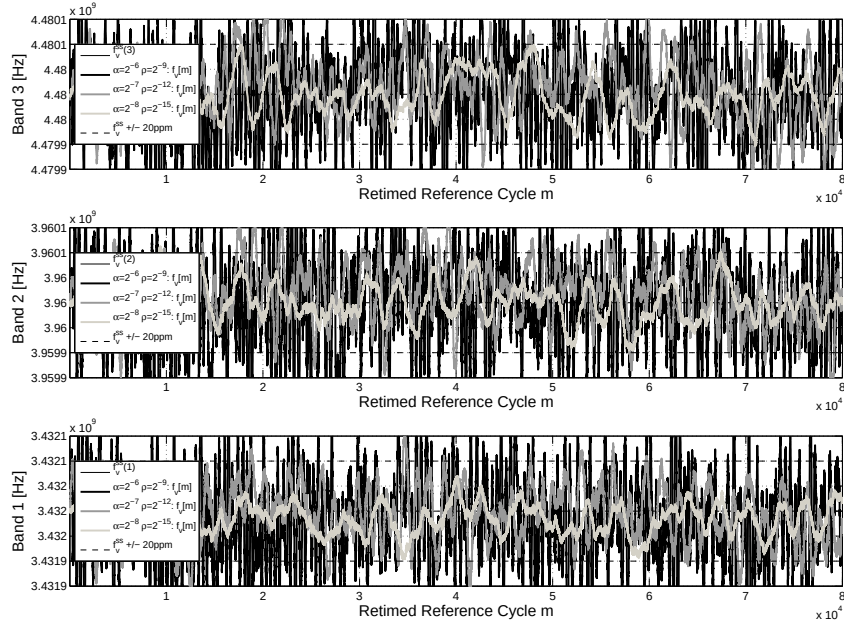


Figure 5.13 Frequency accuracy of the fast frequency-hopping ADPLL and synthesized frequency 3.432 GHz for noise setup A.

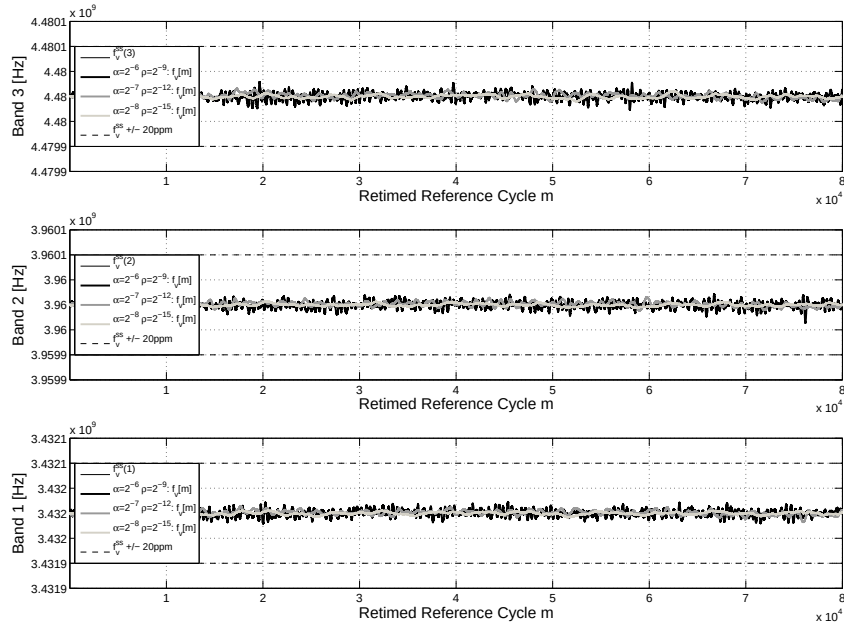


Figure 5.14 Frequency accuracy of the fast frequency-hopping ADPLL and synthesized frequency 3.432 GHz for noise setup B.

5.7 Summary and Discussion

In this chapter we have proposed an ADPLL for fast frequency-hopping applications. In order to comply with the stringent switching requirements of 9.5 ns in MB-OFDM-UWB, a time-interleaved ADPLL operation is proposed. A single ADPLL simulates the operation of three systems by storing and restoring the state of the ADPLL for each of the three frequency bands. Furthermore the digital phase signals are adjusted such that no additional error is introduced. In that way the locking time is reduced to the switching time of the DCO. However, unlike open loop operation of the DCO, i.e., by switching the tuning word only, the closed loop method compensates drifts in the environment such as temperature and voltage. In particular self heating may cause problems in an open loop scenario. The tracking capability of the time-interleaved operation is slowed down by the number of frequency bands, which are operated in interleaved manner, in contrast to regular operation. Additionally we have derived the phase noise performance of the fast frequency-hopping ADPLL extension. The in-band noise is primarily given by the error introduced by the frequency hopping scheme. Storing and restoring the state of the ADPLL introduces an accumulating error with a slope of -20 dB/dec. This imperfection cannot be suppressed by traditional loop filtering, because the TDC or phase detector transfer function is of low-pass nature, cf. Sec. 2.1.5. The magnitude of the error depends on the quality of the phase error signal, which is mainly affected by TDC quantization and oscillator noise. Nevertheless, in contrast to open loop operation, the proposed scheme improves the phase noise performance by mitigating the problematic flicker noise with a slope of -30 dB/dec. In contrast to other low-complexity solutions, i.e., a single PLL, e.g. [170], the time-interleaved ADPLL requires no mixers and produces no severe spurs.

Concluding we can say that the proposed fast frequency-hopping ADPLL operation meets the requirements of the MB-OFDM-UWB standard. Additional mitigation of the noise on the phase error signal may further improve the frequency hopping performance concerning frequency accuracy and phase noise. Therefore methods that aim to improve the phase detector. For instance TDC noise shaping [133] or TDC oversampling [135] may be used to further enhance the performance of the time-interleaved ADPLL. Furthermore, additional filtering techniques that suppress the frequency hopping error are an open research issue.



Summary & Concluding Remarks

In Chapter 1, this thesis was motivated by technology trends that require digital intensive solutions of traditionally analog devices. The demand for single chip solutions [6–10] for high-performance wireless RF transceivers, such as Bluetooth and GSM, has triggered the development of ADPLLs [10, 16, 18, 47–61]. The novel design paradigm for 45 nm CMOS and beyond states that the time resolution of signal transitions is better than the amplitude resolution of voltages [10]. Following these trends, all-digital phase-locked loops (ADPLLs) treat information that is encoded in the zero-crossings of clock signals, which are continuous-time and discrete-amplitude signals and it therefore can be processed with digital CMOS logic.

Thus, ADPLLs can be entirely fabricated in CMOS technology without any analog tuning. Nevertheless, ADPLLs remain mixed-signal systems comprising of analog parts, digital parts, and interfaces between them. Many practical advantages of ADPLLs come at the expense of an increased system complexity that requires a holistic [11] design approach. To exploit the possibility of signal processing in ADPLLs, a mathematical framework and an exact system description for modeling, simulation, and analysis purposes is required. However, little work on the description on ADPLLs can be found in the literature [6, 16]. Therefore, we have introduced in Chapter 2 a z -domain model [71] that emphasizes the basic operation and a mixed-signal model that highlights the mixed-signal nature of the system. These models are used throughout the thesis to evaluate the performance, discuss trade-offs, and develop signal processing methods to enhance the system. As the z -domain model reflects the system operation very accurately, because phase-domain ADPLLs actually aim to implement the phase-domain characteristics of traditional PLLs, it can be used for high-level simulations and algorithm prototyping.

In Chapter 3, we have first analyzed the analog components, which mark the transitions between the analog and the digital signal domain. That is, the time-to-digital converter (TDC) and the digitally controlled oscillator (DCO) architectures from the literature are summarized [34, 42, 94, 109, 111–134] for TDCs and [50, 93, 95, 97–100] for DCOs. The operation and imperfections are discussed for TDCs [94] and for DCOs [93], respectively. In [93] the coarse frequency resolution of the analog DCO is improved by utilizing high-rate $\Sigma\Delta$ dithering of the digital tuning word. A higher sampling rate and increased complexity is traded for higher frequency resolution to finally allow ADPLLs to compete with their analog counterparts. The second part in Chapter 3 reviews a phase-domain ADPLL architecture as described in [10, 16, 18, 47–61] and focuses on the analysis of its periodically time-varying operation in lock [72]. In order to synchronize the slow reference clock domain with the fast DCO output clock domain, the retimed reference architecture introduces a time-varying system clock, denoted as retimed

CHAPTER 6. SUMMARY & CONCLUDING REMARKS

reference clock, by oversampling the reference clock by the DCO output clock. The periodically non-uniform behavior in lock and its impact on the system operation is elaborated and the characteristics of all signals are described analytically. The third part in Chapter 3 proposes an alternative, new ADPLL architecture that operates the entire system with the uniform reference clock and solves the synchronization of the clock domains with a sophisticated metastability avoidance circuit [73]. The synchronous reference architecture, on the one hand, reduces the circuit complexity by eliminating the retimed reference clock and, on the other hand, reduces the number of spurs due to injection pulling by operating the digital logic with an uniform reference clock.

Chapter 4 gives a comprehensive performance analysis of phase-domain ADPLLs. First, the definition of the performance metrics, its derivation, and its computation from discrete observations is discussed. These definitions are used to model oscillator noise on the zero-crossings of the DCO clock signal to reduce the computational burden in an event-driven [49, 60, 61, 85] or high-level [86] simulation environment. Second, all noise sources within the ADPLL and their contribution to the output phase noise spectrum are theoretically described and compared with simulation results. In particular, oscillator noise [49], dithered and noise shaped DCO frequency quantization noise [16, 157], reference clock jitter, and “linearized” TDC quantization noise [6, 110] are considered. In ADPLLs, especially the spurious tones are cumbersome and mainly result from two effects: quantization and parasitic coupling on the die. That is, $\Sigma\Delta$ noise shaping of the DCO increases the DCO resolution such that spurs due to the nonlinear element, i.e., the frequency quantization within the DCO, are prevented. Multi-stage noise shaping (MASH) structures, which are typically used in ADPLLs [35, 47], however, produce a deterministic output sequence with small period for static input signals and consequently spurs in the output spectrum. Moreover, the linear TDC model used in the literature is not valid due to the deterministic periodic input signal of the TDC. The resulting spurs are analyzed with the aid of an open loop signal processing model. The second source for spurs are coupling effects between the digital logic and the analog oscillator [18]. The switching of the digital logic “pulls” the frequency of the oscillator, which is known as injection pulling. In particular, the periodically non-uniform reference clock of the retimed reference architecture in Sec. 3.2 produces a large amount of spurs, whereas the spurious tones in the synchronous reference architecture in Sec. 3.3 are significantly reduced [73].

In Chapter 5, we have proposed a technique that allows the ADPLL to hop between frequency bands within less than one reference cycle in closed loop operation. The ADPLL stores and restores its state to achieve instantaneous frequency-lock and digitally adjusts its phase signals to achieve instantaneous phase-lock. Under the assumption that the time between the recurrence of a frequency band is short, the proposed ADPLL operation achieves a fast frequency hopping in closed loop operation. In particular, the presented scheme is suitable for multiband orthogonal-frequency-division-multiplexing ultra wideband (MB-OFDM-UWB) which supports three bands that repeat periodically in a fast manner. The theoretical performance is evaluated and compared to simulation outcomes.

The performance analysis indicates in accordance with [178] that the TDC is one of the key bottlenecks of ADPLLs. Its coarse quantization and the deterministic input signal of the TDC introduce severe spurs. Increasing the TDC resolution will reduce the power of these spurs but will not prevent their occurrence. Consequently for their mitigation a narrow loop filter is used, which unfortunately increases the locking time and reduces the DCO noise suppression. Methods to randomize the TDC quantization error or even achieve a noise shaping effect, e.g., [34, 132, 133] are an open research issue. Besides high-performance applications, there is a huge demand on PLLs with less stringent specifications, but strict constraints on power and area consumption. For such systems designers aim

to develop ADPLL architectures that meet the requirements without the usage of TDCs [43, 44] in order to reduce the complexity.

Another important open research issue are coupling effects on the die. The resulting spurs significantly impair the system performance, may lead to a violation of the specifications, and therefore degrade the yield of the entire chip. As the degree of integration is constantly increasing, analog shielding for better decoupling of the analog components is becoming difficult and expensive. Therefore the investigation, analysis, and sources of such effects need to be studied and structural modifications, that avoid the generation of these imperfections in the first place, such as [73], are important open research issues.

A

Appendix

A.1 Fourier Transform of a Periodically Non-Uniform Impulse Train

The M -periodic non-uniform impulse train of the retimed reference clock (3.53) is rewritten here again

$$s_{CKR}(t) = \sum_{p=0}^{M-1} s_p(t) \quad (\text{A.1})$$

with

$$s_p(t) = \sum_{k=-\infty}^{\infty} \delta(t - (kM + p + r[p])T_r) \quad (\text{A.2})$$

The transformation of A.1 can be obtained by first developing a Fourier series representation of the form

$$s_p(t) = \sum_{k=-\infty}^{+\infty} c_k e^{j\hat{\omega}_s k t} \quad (\text{A.3})$$

with

$$c_k = \frac{1}{\hat{T}_{ref}} \int_{-\frac{\hat{T}_{ref}}{2}}^{+\frac{\hat{T}_{ref}}{2}} s_m(t) e^{-j\hat{\omega}_s k t} dt. \quad (\text{A.4})$$

The signal $s_p(t)$ has a sampling period of MT_r . Therefore, with substituting

$$\begin{aligned} \hat{T}_{ref} &= MT_r \\ \hat{\omega}_s &= \frac{2\pi}{\hat{T}_{ref}} = \frac{\omega_s}{M} = \frac{2\pi}{MT_r} \end{aligned} \quad (\text{A.5})$$

A.2 Power Spectral Density of an Oscillator Output Signal

the coefficients become

$$\begin{aligned}
c_k &= \frac{1}{MT_r} \int_{-\frac{MT_r}{2}}^{+\frac{MT_r}{2}} s_p(t) e^{-j \frac{2\pi}{MT_r} kt} dt \\
&= \frac{1}{MT_r} \int_{-\frac{MT_r}{2}}^{+\frac{MT_r}{2}} \left(\sum_{n=-\infty}^{\infty} \delta(t - (nM + p + r[p])T_r) \right) e^{-j \frac{2\pi}{MT_r} kt} dt \\
&\quad [\text{the impulse train reduces to a single impulse in the integration interval}] \\
&= \frac{1}{MT_r} \int_{-\frac{MT_r}{2}}^{+\frac{MT_r}{2}} \delta(t - (p + r[p])T_r) e^{-j \frac{2\pi}{MT_r} kt} dt \\
&\quad \left[\text{exploiting the sifting (or masking) property: } \int f(x) \delta(x - a) dx = f(a) \right. \\
&\quad \left. \text{or } \int f(x) \delta(a - x) dx = f(a) \right] \\
&= \frac{1}{MT_r} e^{-j \frac{2\pi}{MT_r} (pT_r + r[p]T_r)k} \\
&= \frac{1}{MT_r} e^{-jk p \frac{2\pi}{M}} e^{-jk \frac{2\pi}{M} r[p]}.
\end{aligned} \tag{A.6}$$

The Fourier transform of the sampling function of the p -th channel is

$$S_p(j\omega) = \frac{2\pi}{MT_r} \sum_{k=-\infty}^{\infty} \delta\left(\omega - k \frac{\omega_s}{M}\right) e^{-jk p \frac{2\pi}{M}} e^{-jk \frac{2\pi}{M} r[p]} \tag{A.7}$$

and the Fourier series of the entire sampling function in (A.1) is

$$S_{CKR}(j\omega) = \frac{2\pi}{MT_r} \sum_{p=0}^{M-1} \sum_{k=-\infty}^{\infty} \delta\left(\omega - k \frac{2\pi}{MT_r}\right) e^{-jk p \frac{2\pi}{M}} e^{-jk \frac{2\pi}{M} r[p]}. \tag{A.8}$$

A.2 Power Spectral Density of an Oscillator Output Signal

The PSD of (4.10) is given by the Fourier transform over the autocorrelation function of (4.10). The autocorrelation function is given by substituting (4.10) into (4.4), i.e.,

$$\begin{aligned}
r_v(\tau) &= \mathcal{E} \left\{ (A \cos(\omega_c t + \varphi_0) - A \sin(\omega_c t + \varphi_0)) \varphi(t) \right. \\
&\quad \left. \times (A \cos(\omega_c(t + \tau) + \varphi_0) - A \sin(\omega_c(t + \tau) + \varphi_0)) \varphi(t + \tau) \right\} \\
&= \underbrace{\mathcal{E} \{ A \cos(\omega_c t + \varphi_0) A \cos(\omega_c(t + \tau) + \varphi_0) \}}_{r_{\cos}(\tau)} \\
&\quad + \underbrace{\mathcal{E} \{ A \sin(\omega_c t + \varphi_0) A \sin(\omega_c(t + \tau) + \varphi_0) \}}_{r_{\cos}(\tau)} \underbrace{\mathcal{E} \{ \varphi(t) \varphi(t + \tau) \}}_{r_{\varphi}(\tau)} \\
&\quad - \underbrace{\mathcal{E} \{ A \cos(\omega_c t + \varphi_0) A \sin(\omega_c(t + \tau) + \varphi_0) \}}_{=0} \underbrace{\mathcal{E} \{ \varphi(t + \tau) \}}_{=0} \\
&\quad - \underbrace{\mathcal{E} \{ A \cos(\omega_c(t + \tau) + \varphi_0) A \sin(\omega_c t + \varphi_0) \}}_{=0} \underbrace{\mathcal{E} \{ \varphi(t) \}}_{=0} \\
&= r_{\cos}(\tau) + r_{\cos}(\tau) r_{\varphi}(\tau)
\end{aligned} \tag{A.9}$$

and consequently the PSD becomes (4.11), where $S_{\cos}(\tau)$ and $S_{\varphi}(\tau)$ are the Fourier transforms of $r_{\cos}(\tau)$ and $r_{\varphi}(\tau)$, respectively.

A.3 Phase Noise Spectrum via Time Sequences

The normalized phase noise power spectral density $\mathcal{L}_\varphi\{\Delta\omega\}$ can be extracted from the time-stamps $t_v[i]$ directly [49]. In a first step we estimate the average period $\hat{T}_v$ from the time observation vector $t_v[i]$ from $i = 1 \dots L$, i.e.,

$$\hat{T}_v = \frac{t_v[L] - t_v[1]}{L - 1}. \quad (\text{A.10})$$

The time deviation vector $\Delta t_v[i]$ is

$$\Delta t_v[i] = t_v[i] - i\hat{T}_v. \quad (\text{A.11})$$

The normalized phase noise vector is given by

$$\Delta\varphi_v[i] = \frac{2\pi\Delta t_v[i]}{\hat{T}_v}. \quad (\text{A.12})$$

The PSD $\mathcal{L}_\varphi\{\Delta\omega\}$ can be estimated from the observation $\Delta\varphi_v[i]$ using the same techniques as described in Sec. 4.1.4.

A.4 Frequency-Domain and Time-Domain Analysis of Frequency-Noise and Phase-Noise

In the following we will derive the theoretical normalized frequency noise spectrum $\mathcal{L}_\omega\{\Delta\omega\}$, the normalized phase noise spectrum $\mathcal{L}_\varphi\{\Delta\omega\}$ due to phase jitter, phase wander (or frequency/period jitter), and flicker noise (filtered frequency/period jitter). These spectra are derived from observations of the clock events or zero-crossings of sinusoidal signals. For simulation purposes the noise may be modelled in a similar fashion on the clock events in order to reduce the computational burden [49]. Oscillator noise may be modelled in the (discrete) time-domain as phase noise on $\Delta T[i]$ or in the (discrete) time-domain as frequency noise $\Delta f[i]$. The underlying process, however, is continuous-time and will be approximated by the discrete-time measurements.

A.4.1 Wander

Wander noise appears as instantaneous variation of the period $T_v[i]$ or frequency $f_v[i]$. Thus the period deviation $\Delta T_{wander}[i]$ is a random process. Justified by the limit theorem $\Delta T_{wander}[i]$ is typically modelled as a zero-mean Gaussian process, i.e.,

$$\Delta T_{wander} \sim \mathcal{N}(0, \sigma_{\Delta T}^2) \quad (\text{A.13})$$

where $\sigma_{\Delta T}^2$ is the variance of the normal distribution denoted with $\mathcal{N}(\mu, \sigma^2)$ with μ being its mean value and σ^2 being its variance. Alternatively, we can model the noise in the frequency signal $f_v[i]$ as

$$\Delta f_{wander} \sim \mathcal{N}(0, \sigma_{\Delta f}^2). \quad (\text{A.14})$$

Note that the actual time-domain distribution is of minor importance as phase noise is characterized in the frequency domain. The frequency variance and the period variance are related by (4.26), i.e., $\sigma_{\Delta f}^2 \approx \sigma_{\Delta T}^2 (f_v^{ss})^4$. The autocorrelation function of the angular frequency deviation (4.27) from the noise process in (A.14) is

$$r_\omega[m] = (2\pi\sigma_{\Delta f})^2 \delta[m] \quad (\text{A.15})$$

A.4 Frequency-Domain and Time-Domain Analysis of Frequency-Noise and Phase-Noise

where $\delta[\cdot]$ denotes the Kronecker Delta. The discrete-time double-sided PSD $S_\omega(e^{j\omega T_v^{ss}})$ is the Fourier transform of the autocorrelation function similar to continuous-time definition in (4.4) and (4.5), i.e.,

$$S_\omega(e^{j\omega T_v^{ss}}) = (2\pi\sigma_{\Delta f})^2. \quad (\text{A.16})$$

The continuous-time spectrum $S_\omega(\omega)$ is given by (4.28), the phase-noise spectrum $S_\varphi(\omega)$ is given by (4.19), and finally the normalized spectra $\mathcal{L}_\omega\{\Delta\omega\}$ and $\mathcal{L}_\varphi\{\Delta\omega\}$ are given by the positive frequencies of $S_\omega(\omega)$ and $S_\varphi(\omega)$, respectively, according to (4.20). That is, the normalized frequency noise spectrum is

$$\mathcal{L}_\omega\{\Delta\omega\} = \frac{(2\pi\sigma_{\Delta f})^2}{f_v^{ss}} \quad (\text{A.17})$$

and the normalized phase noise spectrum becomes with (4.19)

$$\mathcal{L}_\varphi\{\Delta\omega\} = \frac{(2\pi\sigma_{\Delta f})^2}{f_v^{ss}} \left(\frac{1}{\Delta\omega^2} \right). \quad (\text{A.18})$$

In order to relate the spectrum to the result in [49, 60] we use the simplification in (4.26) and express $\mathcal{L}_\varphi\{\Delta\omega\}$ with respect to variance of the period deviations $\sigma_{\Delta T}^2$ as

$$\mathcal{L}_\varphi\{\Delta\omega\} \approx (2\pi\sigma_{\Delta T})^2 (f_v^{ss})^3 \left(\frac{1}{\Delta\omega} \right)^2. \quad (\text{A.19})$$

A.4.2 Jitter

Phase jitter appears as instantaneous fluctuation of the time-instances $t_v[i]$. Thus the time deviation $\Delta t_{jitter}[i]$ is a random process. Due to the central limit theorem $\Delta t_{jitter}[i]$ is typically modelled as a zero-mean Gaussian process, i.e.,

$$\Delta t_{jitter} \sim \mathcal{N}(0, \sigma_{\Delta t}^2). \quad (\text{A.20})$$

The time deviations affect the instantaneous periods in a differentiated manner and may be modelled as

$$\Delta T_v[i-1] = \Delta t_{jitter}[i] - \Delta t_{jitter}[i-1]. \quad (\text{A.21})$$

Note that a discrete time difference is on the one hand an efficient implementation, but on the other hand a rather coarse approximation of a continuous-time differentiation, which will become evident in the spectra, see Fig. A.1. However, equivalently we can model the deviation as frequency deviation as

$$\Delta f_v[i-1] = \Delta f_{jitter}[i] - \Delta f_{jitter}[i-1] \quad (\text{A.22})$$

with

$$\Delta f_{jitter} \sim \mathcal{N}(0, \sigma_{\delta f}^2). \quad (\text{A.23})$$

Next we compute the autocorrelation function of the angular frequency deviation. The time and frequency variances are related by (4.26). The autocorrelation function of the angular frequency

deviation $2\pi\Delta f_{jitter}[i]$ becomes

$$\begin{aligned}
r_\omega[m] &= \mathcal{E} \left\{ 2\pi\Delta f_v[i] \ 2\pi\Delta f_v[i+m] \right\} \\
&= \mathcal{E} \left\{ (2\pi\Delta f_{jitter}[i] - 2\pi\Delta f_{jitter}[i-1]) (2\pi\Delta f_{jitter}[i+m] - 2\pi\Delta f_{jitter}[i-1+m]) \right\} \\
&= \underbrace{\mathcal{E} \left\{ 2\pi\Delta f_v[i] \ 2\pi\Delta f_v[i+m] \right\}}_{(2\pi\sigma_{\Delta f})^2 \delta[m]} + \underbrace{\mathcal{E} \left\{ 2\pi\Delta f_v[i-1] \ 2\pi\Delta f_v[i-1+m] \right\}}_{(2\pi\sigma_{\Delta f})^2 \delta[m]} \\
&\quad - \underbrace{\mathcal{E} \left\{ 2\pi\Delta f_v[i] \ 2\pi\Delta f_v[i-1+m] \right\}}_{(2\pi\sigma_{\Delta f})^2 \delta[m-1]} - \underbrace{\mathcal{E} \left\{ 2\pi\Delta f_v[i-1] \ 2\pi\Delta f_v[i+m] \right\}}_{(2\pi\sigma_{\Delta f})^2 \delta[m+1]} \\
&= (2\pi\sigma_{\Delta f})^2 (-\delta[m-1] + 2\delta[m] - \delta[m+1]).
\end{aligned} \tag{A.24}$$

The Fourier transform of (A.24) is

$$S_\omega \left(e^{j\omega T_v^{ss}} \right) = (2\pi\sigma_{\Delta f})^2 \left(2 - e^{j\omega T_v^{ss}} - e^{-j\omega T_v^{ss}} \right) = (2\pi\sigma_{\Delta f})^2 (2 - 2\cos(\omega T_v^{ss})). \tag{A.25}$$

The frequency noise appears in the range between $\omega T_v^{ss} = 0$ and $\omega T_v^{ss} = \pi$ corresponding to the analog frequencies $\omega = 0$ to $\omega = f_v^{ss}/2$. In order to preserve the information in the low frequencies, we approximate the spectrum in a quadratical manner by $(\omega T_v^{ss})^2$. Figure A.1 illustrates the approximation of the frequency noise spectrum of phase jitter. Using this approximation, converting the discrete spectrum into a continuous spectrum using (4.28) and computing the normalized frequency noise spectrum using (4.20) results in the normalized frequency noise spectrum

$$\mathcal{L}_\omega \{ \Delta\omega \} = \frac{(2\pi\sigma_{\Delta f})^2}{(f_v^{ss})^3} (\Delta\omega)^2. \tag{A.26}$$

The corresponding normalized phase noise spectrum can be obtained by integration, i.e., multiplying the spectrum with $1/(\Delta\omega)^2$ according to (4.19), i.e.,

$$\mathcal{L}_\varphi \{ \Delta\omega \} = \frac{(2\pi\sigma_{\Delta f})^2}{(f_v^{ss})^3} \approx (2\pi\sigma_{\Delta t})^2 f_v^{ss} \tag{A.27}$$

where we have used the approximation $\sigma_{\Delta f}^2 \approx \sigma_{\Delta t}^2 (f_v^{ss})^4$ from (4.26).

A.4.3 Flicker

Flicker noise follows a -10 dB/dec decay in the frequency noise spectrum and a -30 dB/dec decay in the phase noise spectrum. Thus, the normalized frequency noise PSD $\mathcal{L}_\omega \{ \Delta\omega \}$ is given by

$$\mathcal{L}_\omega \{ \Delta\omega \} = \frac{(2\pi\sigma_{\Delta 1/f})^2}{f_v^{ss}} \left(\frac{1}{\Delta\omega} \right). \tag{A.28}$$

Substituting the variance of the period deviations $\sigma_{\Delta 1/T}^2 \approx \sigma_{\Delta 1/f}^2 (f_v^{ss})^4$ according to 4.26 into (A.28) and performing the frequency-to-phase conversion similar to (4.19) yields the phase noise spectrum

$$\mathcal{L}_\varphi \{ \Delta\omega \} \approx (2\pi\sigma_{\Delta 1/T})^3 (f_v^{ss})^3 \left(\frac{1}{\Delta\omega} \right)^3. \tag{A.29}$$

However, to model and generate flicker noise we feed white noise (i.e., frequency jitter) into a filter $h[i]$ that adds additional -10 dB/dec. In a first step we describe a possible generic filter design procedure that may achieve any desired attenuation factor. In a second step we relate the desired output flicker noise spectrum to the spectrum and the variance of the white noise input.

A.4 Frequency-Domain and Time-Domain Analysis of Frequency-Noise and Phase-Noise

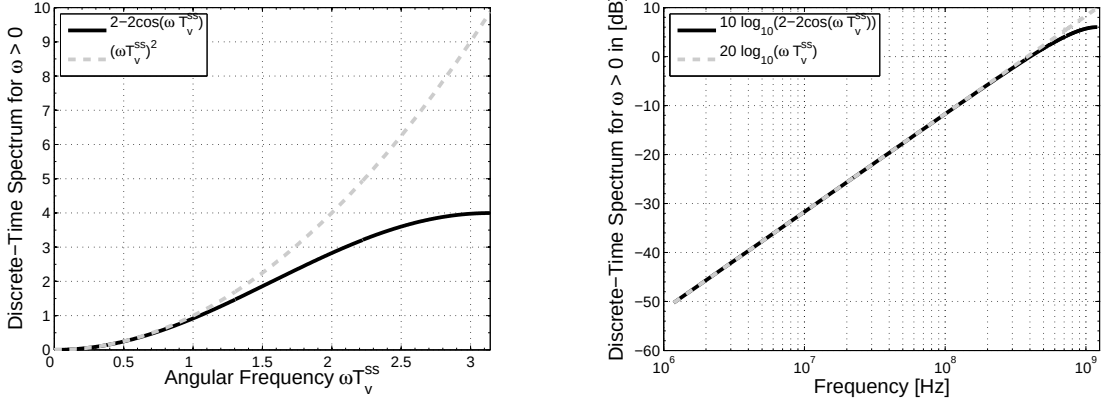


Figure A.1 Approximation (dashed grey) frequency noise spectrum of phase jitter (solid black) in a linear plot (left) and a log-log plot (right).

Generating Colored Noise

In order to achieve a slope of -10 dB/dec of flicker noise or any other colored noise we design a low-pass filter $h[i]$ that follows the required slope between the frequencies $f_{c,0}$ and $f_{c,M-1}$. The input to the filter is a white frequency noise

$$\Delta T_{flicker} \sim \mathcal{N}(0, \sigma_{\Delta T, 1/T}^2) \quad (\text{A.30})$$

which is "colored" by the filter. The computation of the period variance $\sigma_{\Delta T, 1/T}$ or frequency variance $\sigma_{\Delta f, 1/f}$ according to a flicker noise specifications is discussed in App. A.4.3. The presented solution uses a filterbank of M first-order sub-filters $g_k[i]$ for $k = 0, \dots, M-1$, where $f_{c,0}$ to $f_{c,M-1}$ are the 3 dB cut-off frequencies of the sub-filters [49]. The filterbank structure is depicted in Fig. A.2 and follows a certain slope K , which is -10 dB/dec in case of flicker noise. Other attenuation factors, however, are also feasible. Basically we design the sub-filters to have equal gain ratios A and 3dB cut-off frequency ratios F between two consecutive filters. With other words, we place M 1st-order filters with constant gain differences $A_{dB} = 20 \log_{10}(A)$ and constant frequency differences $F_{\log} = 20 \log_{10}(F)$ in a log-log plot as illustrated in Fig. A.3.

The individual sub-filters $g_m[z]$ are given in the z -domain by

$$G_m(z) = \frac{A^{-k}}{A_{cor}} \frac{a_k}{1 - (1 - a_k)z^{-1}} \quad (\text{A.31})$$

where $\frac{A^{-m}}{A_{cor}}$ is the gain of the m -th filter and A_{cor} is a correction factor to adjust the gain of the overall system. The coefficients a_m define the cut-off frequency $f_{c,m}$ of the m -th filter, i.e.,

$$a_m = 2\pi \frac{f_{c,m}}{f_v^{ss}} \quad (\text{A.32})$$

In order to design the sub-filters such that the overall system has a slope of K , we need to choose the number of filters M , the frequency ratio F , the gain ratio A , and the correction factor A_{cor} . To achieve a good linearity of the overall filter, it is a good choice to set the number of filters according to the number of decades the filter shall span, plus one, i.e.,

$$M \approx \log_{10}(f_{c,M-1}) - \log_{10}(f_{c,0}) + 1. \quad (\text{A.33})$$

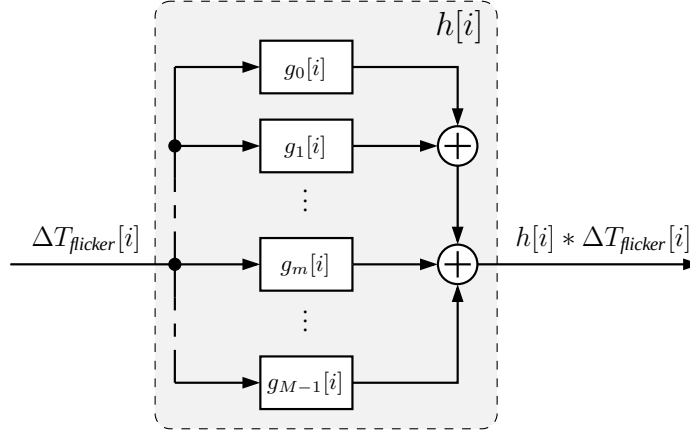


Figure A.2 Filterbank to model colored noise.

The logarithmic frequency offset is obtained by linearly interpolating in the log-domain between $\log_{10}(f_{c,0})$ and $\log_{10}(f_{c,M-1})$, i.e.,

$$F_{\log} = \frac{\log_{10}(f_{c,M-1}) - \log_{10}(f_{c,0})}{M-1}. \quad (\text{A.34})$$

In order to achieve K dB/dec, the A_{dB} gain becomes

$$A_{dB} = K F_{\log}. \quad (\text{A.35})$$

In a final step the correction factor A_{cor} ensures that the filter $h[i]$ has a gain of 0 dB at frequency $f_{c,0}$. The best way to compute A_{cor} is to set it to one, design the filter, and evaluate the gain at $f_{c,0}$ and update the coefficients accordingly. That is,

$$A_{cor} = \left| H \left(e^{j2\pi f_{c,0} T_v^{ss}} \right) \right| \quad (\text{A.36})$$

where $H(e^{j\omega T_v^{ss}})$ is the frequency response of $h[i]$.

Corresponding Input Noise Wander

In order to meet the specifications, i.e., an attenuation of K dB/dec between the frequencies $f_{c,0}$ and $f_{c,M-1}$, we have to feed zero-mean white frequency noise with a variance $\sigma_{\Delta f,1/f}^2$ (or $\sigma_{\Delta T,1/T}^2$ when modelled as noise on the CKV periods) corresponding to the desired phase noise at $f_{c,0}$, i.e., $\mathcal{L}_{\omega}\{2\pi f_{c,0}\}$, into the filter $h[i]$. In practice, however, it is more convenient to define the colored noise by the desired phase noise at $f_{c,M-1}$, i.e., $\mathcal{L}_{\varphi}\{2\pi f_{c,M-1}\}$. That is, in a first step we relate $\mathcal{L}_{\varphi}\{2\pi f_{c,M-1}\}$ to $\mathcal{L}_{\omega}\{2\pi f_{c,0}\}$ and in a second step we relate $\mathcal{L}_{\omega}\{2\pi f_{c,0}\}$ to the required variance $\sigma_{\Delta f,1/f}^2$ of the white frequency noise or, accordingly, $\sigma_{\Delta T,1/T}^2$ of the -20 dB/dec phase wander. The colored noise shall linearly decrease with a slope $(K-20)$ dB/dec in the logarithmic phase noise spectrum. Therefore we can state

$$\mathcal{L}_{\varphi}\{2\pi f_{c,0}\} \text{ in [dB]} = \mathcal{L}_{\varphi}\{2\pi f_{c,0}\} \text{ in [dB]} - (K-20) \log_{10} \left(\frac{f_{c,M-1}}{f_{c,0}} \right). \quad (\text{A.37})$$

or equivalently

$$\mathcal{L}_{\varphi}\{2\pi f_{c,0}\} = \left(\frac{f_{c,M-1}}{f_{c,0}} \right)^{-\frac{K-20}{10}} \mathcal{L}_{\varphi}\{2\pi f_{c,M-1}\}. \quad (\text{A.38})$$

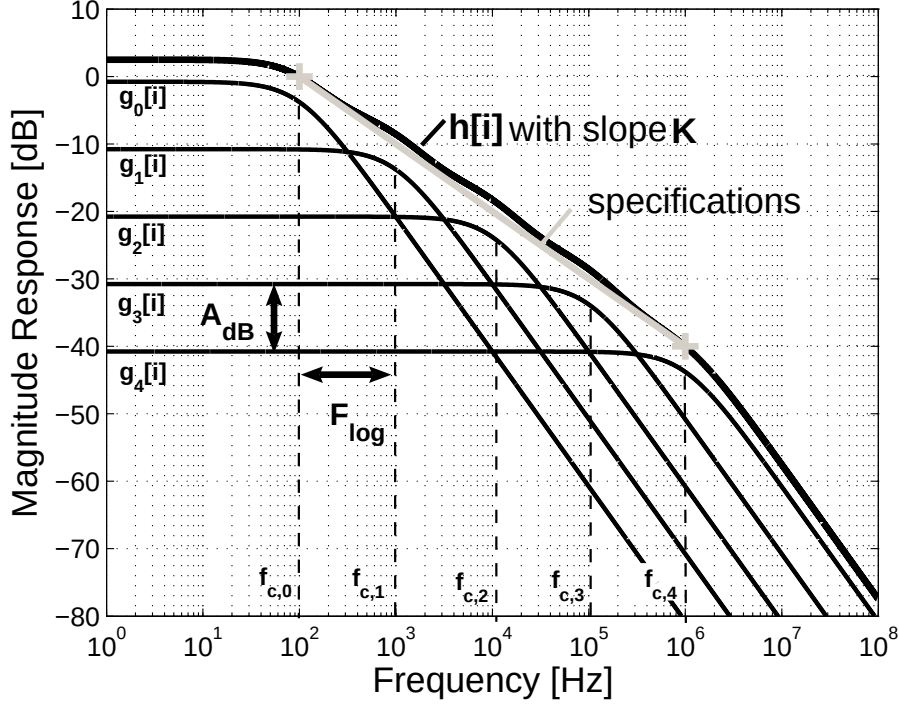


Figure A.3 Magnitude responses of the subfilters and the overall filter-bank.

The variance $\sigma_{\Delta T,1/T}^2$ of the period noise process $\mathcal{L}_\varphi\{2\pi f_{c,0}\}$ can be obtained using Tab. 4.1, i.e.,

$$\sigma_{\Delta T,1/T}^2 = \frac{\left(\frac{f_{c,M-1}}{f_{c,0}}\right)^{-\frac{K-20}{10}} \mathcal{L}_\varphi\{2\pi f_{c,M-1}\}}{(2\pi)^2 (f_v^{ss})^3} (\Delta\omega)^2 \quad \text{A.39}$$

and the frequency noise variance is approximately, $\sigma_{\Delta f,1/f}^2 \approx \sigma_{\Delta T,1/T}^2 (f_v^{ss})^4$ according to (4.26).

Flicker Noise Design Example

The design specifications for this flicker noise example is -125 dBc/Hz at 1 MHz frequency offset for a DCO frequency of $f_v^{ss} = 2.413$ GHz. In a first step we will design the filter $h[i]$ which adds $K = -10$ dB/dec attenuation and in a second step we will choose the variance of period noise $\sigma_{\Delta T,1/T}^2$ (or frequency noise $\sigma_{\Delta f,1/f}^2$).

1. Filter Design:

We choose our sub-filters $g_m[i]$ to range from $f_{c,0} = 100$ Hz to $f_{c,M-1} = 1$ MHz. According to (A.33) the filter order is $M = 5$, according to (A.34) the frequency offset is $F_{\log} = 1$ dec, and according to (A.35) the dB gain difference is $A_{dB} = -10$ dB. After the first filter design we compensate for A_{cor} according to (A.36), which is in this case $A_{cor} = 1.0931$ or 0.7732 dB. The design example in Fig. A.3 uses exactly these parameters. The overall magnitude response of $h[i]$ meets the specifications at $f_{c,0}$ of 0 dB and $f_{c,4}$ of -40 dB with an attenuation of $K = -10$ dB/dec.

2. Noise Variance:

In a second step we compute the variance of the random variable $\Delta T_{\text{flicker}}$ in (A.30). Substituting $\mathcal{L}_\varphi\{2\pi f_{c,M-1}\} = 10^{\frac{-125}{10}}$, $f_v^{ss} = 2.413$ GHz, $f_{c,0} = 100$ Hz, $f_{c,M-1} = 1$ MHz, and $K = -10$ dB/dec into (A.39) gives $\sigma_{\Delta T,1/T}^2 \approx 2.250810^{-25}$ sec. or $\sigma_{\Delta T,1/T} \approx 474$ fs. If the noise is modelled as frequency deviation, then its standard deviation is $\sigma_{\Delta f,1/f} \approx 2.76$ MHz according to (4.26), which corresponds to a white frequency noise with gain $\mathcal{L}_\omega\{2\pi f_{c,0}\} = 50.96$ dB.

Figure A.4 plots the frequency noise spectrum $\mathcal{L}_\omega\{\Delta\omega\}$ and Fig. A.5 plots the phase noise spectrum $\mathcal{L}_\varphi\{\Delta\omega\}$. Both plots show the simulation outputs of the input noise (gray), the output of the sub-filters (light gray), and the flicker noise (black) in comparison to the theoretical values of input noise (dashed black), output of the sub-filters (solid black), and overall output (bold black). First of all we notice that the generated noise matches the specification of -125 dBc at 1 MHz with a slope of -30 dB/Hz. The gain factor of -5 dB at 100 Hz, i.e., $\mathcal{L}_\varphi\{2\pi f_{c,0}\} = -5$ dB, agrees with (A.38). The results of the frequency noise spectrum can be verified using Tab. 4.1 or Fig. 4.4.

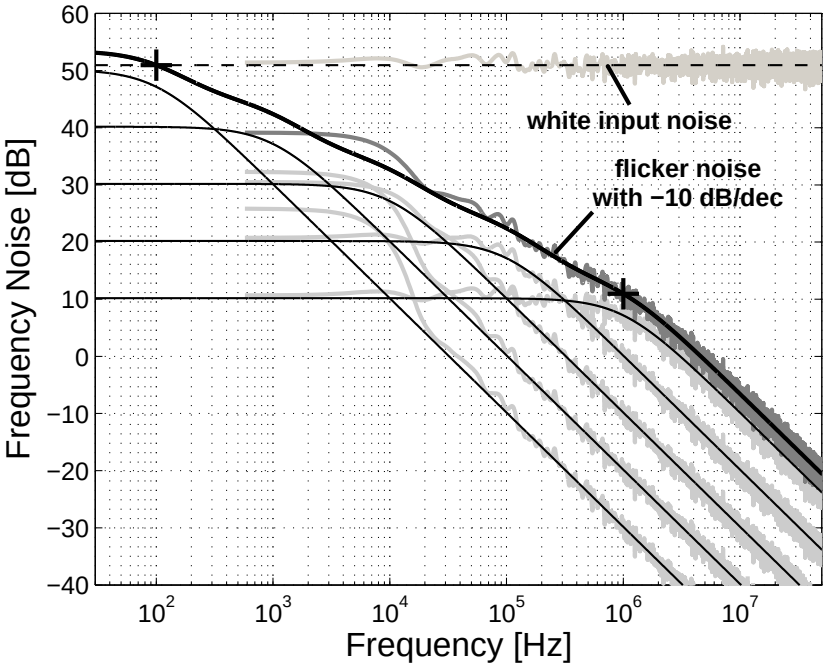


Figure A.4 Generated flicker frequency noise with -10 dB/dec.

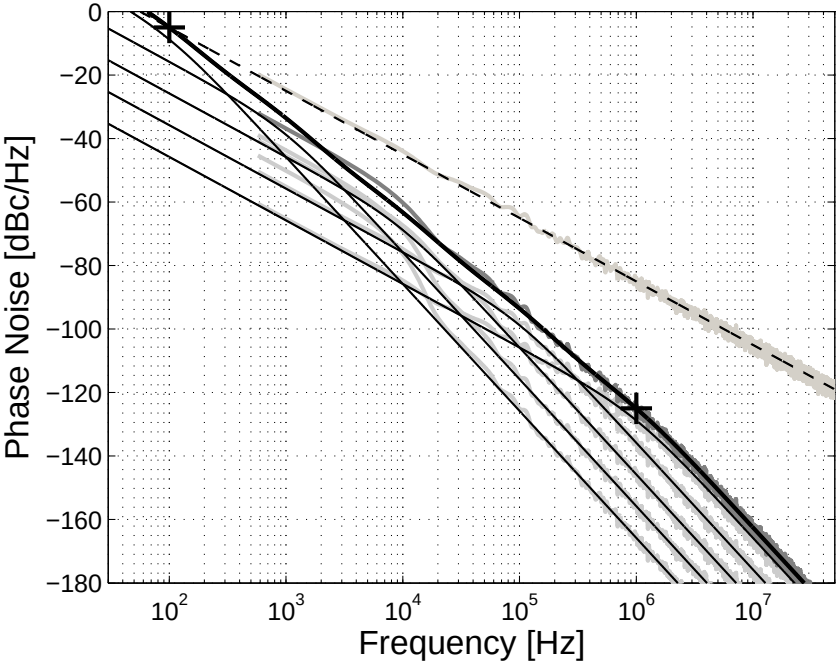


Figure A.5 Generated flicker phase noise with -30 dB/dec.

A.5 Relation between Period Noise Variance and Frequency Noise Variance

Let $\mathbf{x}$ and $\mathbf{y}$ be two random variables which are related by $\mathbf{y} = g(\mathbf{x})$. Under the assumption that the distribution of $\mathbf{x}$ is concentrated around its mean value μ_x , we can express the mean μ_y and variance σ_y^2 of $\mathbf{y}$ by the mean μ_x and variance σ_x of $\mathbf{x}$ [179, page 150]. To be specific we approximate the function $g(x)$ by a polynomial around μ_x

$$g(x) \approx g(\mu_x) + g'(\mu_x)(x - \mu_x) + \cdots + g^{(n)}(\mu_x) \frac{(x - \mu_x)^n}{n!} \quad (\text{A.40})$$

where $g'(\cdot)$ is the first derivative and $g^{(n)}(\cdot)$ is the n th derivative of $g(\cdot)$, $(\cdot)!$ denotes the factorial operator, and n is the order of the polynomial. In particular, if $g(x)$ is approximated as a straight line and all higher order terms are neglected, then μ_y becomes

$$\mu_y = \mathcal{E}\{g(\mathbf{x})\} \approx g(\mu_x) \quad (\text{A.41})$$

and the variance becomes

$$\sigma_y^2 \approx |g'(\mu_x)|^2 \sigma_x^2. \quad (\text{A.42})$$

For the concrete case of frequency and phase noise we can state that $\mathbf{x}$ is the frequency noise, $\mathbf{y}$ is the phase noise (or vice versa), and $g(x) = 1/x$. The mean value of the frequency noise is f_v^{ss} and the variance is σ_f^2 . According to (A.41) the mean period deviation is approximately

$$\mu_T \approx g(f_v^{ss}) = 1/f_v^{ss} = T_v^{ss} \quad (\text{A.43})$$

and according to (A.42) the variance of the period deviations is approximately

$$\sigma_T^2 \approx \left| -\frac{1}{(f_v^{ss})^2} \right|^2 \sigma_f^2 = \frac{\sigma_f^2}{(f_v^{ss})^4}. \quad (\text{A.44})$$

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